

3 R D

DRAWING FOR APPROVAL

CENTRIFUGAL PUMP

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Ship No. : H1933A / 1934A / 1935A / 1936A / 1937A

Ship's Kind : 13000TEU Container Vessel

Please make sure that you will return us one (1) copy of approved drawings with your signature within :

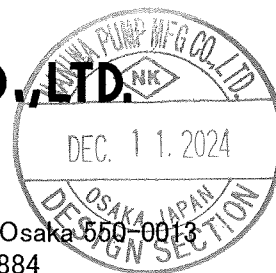
(Date) 10 / (Month) Jan. / (Year) 2025 .

Remarks:

Production will proceed according to the drawings if you fail to return us the drawings within the date as above written, and any changes you may require for afterwards will charge extra.



NANIWA PUMP MFG.CO.,LTD.



Head Office 11-5, Shinmachi 3-chome, Nishi-ku, Osaka 550-0013
TEL 06-6541-6131 FAX 06-6535-1884

Tokyo Office 2F, Nissho-Kan, 1-10, Nihonbashi Kabutocho,
Chuo-ku, Tokyo 103-0026
TEL 03-3669-0661 FAX 03-3669-0675

Mihara Works 476-1, Tanjo, Mihara-ku, Sakai, Osaka 587-0011
TEL 072-361-2946 FAX 072-361-5407

Fukui Works 112, Mikuni-cho Ikegami, Sakai, Fukui 913-0001
TEL 0776-82-2670 FAX 0776-82-2689

DRAWING SEQUENCE

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Ship No. : H1933A / 1934A / 1935A / 1936A / 1937A

No.	SERVICE AS	TYPE	MODEL	BORE	Q'TY
1	NO.3 MAIN COOLING S.W.PUMP	VERTI. CENTRIF. SELF PRIM. SYS.	FEWV-450 with PMC	450	1
2	NO.1 & NO.2 MAIN COOLING S.W.PUMP	VERTICAL CENTRIFUGAL	FEWV-450	450	2
3	NO.1 & NO.2 BALLAST WATER PUMP	VERTI. CENTRIF. SELF PRIM. SYS.	FEWV-350-3 with VK-20NB	350	2
4	NO.1 & NO.2 BILGE, FIRE & G/S PUMP	VERTI. CENTRIF. SELF PRIM. SYS.	FE2V-250-3 with VK-20NB	250	2
5	NO.1 & NO.2 L.T.COOLING F.W.PUMP	VERTICAL CENTRIFUGAL	FEWV-350	350	2
6	NO.1 & NO.2 M/E AIR COOLER L.T.COOLING F.W.PUMP	VERTICAL CENTRIFUGAL	FEWV-350	350	2
7	NO.1 & NO.2 M/E H.T.F.W.PUMP	VERTICAL CENTRIFUGAL	FEV-200E1	200	2
8	M/E JACKET WATER PREHEATER PUMP	HORIZONTAL CENTRIFUGAL	BHR-65	65	1
9	NO.1 & NO.2 L/T.C.W.BOOSTER PUMP	VERTICAL CENTRIFUGAL	FEV-200E1	200	2
10	EMERGENCY FIRE PUMP	VERTI. CENTRIF. SELF PRIM. SYS.	FGV-100-2 with VK-20NC	100	1
11	NO.1 & NO.2 MAIN L.O.PUMP	VERTICAL TANK MOUNT	TOM-300-2	DEL.300	2
12	NO.1 & NO.2 FRESH WATER PUMP	HORIZONTAL CENTRIFUGAL	BHR-50-3	50	2
13	NO.1 & NO.2 HOT WATER CIRCULATING PUMP	HORIZONTAL CENTRIFUGAL	BHR-32	32	2
14	CHEMICAL CLEANING PUMP	HORIZONTAL CENTRIFUGAL	BHR-40	40	1
15	NO.1 & NO.2 HOT GLYCOL WATER CIRCULATING PUMP	HORIZONTAL CENTRIFUGAL	BHR-100	100	2
16	NO.1 & NO.2 COLD GLYCOL WATER CIRCULATING PUMP	HORIZONTAL CENTRIFUGAL	BHR-50	50	2
17	C/H WATER SPRAY PUMP	VERTI. CENTRIF. SELF PRIM. SYS.	FEWV-250-4D with PMC	300 / 250	1

Rule : LR

Paint : 2.5 G8/4

SIGN		
<i>Eicada</i>	<i>Edum</i>	<i>K. SN2BASM</i>



NANIWA PUMP MFG.CO.,LTD.

DRAWING SEQUENCE

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Ship No. : H1933A / 1934A / 1935A / 1936A / 1937A

No.	SERVICE AS	TYPE	MODEL	BORE	Q'TY
18	NO.1 & NO.2 ICER COOLING SEA WATER PUMP	VERTICAL CENTRIFUGAL	FEWV-350-3	350	2
19	NO.1 & NO.2 NaOH TRANSFER PUMP	HORIZONTAL CENTRIFUGAL	BHR-40	40	2
	* SPARE PARTS LIST				
	* ELECTROLYTIC CORROSION PREVENTING ZINC				
	* MECHANICAL SEAL				
	* GAUGE BOARD & GAUGE ROOT				
	* INDICATION LIST OF GAUGE SET POSITION				
	* LUBRICANT LIST				
	* MOTOR OUTLINE DRAWING				
	* ESTIMATED PERFORMANCE CURVE				

Rule : LR

Paint : 2.5 G8/4



NANIWA PUMP MFG.CO.,LTD.



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

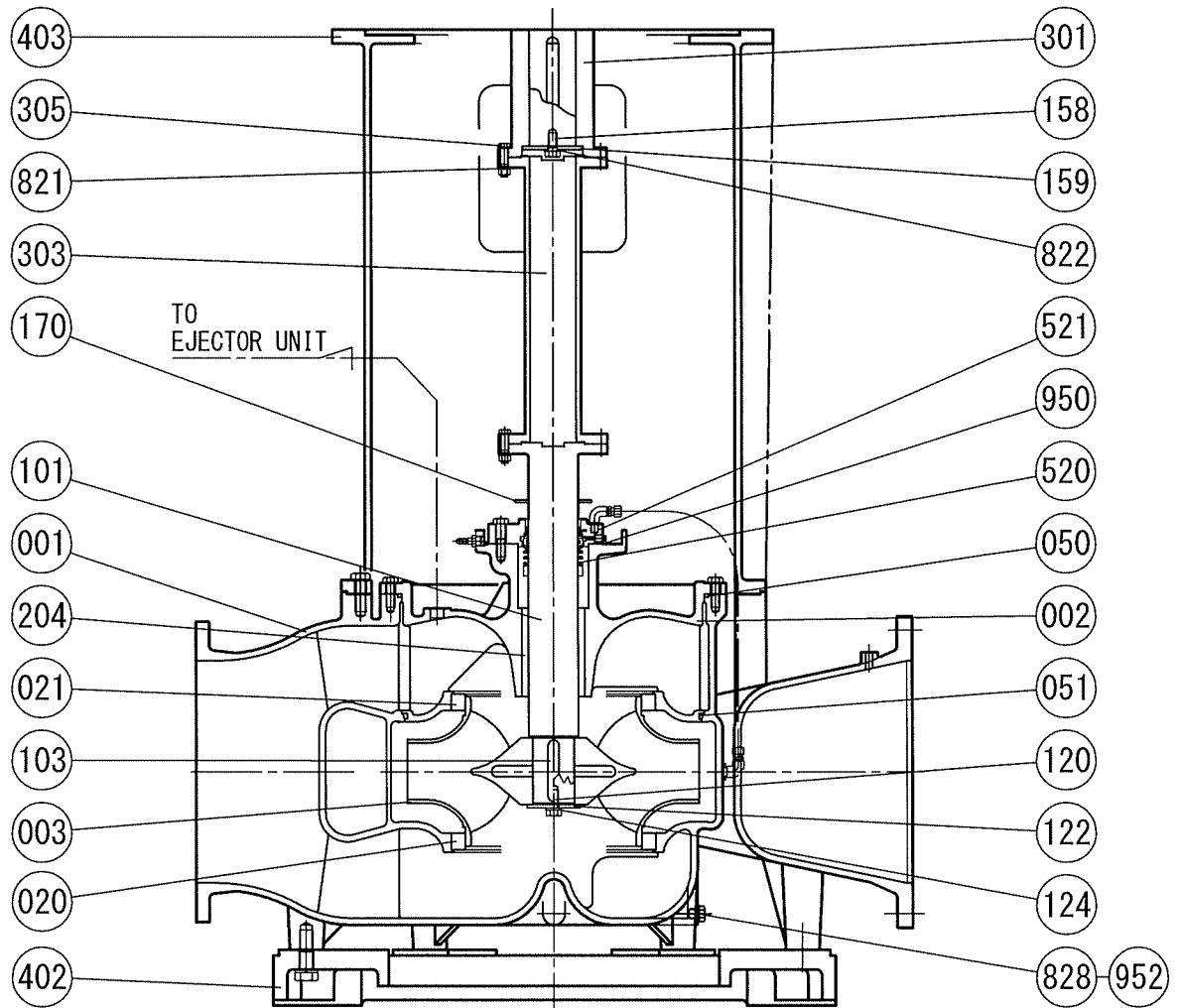
MODEL : FEWV—

DRAWING NO.

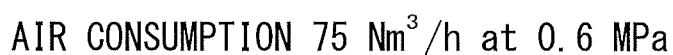
C2DS0201

DATE

12 MAY. 22



159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER BOLT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS316	1	520	MECHANICAL SEAL			1 SET
051	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
050	O-RING	RUBBER		1	402	PUMP BED	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
020	CASING RING	SYNTHETIC RESIN		1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	301	COUPLING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	204	BEARING METAL	CARBON		1
001	CASING	CAST BRONZE	CAC402	1	170	DUST STOPPER	RUBBER		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



3	SOLENOID VALVE	CAST BRONZE	1	6	REGULATOR		1
2	CHECK VALVE	CAST BRONZE	1	5	BRACKET	MILD STEEL	1
1	EJECTOR	CAST BRONZE	1	4	PRESSURE SWITCH		1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY

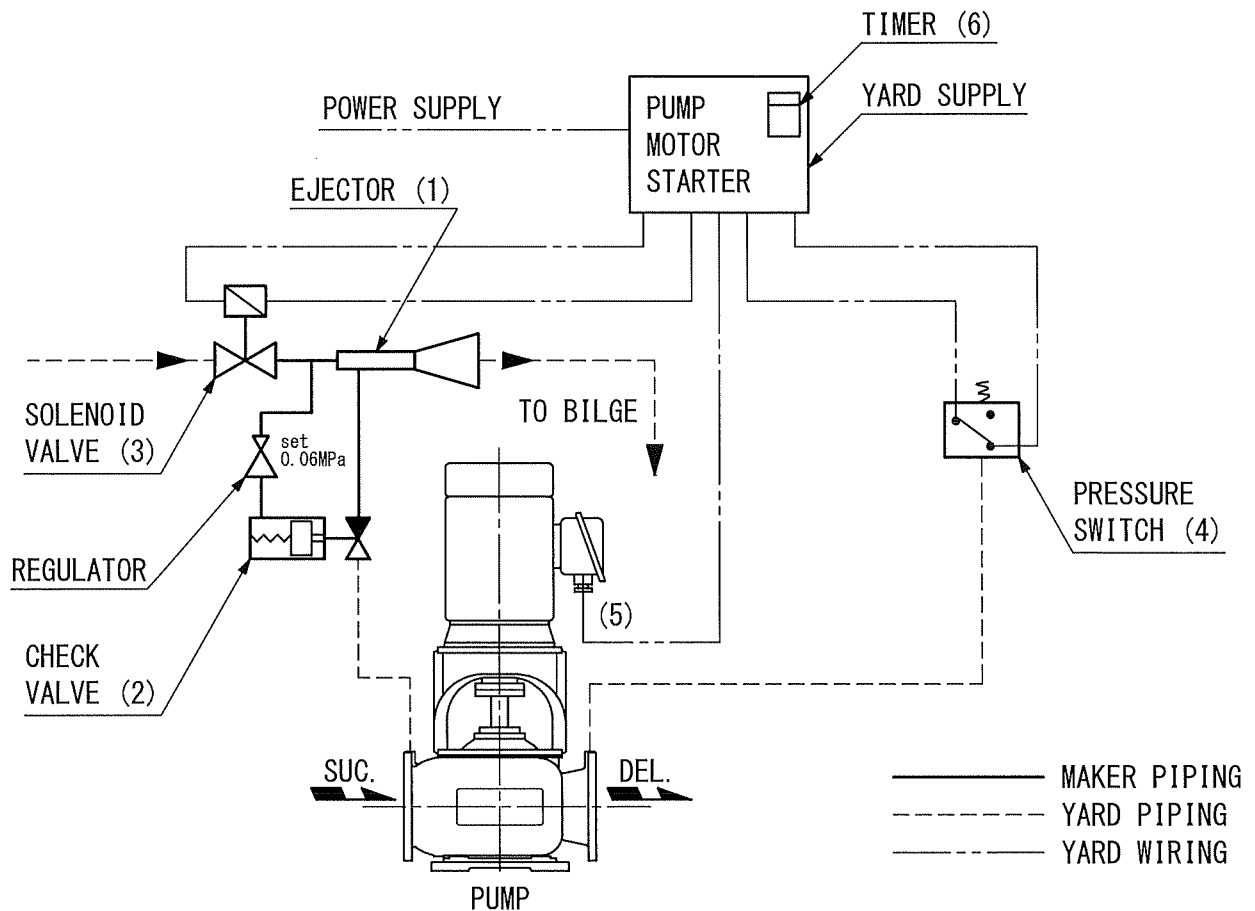


WIRING DIAGRAM

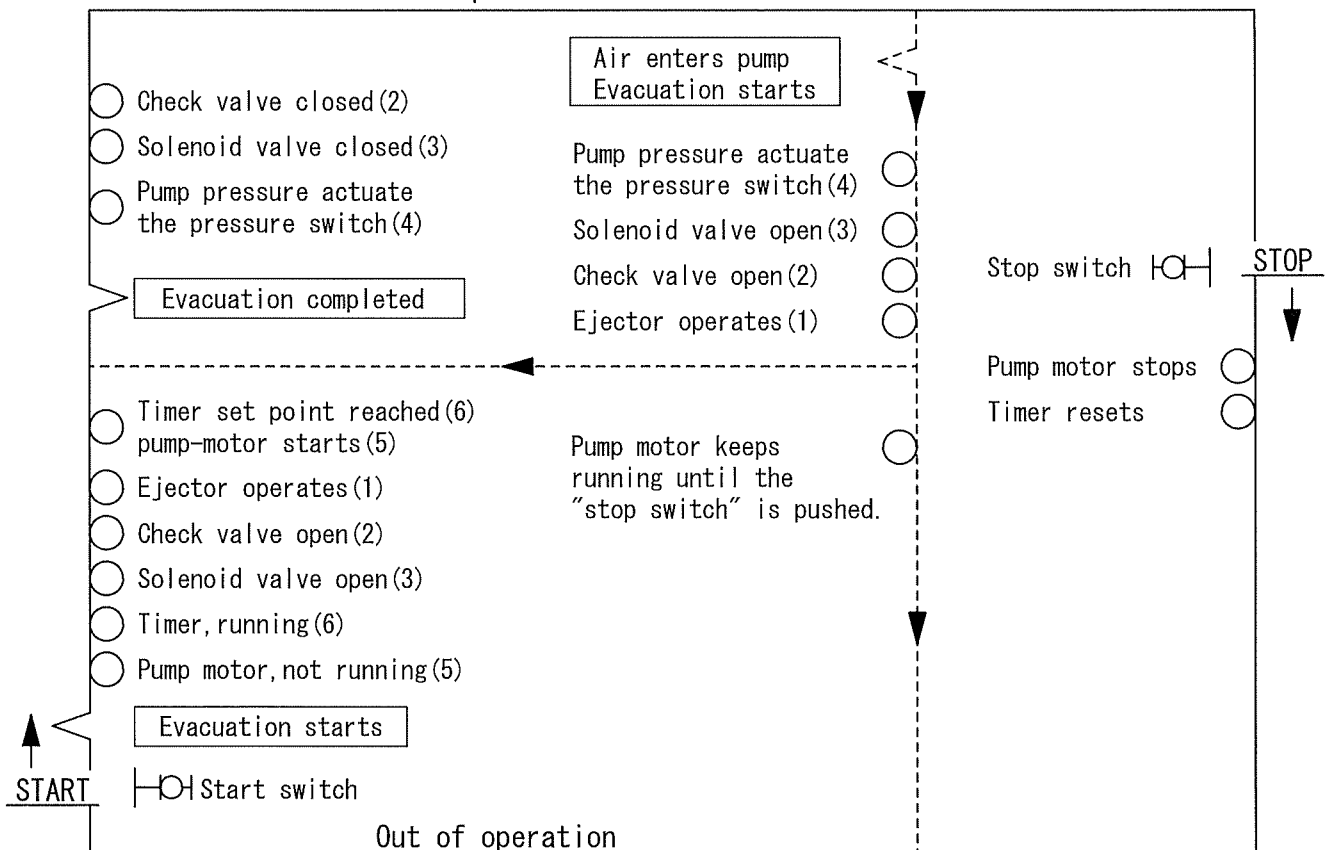
GENERAL DESCRIPTION & FUNCTION DIAGRAM

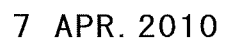
MODEL : PMC

DRAWING NO.
BODG0086
DATE
26 MAR. 2010



In operation







PERFORMANCE CURVE

AIR EJECTOR UNIT

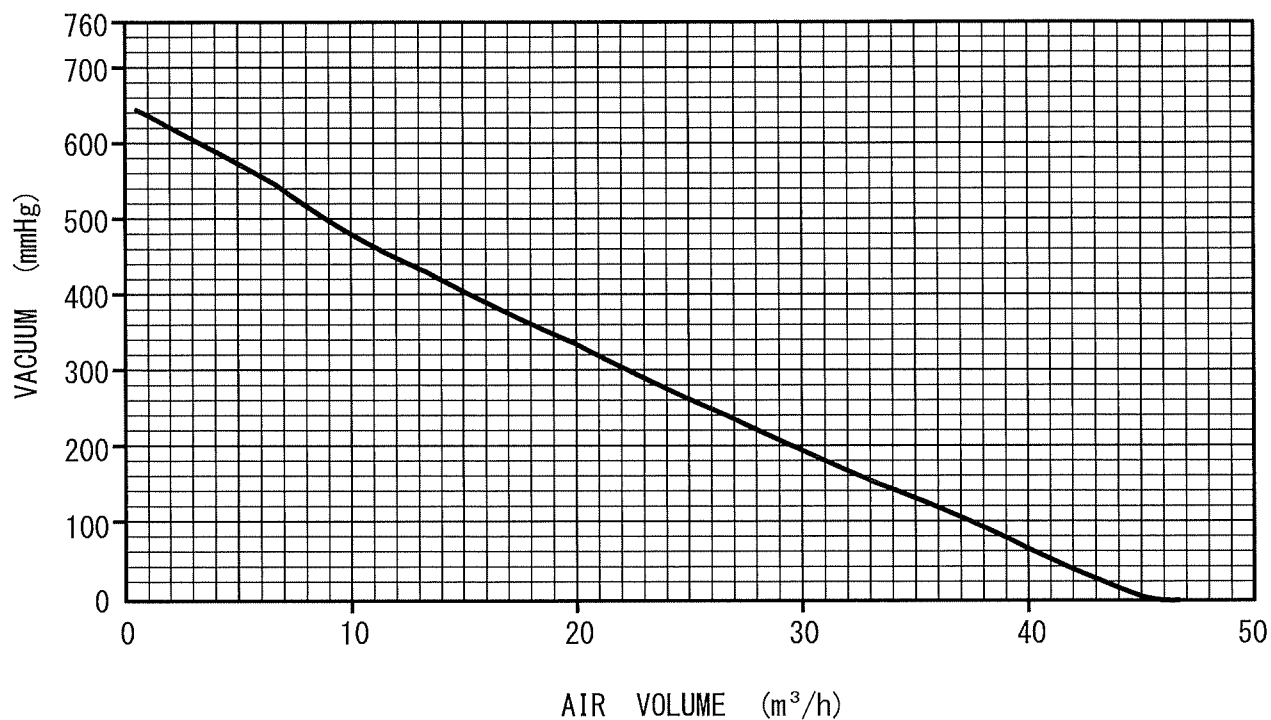
MODEL : PMC

DRAWING NO.

B0DG0093

DATE

7 APR. 2010



Driving air consumption : 75 Nm³/h

Driving air pressure : 0.6 MPa



SECTIONAL DRAWING

PRESSURE SWITCH

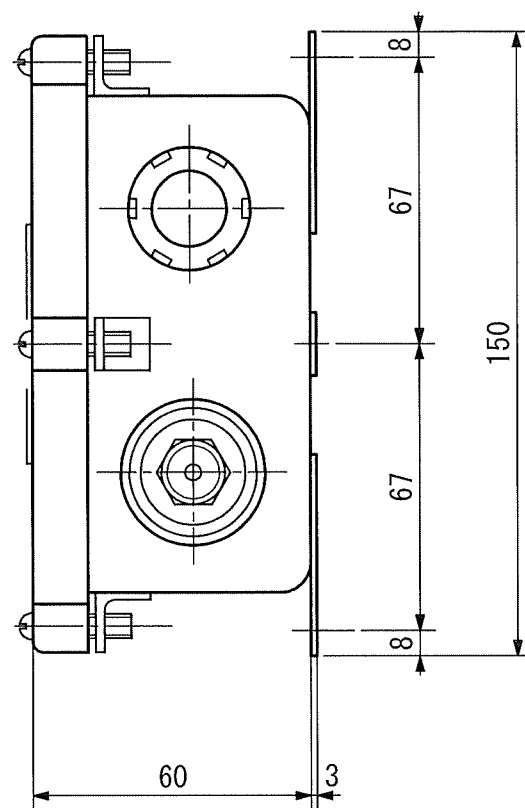
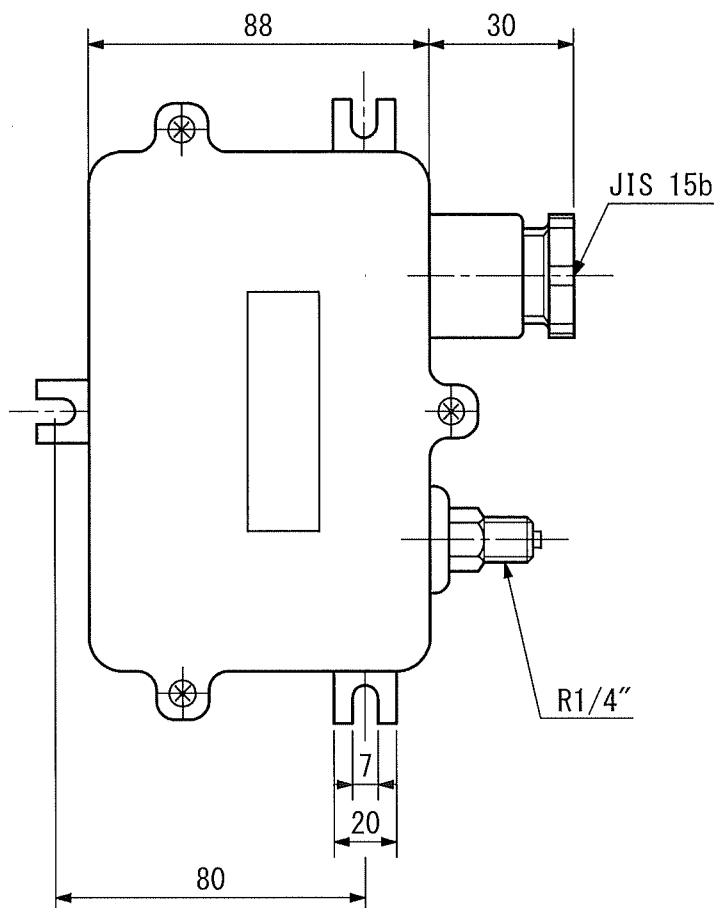
MODEL : SNS-C106W

DRAWING NO.

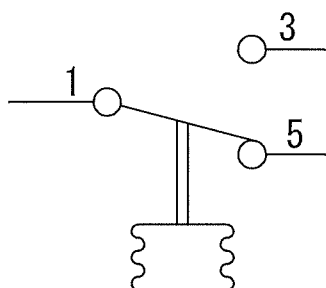
B4DG0947

DATE

20 AUG. 2014



CONTACT TYPE



CONTACT POINTS 1~3
TURNED ON DUE TO PRESSURE INCREASE

CONTACT POINTS 1~5
TURNED ON DUE TO PRESSURE DECREASE

CONTACT POINTS 1
COMMON

RATED CAPACITY

VOLT (V) AMPS (A)	AC			DC	
	125	250	450	24	125
NON INDUCTIVE	10	5	2	5	0.5
INDUCTIVE	8.5	4.5	1	2	0.2

RANGE : -0.06 ~ 0.6 MPa
MECHANICAL DIFFERENTIAL : 0.06 ~ 0.4 MPa
MAX. PERMISSIBLE PRESS. : 1.5 MPa
MASS : 0.4 Kg

SET PRESSURE
PUMP SPECIFIC PRESSURE
MORE THAN 0.5 MPa
→ ON : 0.1 MPa / OFF : 0.2 MPa
0.5 MPa AND LESS
→ ON : 0.05 MPa / OFF : 0.15 MPa



SECTIONAL DRAWING

CHECK VALVE

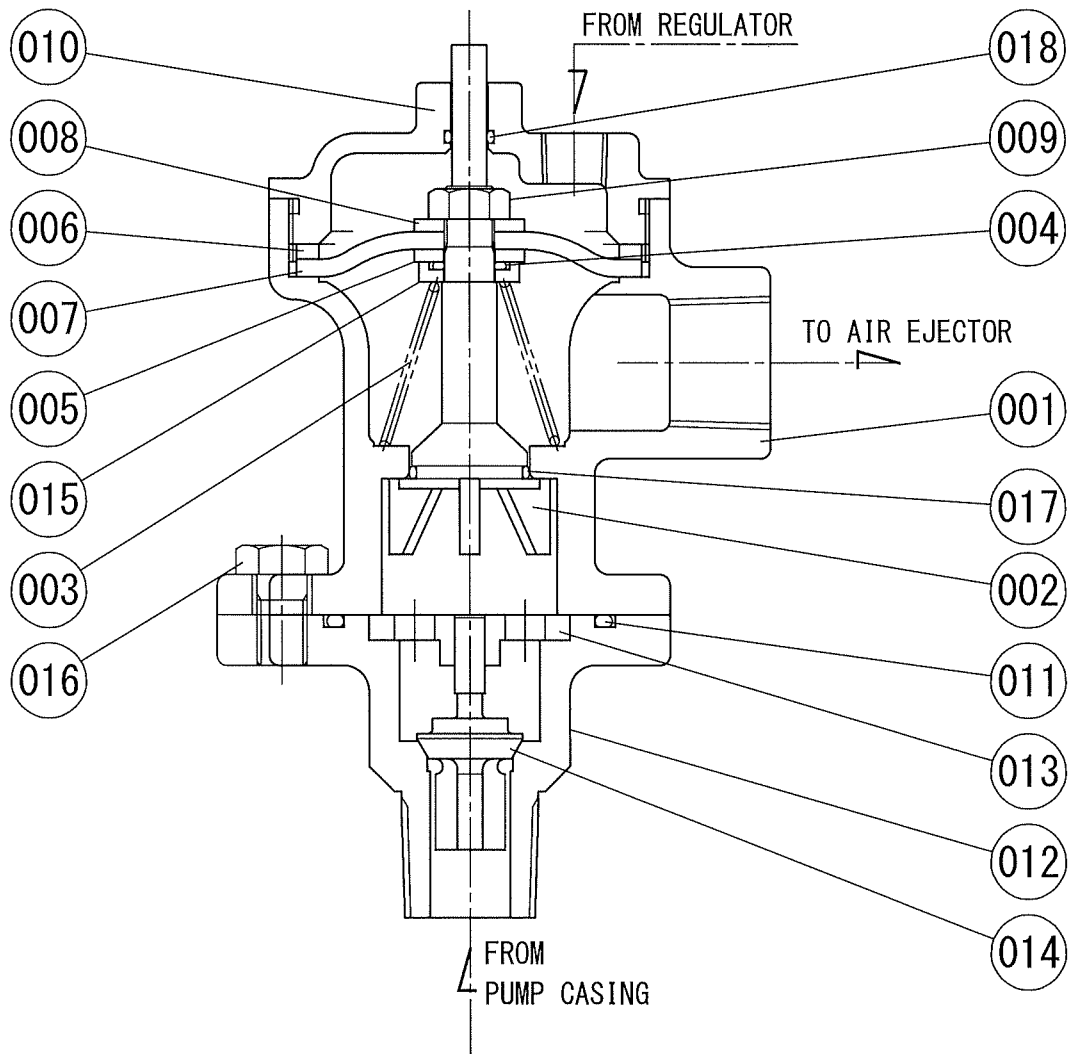
MODEL : PMC

DRAWING NO.

B8DS0126

DATE

9 APR. 2018



009	NUT	COPPER ALLOY ROD	C3604	1	018	O RING	RUBBER		1
008	WASHER	COPPER ALLOY ROD	C3604	1	017	O RING	RUBBER		1
007	DIAPHRAGM	RUBBER		1	016	BOLTS	CARBON STEEL	SWRM10	4
006	DIAPHRAGM RING	COPPER ALLOY ROD	C3604	1	015	SPRING SUPPORT	COPPER ALLOY ROD	C3604	1
005	WASHER	COPPER ALLOY ROD	C3604	1	014	VALVE	CAST BRONZE	CAC406	1
004	O RING	RUBBER		1	013	GUIDE PLATE	CAST BRONZE	CAC406	1
003	SPRING	PHOSPHOR BRONZE	PBW	1	012	VALVE BODY	CAST BRONZE	CAC406	1
002	VALVE SPINDLE	CAST BRONZE	CAC406	1	011	O RING	RUBBER		1
001	BODY	CAST BRONZE	CAC406	1	010	BODY COVER	CAST BRONZE	CAC406	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

SOLENOID VALVE

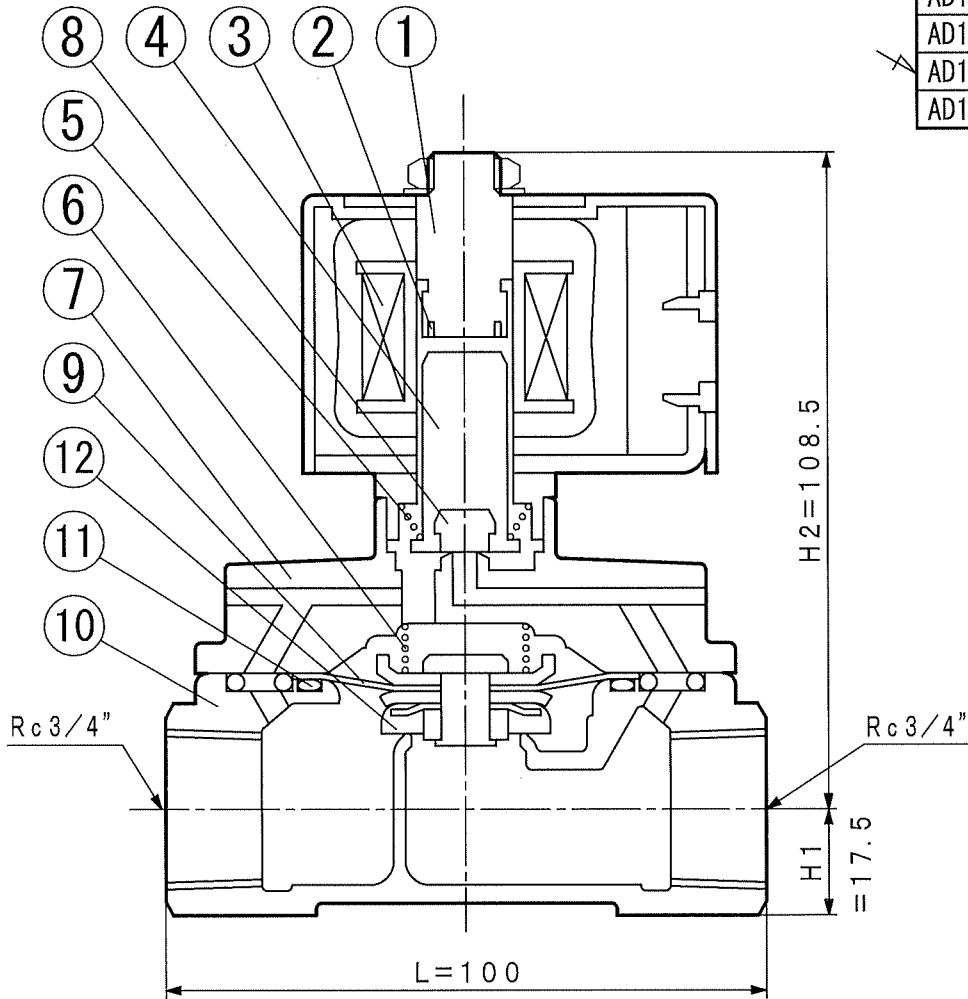
MODEL : AD11-

DRAWING NO.

A9DS0147

DATE

18 JUL. 2009

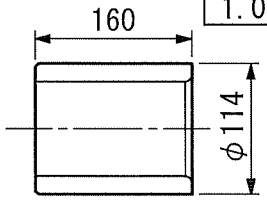
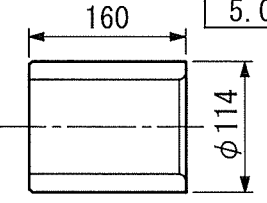
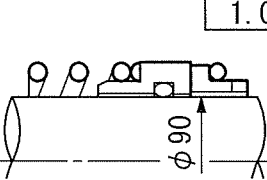
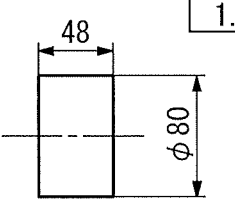


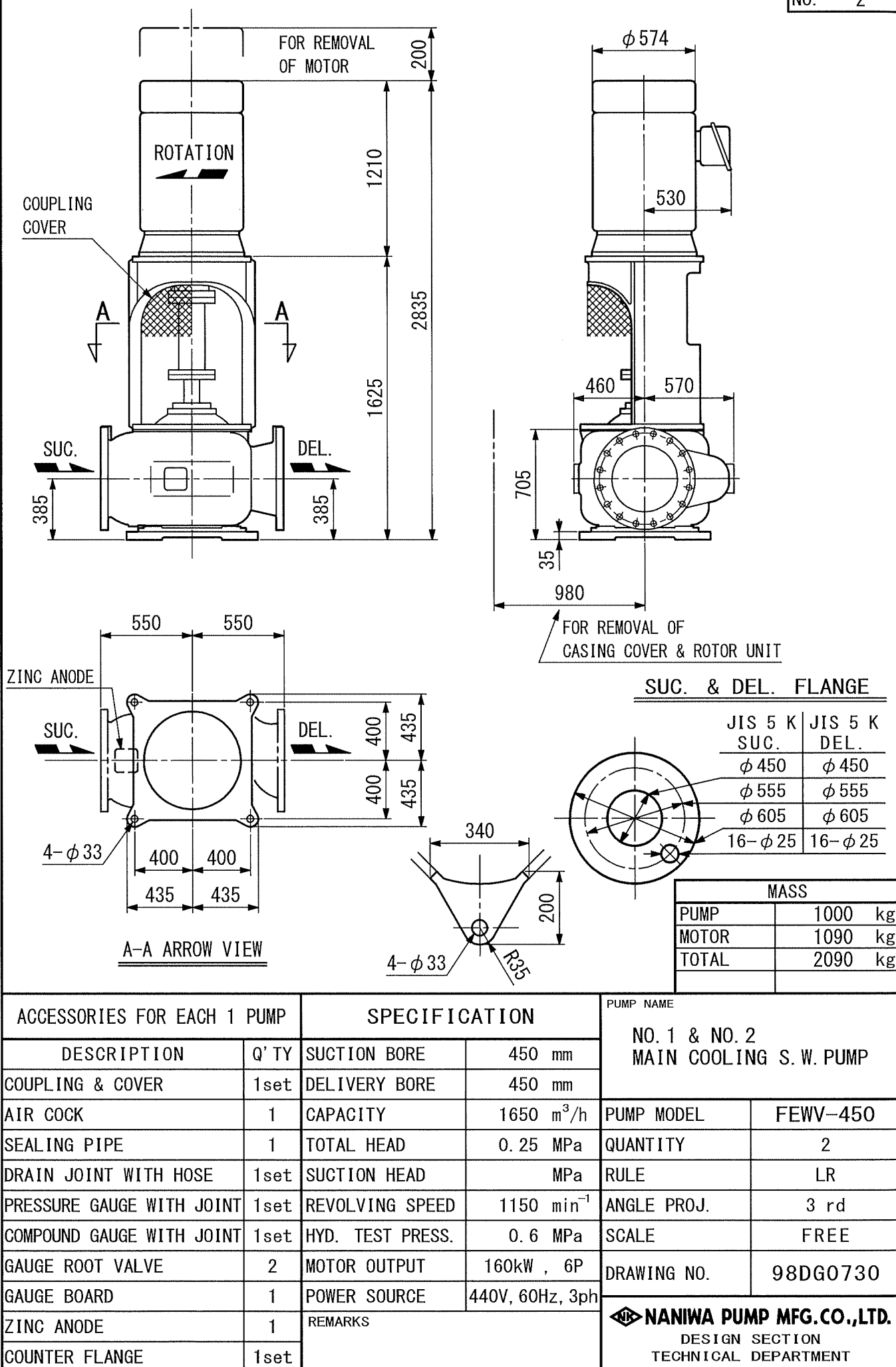
TYPE	L	H1	H2
AD11-8A	45	12	67
AD11-10A	45	12	67
AD11-15A	90	14.5	102.5
AD11-20A	100	17.5	108.5
AD11-25A	110	22	114

AC 220V 60Hz ϕ 1
POWER CAPACITY (THROW) 24VA

TYPE	SIZE	LIQUID	CV ($\Delta P=1\text{psi}$) (gpm)	OPERATING PRESS. (MPa)	LIQUID TEMP. ($^{\circ}\text{C}$)	MAX. TEST PRESS. (MPa)	POWER CONSUMP- TION (W)	MASS (kg)
AD11-8A	Rc 1/4"	WATER AIR PARAFFIN OIL (below 50cSt)	1.5	0.02	-10	2.0	5.7 24VA	0.4
AD11-10A	Rc 3/8"		1.8					0.4
AD11-15A	Rc 1/2"		4.5	?	?			1.2
AD11-20A	Rc 3/4"		9.3	1.0	60			1.2
AD11-25A	Rc 1		12.0					1.9

6	VALVE SPRING	STAINLESS STEEL	1	12	SEAL	RUBBER	1
5	PLUNGER SPRING	STAINLESS STEEL	1	11	O-RING	RUBBER	1
4	PLUNGER	STAINLESS STEEL	1 SET	10	BODY	BRONZE	1
3	COIL		1	9	DIAPHRAGM	RUBBER	4
2	SHADING COIL	COPPER	1	8	SEAL	RUBBER	1
1	CORE	STAINLESS STEEL	1 SET	7	STUFFING	BRONZE	1 SET
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY

SPARE PARTS LIST							PAGE	1, 2			
NO. 1~NO. 3 MAIN COOLING S. W. PUMP							VERTICAL CENTRIFUGAL PUMP		SHIP NO.		
							MODEL: FEWV-450 3 SETS		BOX NO.		
Item NO.	NAME	MASS : kg		MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
		SKETCH			WORK-ING PER PUMP	SPARE PER SHIP	C2DS0201 B7DS0081				
							NO.	PART NO.			
1	BEARING METAL		1.05	CARBON	1	1		204	NO. 3 MAIN COOLING S. W. PUMP		
2	BEARING METAL		5.0	SPECIAL BRONZE	1	1		204	NO. 1 & NO. 2 MAIN COOLING S. W. PUMP		
3	MECHANICAL SEAL		1.0		1 SET	1 SET		520			
4	ZINC ANODE		1.6	ZINC	1	3	B9DS0232	900			





SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

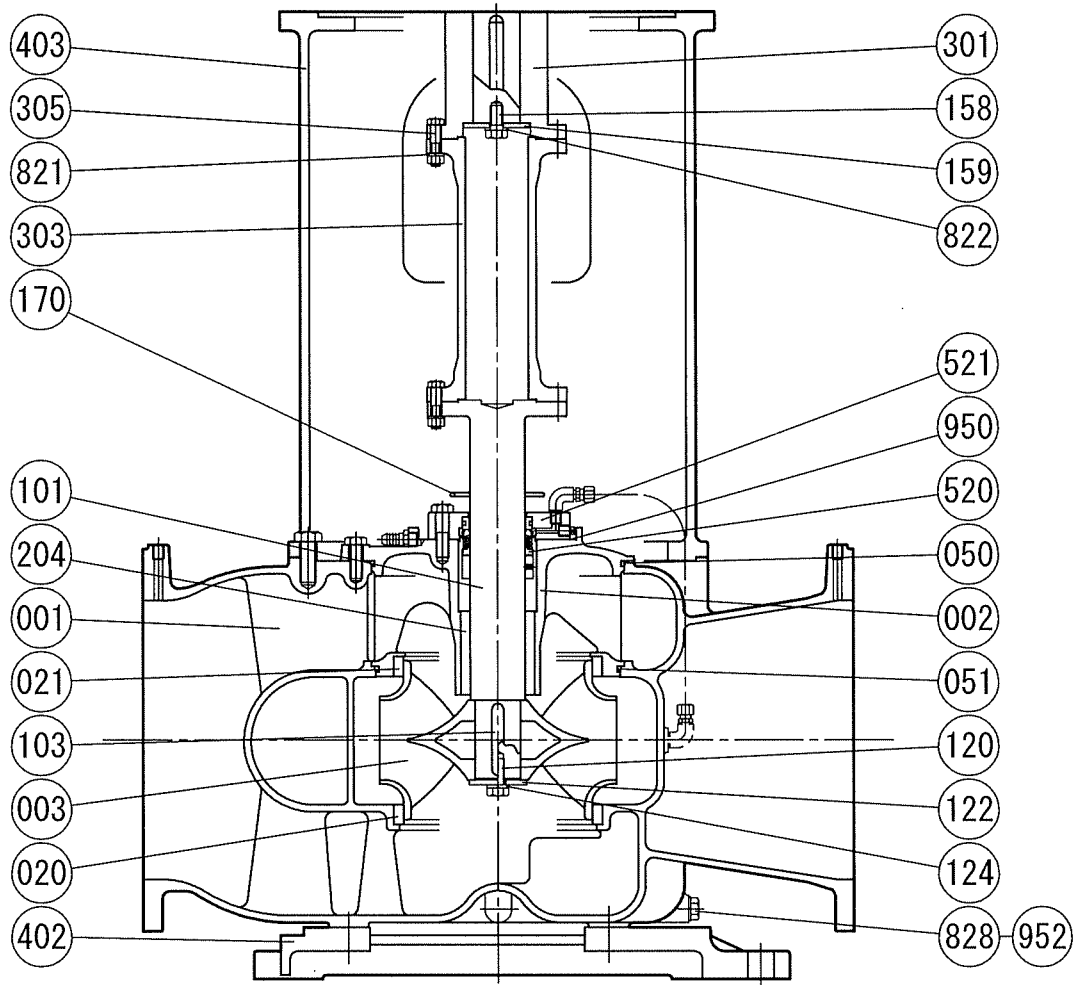
MODEL : FEWV

DRAWING NO.

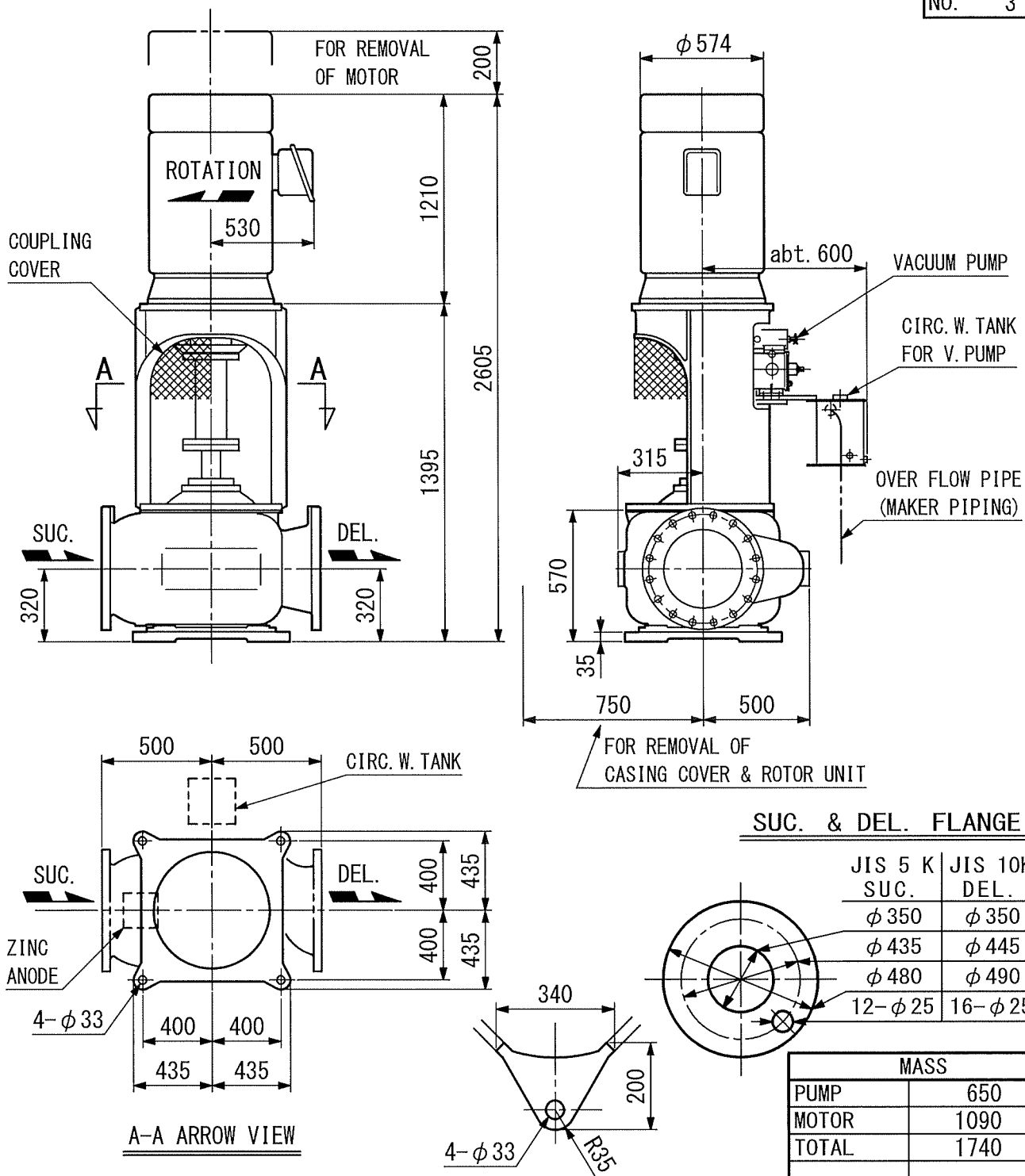
B7DS0081

DATE

7 MAR. 2017



159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER BOLT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS316	1	520	MECHANICAL SEAL			1 SET
051	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
050	O-RING	RUBBER		1	402	PUMP BED	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
020	CASING RING	SYNTHETIC RESIN		1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	301	COUPLING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	204	BEARING METAL	SPECIAL BRONZE		1
001	CASING	CAST BRONZE	CAC402	1	170	DUST STOPPER	RUBBER		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	350 mm	NO.1 & NO.2 BALLAST WATER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	350 mm		
SEALING PIPE	1	CAPACITY	1000 m ³ /h	PUMP MODEL	FEWV-350-3
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.35 MPa	QUANTITY	2
VACUUM PUMP	1set	SUCTION HEAD	-0.05 MPa	RULE	LR
CIRC.W.TANK & PIPING	1set	REVOLVING SPEED	1750 min ⁻¹	ANGLE PROJ.	3 rd
PRESSURE GAUGE WITH JOINT	1set	HYD. TEST PRESS.	0.8 MPa	SCALE	FREE
COMPOUND GAUGE WITH JOINT	1set	MOTOR OUTPUT	145kW , 4P	DRAWING NO.	B1DG0464
GAUGE ROOT VALVE & BOARD	2	POWER SOURCE	440V, 60Hz, 3ph	 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
ZINC ANODE	1	REMARKS			
COUNTER FLANGE	1set				



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

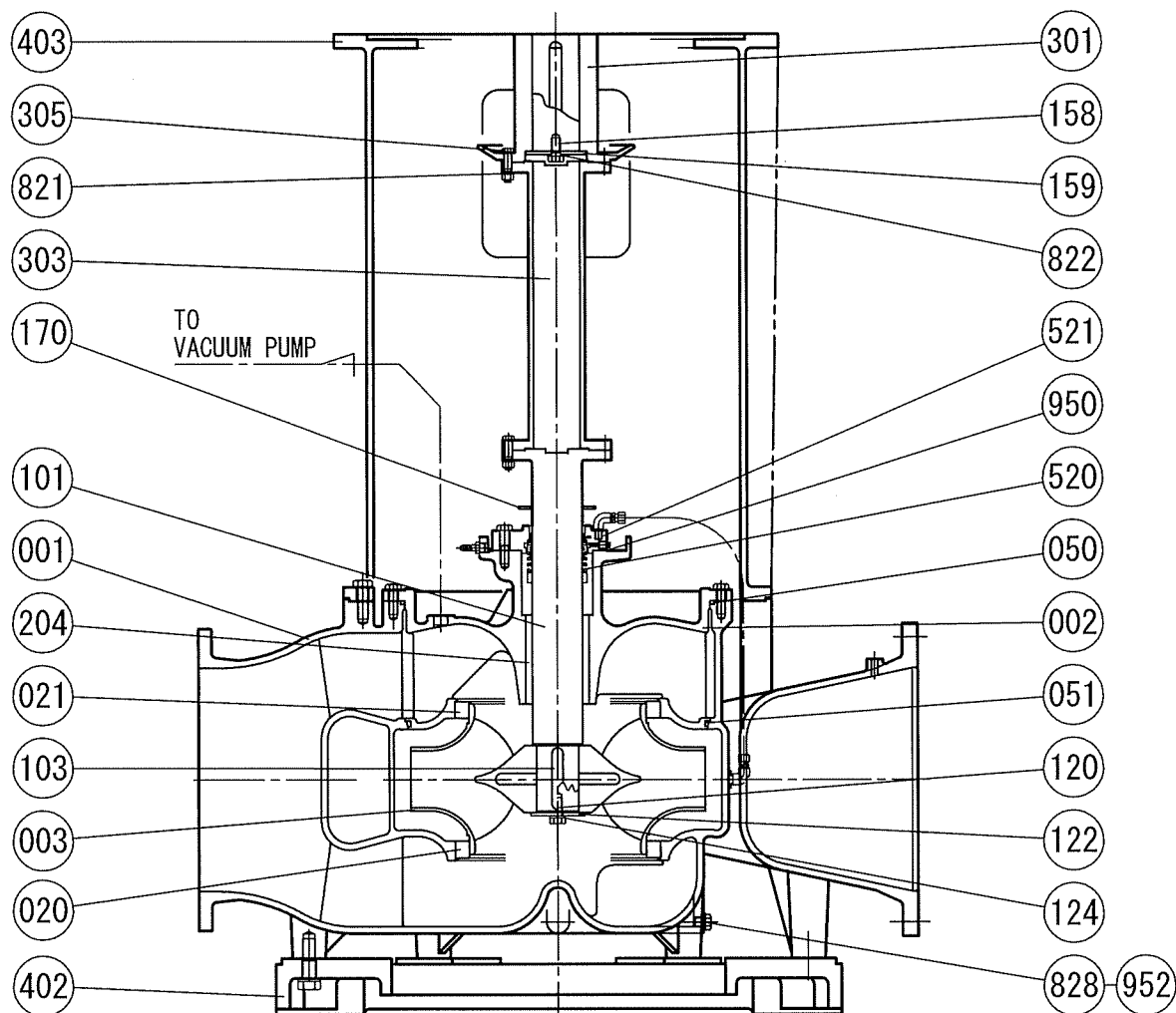
MODEL : FEWV—

DRAWING NO.

C2DS0198

DATE

16 MAY. 22



159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER BOLT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS304	1	520	MECHANICAL SEAL			1 SET
051	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
050	O-RING	RUBBER		1	402	PUMP BED	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
020	CASING RING	SYNTHETIC RESIN		1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	301	COUPLING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	204	BEARING METAL	CARBON		1
001	CASING	CAST BRONZE	CAC402	1	170	DUST STOPPER	RUBBER		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

HORIZONTAL VACUUM PUMP

MODEL : VK-20NB

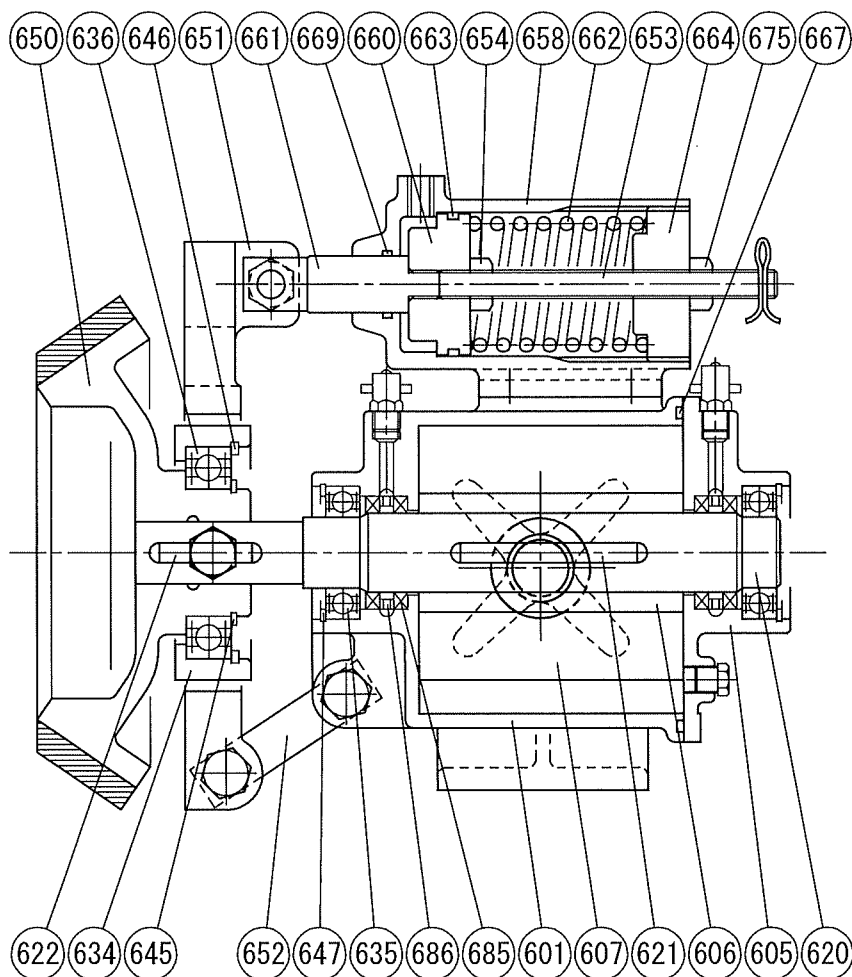
DRAWING NO.

C1DS0306

DATE

6 OCT. 2021

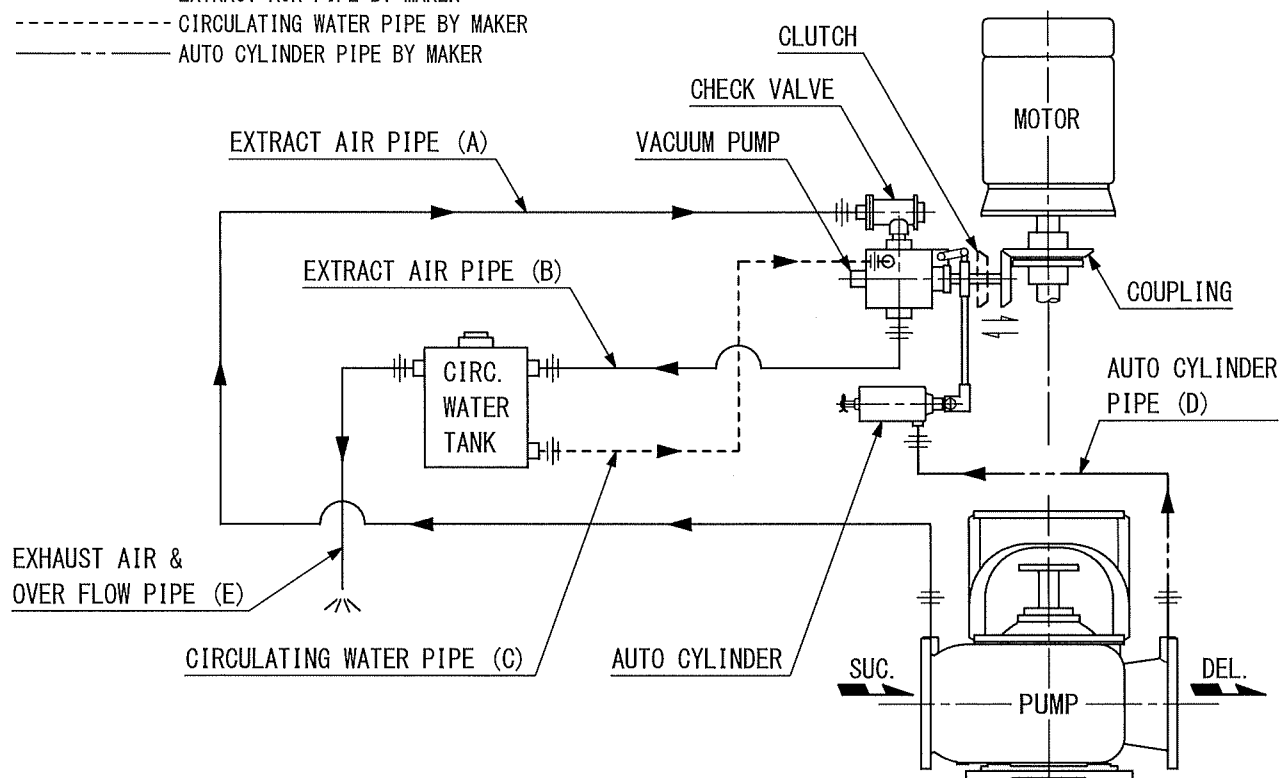
686	SEAL RING	MILD STEEL	SS400	2
685	OIL SEAL	RUBBER		4
675	NUT	MILD STEEL	SS400	1
669	O-RING	RUBBER		1
667	O-RING	RUBBER		1
664	SPRING CARRIER	COPPER ALLOY ROD	C3604	1
663	O-RING	RUBBER		1
662	SPRING	PIANO WIRE	SWP	1
661	PISTON ROD	COPPER ALLOY ROD	C3604	1
660	PISTON	COPPER ALLOY ROD	C3604	1
658	CYLINDER	CAST BRONZE	CAC406	1
654	LOCK NUT	MILD STEEL	SS400	1
653	TIE ROD	MILD STEEL	SS400	1
652	LEVER	MILD STEEL	SS400	1
651	SHIFTER	CAST IRON	FC200	1
650	CLUTCH			1
647	SNAP RING	CARBON TOOL STEEL	SK5	2
646	SNAP RING	CARBON TOOL STEEL	SK5	1
645	SNAP RING	CARBON TOOL STEEL	SK5	1
636	BALL BEARING		SUJ	1
635	BALL BEARING		SUJ	2
634	BEARING CASE	CAST IRON	FC200	1
622	CLUTCH KEY	CARBON STEEL	S45C	1
621	ROTOR KEY	STAINLESS STEEL	SUS304	1
620	SHAFT	STAINLESS STEEL	SUS304	1
607	VANE	BAKELITE		4
606	ROTOR	CAST BRONZE	CAC406	1
605	END COVER	CAST BRONZE	CAC406	1
601	CASING	CAST BRONZE	CAC406	1

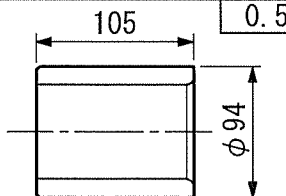
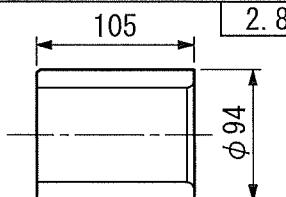
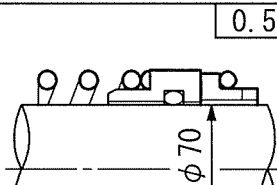
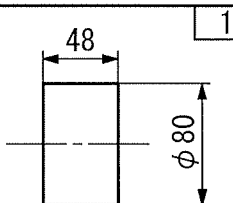


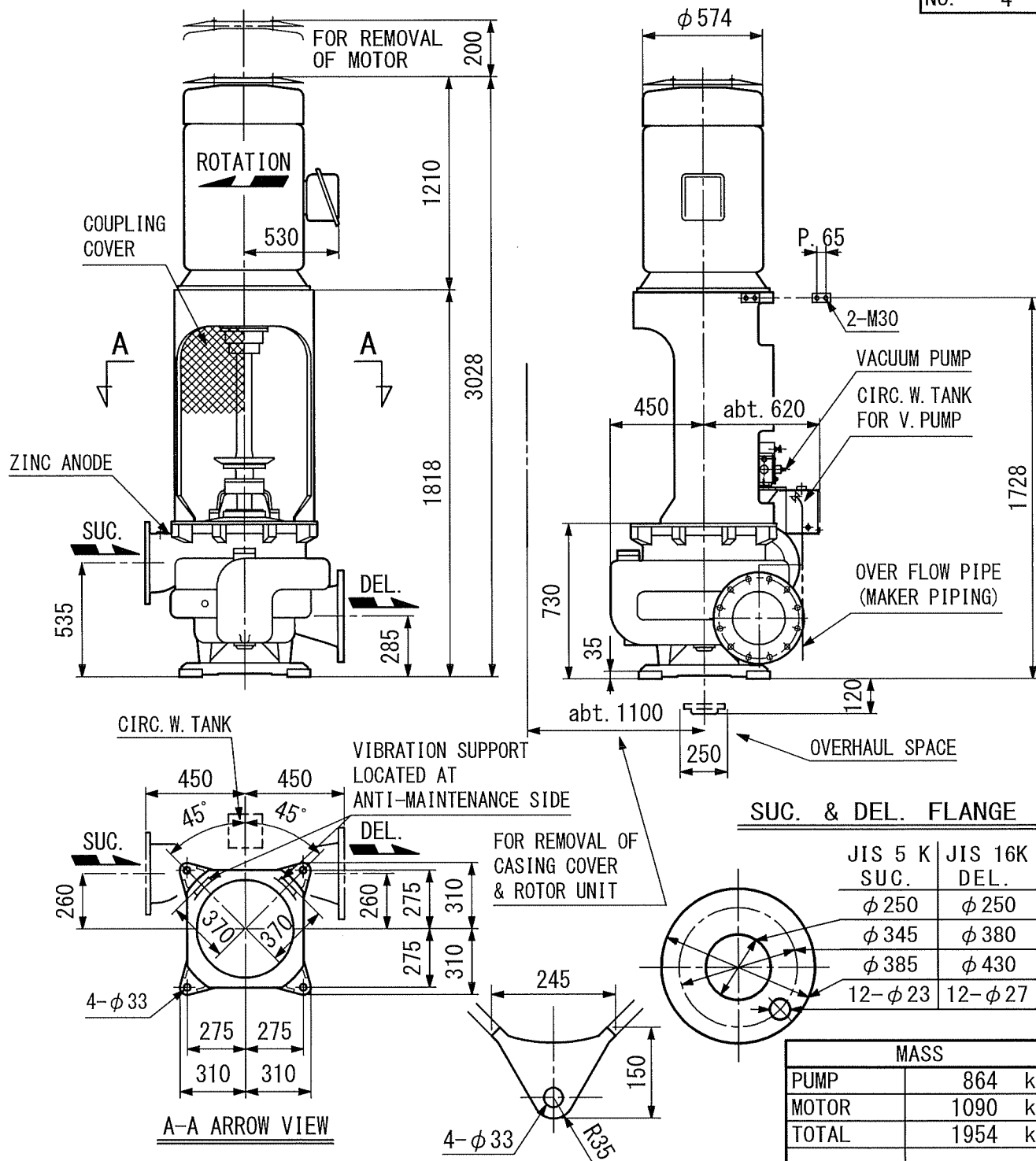
* MARKS OF PIPING

- EXTRACT AIR PIPE BY MAKER
- CIRCULATING WATER PIPE BY MAKER
- AUTO CYLINDER PIPE BY MAKER

PIPE LINE DIAGRAM



SPARE PARTS LIST							PAGE	3, 18	
NO. 1 & NO. 2 BALLAST WATER PUMP NO. 1 & NO. 2 iCER COOLING SEA WATER PUMP			VERTICAL CENTRIFUGAL PUMP MODEL: FEWV-350-3 TOTAL 4 SETS				SHIP NO.		
							BOX NO.		
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO. C2DS0198 B7DS0063		REMARKS
					WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.	
1	BEARING METAL		0.5	CARBON	1	1		204	NO. 1 & NO. 2 BALLAST WATER PUMP
2	BEARING METAL		2.8	SPECIAL BRONZE	1	1		204	NO. 1 & NO. 2 iCER COOLING SEA WATER PUMP
3	MECHANICAL SEAL		0.56		1 SET	1 SET		520	
4	ZINC ANODE		1.6	ZINC	1	4	B9DS0232	900	



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	250 mm	NO. 1 & NO. 2 BILGE, FIRE & G/S PUMP	
GEAR COUPLING & COVER	1set	DELIVERY BORE	250 mm		
SEALING PIPE	1set	CAPACITY	250/510 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	1. 0/0. 3 MPa	PUMP MODEL	FE2V-250-3
VACUUM PUMP	1set	SUCTION HEAD	-0. 05 MPa	QUANTITY	2
CIRC. W. TANK & PIPING	1set	REVOLVING SPEED	1750 min ⁻¹	RULE	LR
PRESSURE GAUGE WITH JOINT	1set	HYD. TEST PRESS.	1. 8 MPa	ANGLE PROJ.	3 rd
COMPOUND GAUGE WITH JOINT	1set	MOTOR OUTPUT	160kW , 4P	SCALE	FREE
GAUGE ROOT VALVE & BOARD	2	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	B3DG0607
ZINC ANODE	1	REMARKS		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
COUNTER FLANGE	1set				



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

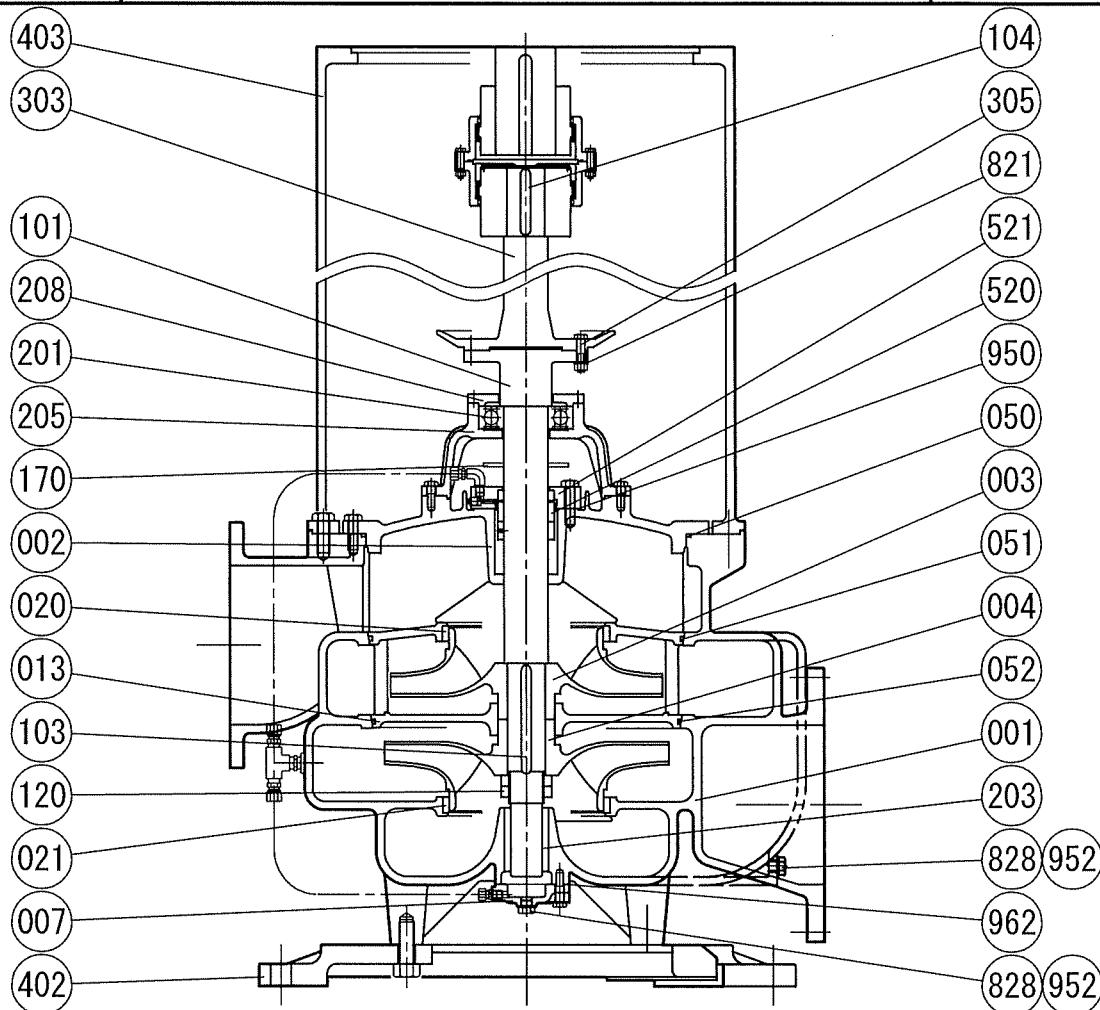
MODEL : FE2V-250-3

DRAWING NO.

C2DS0015

DATE

11 JAN. 2022



170	DUST STOPPER	RUBBER		1					
120	IMPELLER NUT	STAINLESS STEEL	SUS304	2	962	GASKET	NON ASBESTOS FIBER		1
104	COUPLING KEY	CARBON STEEL	S45C	1	952	GASKET	NON ASBESTOS FIBER		2
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
101	SHAFT	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	2
052	O-RING	RUBBER		1	821	SPRING WASHER	HIGH CARBON STEEL	SWRH	1 SET
051	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
050	O-RING	RUBBER		1	520	MECHANICAL SEAL			1 SET
021	CASING RING	SYNTHETIC RESIN		1	403	FRAME	CAST IRON	FC200	1
020	CASING RING	SYNTHETIC RESIN		1	402	PUMP BED	CAST IRON	FC200	1
013	STAGE COVER	CAST BRONZE	CAC402	1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	1 SET
007	END COVER	CAST BRONZE	CAC406	1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
004	IMPELLER	STAINLESS STEEL	SCS14	1	208	BEARING COVER	CAST IRON	FC200	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	205	BEARING HOUSING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	203	SUBMERGED BEARING	CARBON		1
001	CASING	CAST BRONZE	CAC402	1	201	BALL BEARING	SUJ		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

GEAR COUPLING

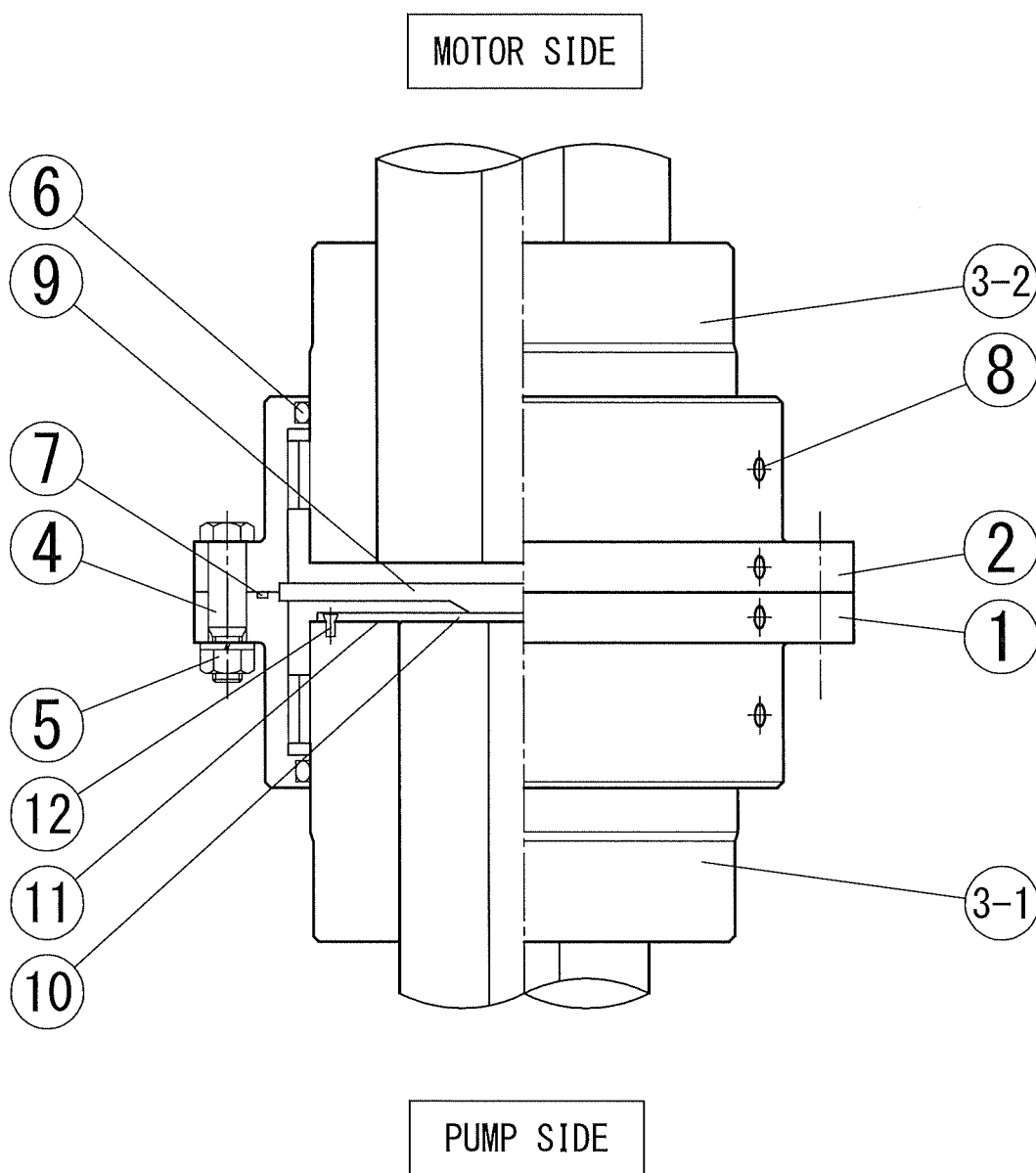
MODEL : FE2V-

DRAWING NO.

B1DS0139

DATE

18 MAY. 2011



6	O-RING	RUBBER		2	12	FLAT HEAD BOLT	MILD STEEL	SS400	8
5	NUT	CARBON STEEL	S45C	1 SET	11	PACKING	NON ASBESTOS FIBER		1
4	COUPLING BOLT	CROM. MOLYB. STEEL	SCM435	1 SET	10	PLATE	CARBON STEEL	S45C	1
3	HUB	CARBON STEEL	S45C	2	9	PLATE	CARBON STEEL	S45C	1
2	SLEEVE (B)	CARBON STEEL	S45C	1	8	OIL PLUG	CARBON STEEL	S45C	8
1	SLEEVE (A)	CARBON STEEL	S45C	1	7	O-RING	RUBBER		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

HORIZONTAL VACUUM PUMP

MODEL : VK-20NB

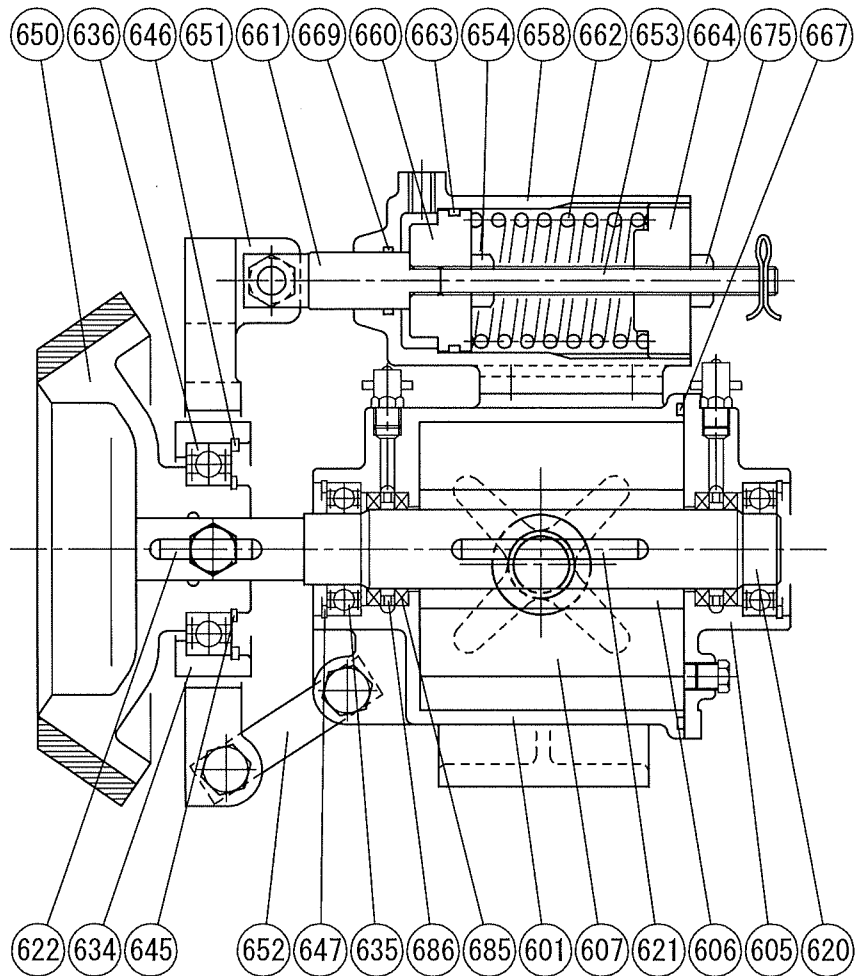
DRAWING NO.

C1DS0306

DATE

6 OCT. 2021

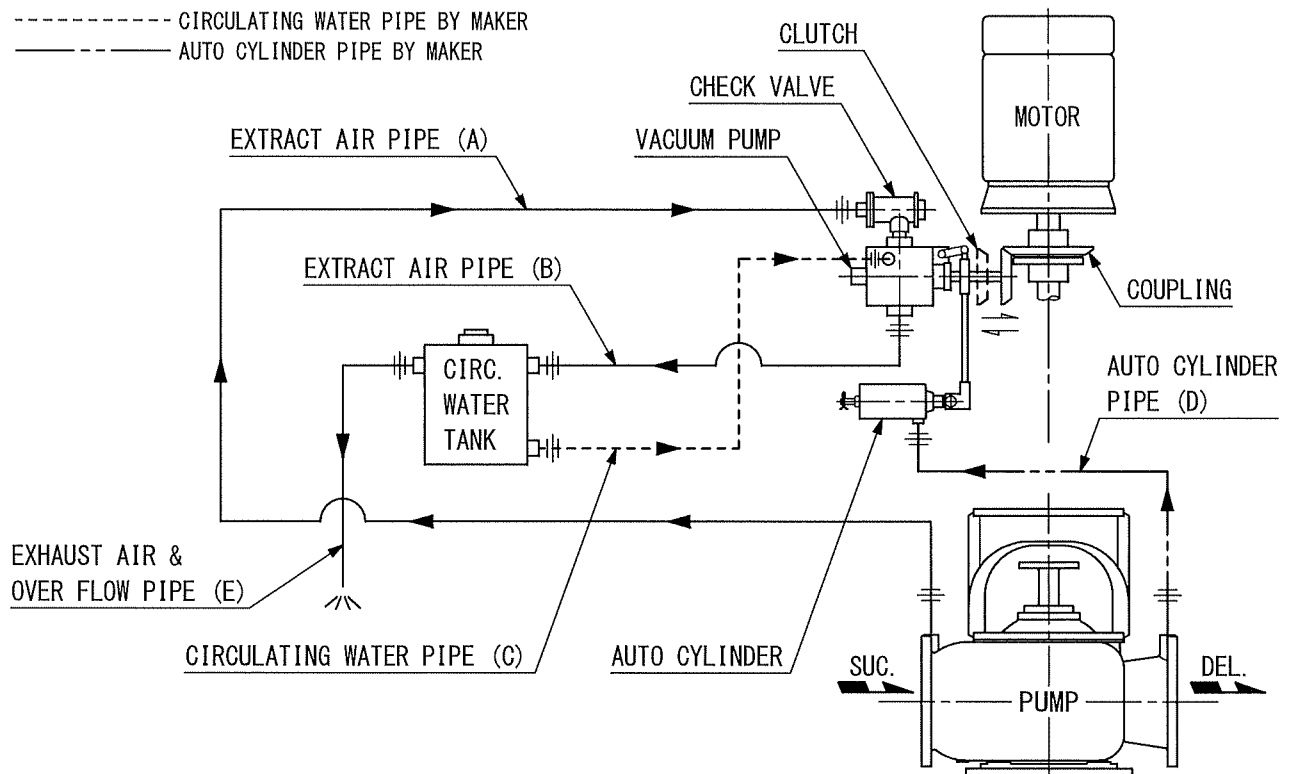
686	SEAL RING	MILD STEEL	SS400	2
685	OIL SEAL	RUBBER		4
675	NUT	MILD STEEL	SS400	1
669	O-RING	RUBBER		1
667	O-RING	RUBBER		1
664	SPRING CARRIER	COPPER ALLOY ROD	C3604	1
663	O-RING	RUBBER		1
662	SPRING	PIANO WIRE	SWP	1
661	PISTON ROD	COPPER ALLOY ROD	C3604	1
660	PISTON	COPPER ALLOY ROD	C3604	1
658	CYLINDER	CAST BRONZE	CAC406	1
654	LOCK NUT	MILD STEEL	SS400	1
653	TIE ROD	MILD STEEL	SS400	1
652	LEVER	MILD STEEL	SS400	1
651	SHIFTER	CAST IRON	FC200	1
650	CLUTCH			1
647	SNAP RING	CARBON TOOL STEEL	SK5	2
646	SNAP RING	CARBON TOOL STEEL	SK5	1
645	SNAP RING	CARBON TOOL STEEL	SK5	1
636	BALL BEARING		SUJ	1
635	BALL BEARING		SUJ	2
634	BEARING CASE	CAST IRON	FC200	1
622	CLUTCH KEY	CARBON STEEL	S45C	1
621	ROTOR KEY	STAINLESS STEEL	SUS304	1
620	SHAFT	STAINLESS STEEL	SUS304	1
607	VANE	BAKELITE		4
606	ROTOR	CAST BRONZE	CAC406	1
605	END COVER	CAST BRONZE	CAC406	1
601	CASING	CAST BRONZE	CAC406	1
PART NO.	NAME OF PART	MATERIAL	Q'TY	

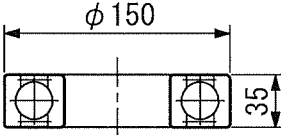
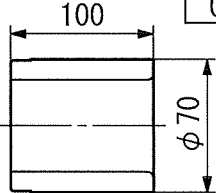
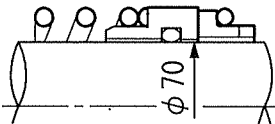
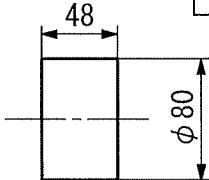


* MARKS OF PIPING

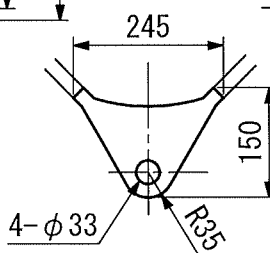
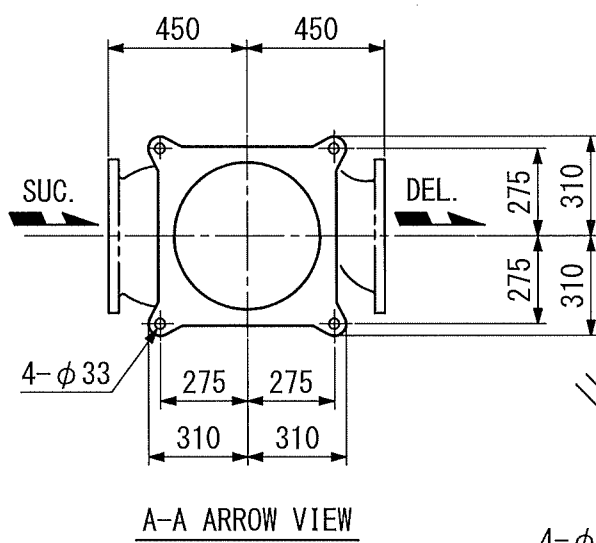
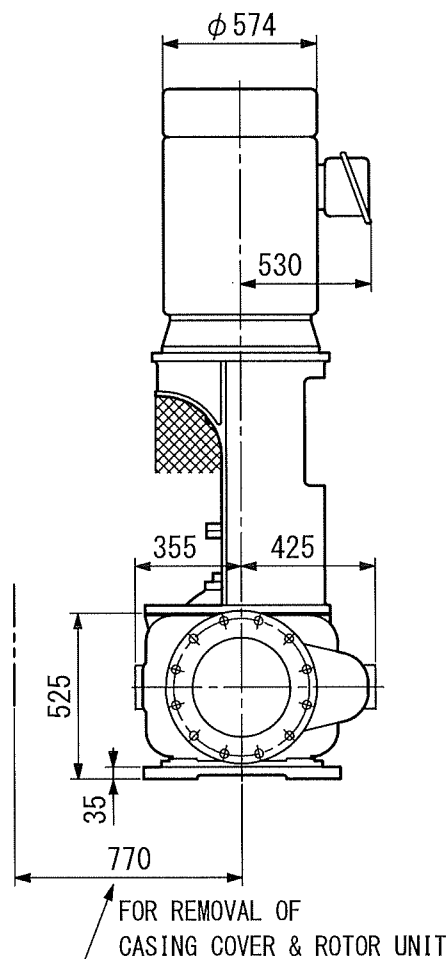
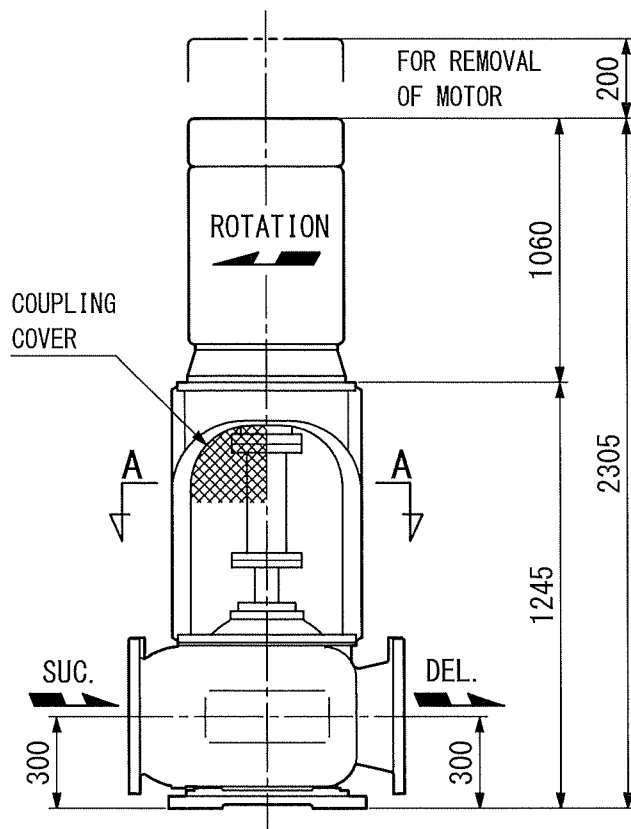
- EXTRACT AIR PIPE BY MAKER
- CIRCULATING WATER PIPE BY MAKER
- AUTO CYLINDER PIPE BY MAKER

PIPE LINE DIAGRAM



SPARE PARTS LIST							PAGE	4
NO. 1 & NO. 2 BILGE, FIRE & G/S PUMP							VERTICAL CENTRIFUGAL PUMP MODEL: FE2V-250-3 2 SETS	
							SHIP NO.	
							BOX NO.	
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS
				WORK -ING PER PUMP	SPARE PER SHIP	C2DS0015 NO.	PART NO.	
1	BALL BEARING	2.52 	SUJ	1	1		201	
2	SUBMERGED BEARING	0.2 	CARBON	1	1		203	
3	MECHANICAL SEAL	0.56 		1 SET	1 SET		520	
4	ZINC ANODE	1.6 	ZINC	1	2	B9DS0232	900	

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		BODP0114



SUC. & DEL. FLANGE

JIS 5 K SUC.	JIS 5 K DEL.
φ 350	φ 350
φ 435	φ 435
φ 480	φ 480
12-φ 25	12-φ 25

MASS	
PUMP	690 kg
MOTOR	810 kg
TOTAL	1500 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME NO. 1 & NO. 2 L. T. COOLING F. W. PUMP NO. 1 & NO. 2 M/E AIR COOLER L. T. COOLING F. W. PUMP		
DESCRIPTION	Q' TY	SUCTION BORE	350 mm		PUMP MODEL	FEWV-350
COUPLING & COVER	1set	DELIVERY BORE	350 mm		QUANTITY	TOTAL 4
AIR COCK	1	CAPACITY	1040 m ³ /h	RULE	LR	
SEALING PIPE	1	TOTAL HEAD	0.3 MPa	ANGLE PROJ.	3 rd	
DRAIN JOINT WITH HOSE	1set	SUCTION HEAD	+0.1 MPa	SCALE	FREE	
PRESSURE GAUGE WITH JOINT	1set	REVOLVING SPEED	1750 min ⁻¹	DRAWING NO.	C3DG0117	
COMPOUND GAUGE WITH JOINT	1set	HYD. TEST PRESS.	0.9 MPa	 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT		
GAUGE ROOT VALVE	2	MOTOR OUTPUT	125kW , 4P			
GAUGE BOARD	1	POWER SOURCE	440V, 60Hz, 3ph			
COUNTER FLANGE	1set	REMARKS WORKING TEMP = 36 °C				



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

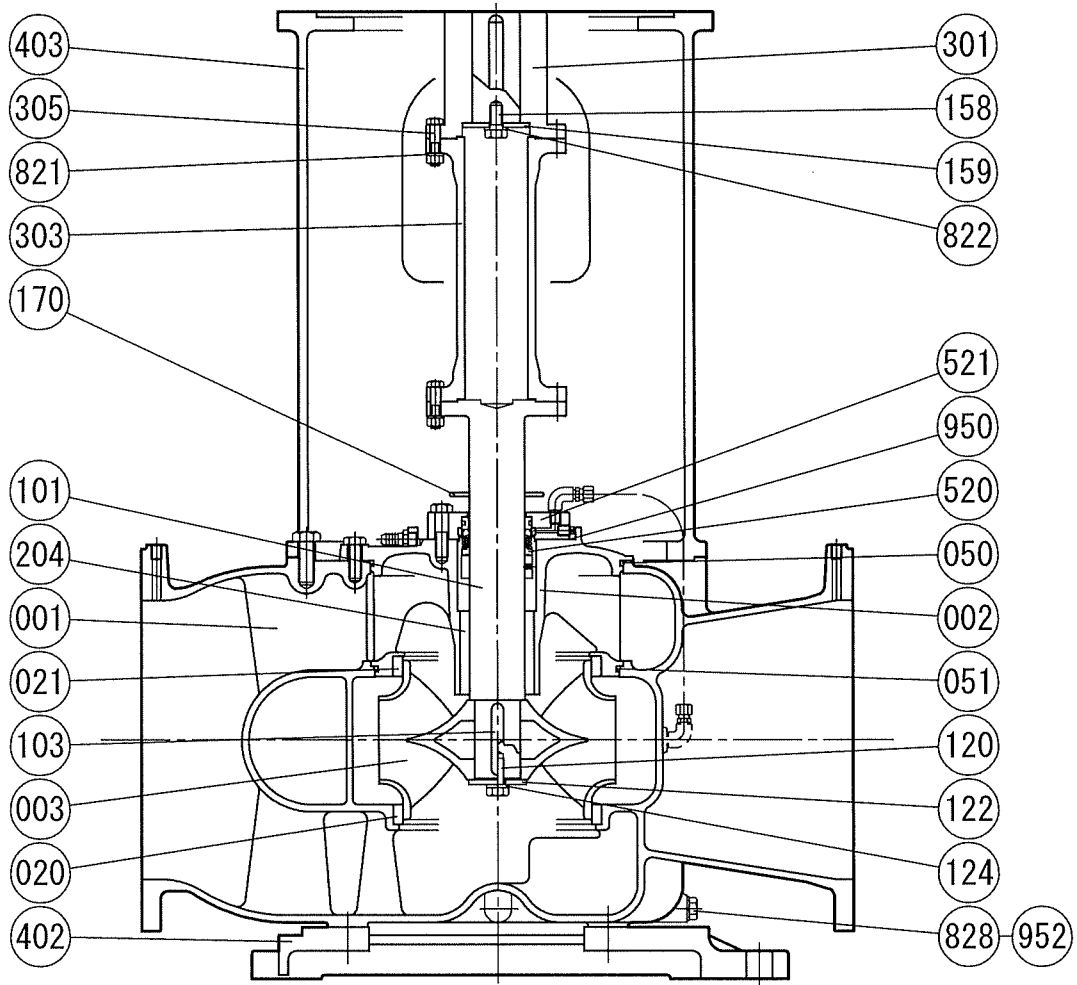
MODEL : FEWV

DRAWING NO.

B7DS0062

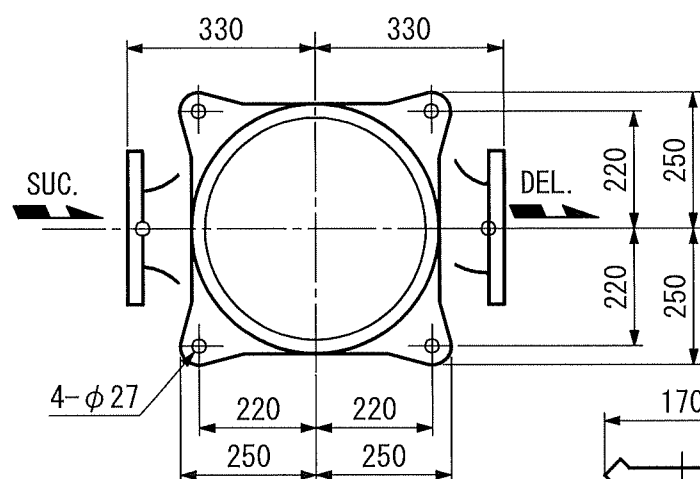
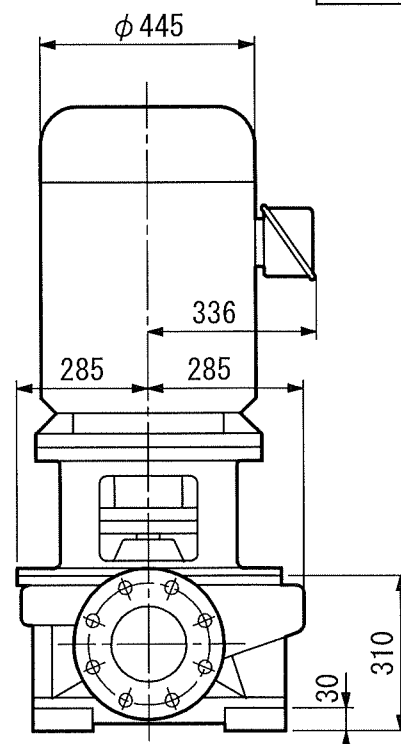
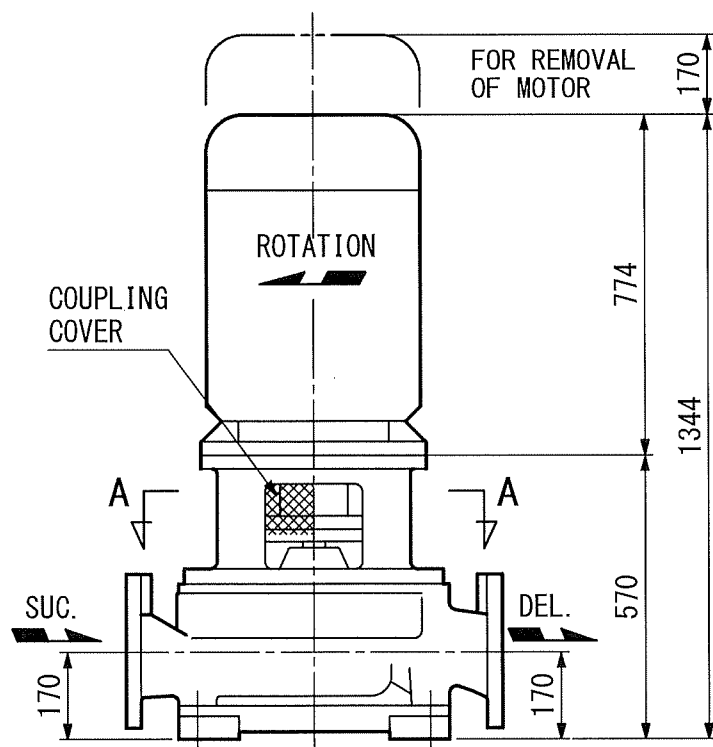
DATE

2 MAR 2017

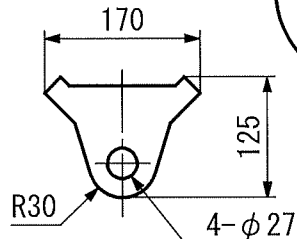


159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER BOLT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS304	1	520	MECHANICAL SEAL			1 SET
051	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
050	O-RING	RUBBER		1	402	PUMP BED	CAST IRON	FC200	1
021	CASING RING	CAST BRONZE	CAC406	1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
020	CASING RING	CAST BRONZE	CAC406	1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
003	IMPELLER	PHOSPHOR BRONZE	CAC502A	1	301	COUPLING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	204	BEARING METAL	SPECIAL BRONZE		1
001	CASING	CAST BRONZE	CAC402	1	170	DUST STOPPER	RUBBER		1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY		

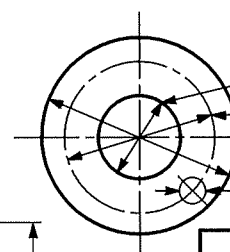
SPARE PARTS LIST							PAGE		5, 6	
NO. 1 & NO. 2 L. T. COOLING F. W. PUMP NO. 1 & NO. 2 M/E AIR COOLER L. T. COOLING F. W. PUMP					VERTICAL CENTRIFUGAL PUMP MODEL: FEWV-350			SHIP NO.		
					TOTAL 4 SETS			BOX NO.		
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS	
					WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.		
1	BEARING METAL		2.8	SPECIAL BRONZE	1	1		204		
2	MECHANICAL SEAL		0.32		1 SET	1 SET		520		
MFR' S NAME & ADDRESS			NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.		
								98DP0332		



A-A ARROW VIEW



SUC. & DEL. FLANGE



JIS 5 K SUC.	JIS 5 K DEL.
$\phi 200$	$\phi 200$
$\phi 280$	$\phi 280$
$\phi 320$	$\phi 320$
8- $\phi 23$	8- $\phi 23$

MASS

PUMP	250	kg
MOTOR	330	kg
TOTAL	580	kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	200 mm	NO. 1 & NO. 2 M/E H. T. F. W. PUMP	
COUPLING & COVER	1set	DELIVERY BORE	200 mm		
AIR COCK	1	CAPACITY	326 m ³ /h		
SEALING PIPE	1set	TOTAL PRESSURE	0.3 MPa	PUMP MODEL	FEV-200E1
PRESSURE GAUGE WITH JOINT	1set	SUCTION PRESSURE	+0.1 MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	1750 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.8 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	45kW , 4P	SCALE	FREE
COOLER	1	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	C2DG0030
COUNTER FLANGE	1set	REMARKS WORKING TEMP. = 100 °C		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

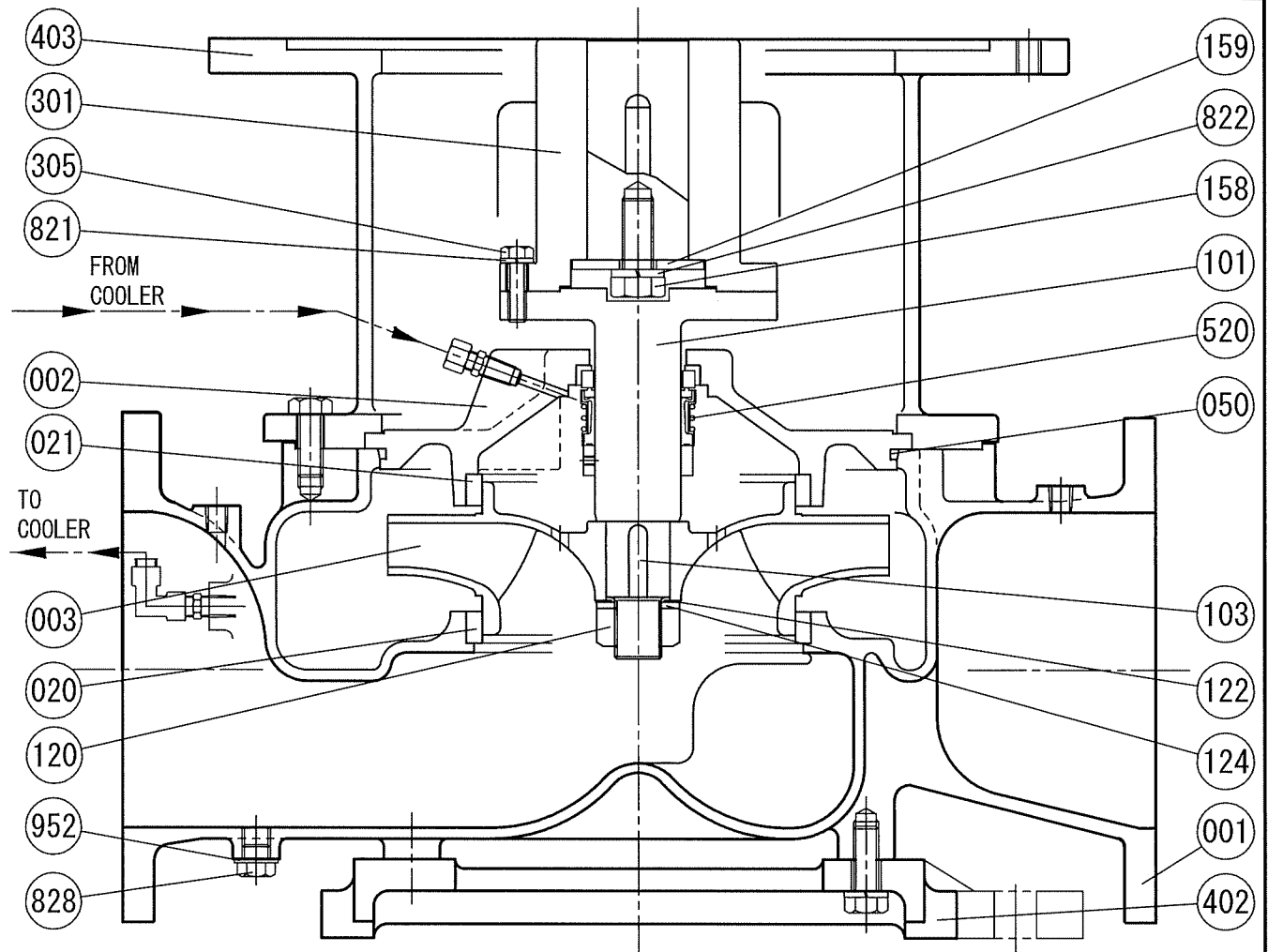
MODEL FEV-200/250E1

DRAWING NO.

C4DS0257

DATE

7 NOV. 2024



158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	IMPELLER WASHER	STAINLESS STEEL	SUS304	1	952	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	822	COUPLING SET WASHER	HIGH CARBON STEEL WIRE	SWRH	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	520	MECHANICAL SEAL			1 SET
021	CASING RING	CAST BRONZE	CAC406	1	403	FRAME	CAST IRON	FC200	1
020	CASING RING	CAST BRONZE	CAC406	1	402	PUMP BED	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	PLATE WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

MECHANICAL SEAL COOLER

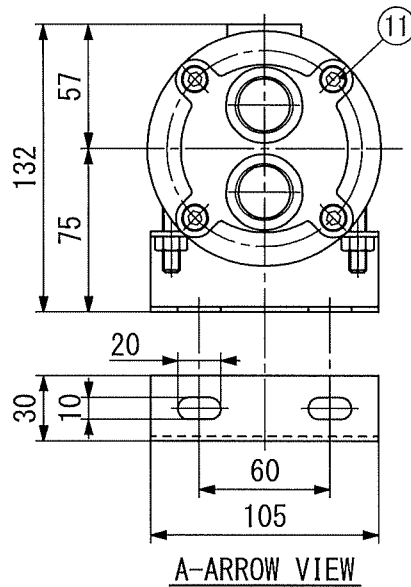
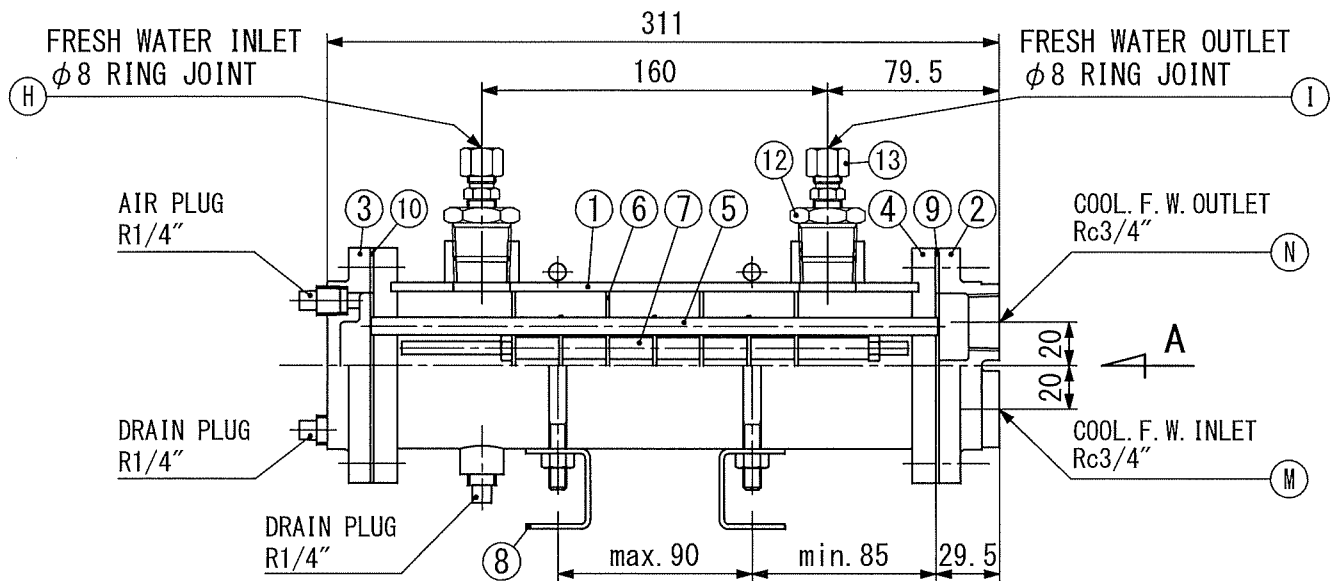
MODEL : NBT-O. 15

DRAWING NO.

C1DS0161

DATE

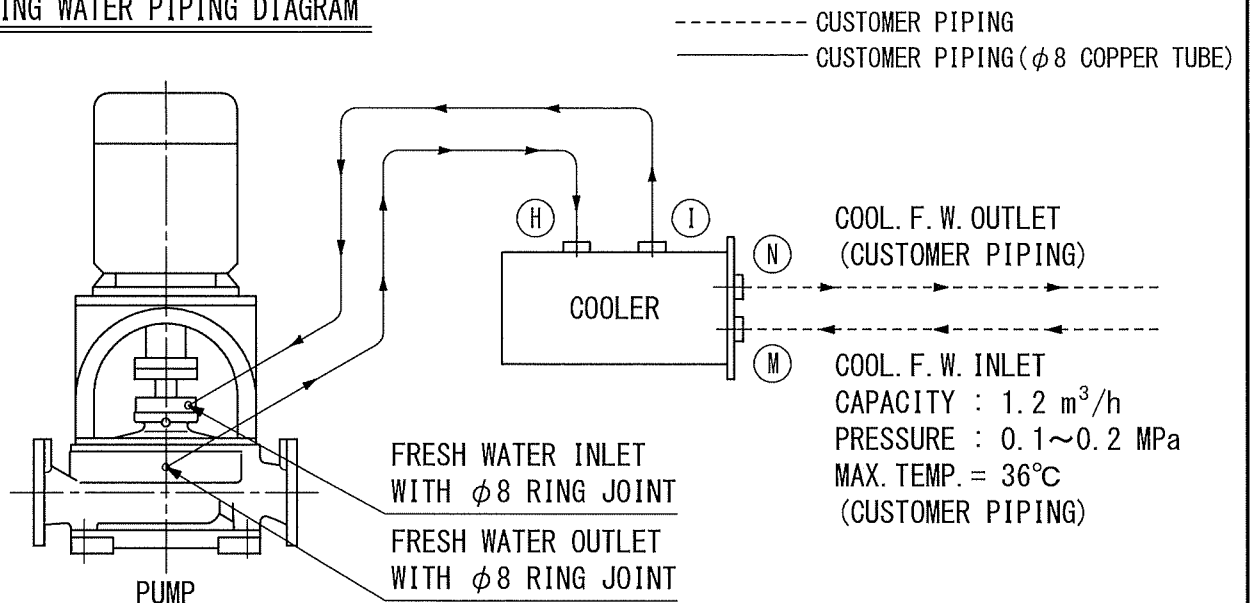
4 JUN 2021

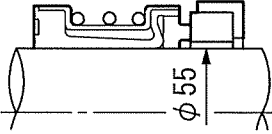


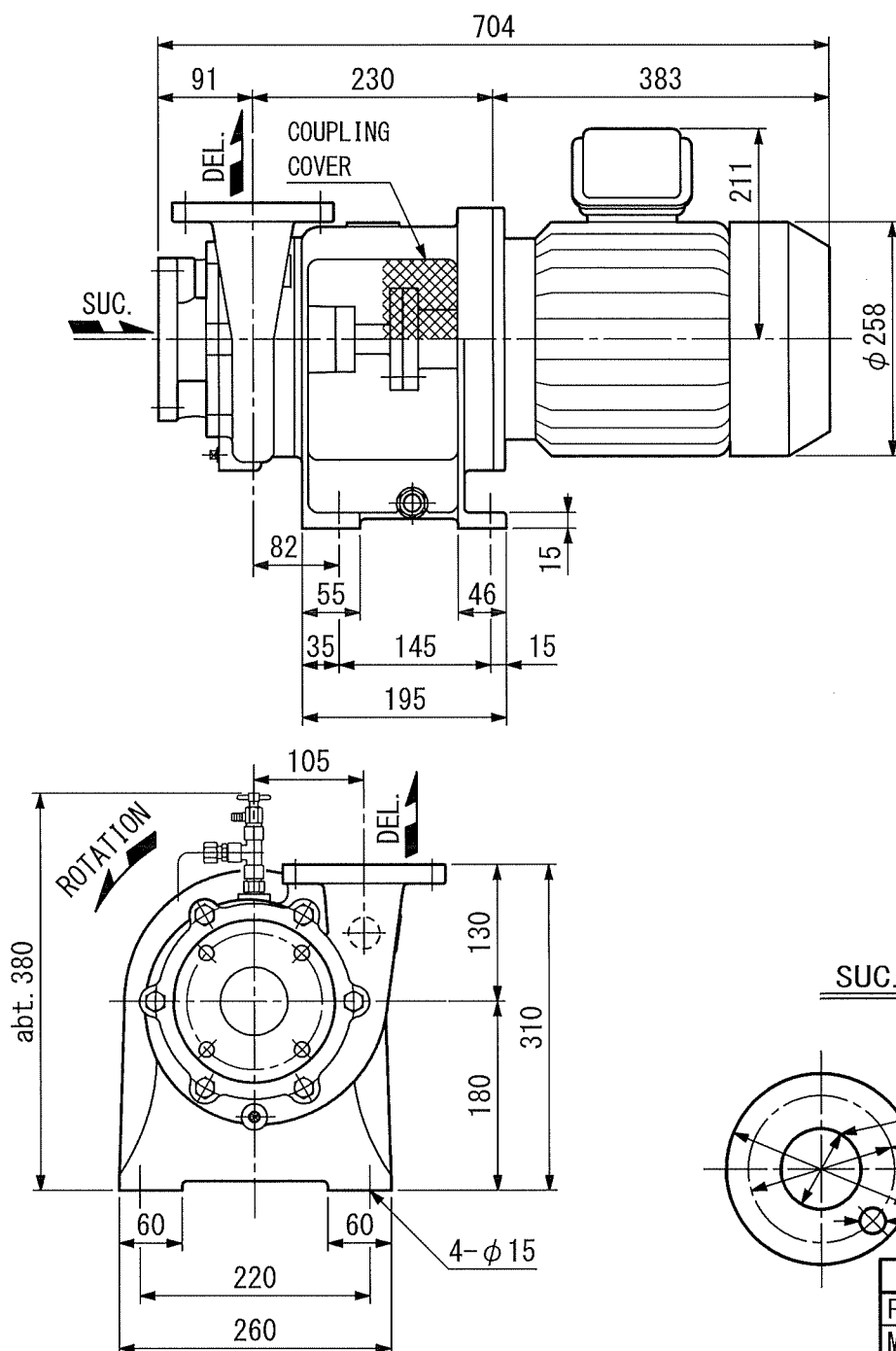
MASS
8 kg

13	RING JOINT	2	COPPER ALLOY ROD
12	BUSHING	2	COPPER ALLOY ROD
11	CAP BOLT	8	CHROMIUM MOLYBDEN. STEEL
10	GASKET	1	NON ASBESTOS
9	GASKET	1	NON ASBESTOS
8	SADDLE, U-BOLT&NUT	2 SETS	MILD STEEL
7	TIE ROD & SPACER	1 SET	MILD STEEL
6	BAFFLE PLATE	1 SET	MILD STEEL
5	COOLING TUBE	26	COPPER
4	TUBE SHEET	2	MILD STEEL
3	RETURN CHAMBER	1	CAST IRON
2	WATER CHAMBER	1	CAST IRON
1	SHELL	1	CARBON STEEL PIPE
NO.	NAME OF PART	Q' TY	MATERIAL

COOLING WATER PIPING DIAGRAM



SPARE PARTS LIST							PAGE		7, 9	
NO. 1 & NO. 2 M/E H. T. F. W. PUMP			VERTICAL CENTRIFUGAL PUMP MODEL: FEV-200E1				SHIP NO.			
NO. 1 & NO. 2 L/T. C. W. BOOSTER PUMP			TOTAL 4 SETS				BOX NO.			
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS	
					WORK -ING PER PUMP	SPARE PER SHIP	NO.	PART NO.		
1	MECHANICAL SEAL		0.24		1 SET	1 SET		520		
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO. C2DP0037		



SUC. & DEL. FLANGE

JIS 5 K SUC.	JIS 5 K DEL.
$\phi 65$	$\phi 65$
$\phi 130$	$\phi 130$
$\phi 155$	$\phi 155$
4- $\phi 15$	4- $\phi 15$

MASS

PUMP	44 kg
MOTOR	54 kg
TOTAL	98 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	65 mm	M/E JACKET WATER PREHEATER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	65 mm		
AIR COCK	1	CAPACITY	35 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.3 MPa	PUMP MODEL	BHR-65
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	QUANTITY	1
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.7 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	5.5kW , 2P	SCALE	FREE
COOLER	1	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	98DG0277
COUNTER FLANGE	1set	REMARKS WORKING TEMP = 100 °C		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

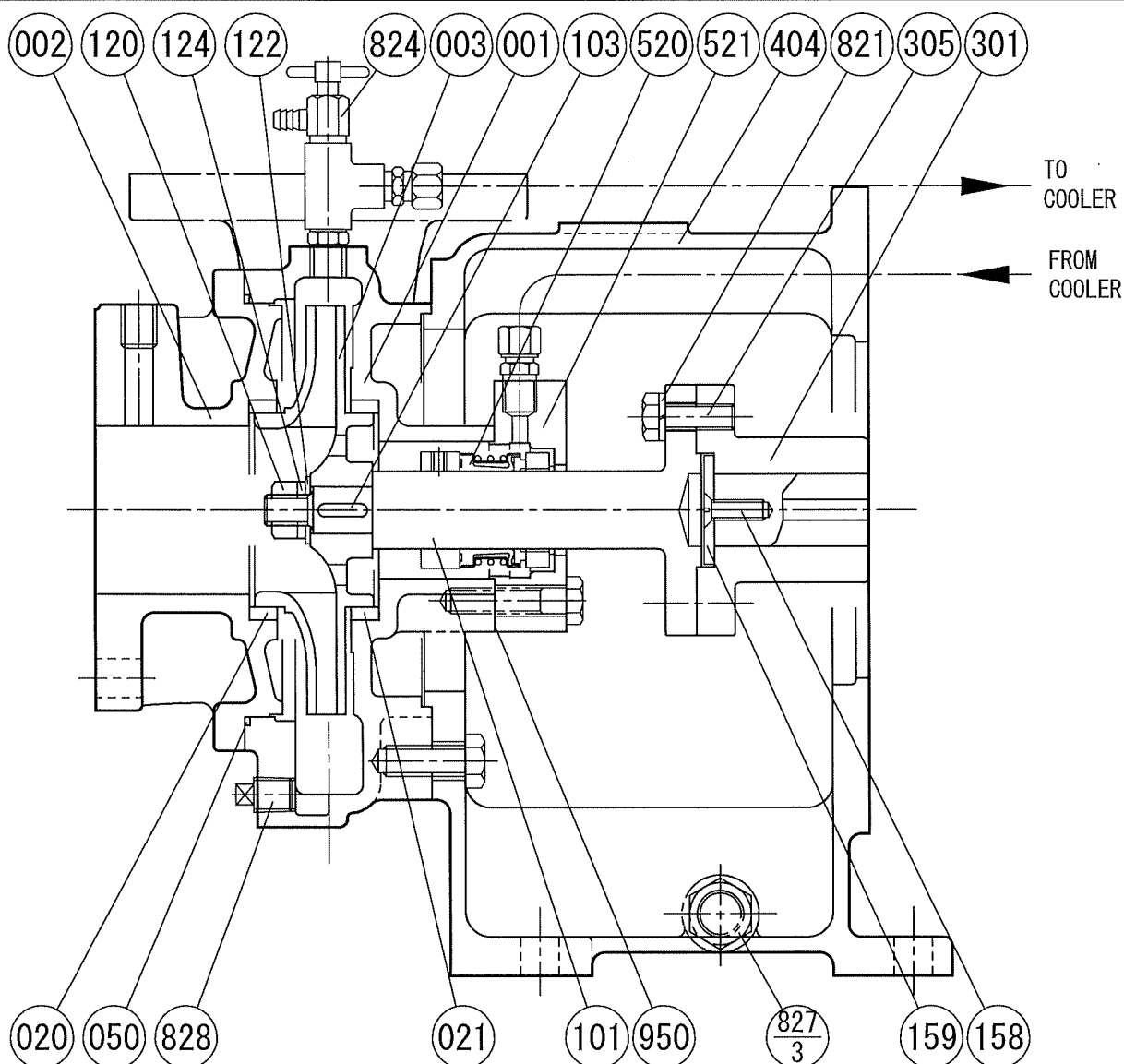
MODEL BHR-65

DRAWING NO.

B8DS0191

DATE

1 JUN. 2018



158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	$\frac{827}{3}$	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
021	CASING RING	CAST BRONZE	CAC406	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY		



SECTIONAL DRAWING

MECHANICAL SEAL COOLER

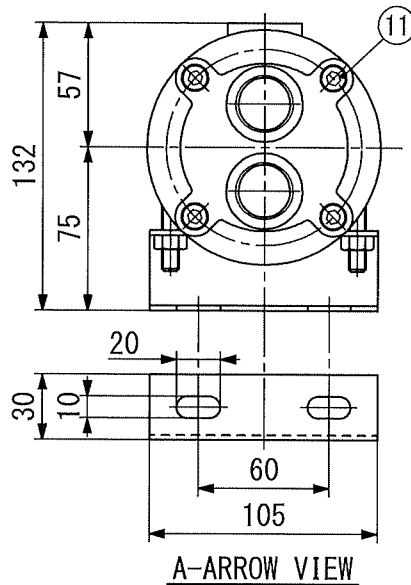
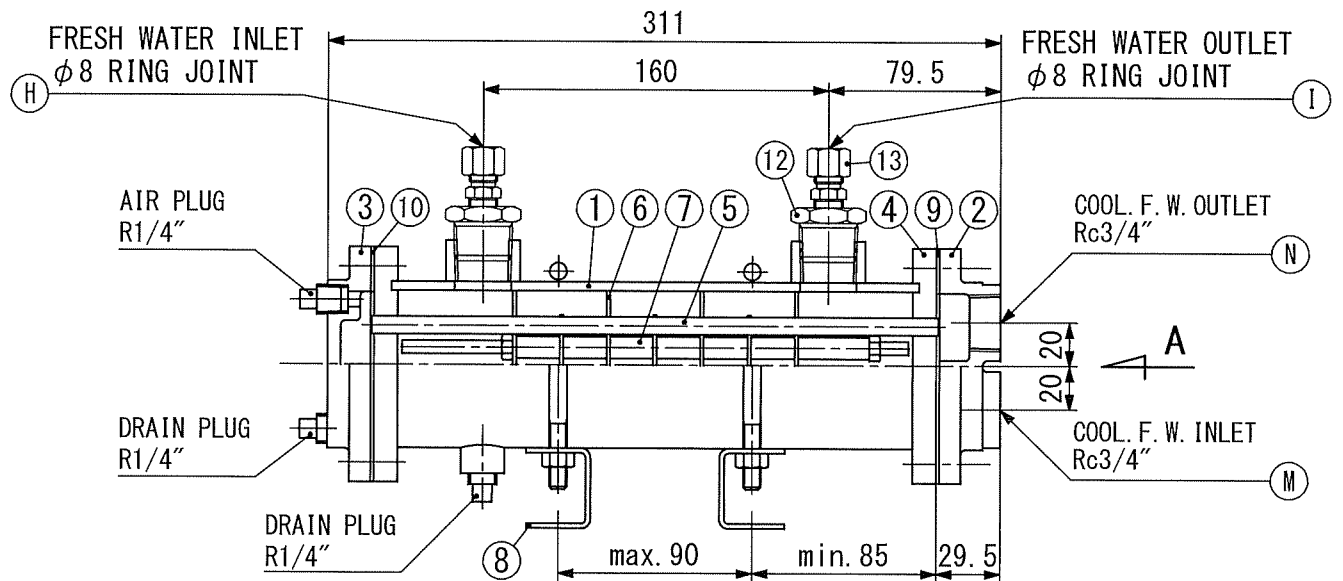
MODEL : NBT-O. 15

DRAWING NO.

C1DS0162

DATE

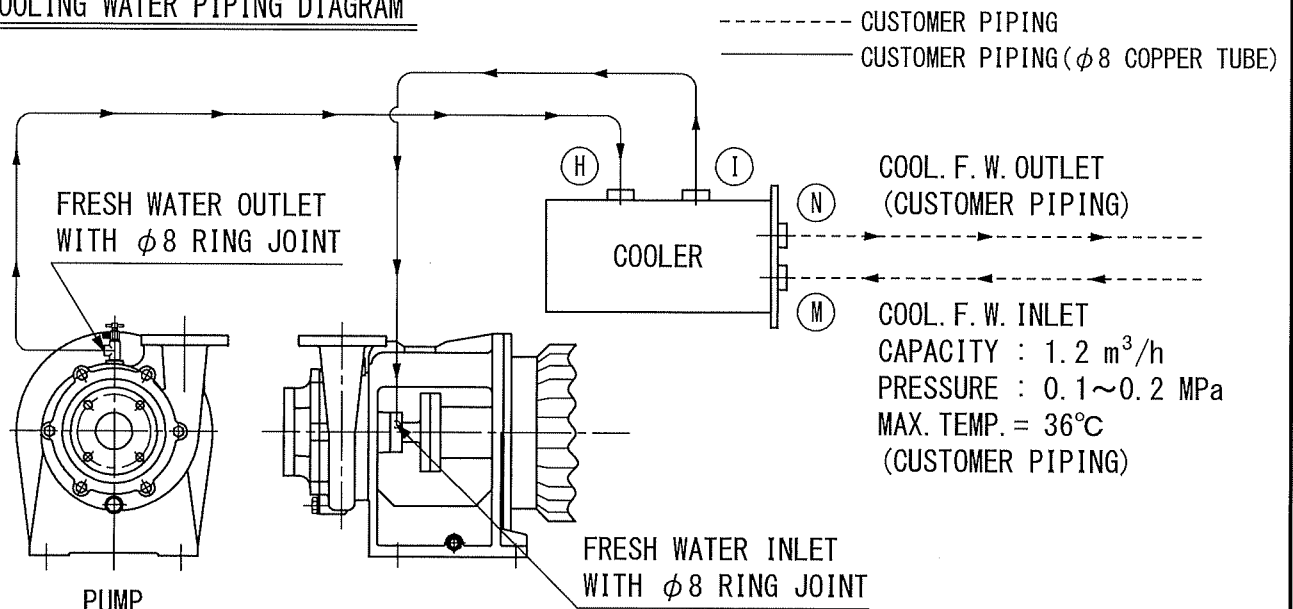
4 JUN 2021



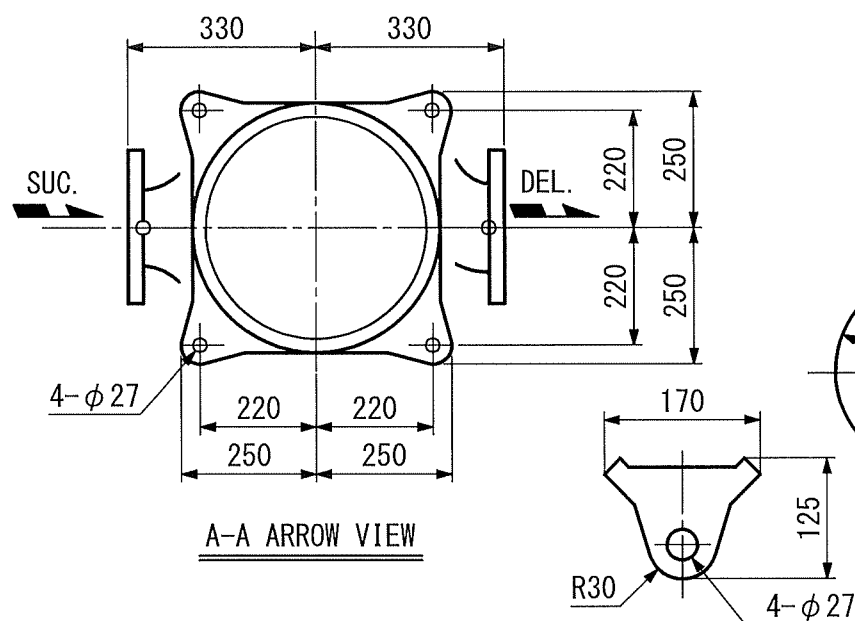
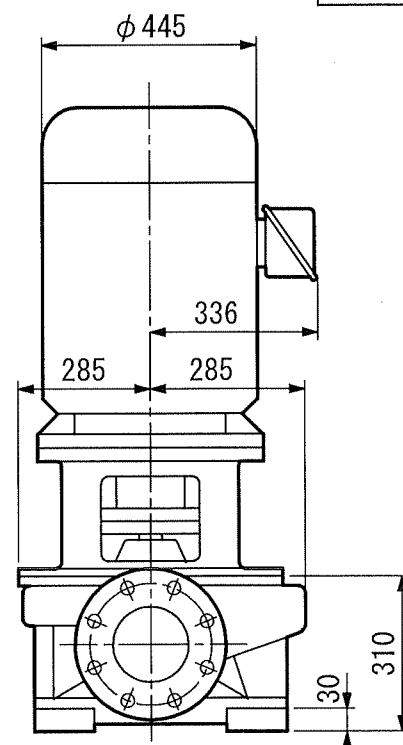
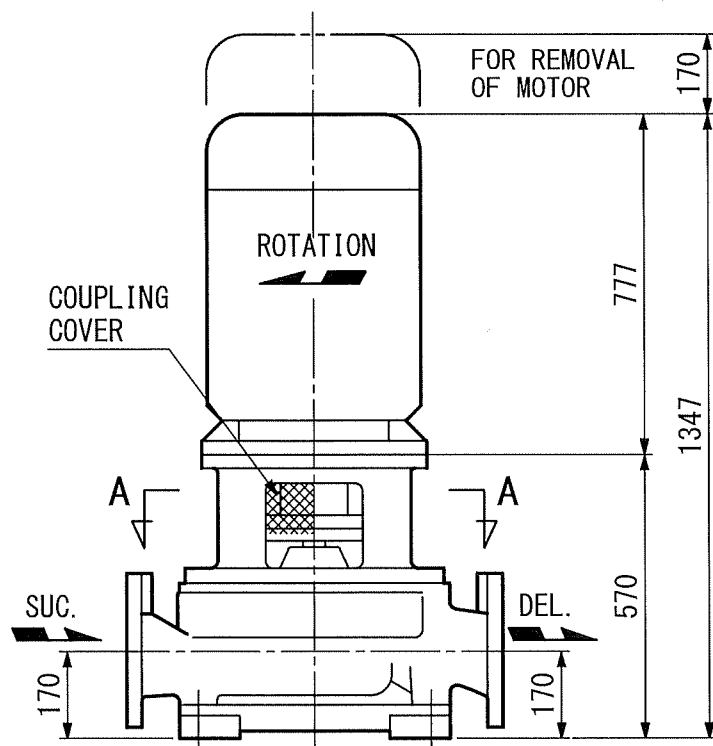
MASS
8 kg

13	RING JOINT	2	COPPER ALLOY ROD
12	BUSHING	2	COPPER ALLOY ROD
11	CAP BOLT	8	CHROMIUM MOLYBDEN. STEEL
10	GASKET	1	NON ASBESTOS
9	GASKET	1	NON ASBESTOS
8	SADDLE, U-BOLT&NUT	2 SETS	MILD STEEL
7	TIE ROD & SPACER	1 SET	MILD STEEL
6	BAFFLE PLATE	1 SET	MILD STEEL
5	COOLING TUBE	26	COPPER
4	TUBE SHEET	2	MILD STEEL
3	RETURN CHAMBER	1	CAST IRON
2	WATER CHAMBER	1	CAST IRON
1	SHELL	1	CARBON STEEL PIPE
NO.	NAME OF PART	Q'TY	MATERIAL

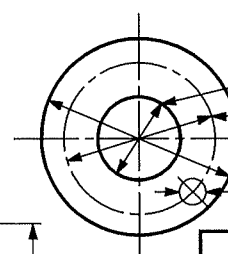
COOLING WATER PIPING DIAGRAM



SPARE PARTS LIST							PAGE		8	
M/E JACKET WATER PREHEATER PUMP							HORIZONTAL CENTRIFUGAL PUMP MODEL: BHR-65 1 SET			
							SHIP NO.			
							BOX NO.			
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
				WORK-ING PER PUMP	SPARE PER SHIP	B8DS0191				
							NO.	PART NO.		
1	MECHANICAL SEAL	0.08		1 SET	1 SET			520		
MFR' S NAME & ADDRESS			NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.		
								98DP0470		



SUC. & DEL. FLANGE



JIS 5 K SUC.	JIS 5 K DEL.
$\phi 200$	$\phi 200$
$\phi 280$	$\phi 280$
$\phi 320$	$\phi 320$
8- $\phi 23$	8- $\phi 23$

MASS

PUMP	250	kg
MOTOR	328	kg
TOTAL	578	kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	200 mm	NO. 1 & NO. 2 L/T. C. W. BOOSTER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	200 mm		
AIR COCK	1	CAPACITY	250 m ³ /h		
SEALING PIPE	1set	TOTAL PRESSURE	0. 4 MPa	PUMP MODEL	FEV-200E1
PRESSURE GAUGE WITH JOINT	1set	SUCTION PRESSURE	+0. 1 MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	1750 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0. 9 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	49kW, 4P	SCALE	FREE
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	C2DG0030
		REMARKS		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		WORKING TEMP. = 36 °C			



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

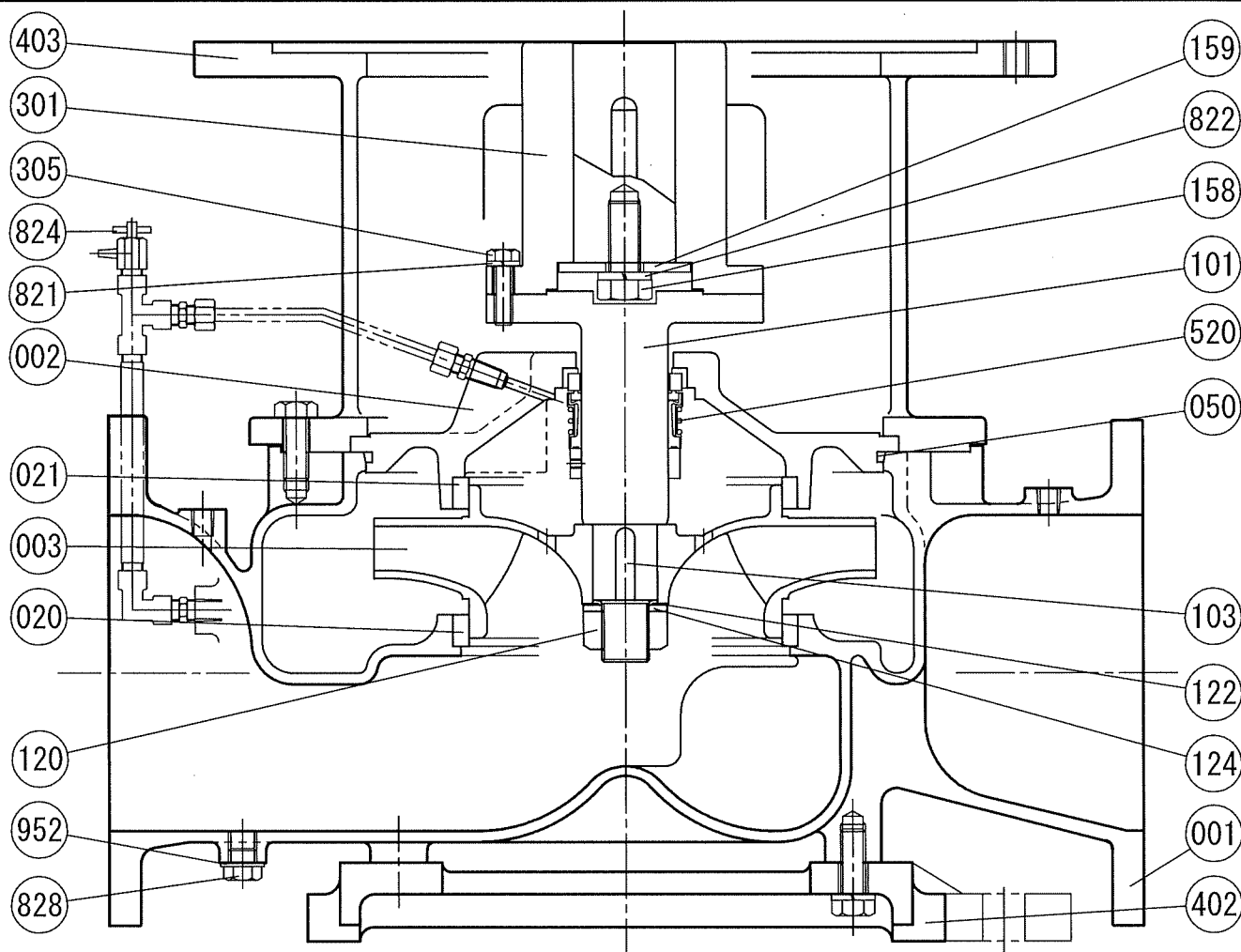
MODEL FEV-200/250E1

DRAWING NO.

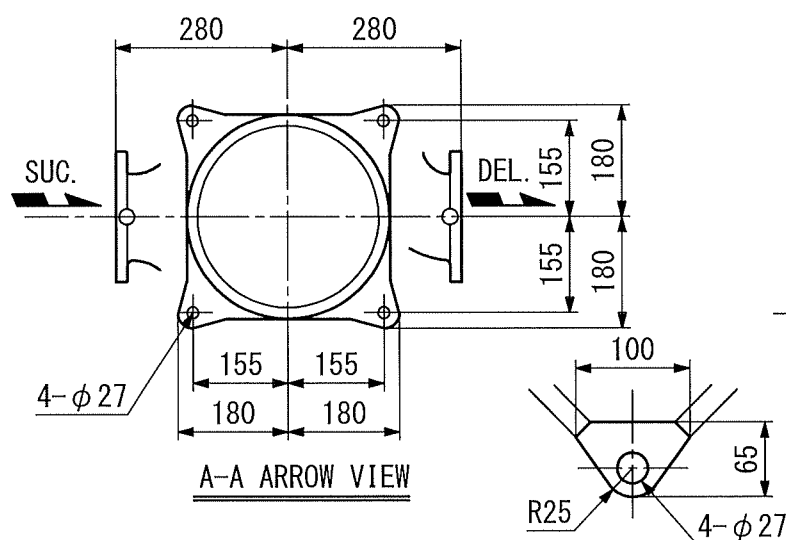
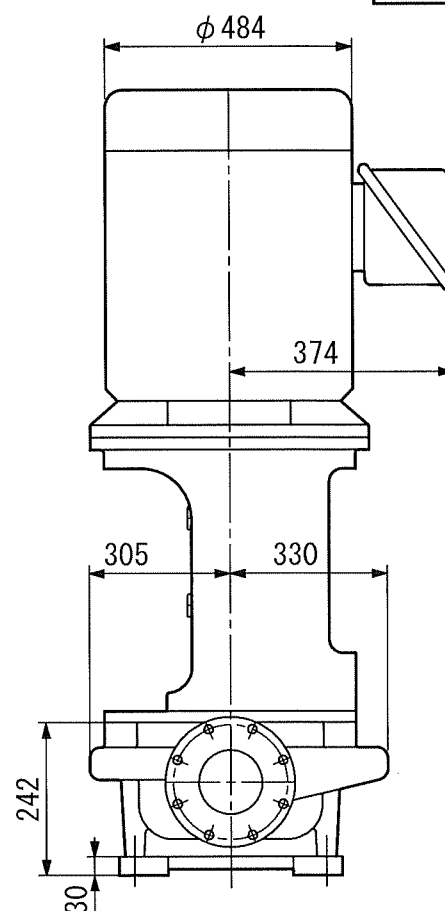
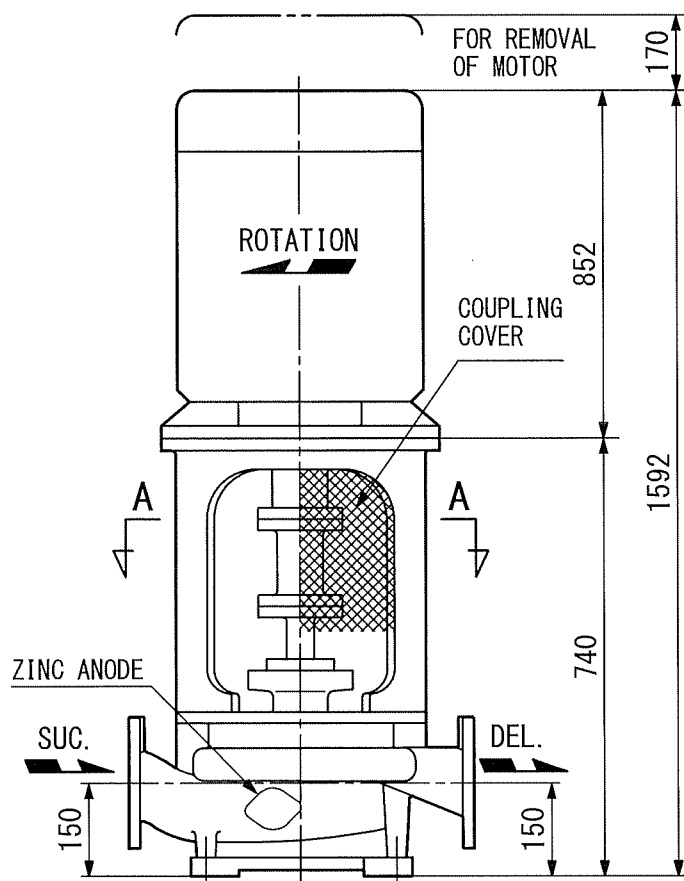
C2DS0153

DATE

18 MAR. 2022



158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	IMPELLER WASHER	STAINLESS STEEL	SUS304	1	952	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	822	COUPLING SET WASHER	HIGH CARBON STEEL WIRE	SWRH	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	520	MECHANICAL SEAL			1 SET
021	CASING RING	CAST BRONZE	CAC406	1	403	FRAME	CAST IRON	FC200	1
020	CASING RING	CAST BRONZE	CAC406	1	402	PUMP BED	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	PLATE WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SUC. & DEL. FLANGE

JIS 10K SUC.	JIS 10K DEL.
$\phi 100$	$\phi 100$
$\phi 175$	$\phi 175$
$\phi 210$	$\phi 210$
8- $\phi 19$	8- $\phi 19$

MASS

PUMP	150	kg
MOTOR	405	kg
TOTAL	555	kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	100 mm	EMERGENCY FIRE PUMP	
COUPLING & COVER	1set	DELIVERY BORE	100 mm		
AIR COCK	1	CAPACITY	72 m ³ /h	PUMP MODEL	FGV-100-2
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	1.0 MPa	QUANTITY	1
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	-0.05 MPa	RULE	LR
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	ANGLE PROJ.	3 rd
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	1.7 MPa	SCALE	FREE
GAUGE BOARD	1	MOTOR OUTPUT	45kW, 2P	DRAWING NO.	B9DGO155
ZINC ANODE	1	POWER SOURCE	440V, 60Hz, 3ph	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
COUNTER FLANGE	1set	REMARKS			



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

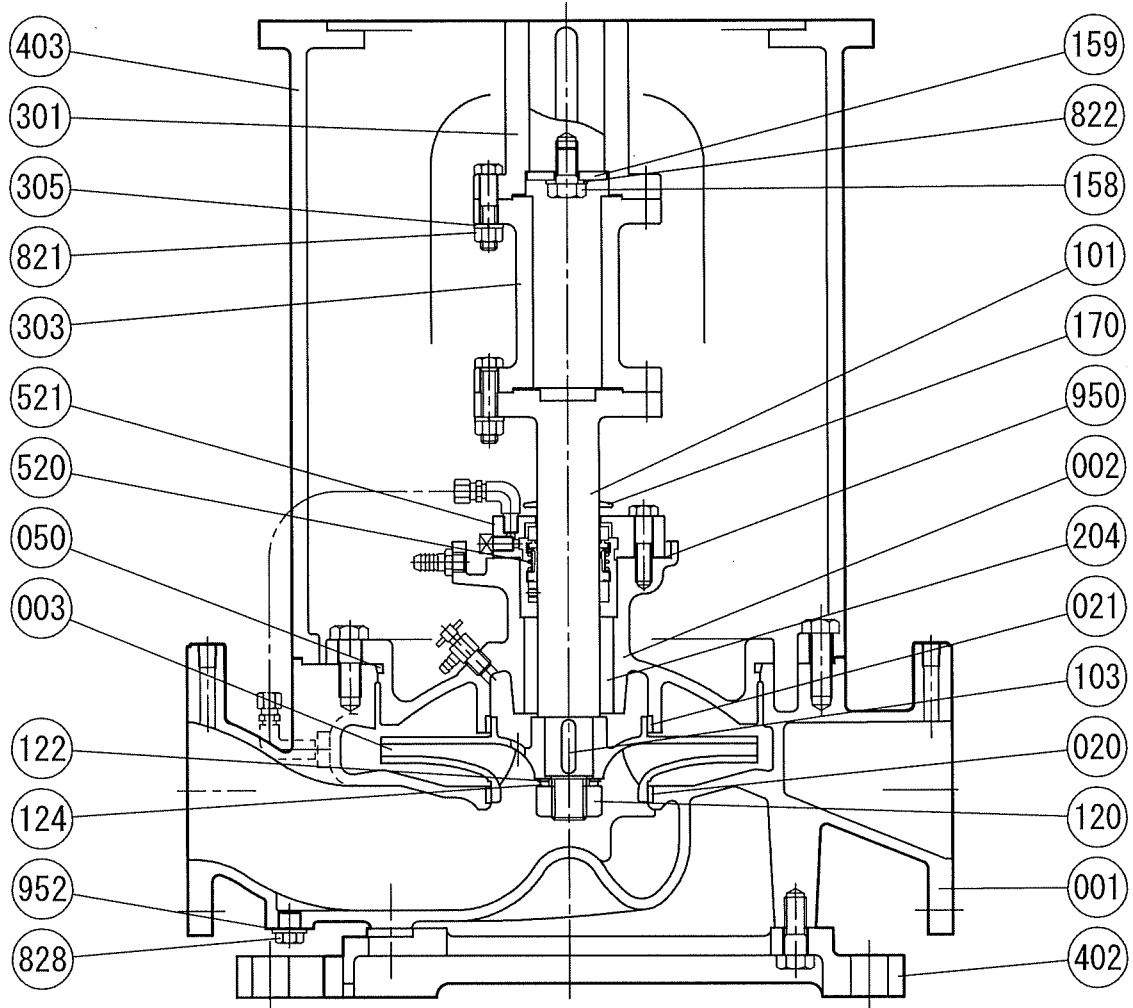
MODEL : FGV-100-2

DRAWING NO.

B8DS0217

DATE

29 JUN. 2018



170	DUST STOPPER	RUBBER		1					
159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER NUT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS304	1	520	MECHANICAL SEAL			1 SET
050	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	402	PUMP BED	CAST IRON	FC200	1
020	CASING RING	SYNTHETIC RESIN		1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
003	IMPELLER	STAINLESS STEEL	SCS14	1	303	COUPLING SPACER	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST BRONZE	CAC402	1	204	BEARING METAL	CARBON		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



SECTIONAL DRAWING

HORIZONTAL VACUUM PUMP

MODEL : VK-20NC

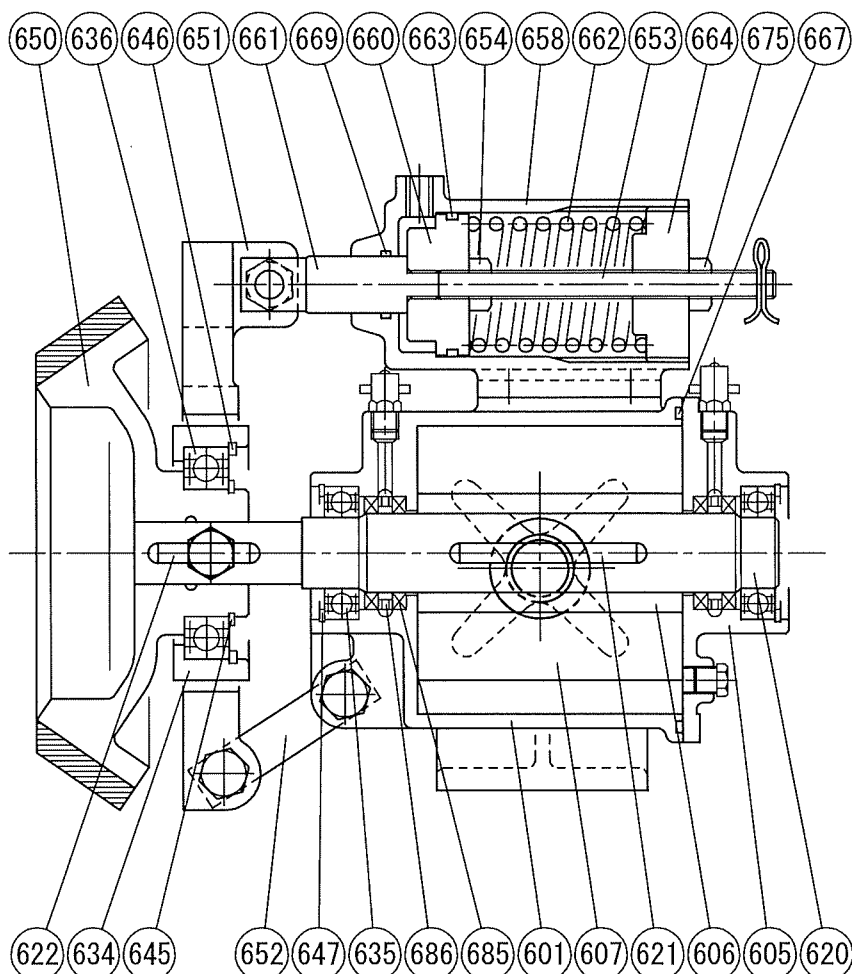
DRAWING NO.

C1DS0308

DATE

6 OCT. 2021

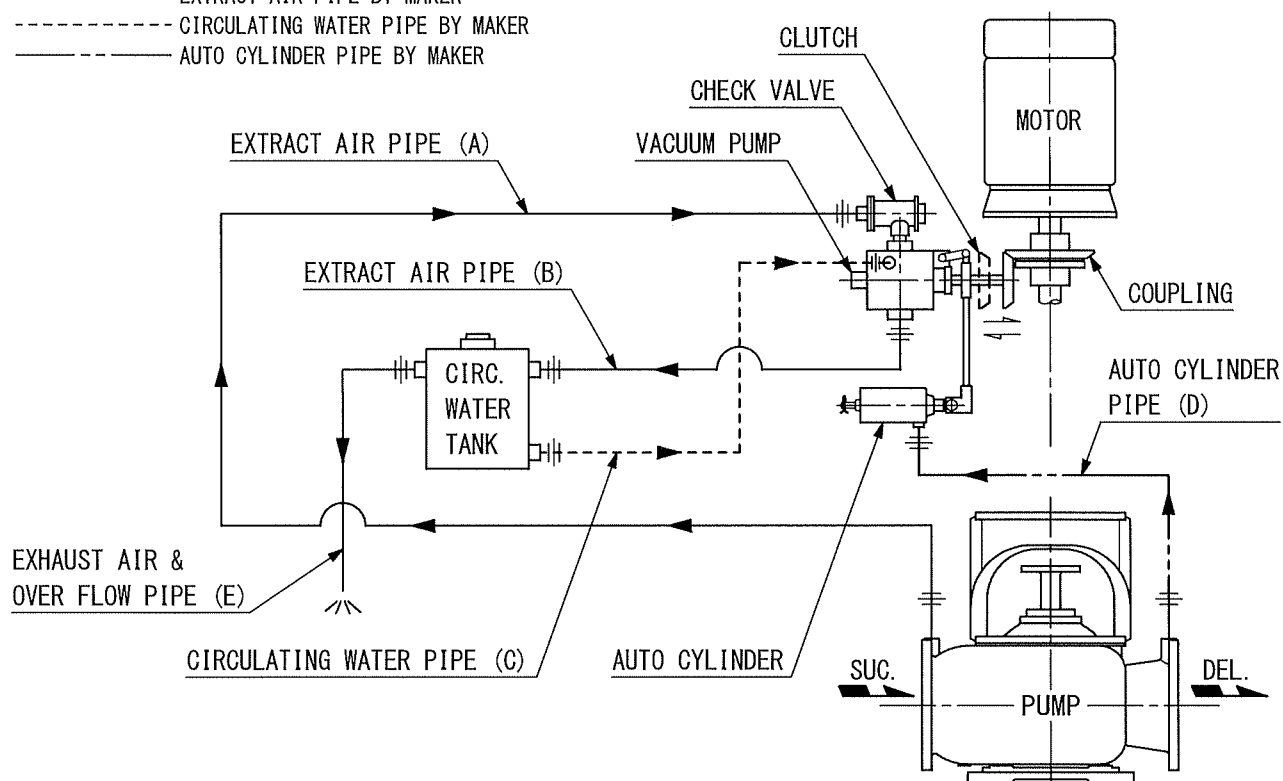
686	SEAL RING	MILD STEEL	SS400	2
685	OIL SEAL	RUBBER		4
675	NUT	MILD STEEL	SS400	1
669	O-RING	RUBBER		1
667	O-RING	RUBBER		1
664	SPRING CARRIER	COPPER ALLOY ROD	C3604	1
663	O-RING	RUBBER		1
662	SPRING	PIANO WIRE	SWP	1
661	PISTON ROD	COPPER ALLOY ROD	C3604	1
660	PISTON	COPPER ALLOY ROD	C3604	1
658	CYLINDER	CAST BRONZE	CAC406	1
654	LOCK NUT	MILD STEEL	SS400	1
653	TIE ROD	MILD STEEL	SS400	1
652	LEVER	MILD STEEL	SS400	1
651	SHIFTER	CAST IRON	FC200	1
650	CLUTCH			1
647	SNAP RING	CARBON TOOL STEEL	SK5	2
646	SNAP RING	CARBON TOOL STEEL	SK5	1
645	SNAP RING	CARBON TOOL STEEL	SK5	1
636	BALL BEARING		SUJ	1
635	BALL BEARING		SUJ	2
634	BEARING CASE	CAST IRON	FC200	1
622	CLUTCH KEY	CARBON STEEL	S45C	1
621	ROTOR KEY	STAINLESS STEEL	SUS304	1
620	SHAFT	STAINLESS STEEL	SUS304	1
607	VANE	BAKELITE		4
606	ROTOR	CAST BRONZE	CAC406	1
605	END COVER	CAST BRONZE	CAC406	1
601	CASING	CAST BRONZE	CAC406	1
PART NO.	NAME OF PART	MATERIAL		Q' TY



* MARKS OF PIPING

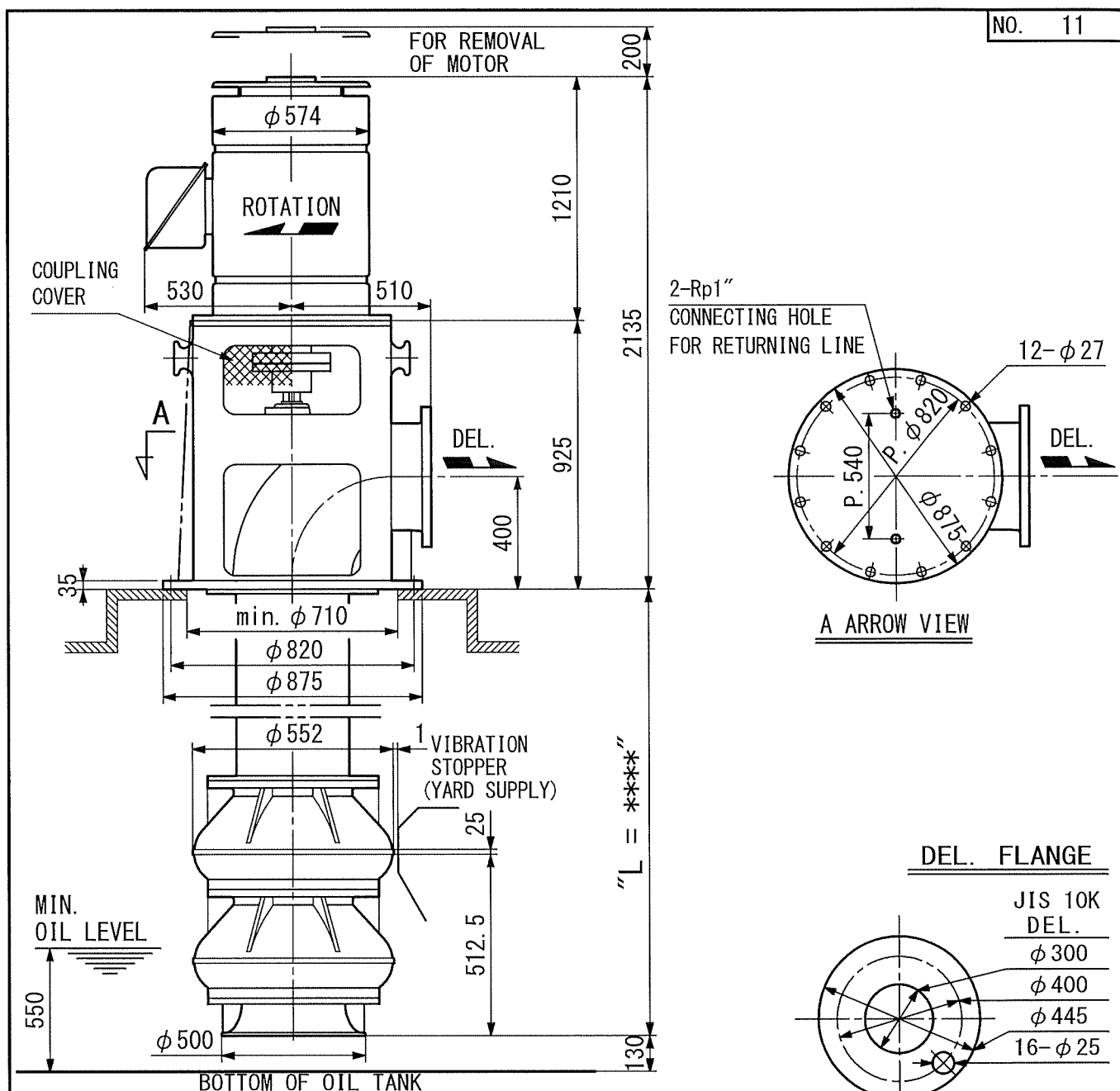
- EXTRACT AIR PIPE BY MAKER
- CIRCULATING WATER PIPE BY MAKER
- AUTO CYLINDER PIPE BY MAKER

PIPE LINE DIAGRAM



SPARE PARTS LIST							PAGE		10	
EMERGENCY FIRE PUMP VERTICAL CENTRIFUGAL PUMP MODEL: FGV-100-2 1 SET							SHIP NO.			
							BOX NO.			
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
				WORK-ING PER PUMP	SPARE PER SHIP	B8DS0217				
		NO.								
1	BEARING METAL		CARBON	1	1		204			
2	MECHANICAL SEAL			1 SET	1 SET		520			
3	ZINC ANODE		ZINC	1	1	B9DS0232	900			

MFR' S NAME & ADDRESS	NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		B8DP0102



1. Please select "L" dimension from the attached sheet.
2. VIBRATION STOPPER (YARD SUPPLY) to be attached if size "L" is longer than 1.8m.
3. LINE BEARING to be attached if size "L" is longer than 2.1m

MASS	
PUMP	1470 kg
MOTOR	1090 kg
TOTAL	2560 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	DELIVERY BORE	300 mm	NO. 1 & NO. 2 MAIN L. O. PUMP	
COUPLING & COVER	1set	CAPACITY	540 m ³ /h		
PRESSURE GAUGE WITH JOINT	1set	TOTAL PRESSURE	0. 69 MPa		
GAUGE ROOT VALVE	1	SUCTION PRESSURE	MPa	PUMP MODEL	TOM-300-2
GAUGE BOARD	1	REVOLVING SPEED	1750 min ⁻¹	QUANTITY	2
COUNTER FLANGE	1set	HYD. TEST PRESS.	1. 5 MPa	RULE	LR
		MOTOR OUTPUT	199kW , 4P	ANGLE PROJ.	3 rd
		POWER SOURCE	440V, 60Hz, 3ph	SCALE	FREE
		REMARKS		DRAWING NO.	C1DG0159
		GUARANT. VIS. FOR POWER 260 mm ² /s		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		GUARANT. VIS. FOR CAPA. 26 mm ² /s			
		OIL TEMP. = 50 °C			

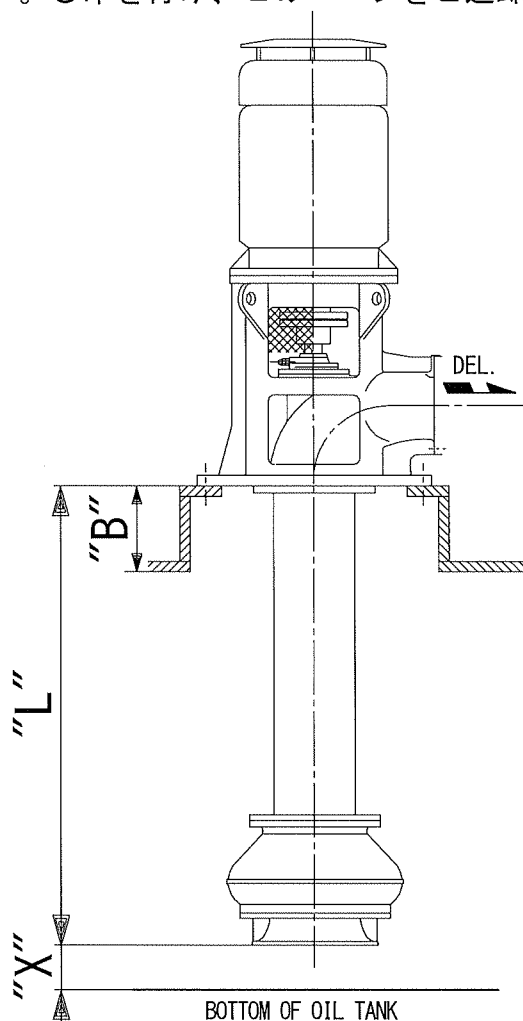
Shipyard : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.
造船所

Ship No. : H1933A/1934A/1935A/1936A/1937A
船番

※Please select "L" dimension by following tables. Please mark "O" and return this page.

“L” 寸法を下の一覧表から選択下さい。○印を付け、このページをご返却ください。

“L”	
	1000
	1100
	1200
	1300
	1400
	1500
	1600
	1700
	1800
	1900
	2000
	2100
	2200
	2300
	2400
	2500
	2600
	2700
	2800
	2900
	3000



1. Please inform us "L" dimension of Pump.
ポンプのL寸法をお知らせ下さい。
2. Please select the suitable length of "L" dimension from above list, because total length of pump is designed in every 100mm. For the dimension adjustment of less than 100mm, please adjust it by "B" dimension of mounting base at hull side.
ポンプの全長は100mm毎に設計されている為、上記のリストから適切な「L」寸法を選択して下さい。
100mm未満の寸法調整は、船体側架台「B」寸法にて調整して下さい。
3. Please keep "X" mm of above for the space between pump and bottom of oil tank.
ポンプとオイルタンク底面の隙間は「X」mm以上確保して下さい。
For "X" dimension, please refer to the outline drawing.
「X」寸法については外形図を参照して下さい。



naniwa pump
Manufacturing Co., Ltd.



SECTIONAL DRAWING

VERTICAL TANK MOUNT CENTRIFUGAL PUMP

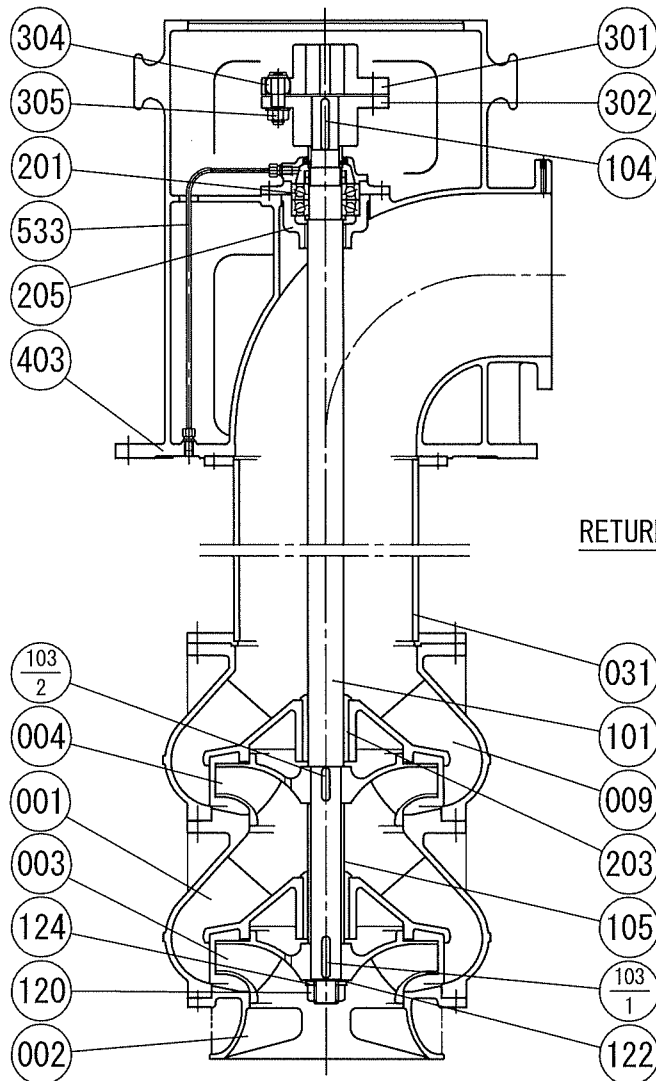
MODEL: TOM-300-2

DRAWING NO.

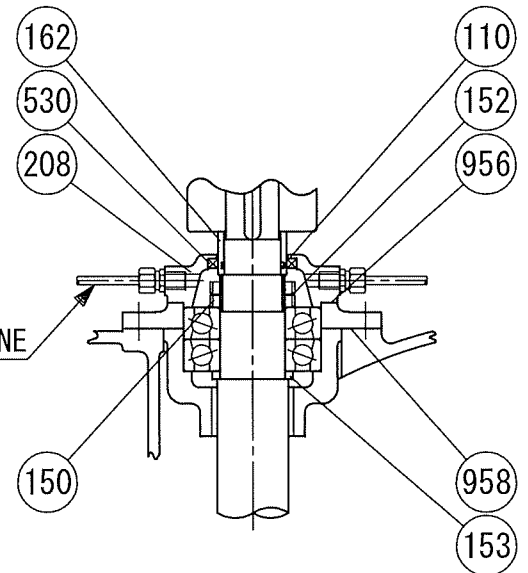
C1DS0213

DATE

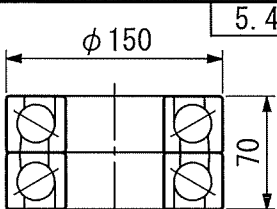
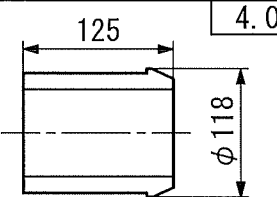
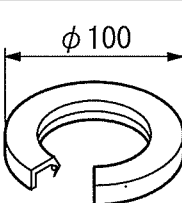
13 JUL. 2021

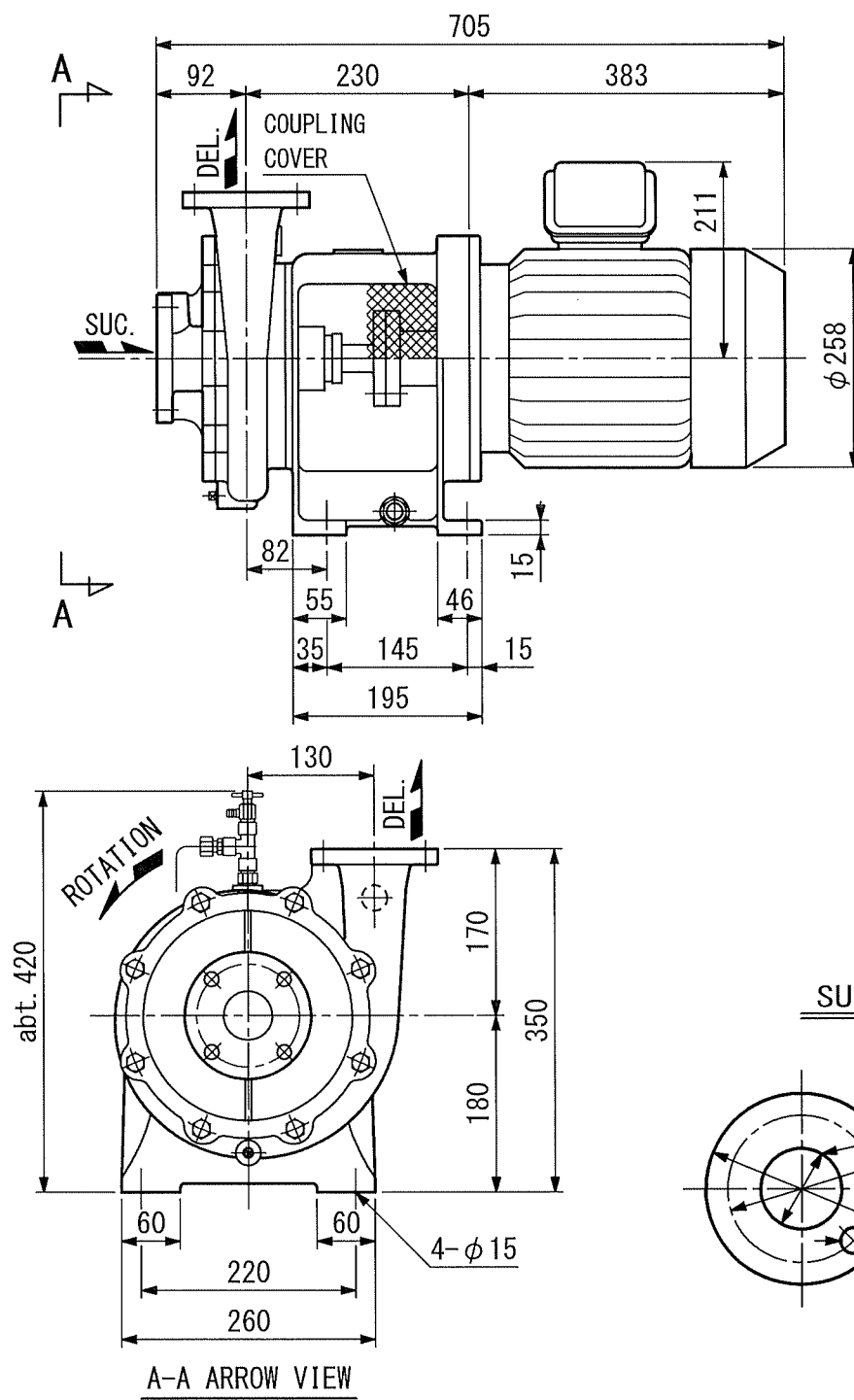


RETURNING LINE



150	BEARING NUT	MILD STEEL	SS400	2	958	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1	956	GASKET	NON ASBESTOS FIBER		1
122	IMPELLER PLATE WASHER	MILD STEEL	SS400	1	533	FLEXIBLE HOSE	STAINLESS STEEL		2
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	530	OIL SEAL	RUBBER		1
110	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
105	SLEEVE	CARBON STEEL	S45C	1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	1 SET
104	COUPLING KEY	CARBON STEEL	S45C	1	304	COUPLING RING	RUBBER		1 SET
103/2	IMPELLER KEY	CARBON STEEL	S45C	1	302	COUPLING	CAST IRON	FC200	1
103/1	IMPELLER KEY	CARBON STEEL	S45C	1	301	COUPLING	CAST IRON	FC200	1
101	SHAFT	CARBON STEEL	S45C	1	208	BEARING COVER	CAST IRON	FC200	1
031	DELIVERY PIPE	CARBON STEEL PIPE		1	205	BEARING HOUSING	CAST IRON	FC200	1
009	DELIVERY CASING	CAST IRON	FC200	1	203	SUBMERGED BEARING	SPECIAL BRONZE		2
004	IMPELLER	CAST BRONZE	CAC406	1	201	BALL BEARING	SUJ		1 SET
003	IMPELLER	CAST BRONZE	CAC406	1	162	COUPLING COLLAR	CARBON STEEL	S45C	1
002	SUCTION COVER	CAST IRON	FC200	1	153	COLLAR	CARBON STEEL	S45C	1
001	CASING	CAST IRON	FC200	1	152	BEARING WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		11		
NO. 1 & NO. 2 MAIN L. O. PUMP							VERTICAL TANK MOUNT CENTRIF. PUMP				
							MODEL: TOM-300-2 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	C1DS0213				
							NO.	PART NO.			
1	BALL BEARING		5.4	SUJ	1 SET	1 SET		201	NO. 7314DF		
2	SUBMERGED BEARING		4.0	SPECIAL BRONZE	2	1		203			
3	OIL SEAL		0.02	RUBBER	1	1		530			



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	50 mm	NO. 1 & NO. 2 FRESH WATER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	50 mm		
AIR COCK	1	CAPACITY	10 m ³ /h	PUMP MODEL	BHR-50-3
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.7 MPa	QUANTITY	2
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	RULE	
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	ANGLE PROJ.	3 rd
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	1.2 MPa	SCALE	FREE
GAUGE BOARD	1	MOTOR OUTPUT	7.5kW , 2P	DRAWING NO.	B1DG0162
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph		
		REMARKS		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		WORKING TEMP = 36 °C			



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

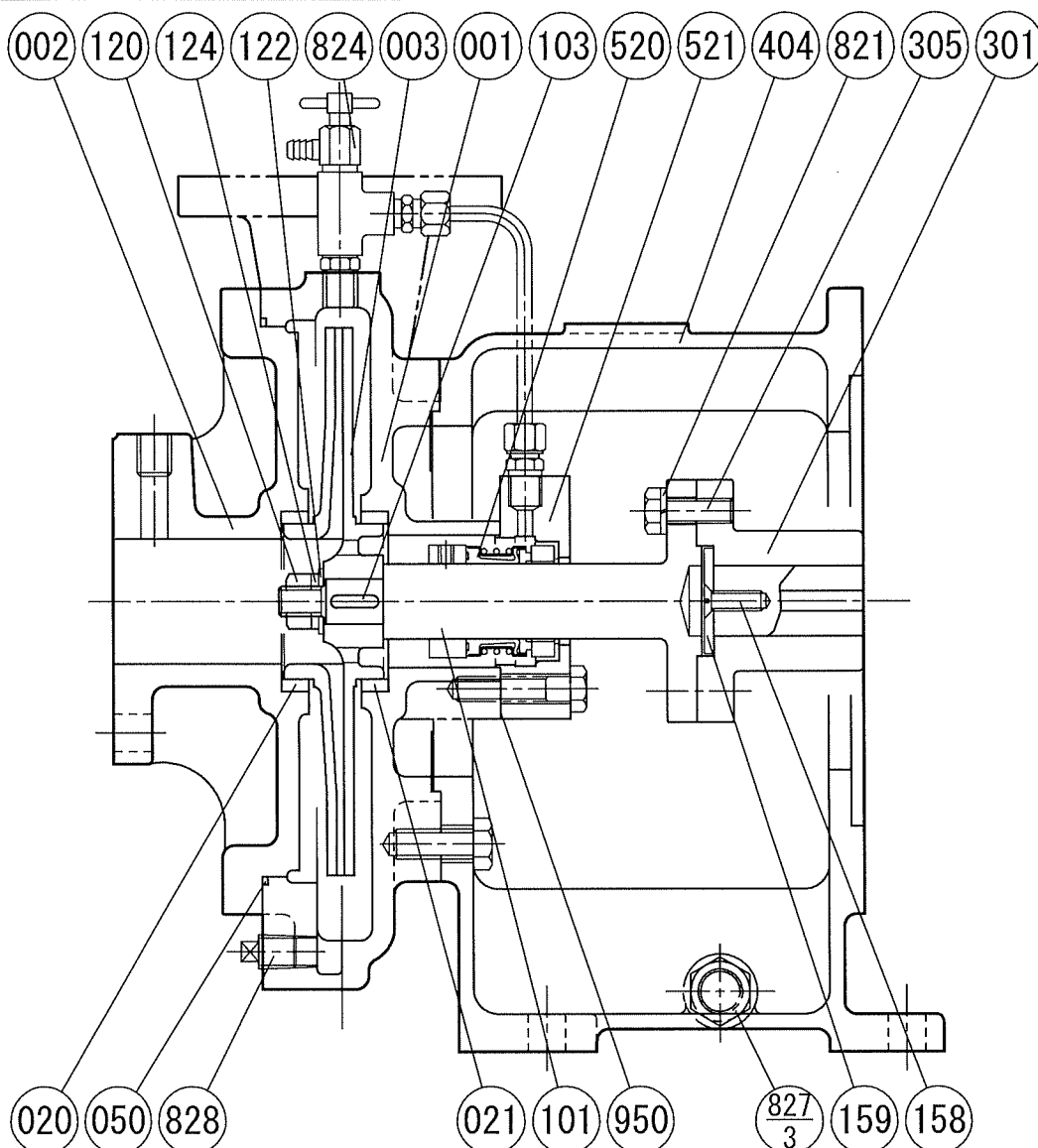
MODEL BHR-50-3

DRAWING NO.

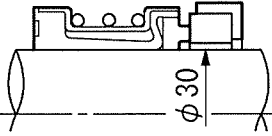
B6DS0522

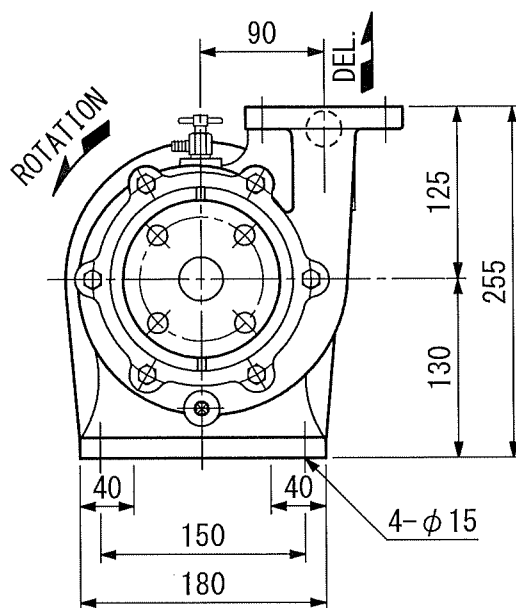
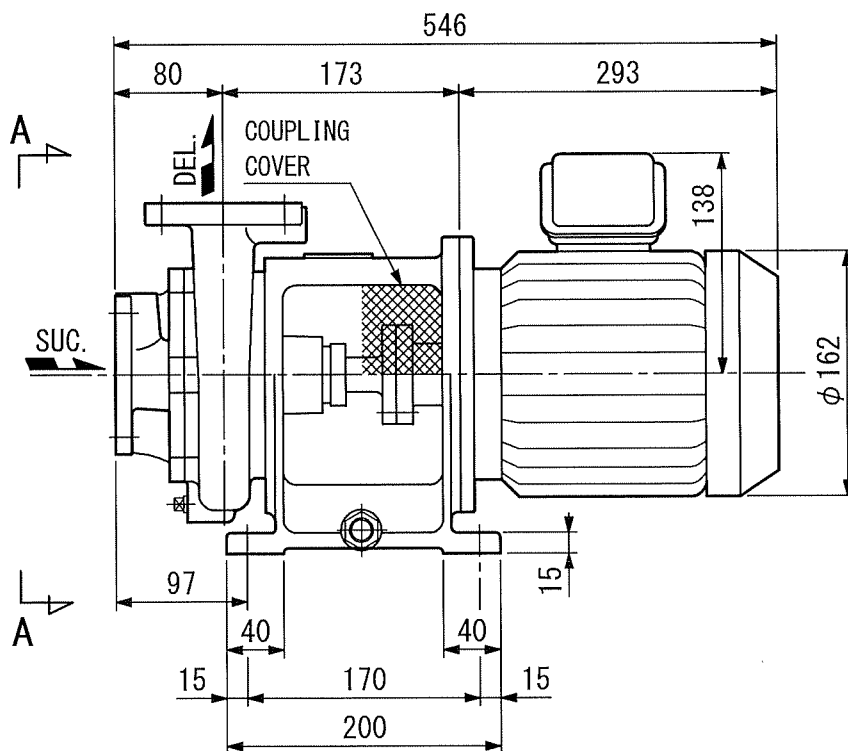
DATE

5 OCT. 2016



158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	827/3	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
021	CASING RING	CAST BRONZE	CAC406	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY		

SPARE PARTS LIST							PAGE		12		
NO. 1 & NO. 2 FRESH WATER PUMP							HORIZONTAL CENTRIFUGAL PUMP MODEL: BHR-50-3 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS			
				WORK-ING PER PUMP	SPARE PER SHIP	B6DS0522					
							NO.	PART NO.			
1	MECHANICAL SEAL	0.08 		1 SET	1 SET			520			
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.			
								99DP0406			



A-A ARROW VIEW

SUC. & DEL. FLANGE

JIS 5 K SUC.	JIS 5 K DEL.
φ 32	φ 32
φ 90	φ 90
φ 115	φ 115
4-φ 15	4-φ 15

MASS

PUMP	28 kg
MOTOR	20 kg
TOTAL	48 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	32 mm	NO. 1 & NO. 2 HOT WATER CIRCULATING PUMP	
COUPLING & COVER	1set	DELIVERY BORE	32 mm		
AIR COCK	1	CAPACITY	2 m ³ /h	PUMP MODEL	BHR-32
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.2 MPa	QUANTITY	2
PRESSURE GAUGE WITH JOINT	2sets	SUCTION HEAD	+0.4 MPa	RULE	
GAUGE ROOT VALVE	2	REVOLVING SPEED	3500 min ⁻¹	ANGLE PROJ.	3 rd
GAUGE BOARD	1	HYD. TEST PRESS.	1.0 MPa	SCALE	FREE
COUNTER FLANGE	1set	MOTOR OUTPUT	0.75kW , 2P	DRAWING NO.	A3DGO242
		POWER SOURCE	440V, 60Hz, 3ph	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		REMARKS			
		WORKING TEMP = 80 °C			



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

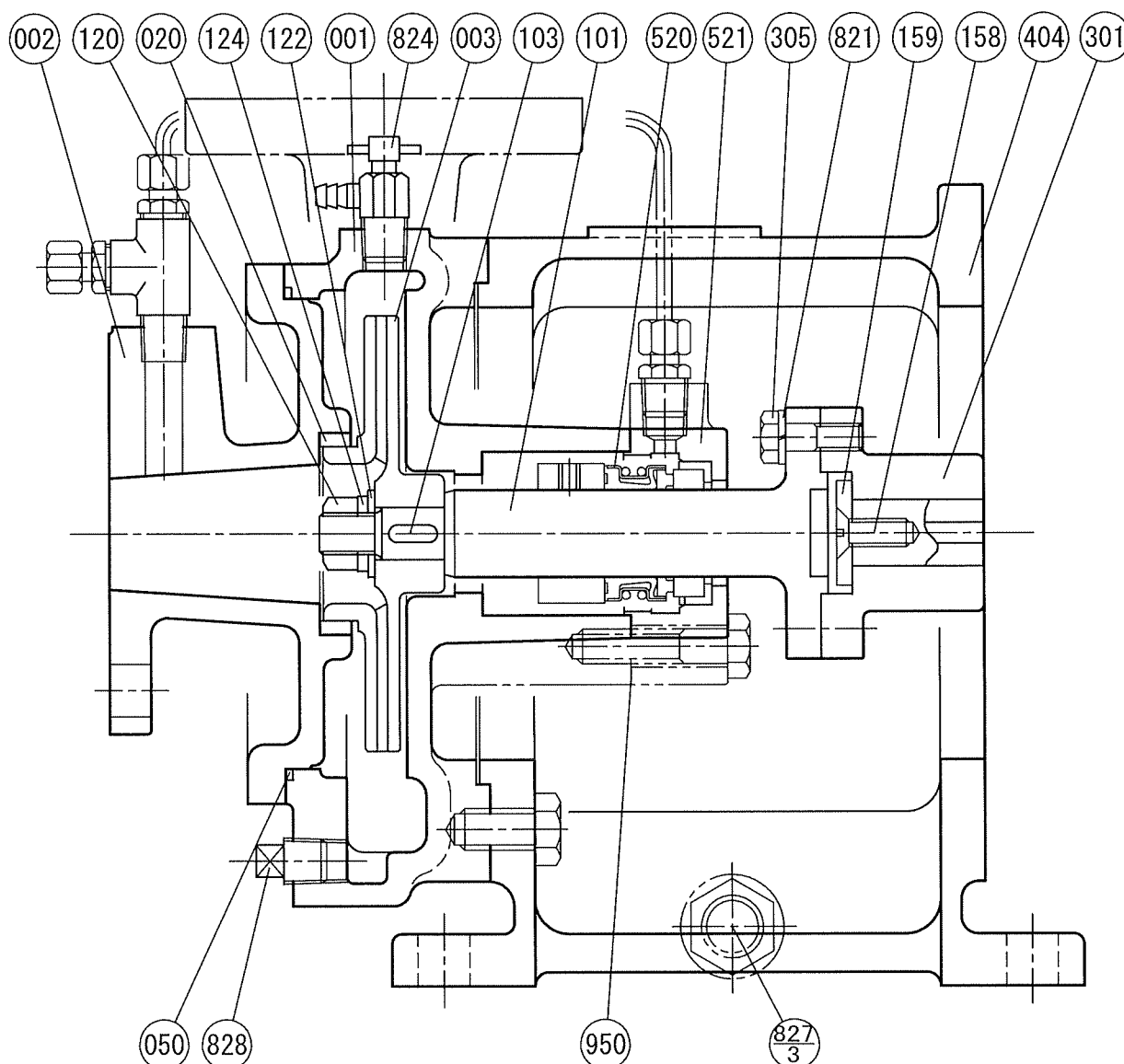
MODEL : BHR-32

DRAWING NO.

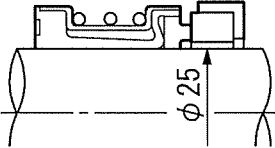
B6DS0493

DATE

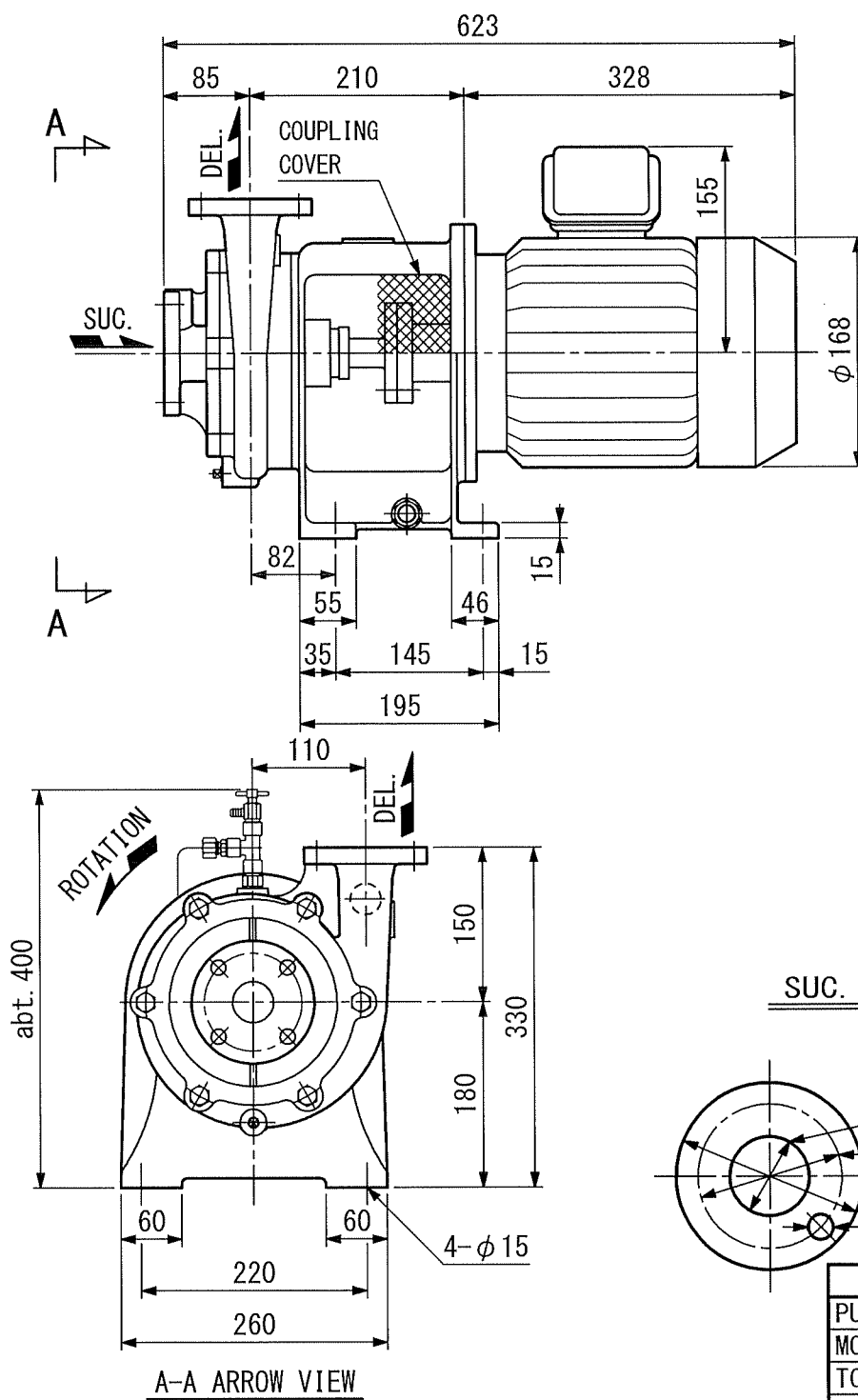
4 OCT. 2016



158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	827/3	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
101	SHAFT	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
050	O RING	RUBBER		1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	PUMP FRAME	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	COUPLING PLATE WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE	13
NO. 1 & NO. 2 HOT WATER CIRCULATING PUMP							HORIZONTAL CENTRIFUGAL PUMP MODEL: BHR-32 2 SETS	
							SHIP NO.	
							BOX NO.	
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS
				WORK- ING PER PUMP	SPARE PER SHIP	B6DS0493		
						NO.	PART NO.	
1	MECHANICAL SEAL	0.06 		1 SET	1 SET		520	

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		98DP0341



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	40 mm	CHEMICAL CLEANING PUMP	
COUPLING & COVER	1set	DELIVERY BORE	40 mm		
AIR COCK	1	CAPACITY	5 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.3 MPa	PUMP MODEL	BHR-40
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	QUANTITY	1
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.6 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	2.2kW , 2P	SCALE	FREE
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	99DG0085
		REMARKS		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		WORKING TEMP = 36 °C			



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

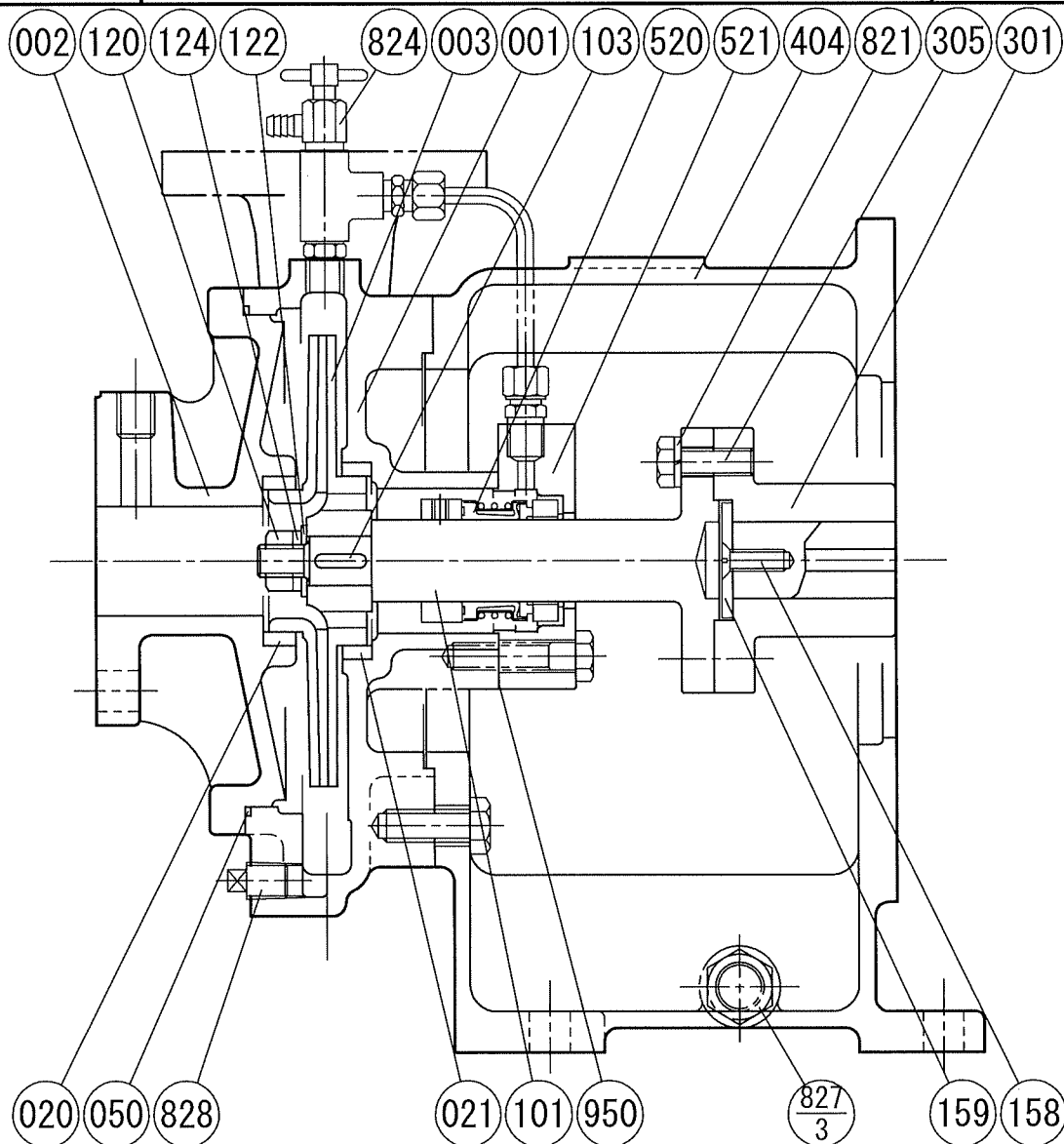
MODEL BHR-40

DRAWING NO.

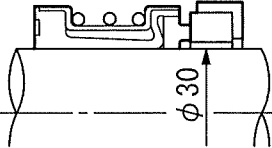
B6DS0500

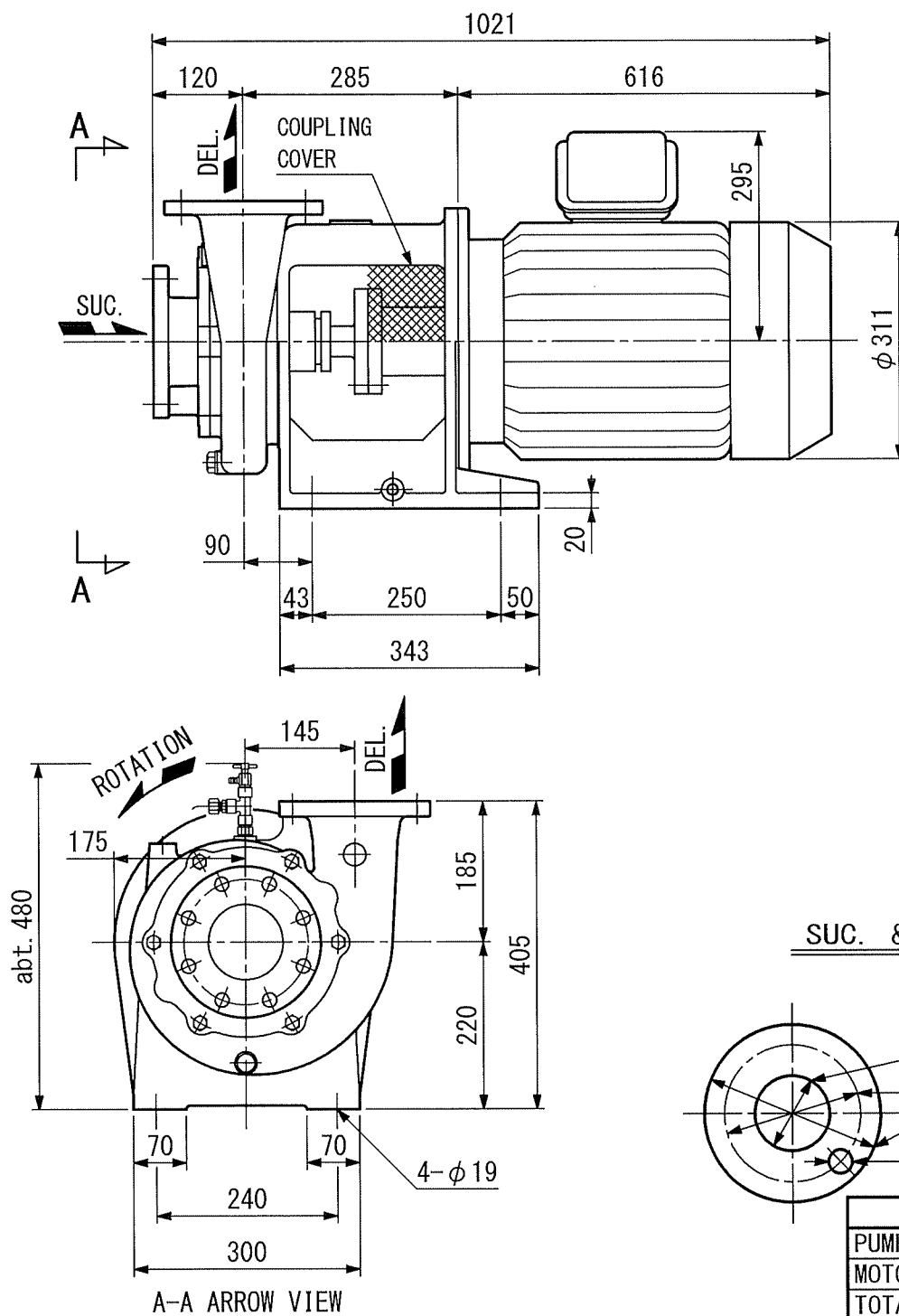
DATE

4 OCT. 2016

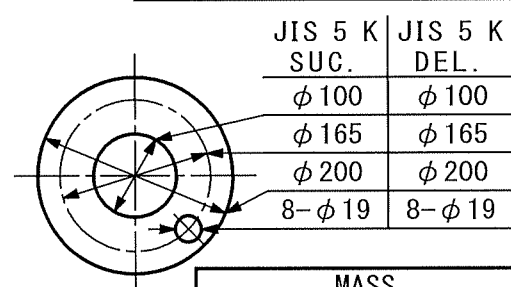


158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	$\frac{827}{3}$	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
021	CASING RING	CAST BRONZE	CAC406	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	CAST BRONZE	CAC406	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		14	
CHEMICAL CLEANING PUMP HORIZONTAL CENTRIFUGAL PUMP MODEL: BHR-40 1 SET							SHIP NO.			
							BOX NO.			
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
				WORK-ING PER PUMP	SPARE PER SHIP	B6DS0500				
							NO.	PART NO.		
1	MECHANICAL SEAL	0.08 		1 SET	1 SET			520		
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN				DRAWING NO.			
							98DP0468			



SUC. & DEL. FLANGE



JIS 5 K SUC.	JIS 5 K DEL.
$\phi 100$	$\phi 100$
$\phi 165$	$\phi 165$
$\phi 200$	$\phi 200$
8- $\phi 19$	8- $\phi 19$

MASS

PUMP	75 kg
MOTOR	206 kg
TOTAL	281 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	100 mm	NO. 1 & NO. 2 HOT GLYCOL WATER CIRCULATING PUMP	
COUPLING & COVER	1set	DELIVERY BORE	100 mm		
AIR COCK	1	CAPACITY	75 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.4 MPa	PUMP MODEL	BHR-100
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.8 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	18.5kW , 2P	SCALE	FREE
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	A7DGO110
		REMARKS	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT		
		WORKING TEMP = -30~100 °C			



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

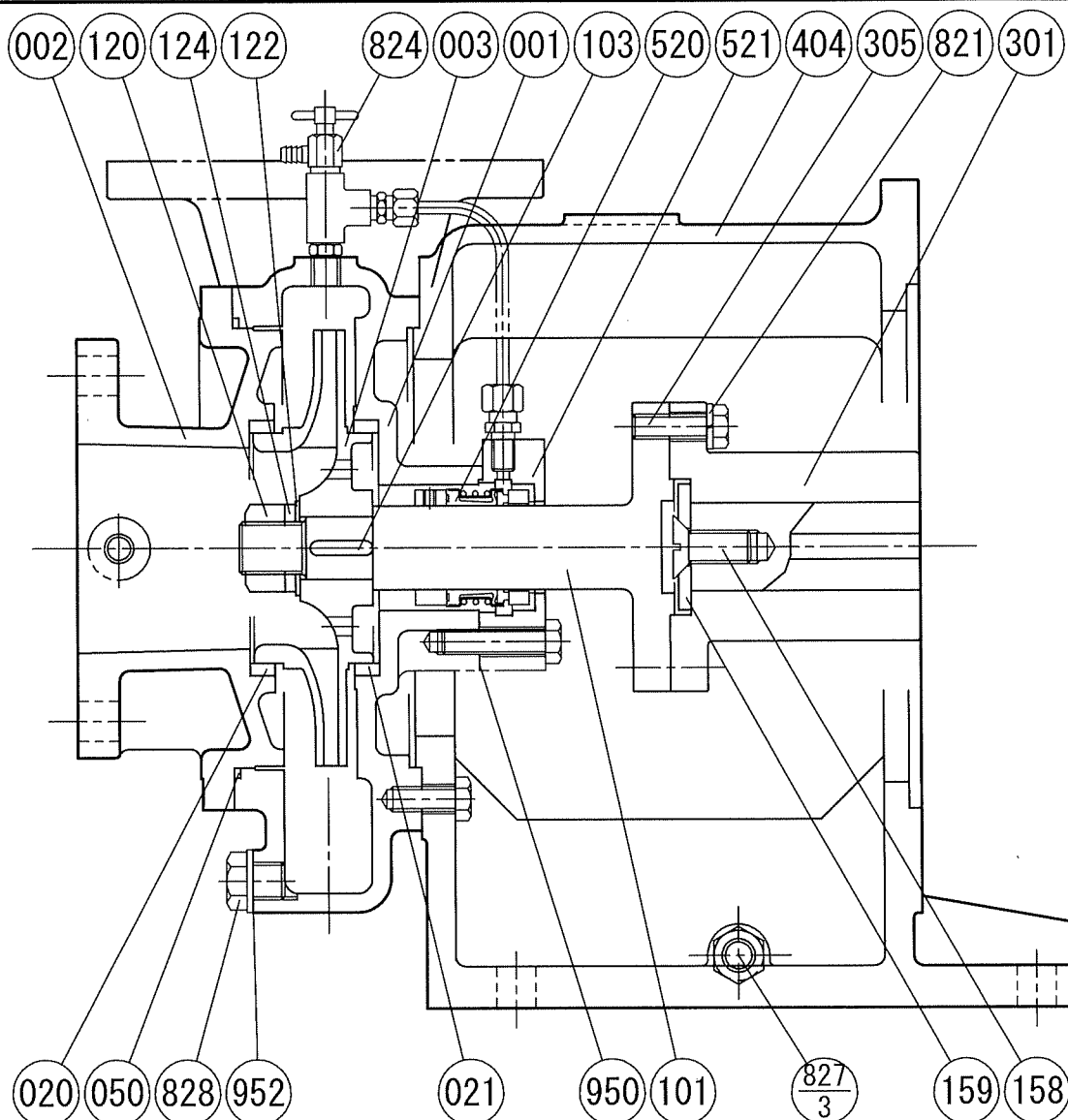
MODEL BHR-100

DRAWING NO.

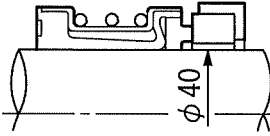
B6DS0363

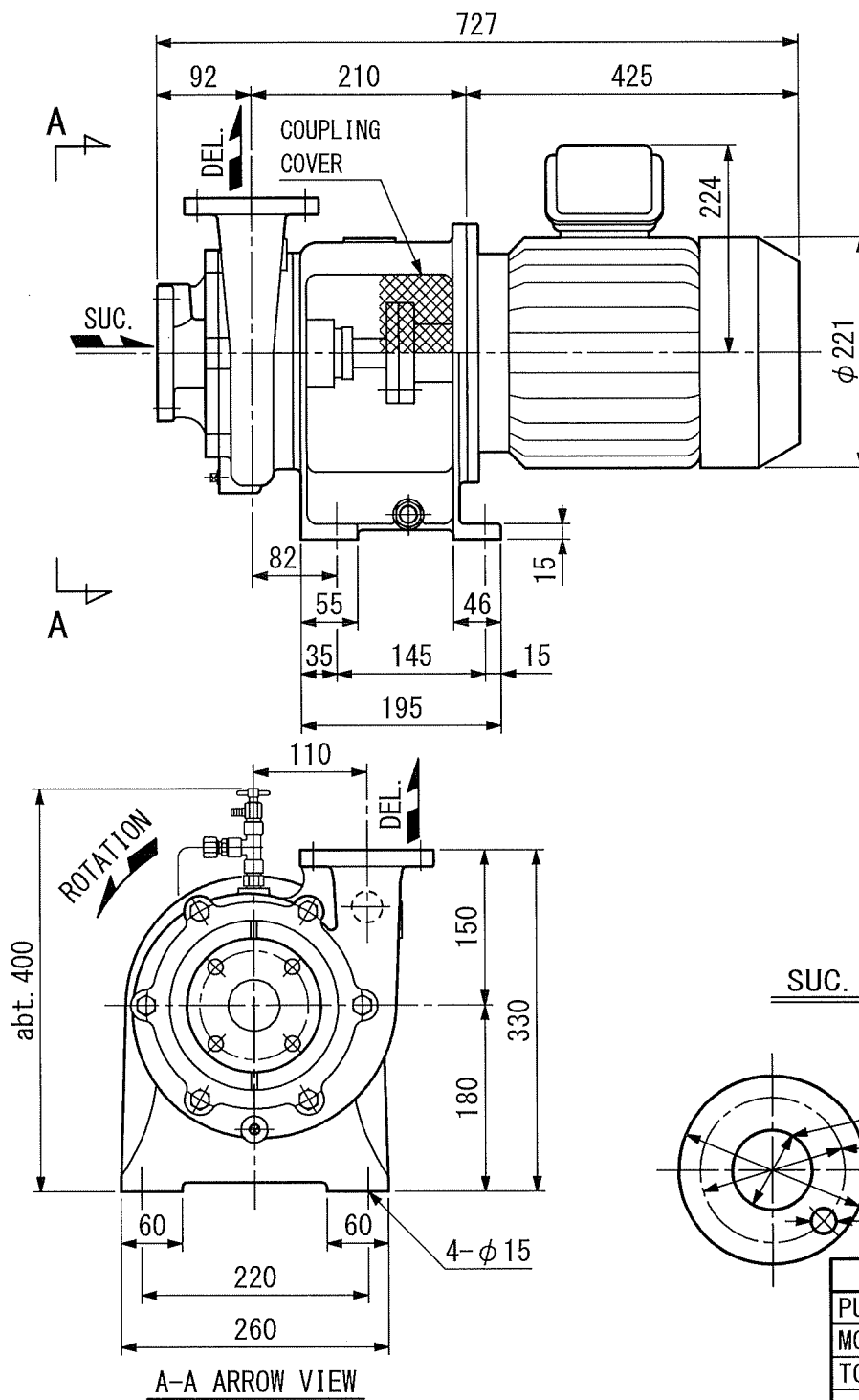
DATE

28 SEP. 2016



158	COUPLING SET BOLT	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	827/3	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
021	CASING RING	CAST BRONZE	CAC406	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	PHOSPHOR BRONZE	CAC502A	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST BRONZE	CAC402	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST BRONZE	CAC402	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		15		
NO. 1 & NO. 2 HOT GLYCOL WATER CIRCULATING PUMP							HORIZONTAL CENTRIFUGAL PUMP				
							MODEL : BHR-100 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	B6DS0363				
								NO.	PART NO.		
1	MECHANICAL SEAL		0.13		1 SET	1 SET		520			
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.			
								98DP0471			



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	50 mm	NO. 1 & NO. 2 COLD GLYCOL WATER CIRCULATING PUMP	
COUPLING & COVER	1set	DELIVERY BORE	50 mm		
AIR COCK	1	CAPACITY	17 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.35 MPa	PUMP MODEL	BHR-50
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.8 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	3.7kW, 2P	SCALE	FREE
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	98DG0762
		REMARKS	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT		
		WORKING TEMP = -30~100 °C			



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

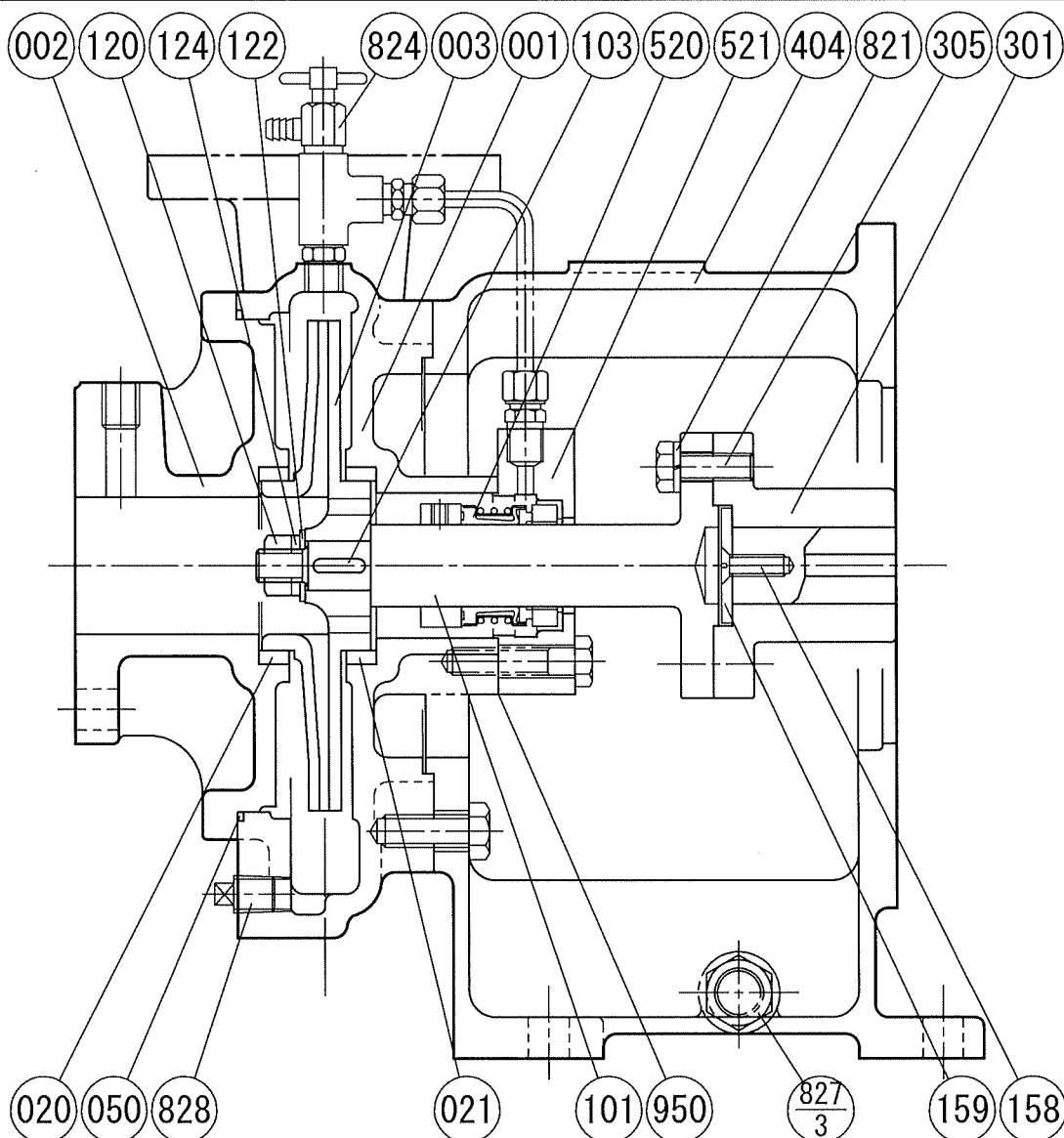
MODEL BHR-50

DRAWING NO.

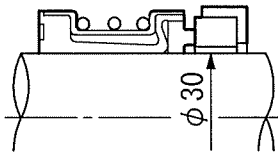
B6DS0526

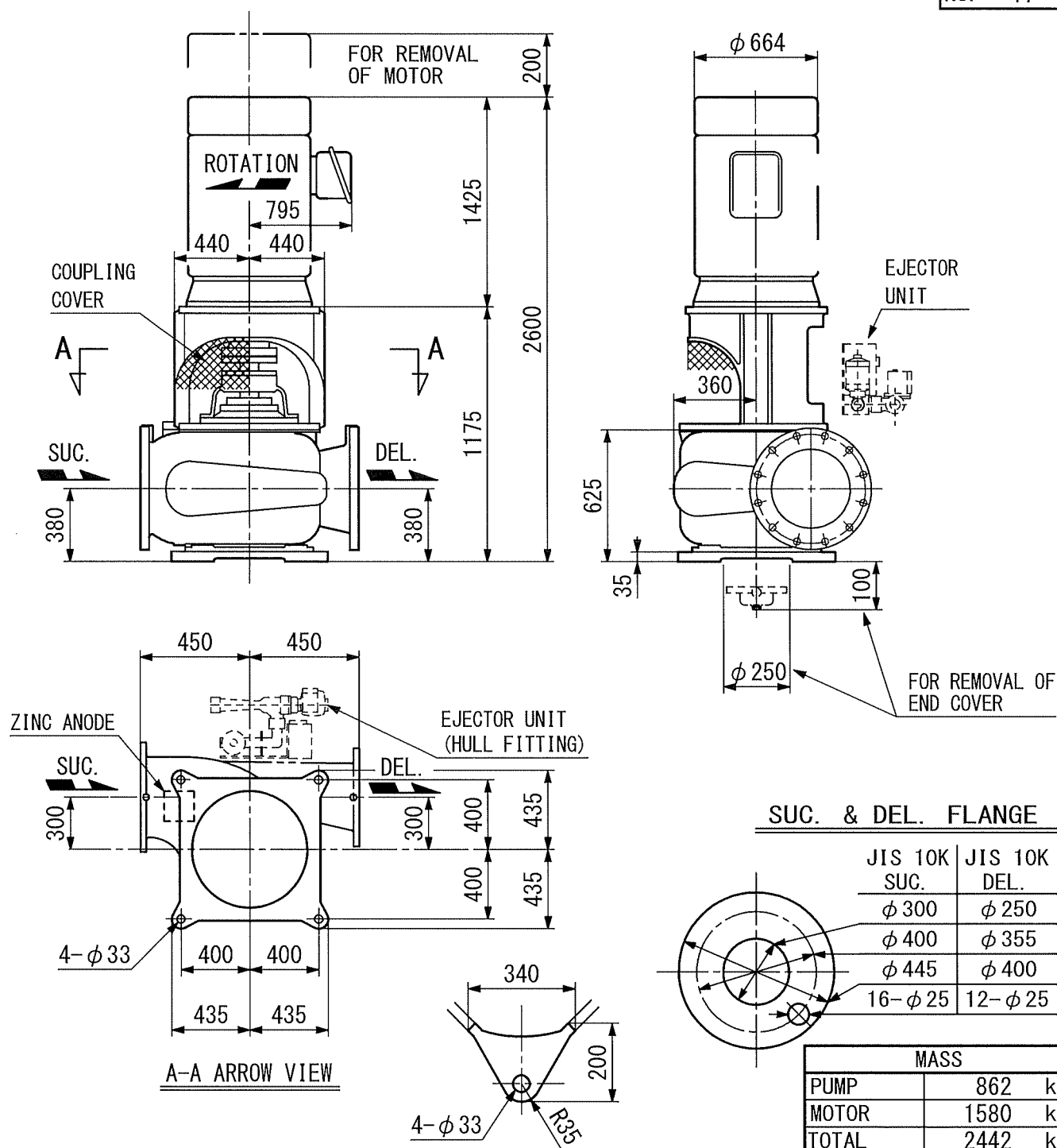
DATE

5 OCT. 2016



158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	950	GASKET	NON ASBESTOS FIBER		1
122	PLATE WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	1	$\frac{827}{3}$	DRAIN JOINT	COPPER ALLOY ROD	C3604	1
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
101	SHAFT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
021	CASING RING	CAST BRONZE	CAC406	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST BRONZE	CAC406	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	PHOSPHOR BRONZE	CAC502A	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	CAST BRONZE	CAC402	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	CAST BRONZE	CAC402	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		16		
NO. 1 & NO. 2 COLD GLYCOL WATER CIRCULATING PUMP							HORIZONTAL CENTRIFUGAL PUMP MODEL: <u>BHR-50</u> 2 SETS				
							SHIP NO.				
		BOX NO.									
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS			
				WORK- ING PER PUMP	SPARE PER SHIP	B6DS0526					
						NO.	PART NO.				
1	MECHANICAL SEAL	0.08 		1 SET	1 SET		520				
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN				DRAWING NO.				
							99DP0474				



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	300 mm	C/H WATER SPRAY PUMP	
COUPLING & COVER	1set	DELIVERY BORE	250 mm		
SEALING PIPE	1set	CAPACITY	500 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	1.0 MPa	PUMP MODEL	FEWV-250-4D
AIR EJECTOR UNIT	1set	SUCTION HEAD	MPa	QUANTITY	1
PRESSURE GAUGE WITH JOINT	1set	REVOLVING SPEED	1750 min ⁻¹	RULE	LR
COMPOUND GAUGE WITH JOINT	1set	HYD. TEST PRESS.	1.9 MPa	ANGLE PROJ.	3 rd
GAUGE ROOT VALVE	2	MOTOR OUTPUT	249kW , 4P	SCALE	FREE
GAUGE BOARD	1	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	B7DG0056
ZINC ANODE	1	REMARKS		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
COUNTER FLANGE	1set				



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

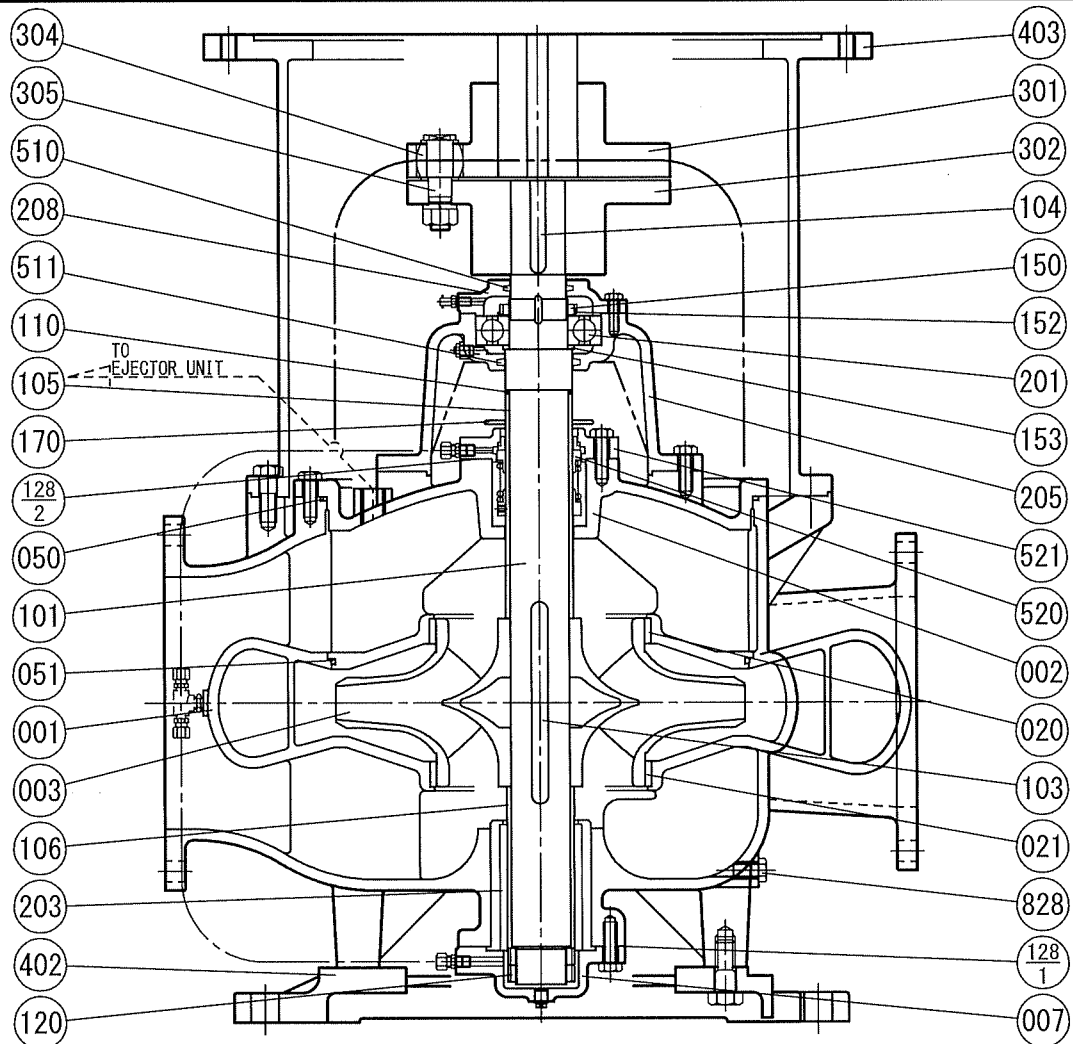
MODEL : FEWV-250-4D

DRAWING NO.

C4DS0256

DATE

4 FEB. 2017



150	BEARING NUT	MILD STEEL	SS400	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	2
128/2	GASKET	NON ASBESTOS FIBER		1	521	SEAL COVER	CAST BRONZE	CAC406	1
128/1	GASKET	NON ASBESTOS FIBER		1	520	MECHANICAL SEAL			1 SET
120	IMPELLER NUT	COPPER ALLOY ROD	C3604	2	511	PACKING RING	FELT		1
110	O-RING	RUBBER		1	510	PACKING RING	FELT		1
106	SHAFT SLEEVE	STAINLESS STEEL	SUS304	1	403	FRAME	CAST IRON	FC200	1
105	SHAFT SLEEVE	STAINLESS STEEL	SUS304	1	402	PUMP BED	CAST IRON	FC200	1
104	COUPLING KEY	CARBON STEEL	S45C	1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	1 SET
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	304	COUPLING RING	RUBBER		1 SET
101	SHAFT	STAINLESS STEEL	SUS304	1	302	COUPLING	CAST IRON	FC200	1
051	O-RING	RUBBER		1	301	COUPLING	CAST IRON	FC200	1
050	O-RING	RUBBER		1	208	BEARING COVER	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	205	BEARING HOUSING	CAST IRON	FC200	1
020	CASING RING	SYNTHETIC RESIN		1	203	SUBMERGED BEARING	CARBON		1
007	END COVER	CAST BRONZE	CAC406	1	201	BALL BEARING	SUJ		1
003	IMPELLER	STAINLESS STEEL	SCS14	1	170	DUST STOPPER	RUBBER		1
002	CASING COVER	CAST BRONZE	CAC402	1	153	BEARING COLLER	CARBON STEEL	S45C	1
001	CASING	CAST BRONZE	CAC402	1	152	BEARING WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



OUTLINE DRAWING

AIR EJECTOR UNIT

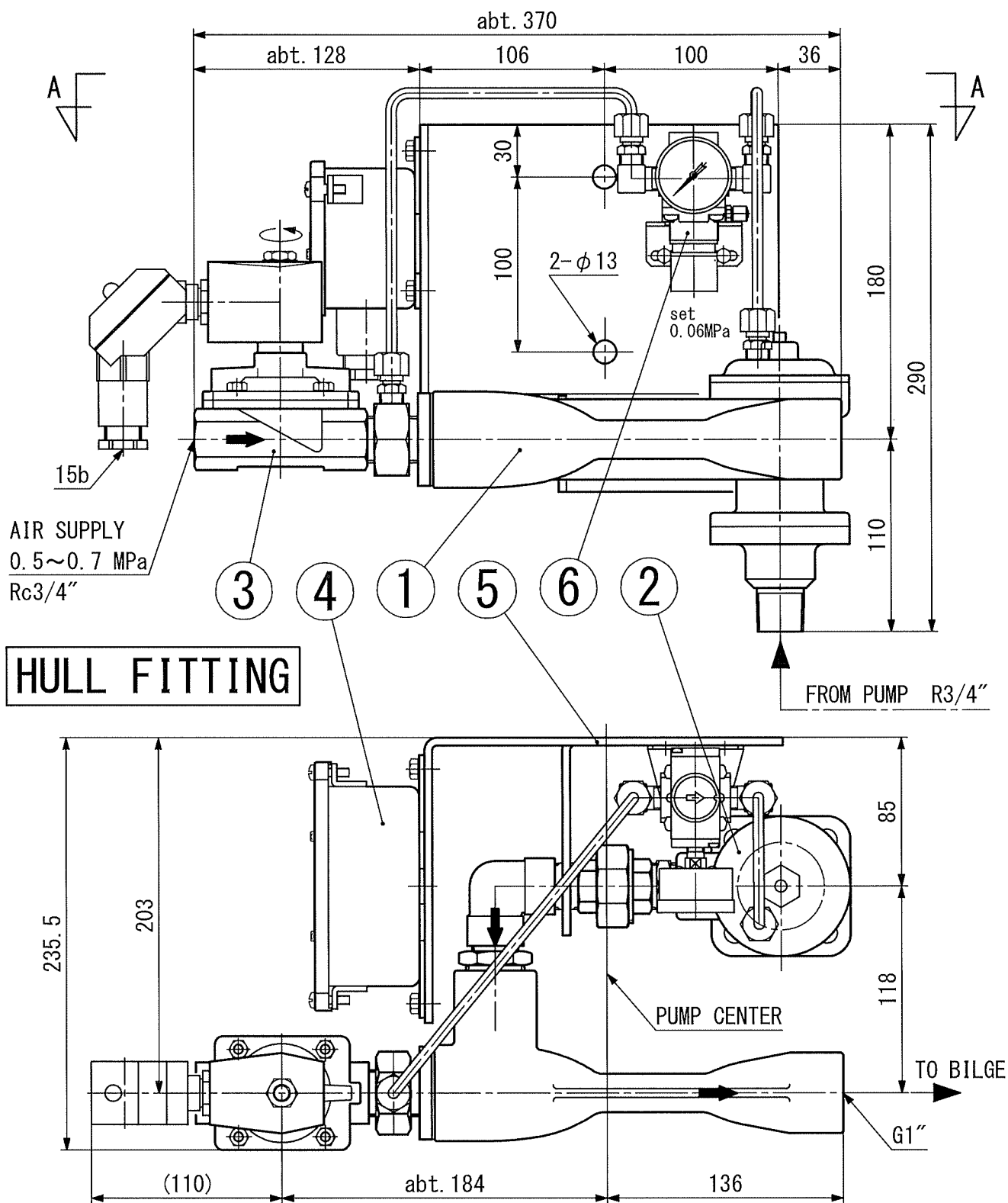
MODEL : PMC

DRAWING NO.

C3DG0247

DATE

26 DEC. 2023



A-A ARROW VIEW

AIR CONSUMPTION 75 Nm³/h at 0.6 MPa

3	SOLENOID VALVE	CAST BRONZE	1	6	REGULATOR		1
2	CHECK VALVE	CAST BRONZE	1	5	BRACKET	MILD STEEL	1
1	EJECTOR	CAST BRONZE	1	4	PRESSURE SWITCH		1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY

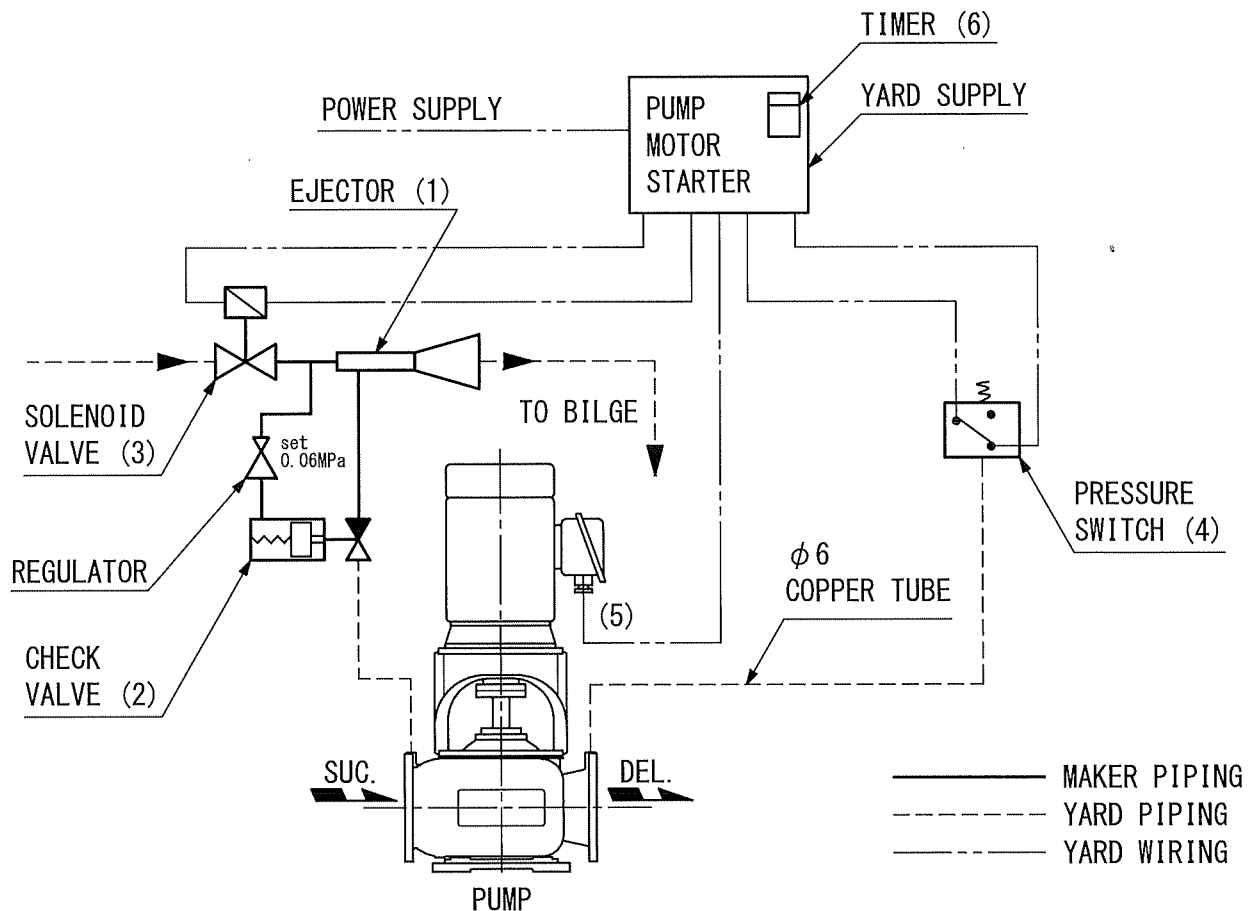


WIRING DIAGRAM

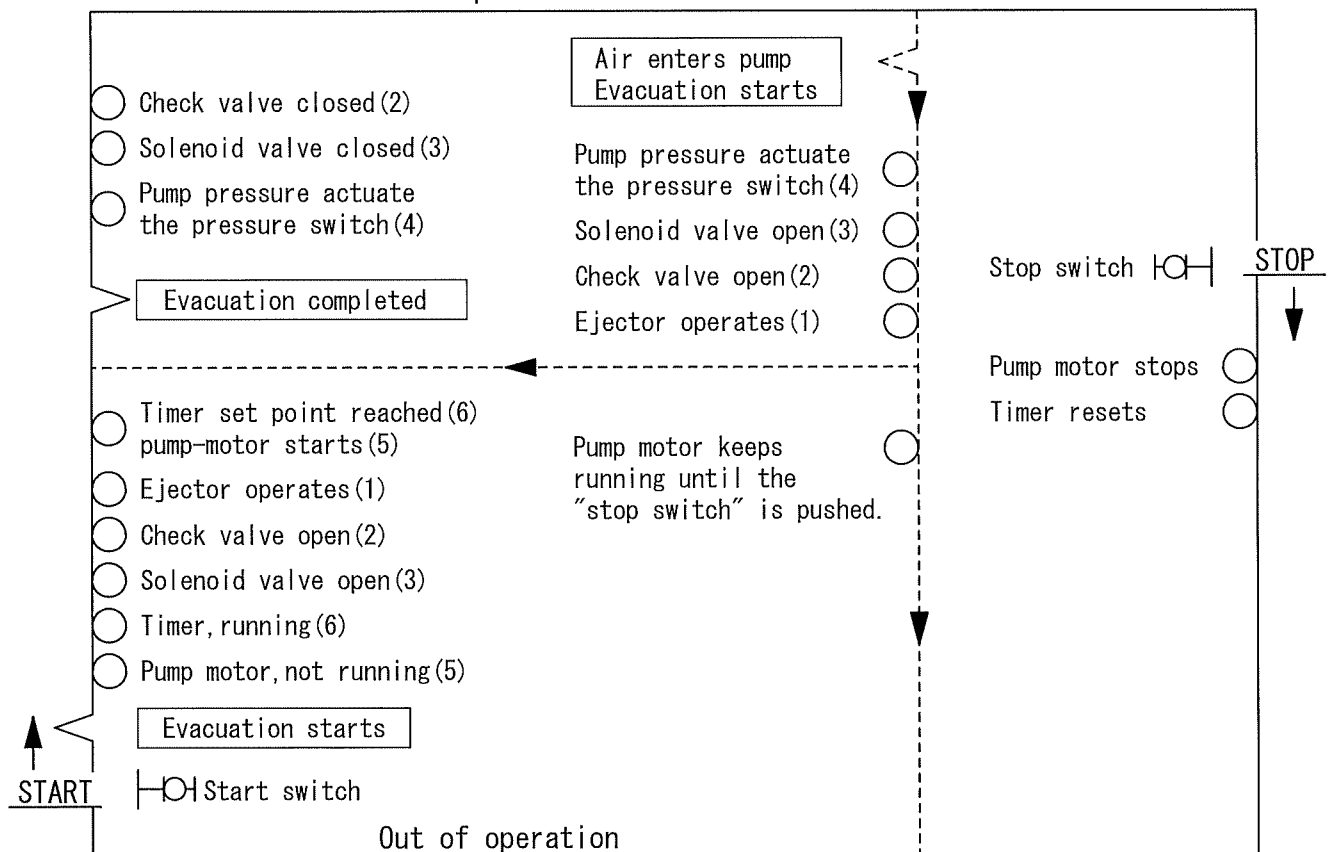
GENERAL DESCRIPTION & FUNCTION DIAGRAM

MODEL : PMC

DRAWING NO.
BODG0086
DATE
26 MAR. 2010



In operation





PERFORMANCE CURVE

AIR EJECTOR UNIT

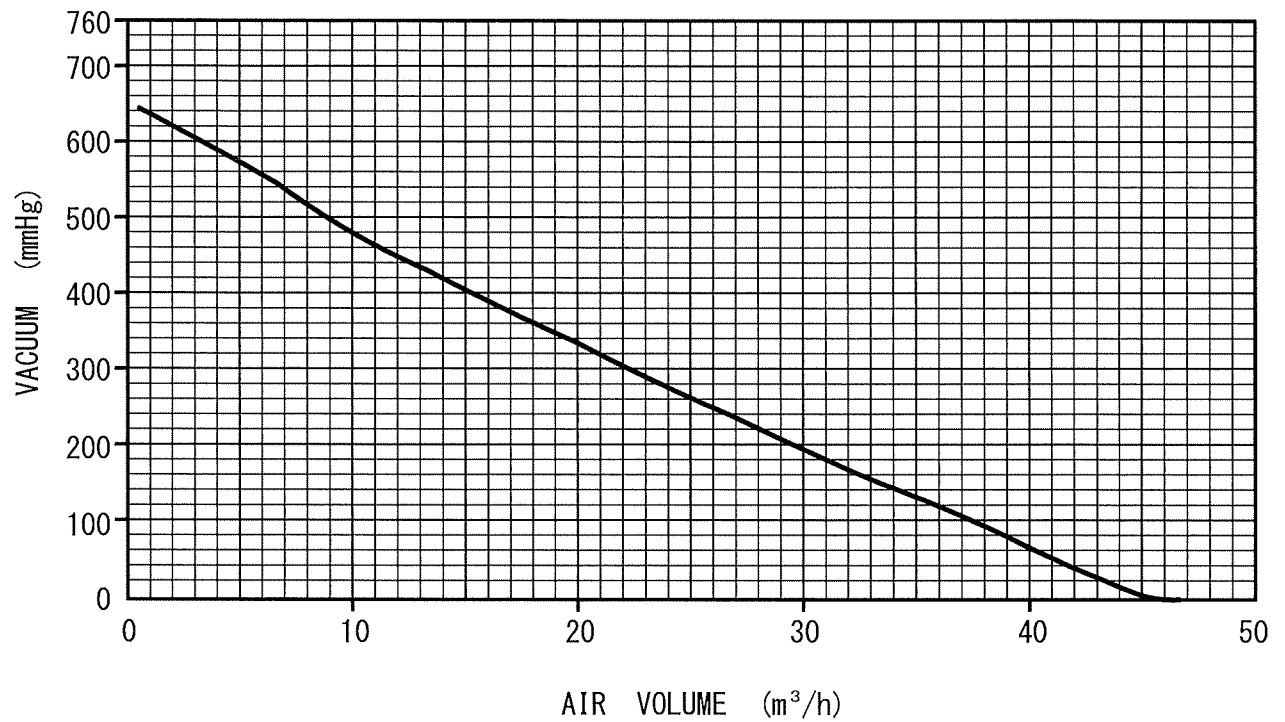
MODEL : PMC

DRAWING NO.

B0DG0093

DATE

7 APR. 2010



Driving air consumption : 75 Nm³/h

Driving air pressure : 0.6 MPa



SECTIONAL DRAWING

PRESSURE SWITCH

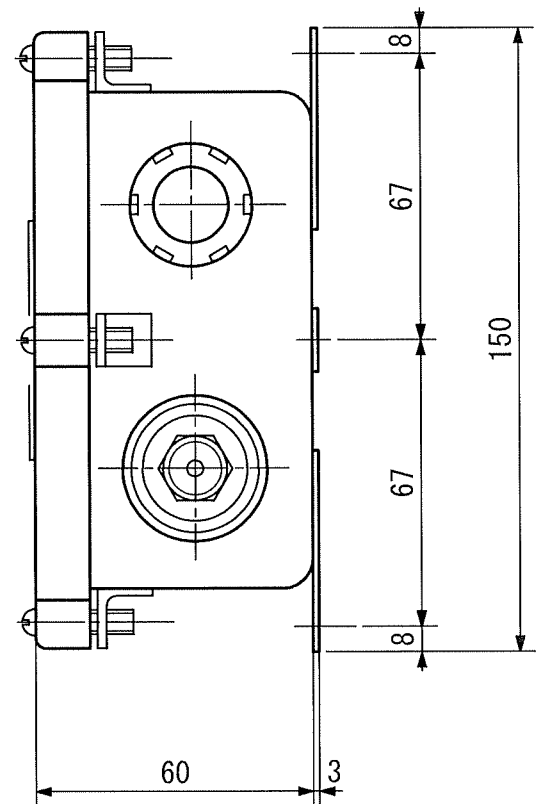
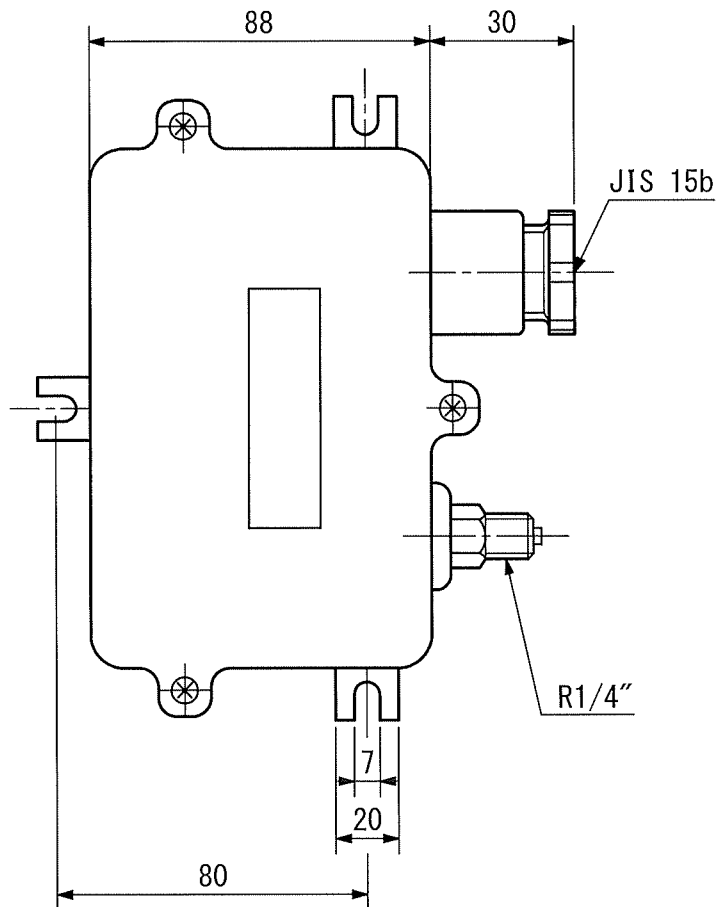
MODEL : SNS-C106W

DRAWING NO.

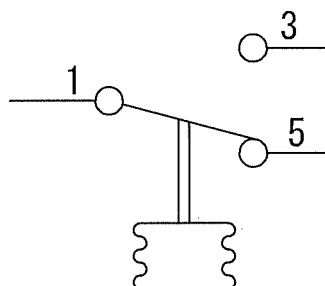
B4DG0947

DATE

20 AUG. 2014



CONTACT TYPE



CONTACT POINTS 1~3
TURNED ON DUE TO PRESSURE INCREASE

CONTACT POINTS 1~5
TURNED ON DUE TO PRESSURE DECREASE

CONTACT POINTS 1
COMMON

RATED CAPACITY

VOLT (V) AMPS (A)	AC			DC	
	125	250	450	24	125
NON INDUCTIVE	10	5	2	5	0.5
INDUCTIVE	8.5	4.5	1	2	0.2

RANGE : -0.06 ~ 0.6 MPa

MECHANICAL DIFFERENTIAL : 0.06 ~ 0.4 MPa

MAX. PERMISSIBLE PRESS. : 1.5 MPa

MASS : 0.4 Kg

SET PRESSURE

PUMP SPECIFIC PRESSURE

MORE THAN 0.5 MPa

→ ON : 0.1 MPa / OFF : 0.2 MPa

0.5 MPa AND LESS

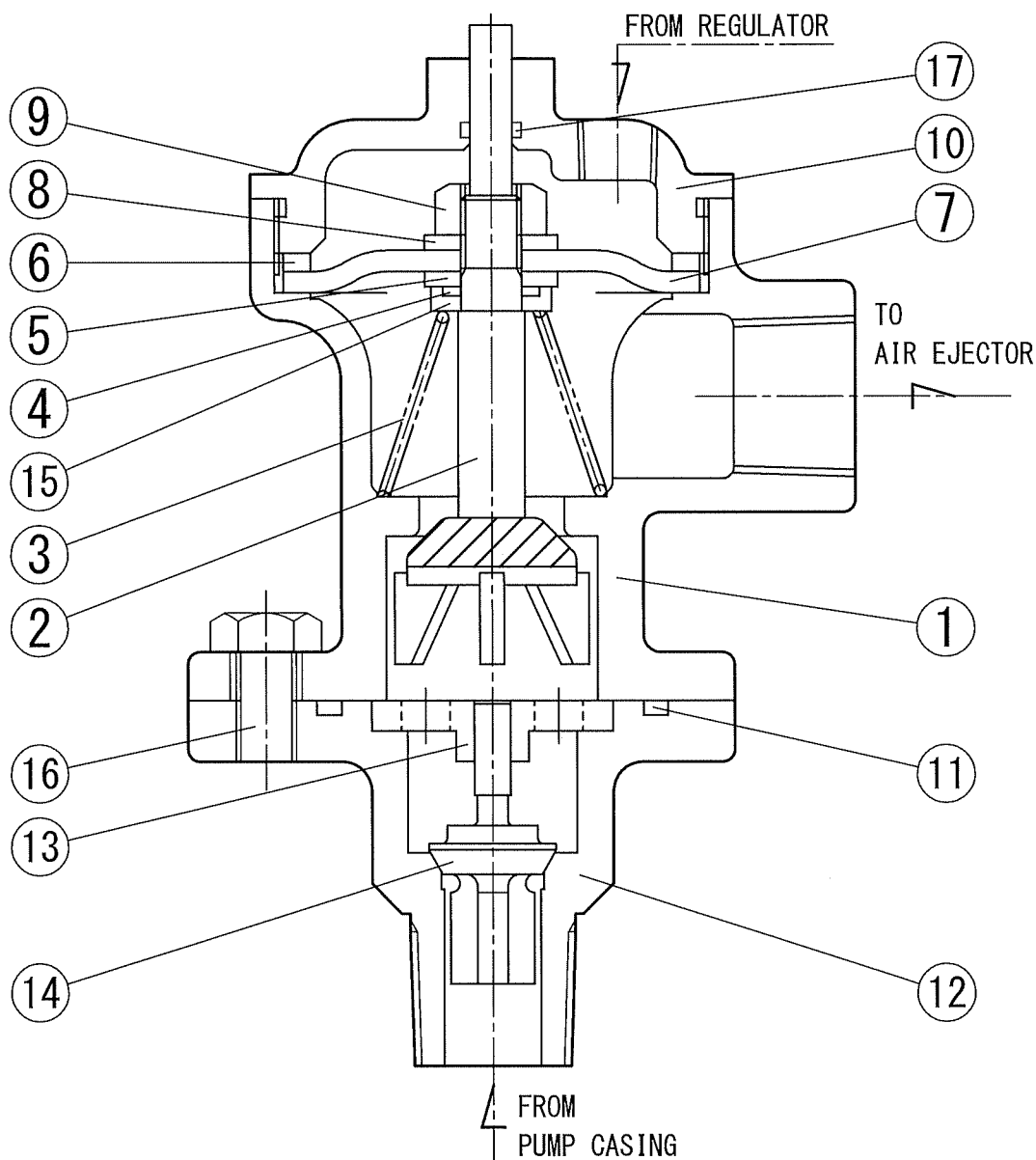
→ ON : 0.05 MPa / OFF : 0.15 MPa



SECTIONAL DRAWING

CHECK VALVE

DRAWING NO.
C4DS0163
DATE
20 JUL. 2024



9	NUT	COPPER ALLOY ROD	C3604	1					
8	WASHER		C3604	1	17	O RING	RUBBER		1
7	DIAPHRAGM	RUBBER		1	16	BOLTS	CARBON STEEL	SWRM10	4
6	DIAPHRAGM RING	COPPER ALLOY ROD	C3604	1	15	SPRING SUPPORT	COPPER ALLOY ROD	C3604	1
5	WASHER	COPPER ALLOY ROD	C3604	1	14	VALVE	CAST BRONZE	CAC406	1
4	O RING	RUBBER		1	13	GUIDE PLATE	CAST BRONZE	CAC406	1
3	SPRING	PHOSPHOR BRONZE	PBW	1	12	VALVE BODY	CAST BRONZE	CAC406	1
2	VALVE SPINDLE	CAST BRONZE & RUBER	CAC406	1	11	O RING	RUBBER		1
1	BODY	CAST BRONZE	CAC406	1	10	BODY COVER	CAST BRONZE	CAC406	1
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY		



SECTIONAL DRAWING

SOLENOID VALVE

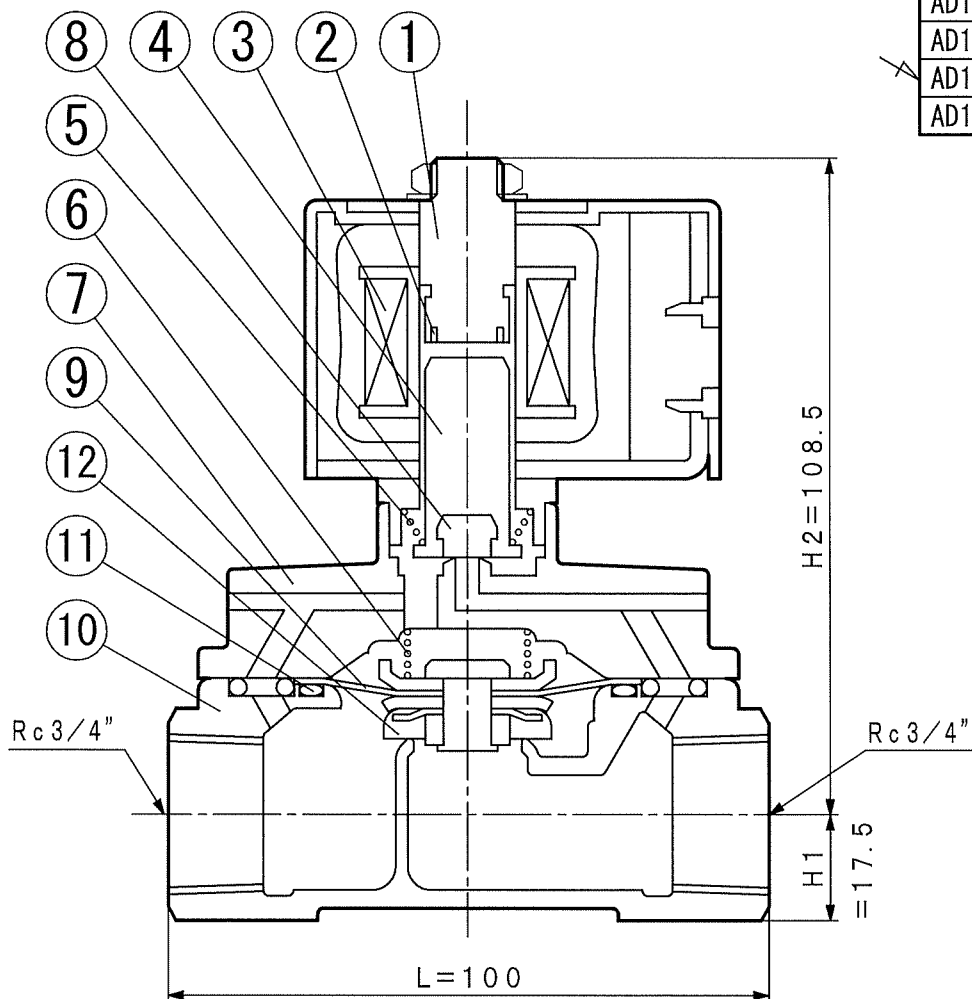
MODEL : AD11—

DRAWING NO.

A9DS0147

DATE

18 JUL. 2009

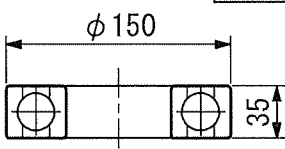
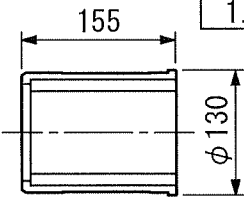
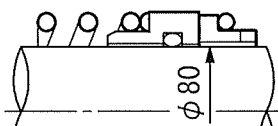
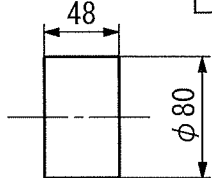


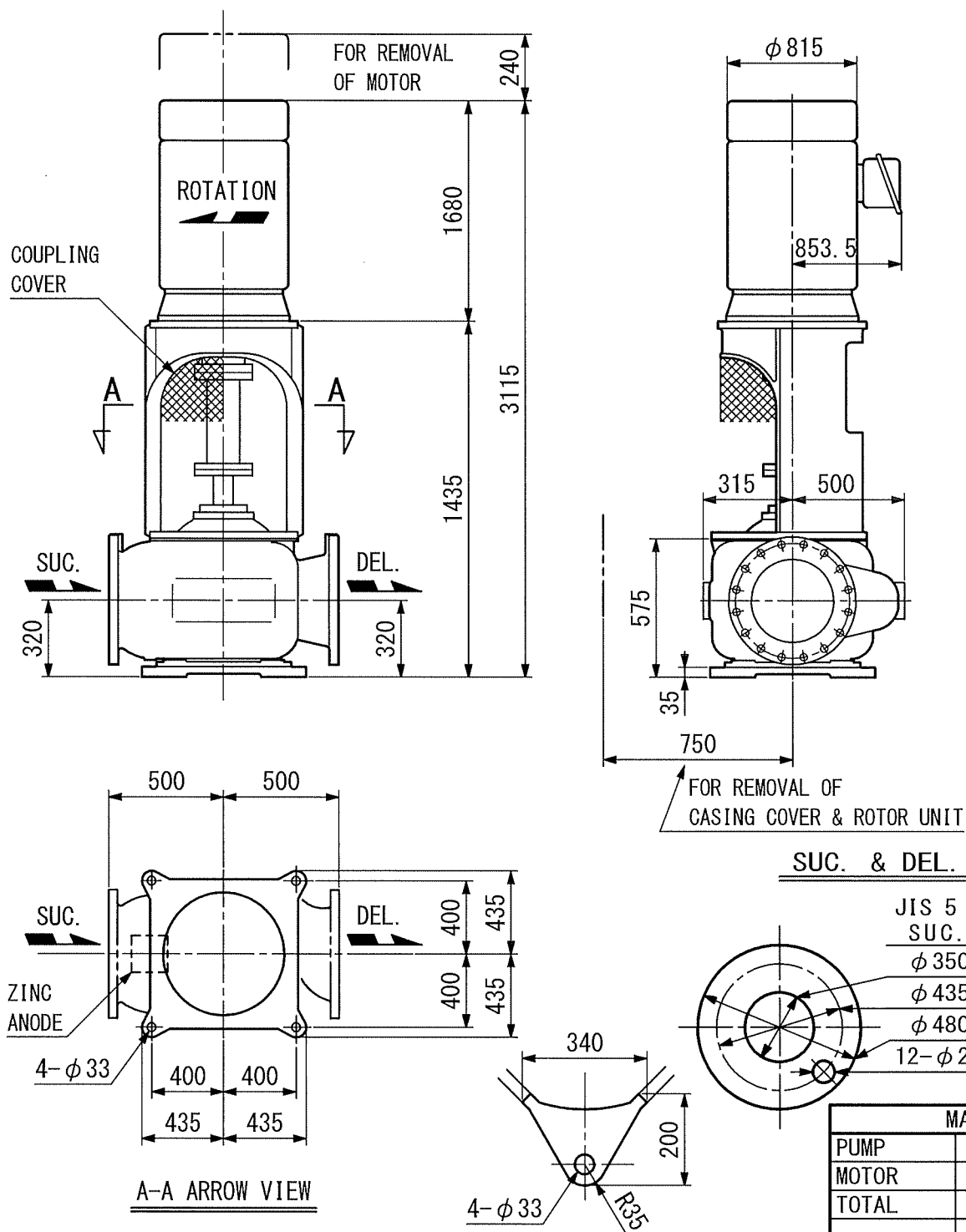
TYPE	L	H1	H2
AD11-8A	45	12	67
AD11-10A	45	12	67
AD11-15A	90	14.5	102.5
AD11-20A	100	17.5	108.5
AD11-25A	110	22	114

AC 220V 60Hz ϕ 1
POWER CAPACITY (THROW) 24VA

TYPE	SIZE	LIQUID	CV ($\Delta P=1\text{psi}$) (gpm)	OPERATING PRESS. (MPa)	LIQUID TEMP. ($^{\circ}\text{C}$)	MAX. TEST PRESS. (MPa)	POWER CONSUMP- TION (W)	MASS (kg)
AD11-8A	Rc 1/4"	WATER AIR PARAFFIN OIL (below 50cSt)	1.5	0.02	-10	2.0	5.7 24VA	0.4
AD11-10A	Rc 3/8"		1.8					0.4
AD11-15A	Rc 1/2"		4.5	1	60			1.2
AD11-20A	Rc 3/4"		9.3	1.0				1.2
AD11-25A	Rc 1		12.0					1.9

6	VALVE SPRING	STAINLESS STEEL	1	12	SEAL	RUBBER	1
5	PLUNGER SPRING	STAINLESS STEEL	1	11	O-RING	RUBBER	1
4	PLUNGER	STAINLESS STEEL	1 SET	10	BODY	BRONZE	1
3	COIL		1	9	DIAPHRAGM	RUBBER	4
2	SHADING COIL	COPPER	1	8	SEAL	RUBBER	1
1	CORE	STAINLESS STEEL	1 SET	7	STUFFING	BRONZE	1 SET
PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY

SPARE PARTS LIST							PAGE		17	
C/H WATER SPRAY PUMP MODEL : FEWV-250-4D 1 SET							SHIP NO.			
							BOX NO.			
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS	
					WORK-ING PER PUMP	SPARE PER SHIP	C4DS0256			
							NO.	PART NO.		
1	BALL BEARING		2.56	SUJ	1	1		201		
2	SUBMERGED BEARING		1.8	CARBON	1	1		203		
3	MECHANICAL SEAL		0.7		1 SET	1 SET		520		
4	ZINC ANODE		1.6	ZINC	1	1	B9DS0232	900		



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	350 mm	NO. 1 & NO. 2 iCER COOLING SEA WATER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	350 mm		
AIR COCK	1	CAPACITY	1150 m ³ /h		
SEALING PIPE	1	TOTAL HEAD	0.6 MPa	PUMP MODEL	FEWV-350-3
DRAIN JOINT WITH HOSE	1set	SUCTION HEAD	MPa	QUANTITY	2
PRESSURE GAUGE WITH JOINT	1set	REVOLVING SPEED	1750 min ⁻¹	RULE	LR
COMPOUND GAUGE WITH JOINT	1set	HYD. TEST PRESS.	1.2 MPa	ANGLE PROJ.	3 rd
GAUGE ROOT VALVE	2	MOTOR OUTPUT	299kW , 4P	SCALE	FREE
GAUGE BOARD	1	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	C4DG0243
ZINC ANODE	1	REMARKS		NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
COUNTER FLANGE	1set				



SECTIONAL DRAWING

VERTICAL CENTRIFUGAL PUMP

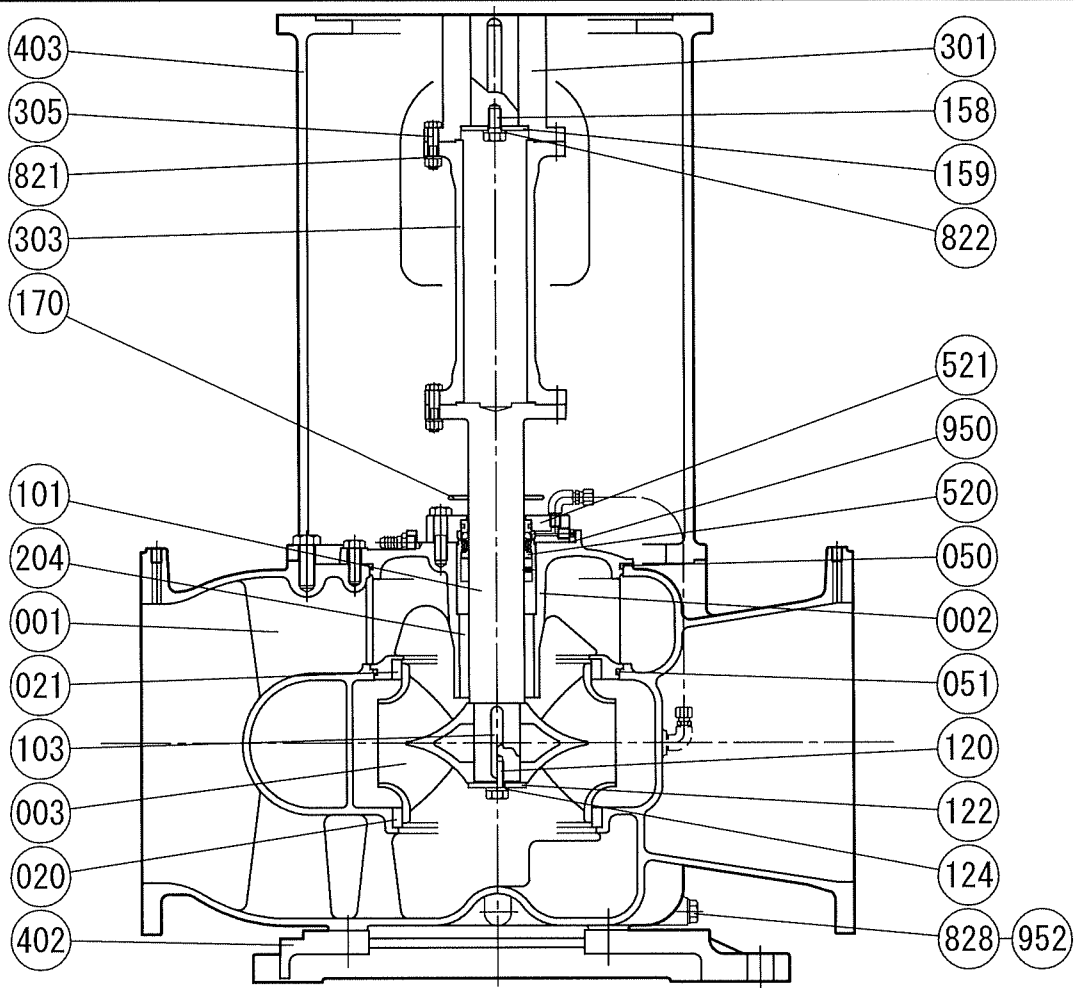
MODEL : FEWV

DRAWING NO.

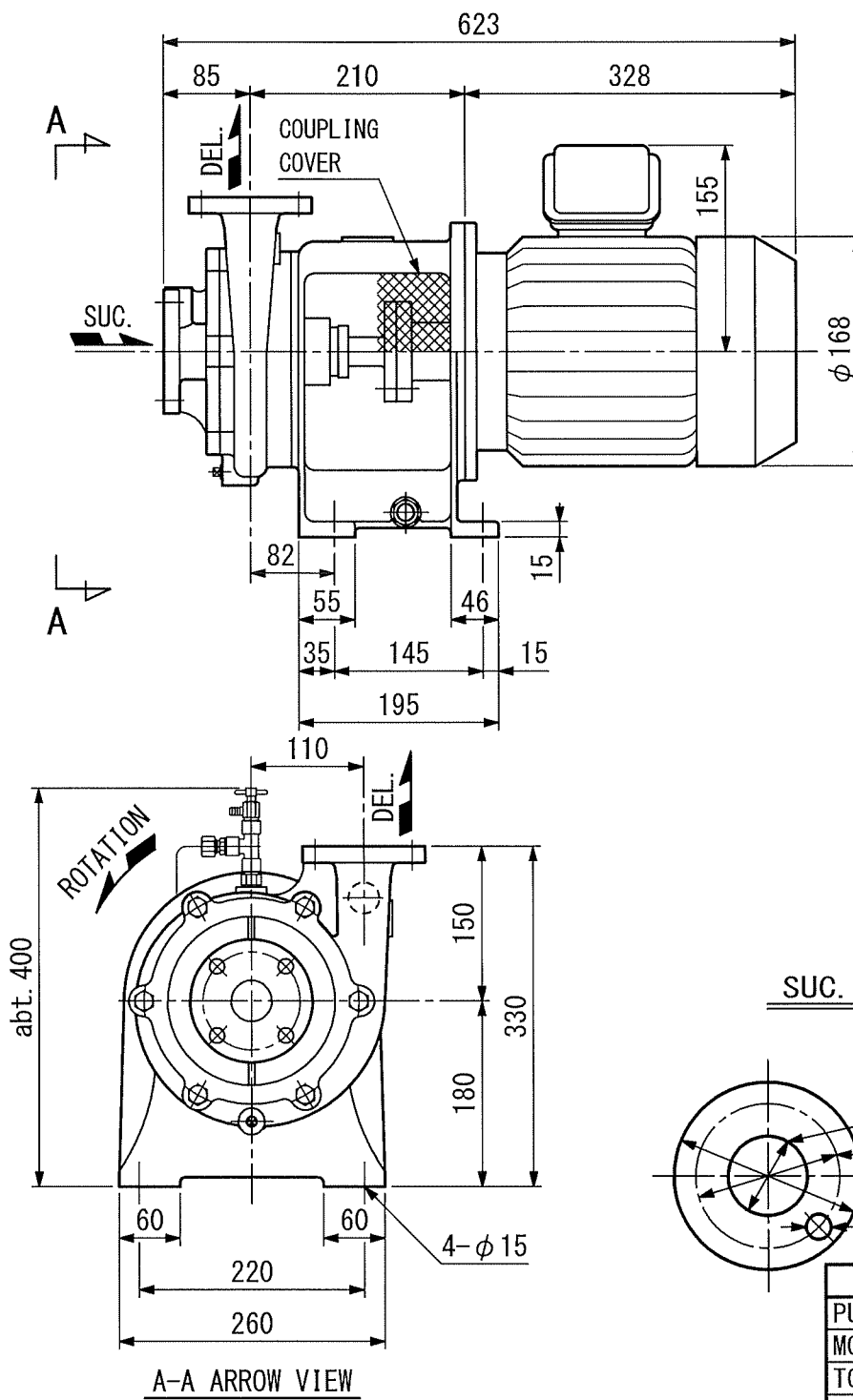
B7DS0063

DATE

2 MAR 2017



159	COUPLING PLATE WASHER	MILD STEEL	SS400	1	952	GASKET	NON ASBESTOS FIBER		1
158	COUPLING SET BOLT	MILD STEEL	SS400	1	950	GASKET	NON ASBESTOS FIBER		1
124	SPRING WASHER	STAINLESS STEEL	SUS304	1	828	DRAIN PLUG	COPPER ALLOY ROD	C3604	1
122	IMPELLER PLATE WASHER	STAINLESS STEEL	SUS304	1	822	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1
120	IMPELLER BOLT	STAINLESS STEEL	SUS304	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	2 SETS
103	IMPELLER KEY	STAINLESS STEEL	SUS304	1	521	SEAL COVER	CAST BRONZE	CAC406	1
101	SHAFT	STAINLESS STEEL	SUS304	1	520	MECHANICAL SEAL			1 SET
051	O-RING	RUBBER		1	403	FRAME	CAST IRON	FC200	1
050	O-RING	RUBBER		1	402	PUMP BED	CAST IRON	FC200	1
021	CASING RING	SYNTHETIC RESIN		1	305	COUPLING BOLT & NUT	MILD STEEL	SS400	2 SETS
020	CASING RING	SYNTHETIC RESIN		1	303	COUPLING SPACER	DUCTILE CAST IRON	FCD400	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	301	COUPLING	CAST IRON	FC200	1
002	CASING COVER	CAST BRONZE	CAC402	1	204	BEARING METAL	SPECIAL BRONZE		1
001	CASING	CAST BRONZE	CAC402	1	170	DUST STOPPER	RUBBER		1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	40 mm	NO. 1 & NO. 2 NaOH TRANSFER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	40 mm		
AIR COCK	1	CAPACITY	1.0 m ³ /h		
DRAIN JOINT WITH HOSE	1set	TOTAL HEAD	0.35 MPa	PUMP MODEL	BHR-40
PRESSURE GAUGE WITH JOINT	1set	SUCTION HEAD	MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	REVOLVING SPEED	3500 min ⁻¹	RULE	LR
GAUGE ROOT VALVE	2	HYD. TEST PRESS.	0.6 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD	1	MOTOR OUTPUT	2.2kW , 2P	SCALE	FREE
COUNTER FLANGE	1set	POWER SOURCE	440V, 60Hz, 3ph	DRAWING NO.	99DG0085
		REMARKS WORKING TEMP = 50 °C		 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	



SECTIONAL DRAWING

HORIZONTAL CENTRIFUGAL PUMP

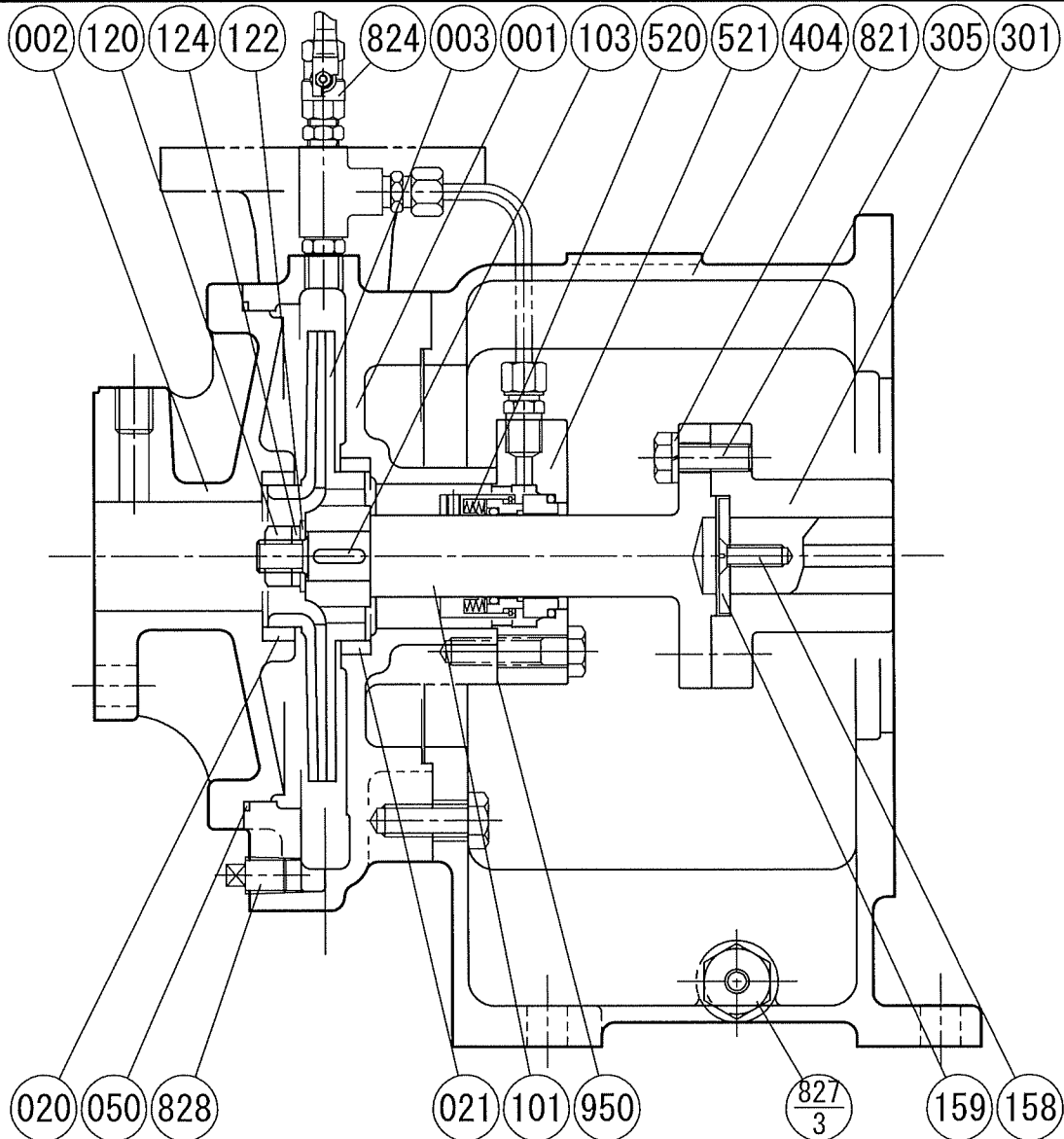
MODEL BHR-40

DRAWING NO.

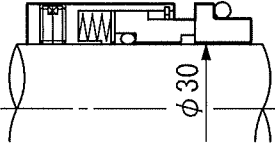
C3DS0114

DATE

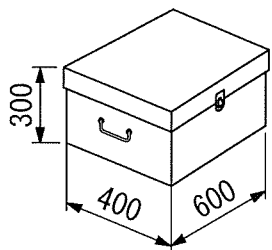
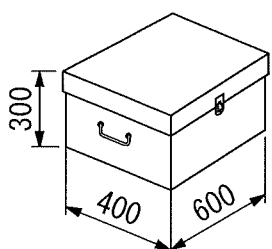
7 JUN. 2023

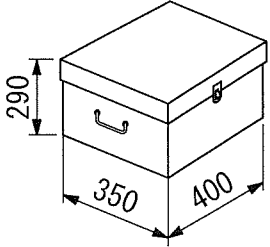


158	COUPLING SET BOLT	MILD STEEL	SS400	1					
124	SPRING WASHER	STAINLESS STEEL	SUS316	1	950	GASKET	PTFE		1
122	PLATE WASHER	STAINLESS STEEL	SUS316	1	828	DRAIN PLUG	STAINLESS STEEL	SUS316	1
120	IMPELLER NUT	STAINLESS STEEL	SUS316	1	827/3	DRAIN JOINT	STAINLESS STEEL	SUS316	1
103	IMPELLER KEY	STAINLESS STEEL	SUS316	1	824	AIR COCK	STAINLESS STEEL	SCS14	1
101	SHAFT	STAINLESS STEEL	SUS316	1	821	SPRING WASHER	HIGH CARBON STEEL WIRE	SWRH	1 SET
050	O-RING	RUBBER		1	521	SEAL COVER	STAINLESS STEEL	SCS14	1
021	CASING RING	CAST IRON	FC200	1	520	MECHANICAL SEAL			1 SET
020	CASING RING	CAST IRON	FC200	1	404	FRAME	CAST IRON	FC200	1
003	IMPELLER	STAINLESS STEEL	SCS14	1	305	COUPLING BOLT	MILD STEEL	SS400	1 SET
002	CASING COVER	STAINLESS STEEL	SCS14	1	301	COUPLING	CAST IRON	FC200	1
001	CASING	STAINLESS STEEL	SCS14	1	159	COUPLING SET WASHER	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		19		
NO. 1 & NO. 2 NaOH TRANSFER PUMP							HORIZONTAL CENTRIFUGAL PUMP MODEL: BHR-40 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS			
				WORK- ING PER PUMP	SPARE PER SHIP	C3DS0114					
1	MECHANICAL SEAL	0.16 		1 SET	1 SET		520				
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.			
								98DP0468			

ADDITIONAL SPARE PARTS LIST						PAGE		
						SHIP NO.		
						BOX NO.		
Item NO.	NAME	MASS : kg SKETCH	MATERIAL	SUPPLY		DRAWING NO.		REMARKS
				WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.	
1	MECHANICAL SEAL	1.0		1 SET	3 SETS		520	NO. 1~NO. 3 MAIN COOLING S. W. PUMP
2	MECHANICAL SEAL	0.56		1 SET	2 SETS		520	NO. 1 & NO. 2 BALLAST WATER PUMP
3	MECHANICAL SEAL	0.56		1 SET	2 SETS		520	NO. 1 & NO. 2 BILGE, FIRE & G/S PUMP
4	MECHANICAL SEAL	0.14		1 SET	1 SET		520	EMERGENCY FIRE PUMP
5	MECHANICAL SEAL	0.56		1 SET	2 SETS		520	NO. 1 & NO. 2 ICER COOLING SEA WATER PUMP
MFR' S NAME & ADDRESS		NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.	
							CODP0045	

SPARE PARTS BOX				PAGE	
				SHIP NO.	
SIZE	MATERIAL	SERVICE AS	MODEL	Q' TY	
SPARE PARTS BOX 	MILD STEEL	MAIN COOLING S. W. PUMP BALLAST WATER PUMP ICER COOLING SEA WATER PUMP BILGE, FIRE & G/S PUMP L. T. COOLING F. W. PUMP M/E AIR COOLER L. T. COOLING F. W. PUMP SPARE PARTS TOTAL MASS abt. 45 kg	FEWV-450 FEWV-350-3 FE2V-250-3 FEWV-350	1	
SPARE PARTS BOX 	MILD STEEL	M/E H. T. F. W. PUMP L/T. C. W. BOOSTER PUMP M/E JACKET WATER PREHEATER PUMP EMERGENCY FIRE PUMP MAIN L. O. PUMP FRESH WATER PUMP HOT WATER CIRCULATING PUMP CHEMICAL CLEANING PUMP HOT GLYCOL WATER CIRCULATING PUMP COLD GLYCOL WATER CIRCULATING PUMP C/H WATER SPRAY PUMP NaOH TRANSFER PUMP SPARE PARTS TOTAL MASS abt. 35 kg	FEV-200E1 BHR-65 FGV-100-2 TOM-300-2 BHR-50-3 BHR-32 BHR-40 BHR-100 BHR-50 FEWV-250-4D BHR-40	1	
COLOR PAINT : 7.5BG 7/2					
MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN			DRAWING NO. A1DP0343	

SPECIAL SPARE PARTS BOX			PAGE	
			SHIP NO.	
SIZE	MATERIAL	SERVICE AS	MODEL	Q' TY
SPARE PARTS BOX 	MILD STEEL	SPECIAL SPARE PARTS MAIN COOLING S.W. PUMP BALLAST WATER PUMP BILGE, FIRE & G/S PUMP EMERGENCY FIRE PUMP iCER COOLING SEA WATER PUMP SPARE PARTS TOTAL MASS abt. 15 kg	FEWV-450 FEWV-350-3 FE2V-250-3 FGV-100-2 FEWV-350-3	1
COLOR PAINT : 7.5BG 7/2				
MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN		DRAWING NO. A1DP0343	



SECTIONAL DRAWING

MECHANICAL SEAL

DRAWING NO.
98DS0226
DATE
18 MAY 1998

FIG. 1

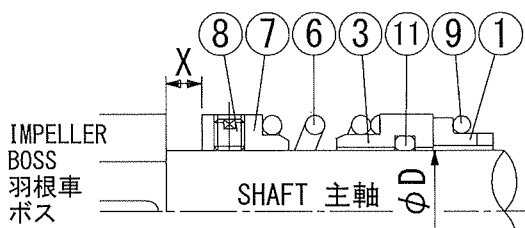


FIG. 2

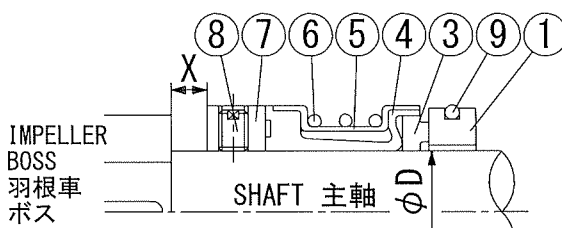
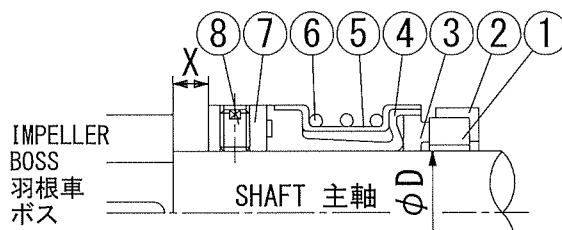


FIG. 3



NO.	PUMP NAME	MODEL	SHAFT DIA ϕ	SET DIMENSION X	FIG NO.	PART NO.	NAME OF PART	Q' TY
1, 2	MAIN COOLING S. W. PUMP	FEWV-450	90	241	1	1	FLOATING SEAT	1
3	BALLAST WATER PUMP	FEWV-350-3	70	269	1	2	SEAT CAP	1
4	BILGE, FIRE & G/S PUMP	FE2V-250-3	70	187	1	3	SEAL RING	1
5	L. T. COOLING F. W. PUMP	FEWV-350	70	153	2	4	BELLOWS	1
6	M/E AIR COOLER L. T. COOLING F. W. PUMP	FEWV-350	70	153	2	5	RETAINER	1
7	M/E H. T. F. W. PUMP	FEV-200E1	55	29.5	3	6	COIL SPRING	1
8	M/E JACKET WATER PREHEATER PUMP	BHR-65	30	18.5	3	7	COLLAR	1
9	L/T. C. W. BOOSTER PUMP	FEV-200E1	55	29.5	3	8	SET SCREW	1SET
10	EMERGENCY FIRE PUMP	FGV-100-2	45	84	3	9	O-RING	1
12	FRESH WATER PUMP	BHR-50-3	30	18	3	11	O-RING	1
13	HOT WATER CIRCULATING PUMP	BHR-32	25	21	3			
14	CHEMICAL CLEANING PUMP	BHR-40	30	18	3			
15	HOT GLYCOL WATER CIRCULATING PUMP	BHR-100	40	21	3			
16	COLD GLYCOL WATER CIRCULATING PUMP	BHR-50	30	18	3			
17	C/H WATER SPRAY PUMP	FEWV-250-4D	80	122.5	1			
18	ICER COOLING SEA WATER PUMP	FEWV-350-3	70	269	1			



GAUGE BOARD

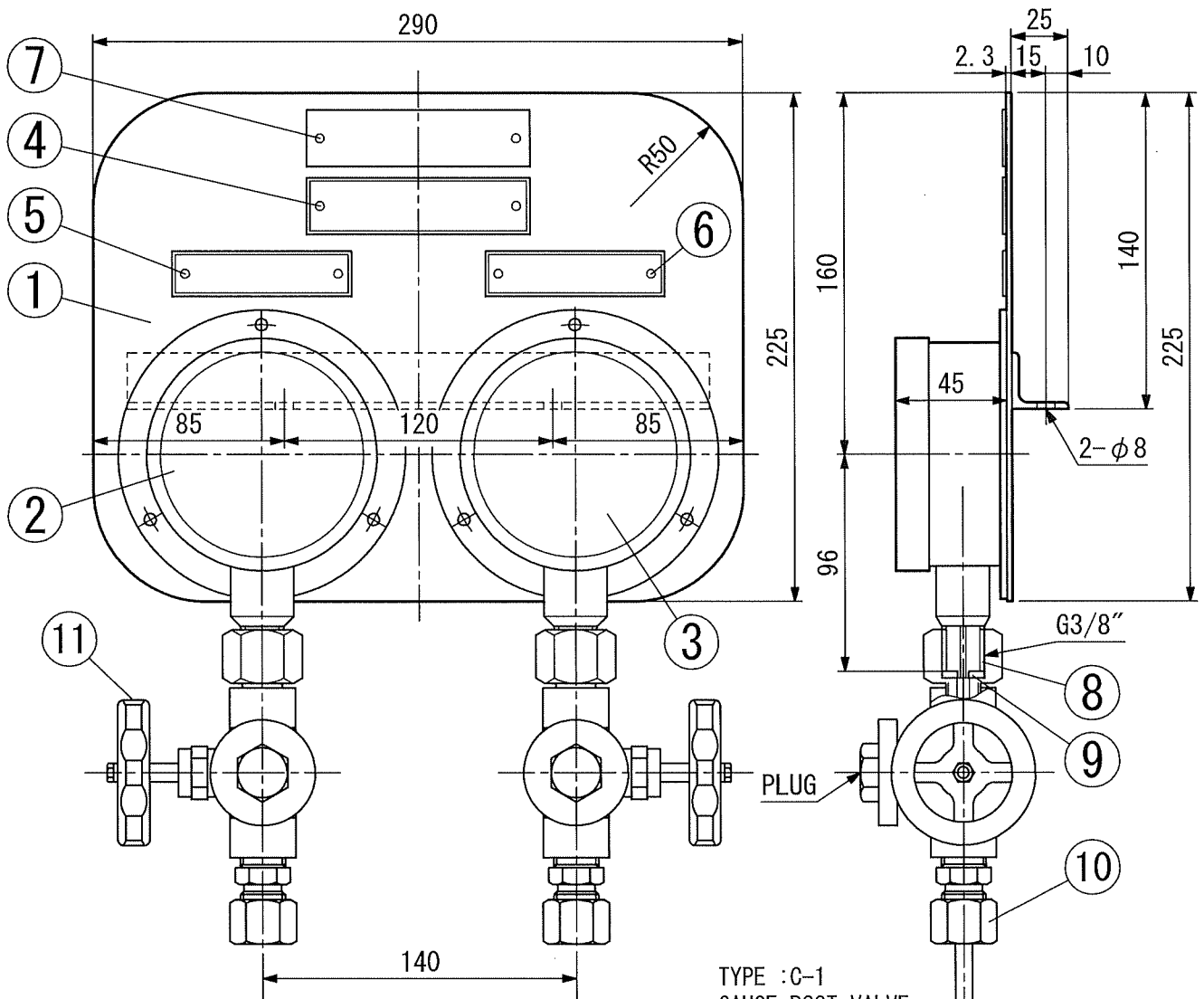
MODEL : BVS- ϕ 100x3/8"

DRAWING NO.

C4DH0023

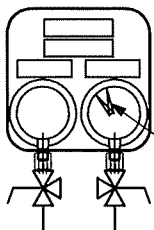
DATE

13 JUN. 2024

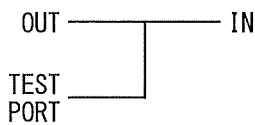


SILICONE OIL FILLED TYPE

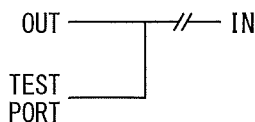
GAUGE BOARD



【3-WAY TEST VALVE OPEN】

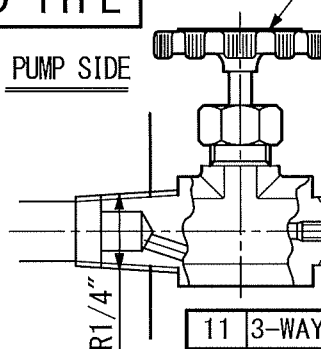


【3-WAY TEST VALVE CLOSED】



GREEN MARK
(TOTAL PRESSURE)

PUMP SIDE



R1/4"

TYPE : C-1
GAUGE ROOT VALVE

ϕ 6 COPPER TUBE

GAUGE ROOT VALVE
3.0 MPa and less.

REMARKS	
UNIT	MPa
MASS	1.4 kg

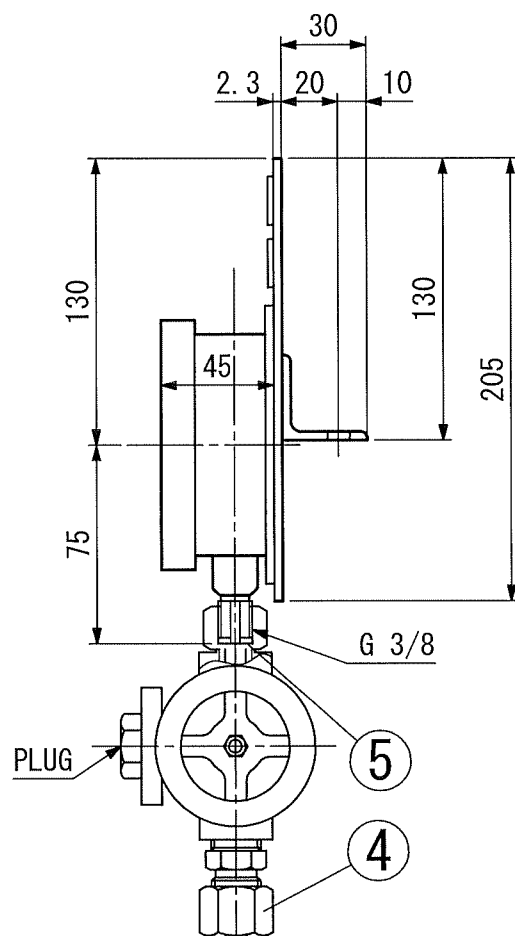
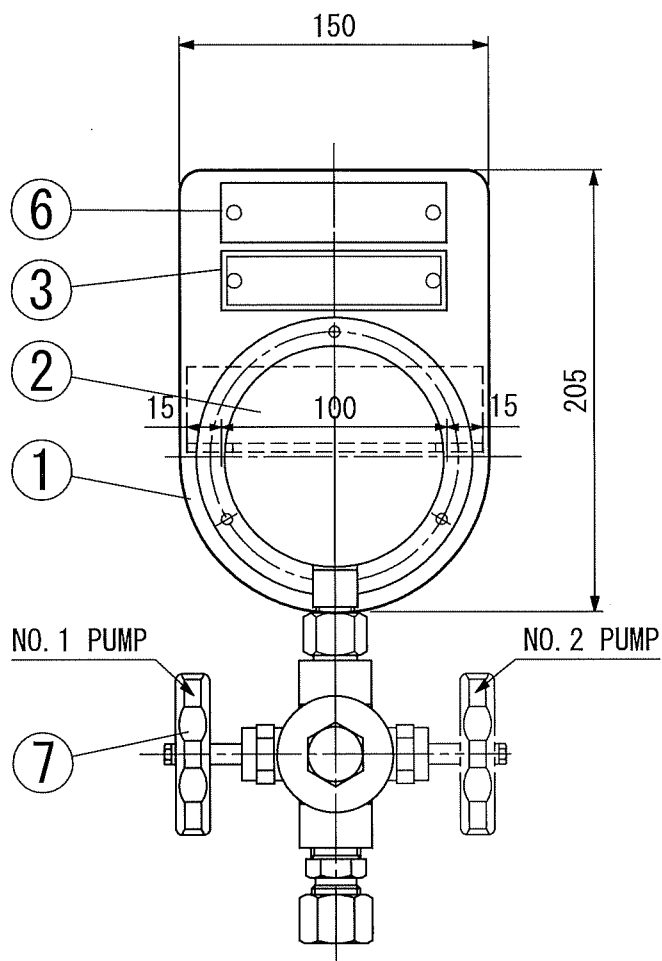
11	3-WAY TEST VALVE	COPPER ALLOY	2
10	RING JOINT	COPPER ALLOY	2
9	GASKET		2
8	BUSH	COPPER ALLOY	2
7	NAME COMPANY PLATE	STAINLESS STEEL	1
6	DEL. NAME PLATE	COPPER ALLOY	1
5	SUC. NAME PLATE	COPPER ALLOY	1
4	PUMP NAME PLATE	COPPER ALLOY	1
3	DELIVERY GAUGE		1
2	SUCTION GAUGE		1
1	GAUGE BOARD	MILD STEEL	1
No.	NAME OF PART	MATERIAL	Q' TY



GAUGE BOARD

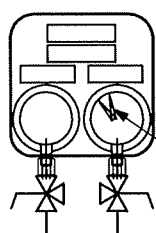
MODEL : BVS- $\phi 100 \times 3/8''$

DRAWING NO.
C4DH0013
DATE
13 MAY 2024



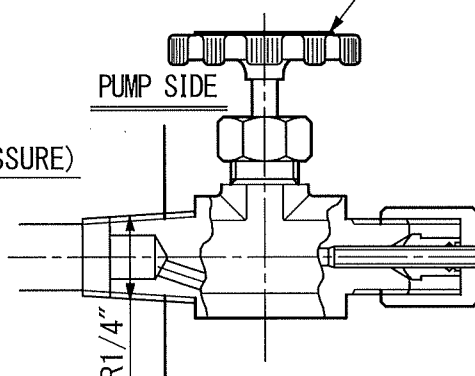
SILICONE OIL FILLED TYPE

GAUGE BOARD



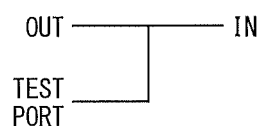
GREEN MARK
(TOTAL PRESSURE)

TYPE : C-1
GAUGE ROOT VALVE



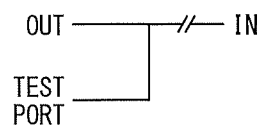
$\phi 6$ COPPER TUBE

【3-WAY TEST VALVE OPEN】



GAUGE ROOT VALVE
3.0 MPa and less.

【3-WAY TEST VALVE CLOSED】



REMARKS	
UNIT	MPa
MASS	1.4 kg

NO.	NAME OF PART	MATERIAL	Q' TY
7	3-WAY TEST VALVE	COPPER ALLOY	1
6	NAME COMPANY PLATE	STAINLESS STEEL	1
5	GASKET		1
4	RING JOINT	COPPER ALLOY	1
3	PUMP NAME PLATE	COPPER ALLOY	1
2	DELIVERY GAUGE		1
1	GAUGE BOARD	MILD STEEL	1

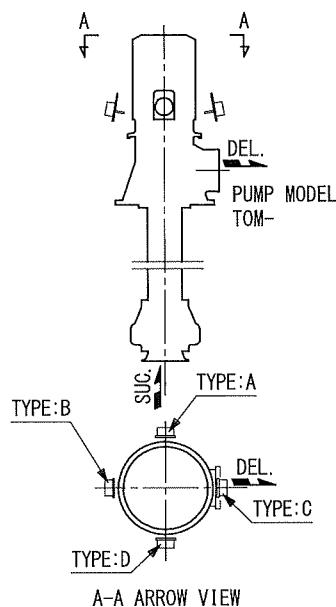
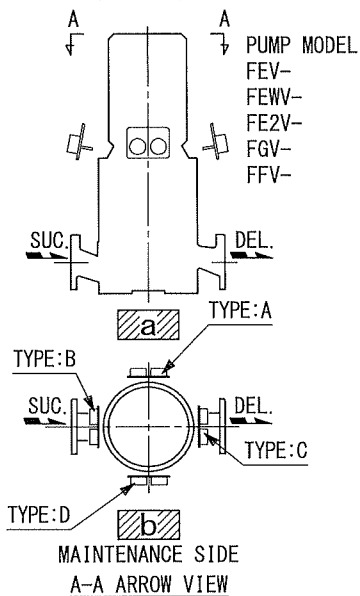


INDICATION LIST OF GAUGE SET POSITION

DRAWING NO.
98DG0107
DATE
20 FEB. 1998

VERTICAL PUMP TYPE

CUSTOMER : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.
SHIP No. : H1933A/1934A/1935A/1936A/1937A



Please indicate your required position in the blank.

- Remarks :
1. Please indicate "H" for plant fitting Gauge Boards. In this case, setting and piping for Gauge Boards are worked by customer. Therefore the Gauge Boards will be sent separately from pump.
 2. Gauge Board size and the setting method are in accordance with maker's standard.
 3. Terminal Box cannot adopt same position as Gauge Board.
 4. Terminal Box cannot adopt same position as Vacuum Pump or Ejector Unit.(If fitted).
 5. In case of Vacuum Pump (VK-20NB or VK-20NC) or Ejector Unit (PMC) is fitted, its position to be anti-maintenance side only.
 6. For pump model FBV, FBSV, FBWV, FB2V, FBCV, the positions SUC./ DEL.. Maintenance (b) and Vacuum Pump (A) is fixed and cannot change according to the constraintion.

SERVICE AS	PUMP MODEL	GAUGE SET POSITION	MAINTENANCE SIDE	TERMINAL BOX SET POSITION
NO. 3 MAIN COOLING S. W. PUMP	FEWV-450 + PMC	H	b	C
NO. 1 MAIN COOLING S. W. PUMP	FEWV-450	H	b	A
NO. 2 MAIN COOLING S. W. PUMP	FEWV-450	H	b	A
NO. 1 BALLAST WATER PUMP	FEWV-350-3 + VK	H	b	C
NO. 2 BALLAST WATER PUMP	FEWV-350-3 + VK	H	b	C
NO. 1 BILGE, FIRE & G/S PUMP	FE2V-250-3 + VK	H	b	C
NO. 2 BILGE, FIRE & G/S PUMP	FE2V-250-3 + VK	H	b	C
NO. 1 L. T. COOLING F. W. PUMP	FEWV-350	H	b	A
NO. 2 L. T. COOLING F. W. PUMP	FEWV-350	H	b	A
NO. 1 M/E AIR COOLER L. T. COOLING F. W. PUMP	FEWV-350	H	b	A
NO. 2 M/E AIR COOLER L. T. COOLING F. W. PUMP	FEWV-350	H	b	A
NO. 1 M/E H. T. F. W. PUMP	FEV-200E1	H	b	A
NO. 2 M/E H. T. F. W. PUMP	FEV-200E1	H	b	A
NO. 1 L/T. C. W. BOOSTER PUMP	FEV-200E1	H	b	A
NO. 2 L/T. C. W. BOOSTER PUMP	FEV-200E1	H	b	A
EMERGENCY FIRE PUMP	FGV-100-2 + VK	H	b	C
NO. 1 MAIN L. O. PUMP	TOM-300-2	H		A
NO. 2 MAIN L. O. PUMP	TOM-300-2	H		A
C/H WATER SPRAY PUMP	FEWV-250-4D + PMC	H	b	C
NO. 1 ICER COOLING SEA WATER PUMP	FEWV-350-3	H	b	A
NO. 2 ICER COOLING SEA WATER PUMP	FEWV-350-3	H	b	A

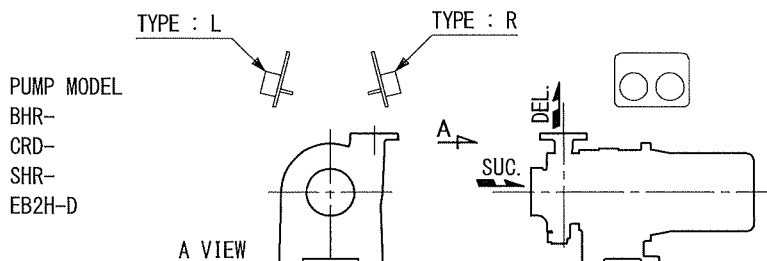


INDICATION LIST OF GAUGE SET POSITION

DRAWING NO.
98DG0106
DATE
20 FEB. 1998

HORIZONTAL PUMP TYPE

CUSTOMER : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.
SHIP No. : H1933A/1934A/1935A/1936A/1937A



Please indicate your required position in the blank.

*Terminal Box position : Actual position is as indicated in pump's outline drawing.

Remarks : 1. Please indicate "H" for **plant fitting** Gauge Boards. In this case, setting and piping for Gauge Boards are worked by customer. Therefore the Gauge Boards will be sent separately from pump.
2. Gauge Board size and the setting method are in accordance with maker's standard.

SERVICE AS	PUMP MODEL	GAUGE SET POSITION	NOTE
M/E JACKET WATER PREHEATER PUMP	BHR-65	H	
NO. 1 FRESH WATER PUMP	BHR-50-3	H	
NO. 2 FRESH WATER PUMP	BHR-50-3	H	
NO. 1 HOT WATER CIRCULATING PUMP	BHR-32	H	
NO. 2 HOT WATER CIRCULATING PUMP	BHR-32	H	
CHEMICAL CLEANING PUMP	BHR-40	H	
NO. 1 HOT GLYCOL WATER CIRCULATING PUMP	BHR-100	H	
NO. 2 HOT GLYCOL WATER CIRCULATING PUMP	BHR-100	H	
NO. 1 COLD GLYCOL WATER CIRCULATING PUMP	BHR-50	H	
NO. 2 COLD GLYCOL WATER CIRCULATING PUMP	BHR-50	H	
NO. 1 NaOH TRANSFER PUMP	BHR-40	H	
NO. 2 NaOH TRANSFER PUMP	BHR-40	H	



LUBRICANT LIST

DRAWING NO.

B6DG0341

DATE

5 OCT. 2016

SERVICE AS	MODEL	LUBRICANT POINT	QUANTITY / PUMP	
			GREASE (g)	OIL (L)
BALLAST WATER PUMP	FEWV-350-3 with VK-20NB	VACUUM PUMP OIL SEAL	4	---
BILGE, FIRE & G/S PUMP	FE2V-250-3 with VK-20NB	VACUUM PUMP OIL SEAL	4	---
EMERGENCY FIRE PUMP	FGV-100-2 with VK-20NC	VACUUM PUMP OIL SEAL	4	---
C/H WATER SPRAY PUMP	FEWV-250-4D	BALL BEARING	40	---

Recommended Grease (NLGI No.2)

Soap base : Lithium
 Consistency : 265 ~ 295
 Dropping point : 170 ~ 190 °C
 Range of working temp : 0 ~ 100 °C

Recommended Lubricating Oil (ISO VG32~68)

Type of Oil : Turbine Oil No.ISO VG32~68
 Flash Point : More Than 180 ~ 190 °C
 Viscosity : 38 ~ 90 mm² /s and less (at 37.8°C)
 : 4.6 ~ 8.0 mm² /s and less (at 98.9°C)
 Pour point : -5 ~ -7.5 °C and less
 Specific Gravity : 0.87~0.88

Maker	Grease Brand	Turbine Oil Brand
Idemitsu Kosan	Daphne Grease MP No.2 Daphne Eponex SR No.2	Daphne Super Turbine Oil 32,46 and 68
ENEOS	Multinoc Grease NO.2 Epnoc Grease AP-2	FBK Turbine 32,46,56 and 68 RIX Turbine 32,46,56 and 68
COSMO Oil Co.	Dinamax No.2 and EP-2	Special A Turbine Oil 32,46,56 and 68
EMG Lubricants	Mobil Grease XHP 222	Mobile DTE Oil Light, Medium & Heavy Medium
Shell Lubricants Japan	Shell Alvania EP Grease2	Shell Turbo Oil T32, 46,56 and 68
Castrol (BP)	Spheerol SX2	Perfecto X 32,46 and 68
Gulf	GulfSea Hyperbar CS2	GulfSea Turbine Oil 32,46 and 68
Chevron	Multifak EP2 Dura-Lith EP2	Regal R&O 32,46 and 68 Turbine Oil GST 32,46 and 68
Total Lubmarine	Ceran XM220	Preslia 32,46 and 68
SK	ZIC Crown Grease EP	SK Turbine Oil 32,46 and 68



LUBRICANT LIST

GEAR COUPLING

DRAWING NO.

B3DG0026

DATE

23 JAN. 2013

SERVICE AS	MODEL	LUBRICANT POINT	QUANTITY / PUMP	
			GREASE (g)	OIL (L)
BILGE, FIRE & G/S PUMP	FE2V-200-3	GEAR COUPLING	120 (UPPER) 150 (LOWER)	---

Recommendable lubricants

Use oil or grease for lubrication.

Table 1 show recommendable lubricants at applicable temperature from 0 to 80°C.

For application at rather low temperature, use the lubricants of lower viscosity by 1 to 2 grade than those shown in the tables.

Maker	Grease Brand	Oil Brand	Oil Brand
	NS, NC TYPE	NS TYPE	NC TYPE
Idemitsu Kosan	Daphne Eponex Grease EP	Daphne Super gear oil 680	Daphne Super gear oil 4600
Exxon Mobil	Pen-o-led EP grease Gemico grease ME-1 Mobilith SHC 460	Spartan EP 680 General SP Gearol 680 Mobilgear 636	Mobilgear MM, QQ
Shell Lubricants Japan	Gadus S2 V220 1	Omala oil 680	Omala oil 1500
JX Nippon Oil & Energy	Epnoc grease AP1 Diamond Multi Purpose EP1 grease	Bonnoc SP 680 Diamond Gearlube SP 680	Bonnoc SP 3800 Diamond Gearlube SP 1800, 3800
Cosmo oil	Cosmo grease Dynamax EP1	Cosmo gear SE 680	Cosmo gear SE 4600
BP / Castrol	Spheerol EPL1	Molub-alloy gear oil GO170W	Molub-alloy gear oil GO1500
Kyoudo yushi	Molylex M1		

Table 1

MOTORS

Hull No.: H1933A/34A/35A/36A/37A

TITLE:

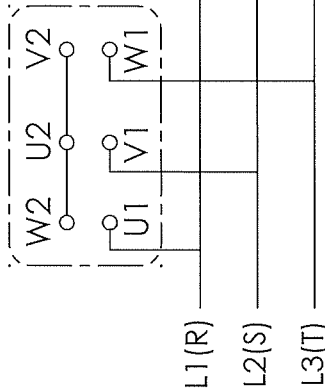
SPECIFICATION OF THREE PHASE INDUCTION MOTOR

CLASS	LR
DATE	2024.12.9
DWG.NO.	

INDEX

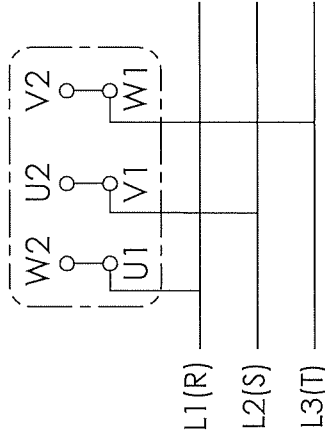
	Description	Motor Type	Voltage	QTY	Mount.	Starting method	Rating (kW,P)		Rated Current (A)	Starting Current (A)	Element space heater
01	M/E H.T.F.W. PUMP	HMC2 225M-4	440V	2	V1	DOL	45	4	70	521.5	1x 230V 80W
02	M/E JACKET WATER PREHEATER PUMP	HMC2 132S1-2	440V	1	B5	DOL	5.5	2	9	69.3	1x 230V 50W
03	L/T.C.W. BOOSTER PUMP	HMC2 225M-4	440V	2	V1	DOL	49	4	76	562.4	1x 230V 80W
04	EMERGENCY FIRE PUMP	HMC2 250M-2	440V	1	V1	Y-D	45	2	72.5	696.0	1x 230V 80W
05	FRESH WATER PUMP	HMC2 132S2-2	440V	2	B5	DOL	7.5	2	12.2	88.5	
06	HOT WATER CIRCULATING PUMP	HMC2 80 1-2	440V	2	B5	DOL	0.75	2	1.56	10.4	
07	CHEMICAL CLEANING PUMP	HMC2 90L-2	440V	1	B5	DOL	2.2	2	4	30.6	
08	GREY WATER DISCHARGE PUMP	HMC2 132M1-6	440V	1	B3	DOL	3.7	6	7.7	55.1	
09	SEWAGE DISCHARGE PUMP	HMC2 132M1-6	440V	1	B3	DOL	3.7	6	7.7	55.1	
10	E/R SEWAGE DISCHARGE PUMP	HMC2 112M2-6	440V	1	B3	DOL	2.2	6	4.85	25.5	
11	HOT GLYCOL WATER CIRCULATING PUMP	HMCX 160L-2	440V	2	B5	DOL	18.5	2	28.4	252.8	
12	COLD GLYCOL WATER CIRCULATING PUMP	HMCX 112M1-2	440V	2	B5	DOL	3.7	2	6.1	47.9	
13	NaOH TRANSFER PUMP	HMC2 90L-2	440V	2	B5	DOL	2.2	2	4	30.6	

Y

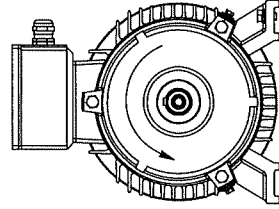


CounterClockwise

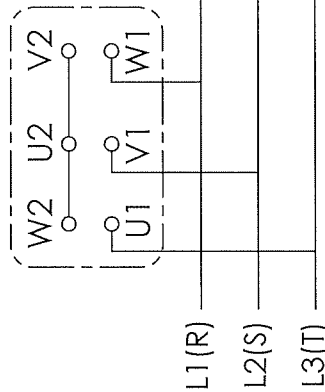
△



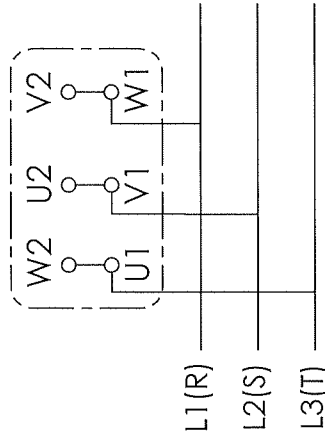
CounterClockwise



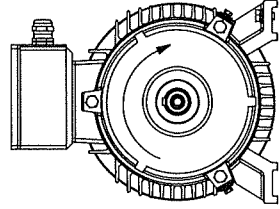
CounterClockwise



clockwise



clockwise



Clockwise

 HOYER EXCEEDING EXPECTATIONS		Hoyer Motors Over Heistenveld 42, DK-8370 Hadsten Tel : +45 86962111 Fax : +45 86961779 www.hoyermotors.com		DEBUR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION 02
		TITLE wiring diagram				
DRAWN _____ CHK'D _____ APP'VD _____ MFG _____ Q.A. _____	NAME _____ SIGNATURE _____ _____ _____ _____ _____	MATERIAL: _____ _____ _____ _____ _____		DWG NO. Drawing No.		A4
		WEIGHT: kg _____		SCALE: 1:10 SHEET 1 OF 1		

Data Sheet

Itemnumber....: TK42251217

POS.1 M/E H.T.F.W. PUMP

Version Type.....: HMC2 225M-4 Design.....: Induction motor Standard series.....: IEC 60034 Phase / Voltage range.....: 3~ / Low	Mechanical design Frame size.....: 225 Mounting.....: V1 Rain cap.....: Yes Protection class.....: IP55 Cooling method.....: IC411/TEFC External grounding.....: Yes Drain hole.....: Yes Frame material.....: Cast Iron Material approval.....: None Shaft.....: IEC standard Key.....: Closed key Balancing.....: Half key balancing Vibration class.....: Grade-A Weight (kg).....: 330	Bearing DE Bearing.....: 6313-ZZ/C3 NDE Bearing.....: 6312-ZZ/C3 Fixed bearing.....: DE
Electrical design Efficiency.....: IE2 Pole.....: 4 Power at 50 Hz (kW).....: 45,0 Hz.....: 60 (50 Hz output) Voltage.....: 440VD Winding voltage.....: 400VD/690VY 50 Hz Power output (kW).....: 45,0 Duty.....: S1 Insulation class.....: F Temperature rise.....: B	Environment condition Ambient temp. min. (°C)...: -20 Ambient temp. max. (°C)...: 45 Altitude (mtr up to).....: 1000	Terminal box Tbox position.....: Top Cable entry direction.....: Right (from DE) Cable entry Main.....: 2 x M50x1,5 blindcaps Accessory.....: 2 x M20x1,5 blindcaps Terminal board thread.....: 6-M8
Motor protection Thermal protection main....: PTC 3x 155 dgr Thermal protection second..: None Space heater.....: 1x 230V 80W Temperature detector.....: No SPM.....: No IR wire.....: No Tropical insulation.....: Yes		Test values Rotor inertia (kgm²).....: 0,57 Noise level (dB(A)).....: 68 No load current (A).....: 23,0 Winding resist. (ohm).....: 0,16 Starting time (sec.).....: 0,13 Temp. rise winding (K).....: 75 Temp. rise surface (K).....: 58
Explosion protection According to.....: None Type of protection.....: None		
General Starting method.....: DOL Direction of rotation.....: CCW Painting.....: Munsell 2.5 G 8/4 Nameplate.....: P2 50Hz Marine Special packing.....: No Special requirements.....: No		

Rated power (kW)	45,0
Frequency (Hz)	60
Voltage (V)	440
Connection	D
Full load current (A)	70
Speed (rpm)	1770
Power factor cos(phi)	0,91
Efficiency, 100/75/50 (%)	92,6/ 92,5 /92,0
Ist/In	7,45
Full load torque (Nm)	240
Tst/Tn	2,48
Tmax/Tn	3,25
Duty	S1
Ambient temp. (°C)	45

Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz

Data Sheet

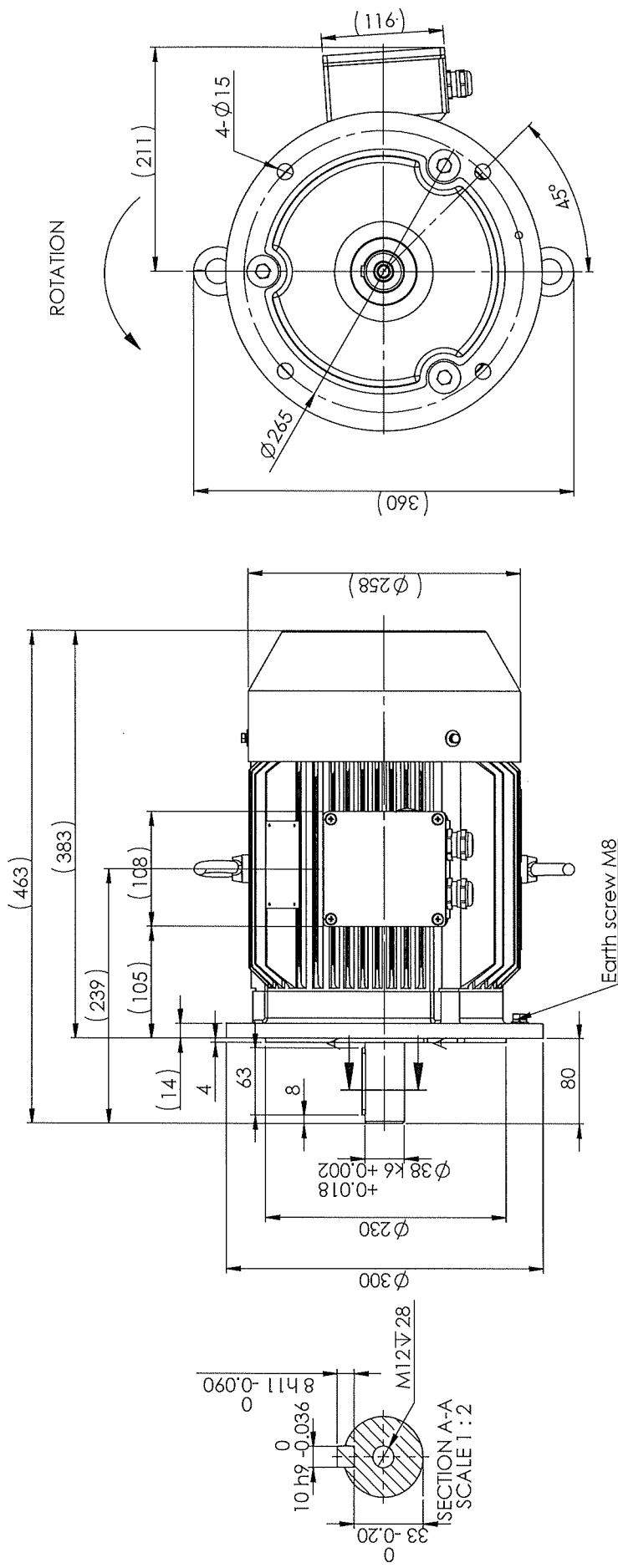
Itemnumber....: TK21320218

POS.2 M/E JACKET WATER PREHEATER PUMP

Version Type.....: HMC2 132S1-2 Design.....: Induction motor Standard series.....: IEC 60034-1 Phase / Voltage range.....: 3~ / Low	Mechanical design Frame size.....: 132 Mounting.....: B5 Rain cap.....: No Protection class.....: IP55 Cooling method.....: IC411/TEFC External grounding.....: Yes Drain hole.....: Yes Frame material.....: Cast Iron Material approval.....: None Shaft.....: IEC standard Key.....: Closed key Balancing.....: Half key balancing Vibration class.....: Grade-A Weight (kg).....: 54,0	Bearing DE Bearing.....: 6308-ZZ/C3 NDE Bearing.....: 6308-ZZ/C3 Fixed bearing.....: D-end
Electrical design Efficiency.....: IE2 Pole.....: 2 Power at 50 Hz (kW).....: 5,5 Hz.....: 60 (50 Hz output) Voltage.....: 440VD Winding voltage.....: 400VD/690VY 50 Hz Power output (kW).....: 5,5 Duty.....: S1 Insulation class.....: F Temperature rise.....: B	Environment condition Ambient temp. min. (°C)...: -20 Ambient temp. max. (°C)...: 45 Altitude (mtr up to).....: 1000	Terminal box Tbox position.....: Top Cable entry direction.....: Right (from DE) Cable entry Main.....: 2 x M25x1,5 blindcaps Accessory.....: 1 x M20x1,5 blindcaps Terminal board thread.....: 6-M5
Motor protection Thermal protection main....: PTC 3x 155 dgr Thermal protection second..: None Space heater.....: 1 X 50W Space heater power.....: 230V Temperature detector.....: No SPM.....: No IR wire.....: No Tropical insulation.....: Yes		Test values Rotor inertia (kgm²).....: 0,017 Noise level (dB(A)).....: 68 No load current (A).....: 2,90 Winding resist. (ohm).....: 2,23 Starting time (sec.).....: 0,12 Temp. rise winding (K).....: 56 Temp. rise surface (K).....: 49
Explosion protection According to.....: None		
General Starting method.....: DOL Direction of rotation.....: CCW Painting.....: Munsell 2.5 G 8/4 Nameplate.....: P2 50Hz Marine Special packing.....: No Special requirements.....: No		

Rated power (kW)	5,50
Frequency (Hz)	60
Voltage (V)	440
Connection	D
Full load current (A)	9,00
Speed (rpm)	3500
Power factor cos(phi)	0,92
Efficiency, 100/75/50 (%)	86,1/ 87,1 /87,1
Ist/In	7,70
Full load torque (Nm)	15,0
Tst/Tn	2,26
Tmax/Tn	3,50
Duty	S1
Ambient temp. (°C)	45

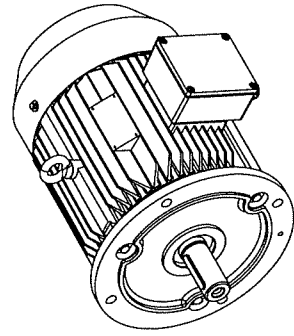
Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz



Fa=134N

Remark:

- 1.Cable glands are 2xM25X1.5 and Blind cap is M20X1.5
- 2.Eye bolts are M10



HOYER		Hoyer Motors Over Hadstenvej 42, DK-8370 Hadsen Tel: +45 86982111 Fax: +45 86981779 www.hoyermotors.com		DO NOT SCALE DRAWING	REVISION	02
DESIGNED	NAME	SIGNATURE	DATE	POS.2 M/E JACKET WATER PREHEATER PUMP		
DRAWN	TH		20170622	TITLE		
CHKD				HMC2-132S-B5		
APPRD				DWG NO.		
WFG				70519		
QA				MATERIAL: cast iron		
				WEIGHT: Kg		
				SCALE: 1:10		
				SHEET 1 OF 1		
				A3		

Data Sheet

Itemnumber....: TK42251217

POS.3 L/T.C.W. BOOSTER PUMP

Version

Type.....: HMC2 225M-4
Design.....: Induction motor
Standard series.....: IEC 60034
Phase / Voltage range.....: 3~ / Low

Electrical design

Efficiency.....: IE2
Pole.....: 4
Power at 50 Hz (kW).....: 45,0
Hz.....: 60 (50 Hz output)
Voltage.....: 440VD
Winding voltage.....: 400VD/690VY 50 Hz
Power output (kW).....: 45,0
Duty.....: S1
Insulation class.....: F
Temperature rise.....: B

Motor protection

Thermal protection main....: PTC 3x 155 dgr
Thermal protection second..: None
Space heater.....: 1x 230V 80W
Temperature detector.....: No
SPM.....: No
IR wire.....: No
Tropical insulation.....: Yes

Explosion protection

According to.....: None
Type of protection.....: None

General

Starting method.....: DOL
Direction of rotation.....: CCW
Painting.....: Munsell 2.5 G 8/4
Nameplate.....: P2 50Hz Marine
Special packing.....: No
Special requirements.....: No

Mechanical design

Frame size.....: 225
Mounting.....: V1
Rain cap.....: Yes
Protection class.....: IP55
Cooling method.....: IC411/TEFC
External grounding.....: Yes
Drain hole.....: Yes
Frame material.....: Cast Iron
Material approval.....: None
Shaft.....: IEC standard
Key.....: Closed key
Balancing.....: Half key balancing
Vibration class.....: Grade-A
Weight (kg).....: 330

Environment condition

Ambient temp. min. (°C)...: -20
Ambient temp. max. (°C)...: 45
Altitude (mtr up to).....: 1000

Bearing

DE Bearing.....: 6313-ZZ/C3
NDE Bearing.....: 6312-ZZ/C3
Fixed bearing.....: DE

Terminal box

Tbox position.....: Top
Cable entry direction.....: Right (from DE)
Cable entry
Main.....: 2 x M50x1,5
blindcaps
Accessory.....: 2 x M20x1,5
blindcaps
Terminal board thread.....: 6-M8

Test values

Rotor inertia (kgm²).....: 0,57
Noise level (dB(A)).....: 68
No load current (A).....: 23,0
Winding resist. (ohm).....: 0,16
Starting time (sec.).....: 0,13
Temp. rise winding (K).....: 75
Temp. rise surface (K).....: 58

Rated power (kW): 49,0
Frequency (Hz): 60
Voltage (V): 440
Connection: D
Full load current (A): 76,0
Speed (rpm): 1770
Power factor cos(phi): 0,91
Efficiency, 100/75/50 (%) : 92,6/ 92,5 /92,0
Ist/In: 7,40
Full load torque (Nm): 270
Tst/Tn: 2,24
Tmax/Tn: 2,90
Duty: S1
Ambient temp. (°C): 45

Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz

Data Sheet

Itemnumber....: TK22500217

POS.4 EMERGENCY FIRE PUMP

Version

Type.....: HMC2 250M-2
Design.....: Induction motor
Standard series.....: IEC 60034
Phase / Voltage range.....: 3~ / Low

Electrical design

Efficiency.....: IE2
Pole.....: 2
Power at 50 Hz (kW).....: 55,0
Hz.....: 60 (50 Hz output)
Voltage.....: 440VD
Winding voltage.....: 400VD/690VY 50 Hz
Power output (kW).....: 55,0
Duty.....: S1
Insulation class.....: F
Temperature rise.....: B

Motor protection

Thermal protection main....: PTC 3x 155 dgr
Thermal protection second..: None
Space heater.....: 1x 230V 80W
Temperature detector.....: No
SPM.....: No
IR wire.....: No
Tropical insulation.....: Yes

Explosion protection

According to.....: None
Type of protection.....: None

General

Starting method.....: Y/D
Direction of rotation.....: CCW
Painting.....: Munsell 2.5 G 8/4
Nameplate.....: P2 50Hz Marine
Special packing.....: No
Special requirements.....: No

Mechanical design

Frame size.....: 250
Mounting.....: V1
Rain cap.....: Yes
Protection class.....: IP55
Cooling method.....: IC411/TEFC
External grounding.....: Yes
Drain hole.....: Yes
Frame material.....: Cast Iron
Material approval.....: None
Shaft.....: IEC standard
Key.....: Closed key
Balancing.....: Half key balancing
Vibration class.....: Grade-A
Weight (kg).....: 405

Environment condition

Ambient temp. min. (°C)...: -20
Ambient temp. max. (°C)...: 45
Altitude (mtr up to).....: 1000

Bearing

DE Bearing.....: 6313/C3
NDE Bearing.....: 6313/C3
Fixed bearing.....: DE

Terminal box

Tbox position.....: Top
Cable entry direction.....: Right (from DE)
Cable entry
Main.....: 2 x M63x1,5
blindcaps
Accessory.....: 2 x M20x1,5
blindcaps
Terminal board thread.....: 6-M10

Test values

Rotor inertia (kgm²).....: 0,34
Noise level (dB(A)).....: 79
No load current (A).....: 26,0
Winding resist. (ohm).....: 0,078
Starting time (sec.).....: 0,29
Temp. rise winding (K).....: 68
Temp. rise surface (K).....: 49

Rated power (kW)	64,0	45,0
Frequency (Hz)	60	60
Voltage (V)	440	440
Connection	D	D
Full load current (A)	102	72,5
Speed (rpm)	3550	3560
Power factor cos(phi)	0,89	0,88
Efficiency, 100/75/50 (%)	92,9/ 92,8 /92,0	93,0/ 92,5 /91,5
Ist/In	6,20	9,60
Full load torque (Nm)	172	121
Tst/Tn	1,66	2,72
Tmax/Tn	2,75	4,55
Duty	S1	S1
Ambient temp. (°C)	45	45

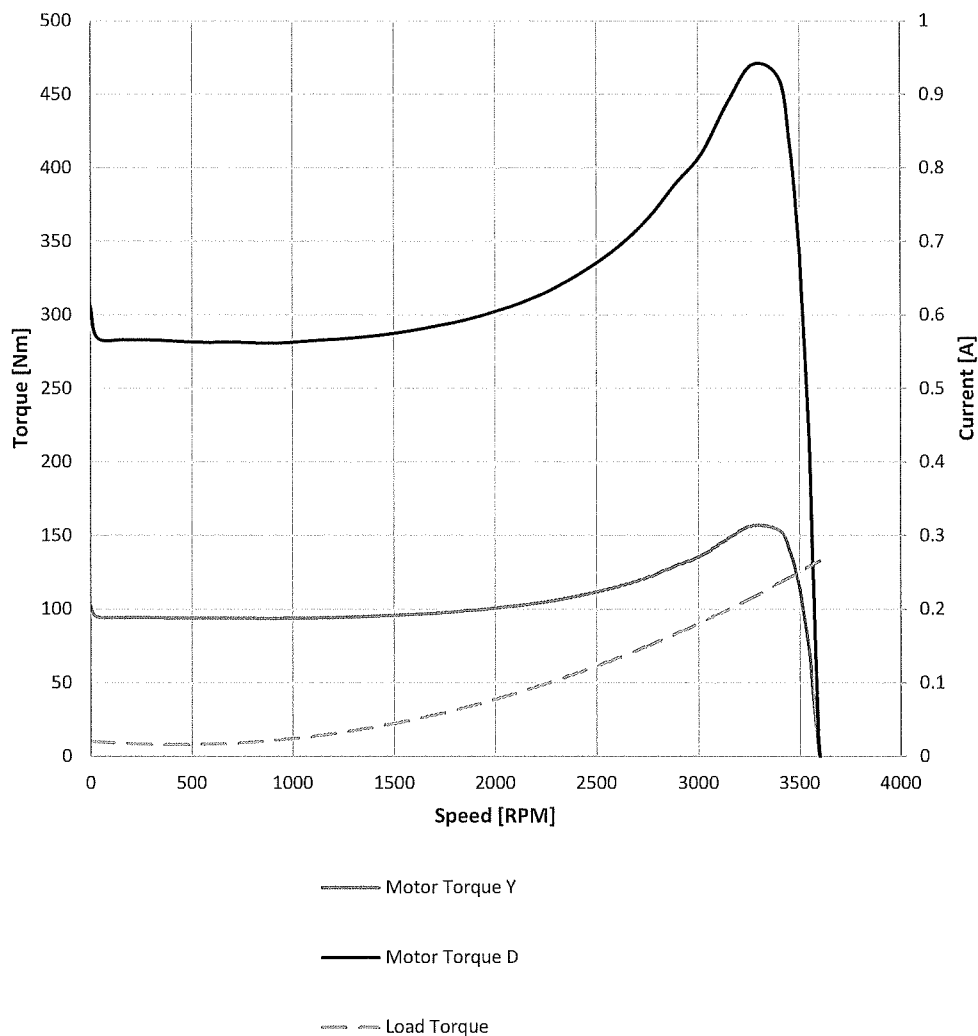
Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz

Star/Delta Torque and Current vs. Speed Curve

HOYER

100% QUALITY ASSURANCE

Pump		Motor	
Use	EMERGENCY FIRE PUMP	Type	HMC2 250M-2
Power [kW]	45	Power [kW]	64
Speed [RPM]	3500	Voltage [V]	440
GD ² [N·m ²]	4.9	Frequency [Hz]	60
Date	11/29/2024	Full load current [A]	102
Comments		Speed [RPM]	3550
		Inertia [N·m ²]	2.45
		Poles	2
		Item number	-
		Serialnumber	-



The curves are created by: CAC App. by: GUG Data are: Calc.

Remarks:

Data are not binding. Hoyer reserves the right to implement changes without notice. The extent of data are not equivalent to nameplate on motors.

Data Sheet

Itemnumber....: TK21321218

POS.5 FRESH WATER PUMP

Version

Type.....: HMC2 132S2-2
Design.....: Induction motor
Standard series.....: IEC 60034
Phase / Voltage range.....: 3~ / Low

Electrical design

Efficiency.....: IE2
Pole.....: 2
Power at 50 Hz (kW).....: 7,50
Hz.....: 60 (50 Hz output)
Voltage.....: 440VD
Winding voltage.....: 400VD/690VY 50 Hz
Power output (kW).....: 7,50
Duty.....: S1
Insulation class.....: F
Temperature rise.....: B

Motor protection

Thermal protection main....: None
Thermal protection second...: None
Space heater.....: None
Temperature detector.....: No
SPM.....: No
IR wire.....: No
Tropical insulation.....: Yes

Explosion protection

According to.....: None
Type of protection.....: None

General

Starting method.....: DOL
Direction of rotation.....: CCW
Painting.....: Munsell 2.5 G 8/4
Nameplate.....: P2 50Hz Marine
Special packing.....: No
Special requirements.....: No

Mechanical design

Frame size.....: 132
Mounting.....: B5
Rain cap.....: No
Protection class.....: IP55
Cooling method.....: IC411/TEFC
External grounding.....: Yes
Drain hole.....: Yes
Frame material.....: Cast Iron
Material approval.....: None
Shaft.....: IEC standard
Key.....: Closed key
Balancing.....: Half key balancing
Vibration class.....: Grade-A
Weight (kg).....: 59,0

Environment condition

Ambient temp. min. (°C)...: -20
Ambient temp. max. (°C)...: 45
Altitude (mtr up to).....: 1000

Bearing

DE Bearing.....: 6308-ZZ/C3
NDE Bearing.....: 6308-ZZ/C3
Fixed bearing.....: DE

Terminal box

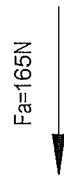
Tbox position.....: Top
Cable entry direction.....: Right (from DE)
Cable entry
Main.....: 2 x M25x1,5
blindcaps
Accessory.....: 1 x M20x1,5
blindcaps
Terminal board thread.....: 6-M5

Test values

Rotor inertia (kgm²).....: 0,021
Noise level (dB(A)).....: 72
No load current (A).....: 4,55
Winding resist. (ohm).....: 1,94
Starting time (sec.).....: 0,12
Temp. rise winding (K).....: 74

Rated power (kW): 7,50
Frequency (Hz): 60
Voltage (V): 440
Connection: D
Full load current (A): 12,2
Speed (rpm): 3510
Power factor cos(phi): 0,91
Efficiency, 100/75/50 (%) : 87,3/ 87,8 /87,4
Ist/In: 7,25
Full load torque (Nm): 20,4
Tst/Tn: 2,18
Tmax/Tn: 3,30
Duty: S1
Ambient temp. (°C): 45

Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz



END PLAY $\leq 0.5\text{mm}$

- | | | | | | | | | | | | |
|---|--|---|--|-----------------------------------|--|----------------------|--|--|--|--------------|--|
|  HOYER
<small>CASTING SOLUTIONS</small> | | Hoyer Motors
Over Hagedorn 42, DK-4370 Hadsten
Denmark
Tel: +45 8681 770
Fax: +45 8681 775
www.hoyermotors.com | | DEBUR AND
BREAK SHARP
EDGES | | DO NOT SCALE DRAWING | | REVISION | | 02 | |
| | | | | | | | | | | | |
| DRAWN | | TH | | DATE | | 20170622 | | POS.5 FRESH WATER PUMP

HMC2-132S-B5

TITLE: | | | |
| CHK'D | | | | | | | | | | | |
| APP'D | | | | | | | | | | | |
| MFG | | | | | | | | | | | |
| G.A. | | | | | | | | | | | |
| | | | | MATERIAL: | | cast iron | | DWG NO. | | 70519 | |
| | | | | | | | | SCALE:1:10 | | SHEET 1 OF 1 | |
| | | | | | | WEIGHT: Kg | | A3 | | | |

Data Sheet

Itemnumber....: 3220800209

POS.6 HOT WATER CIRCULATING PUMP

Version Type.....: HMC2 80 1-2 Design.....: Induction motor Standard series.....: IEC 60034 Phase / Voltage range.....: 3~ / Low Electrical design Efficiency.....: IE2 Pole.....: 2 Hz.....: 60 (60 Hz output) Voltage.....: 440VY Vibration class.....: Grade-A Duty.....: S1 Insulation class.....: F Temperature rise.....: B Motor protection Thermal protection main....: None Thermal protection second...: None Space heater.....: None Temperature detector.....: No SPM.....: No IR wire.....: No Tropical insulation.....: Yes Explosion protection According to.....: None Type of protection.....: None General Starting method.....: DOL Direction of rotation.....: CCW Painting.....: Munsell 2.5 G 8/4 Nameplate.....: P2 50Hz Marine Special packing.....: No Special requirements.....: No	Mechanical design Frame size.....: 80 Mounting.....: B5 Rain cap.....: No Protection class.....: IP55 Cooling method.....: IC411/TEFC External grounding.....: Yes Drain hole.....: Yes Frame material.....: Cast Iron Material approval.....: None Shaft.....: IEC Standard Key.....: Closed key Balancing.....: Half key balancing Weight (kg).....: 20,00 Environment condition Ambient temp. min. (°C)...: -20 Ambient temp. max. (°C)...: 45 Altitude (mtr up to).....: 1000	Bearing DE Bearing.....: 6204-ZZ/C3 NDE Bearing.....: 6204-ZZ/C3 Fixed bearing.....: DE Terminal box Thox position.....: Top Cable entry direction.....: Right (from DE) Cable entry Main.....: 1 x M20x1,5 Ø8-13mm metal Accessory.....: 1 x M20x1,5 Metal blindcap Terminal board thread.....: 6-M4 Test values Rotor inertia (kgm²).....: 0,0009
---	--	--

Rated power (kW)	0,75
Frequency (Hz)	60
Voltage (V)	440
Connection	Y
Full load current (A)	1,56
Speed (rpm)	3450
Power factor cos(phi)	0,82
Efficiency, 100/75/50 (%) : 75,8/ 76,1 /73,4	
Ist/In	6,65
Full load torque (Nm)	2,06
Tst/Tn	2,85
Tmax/Tn	3,00
Duty	S1
Ambient temp. (°C)	45

IE2 efficiency @400V 50Hz

Data Sheet

Itemnumber....: TK20901218

POS.7 CHEMICAL CLEANING PUMP

Version		Mechanical design		Bearing	
Type.....	HMC2 90L-2	Frame size.....	90	DE Bearing.....	6205-ZZ/C3
Design.....	Induction motor	Mounting.....	B5	NDE Bearing.....	6205-ZZ/C3
Standard series.....	IEC 60034-1	Rain cap.....	No	Fixed bearing.....	D-end
Phase / Voltage range.....	3~ / Low	Protection class.....	IP55		
Electrical design		Cooling method.....	IC411/TEFC		
Efficiency.....	IE2	External grounding.....	Yes	Terminal box	
Pole.....	2	Drain hole.....	Yes	Tbox position.....	Top
Power at 50 Hz (kW).....	2,20	Frame material.....	Cast Iron	Cable entry direction.....	Right (from DE)
Hz.....	60 (50 Hz output)	Material approval.....	None	Cable entry	
Voltage.....	440VD	Shaft.....	IEC standard	Main.....	1xM20x1,5 blindcaps
Winding voltage.....	230VD/400VY 50 Hz	Key.....	Closed key	Accessory.....	1xM20x1,5 blindcaps
Power output (kW).....	2,20	Balancing.....	Half key balancing	Terminal board thread.....	6-M4
Duty.....	S1	Vibration class.....	Grade-A		
Insulation class.....	F	Weight (kg).....	33,0		
Temperature rise.....	B				
Motor protection		Environment condition		Test values	
Thermal protection main....	None	Ambient temp. min. (°C)...	-20	Rotor inertia (kgm²).....	0,0030
Thermal protection second..	None	Ambient temp. max. (°C)...	45	Noise level (dB(A)).....	66
Space heater.....	None	Altitude (mtr up to).....	1000	No load current (A).....	1,94
Temperature detector.....	No			Winding resist. (ohm).....	2,15
SPM.....	No			Starting time (sec.).....	0,05
IR wire.....	No			Temp. rise winding (K).....	54
Tropical insulation.....	Yes			Temp. rise surface (K).....	39
Explosion protection					
According to.....	None				
General					
Starting method.....	DOL				
Direction of rotation.....	CCW				
Painting.....	Munsell 2.5 G 8/4				
Nameplate.....	P2 50Hz Marine				
Special packing.....	No				
Special requirements.....	No				
Rated power (kW)	2,20				
Frequency (Hz)	60				
Voltage (V)	440				
Connection	Y				
Full load current (A)	4,00				
Speed (rpm)	3440				
Power factor cos(phi)	0,87				
Efficiency, 100/75/50 (%) . .	82,0/ 83,8 /83,6				
Ist/In	7,65				
Full load torque (Nm)	5,80				
Tst/Tn	2,80				
Tmax/Tn	3,05				
Duty	S1				
Ambient temp. (°C)	45				
Special features	Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz				

Data Sheet

HOYER

MOTORS

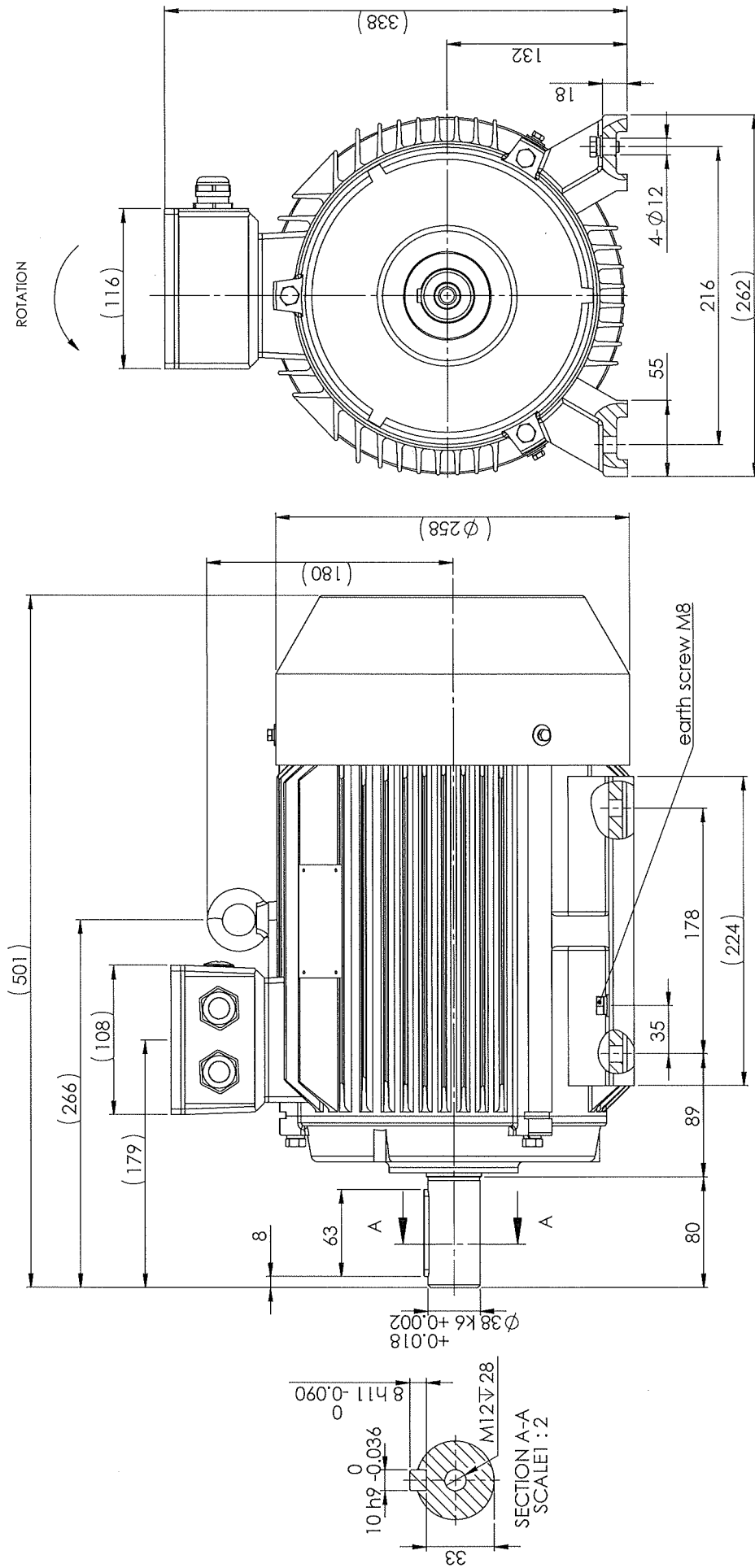
POS.8 GREY WATER DISCHARGE PUMP

Type		: HMC2 132M1-6
Mounting		: B3
Protection class		: IP55
Insulations class		: F
Temperature rise		: B
Standard		: IEC 60034-1
Test standard		: IEC 60034-2-1
Altitude above s.l. max.(m)	..	: 1000
Cooling method		: IC411/TEFC
Poles		: 6
Painting		: Munsell 2.5 G 8/4
DE. Bearing		: 6308-ZZ/C3
NDE. Bearing		: 6308-ZZ/C3
Bearing fixed in		: D-end
Weight (kg)		: 66.0

Rotor inertia (kgm ²)	...	0,044
Noise level (dB(A))	...	59
Vibration class	...	Grade-A
No load current (A)	...	5,35
Winding resistance (ohm)	...	3,62
Starting time (sec.)	...	0,05
Temperature rise winding	...	77
Temperature rise surface	...	84
Tbox position	...	Top
Cable entry main	...	2 x M25x1,5 Ø11-17mm
Cable entry accessory	...	1 x M20x1,5
Terminal board thread	...	6-M5
Thermal protection	...	
Space heater	...	
Starting method	...	DOL

Rated power (kW)	 :	3,70
Frequency (Hz)	 :	60
Voltage (V)	 :	440
Connection	 :	D
Full load current (A)	 :	7,70
Speed (rpm)	 :	1150
Power factor cos(phi)	 :	0,78
Efficiency, 100/75/50 (%)	... :	83,5 83,0 81,2
Ist/In	 :	7,15
Full load torque (Nm)	 :	31,0
Tst/Tn	 :	2,90
Tmax /Tn	 :	2,90
Duty	 :	S1
Ambient temp. (°C)	 :	45

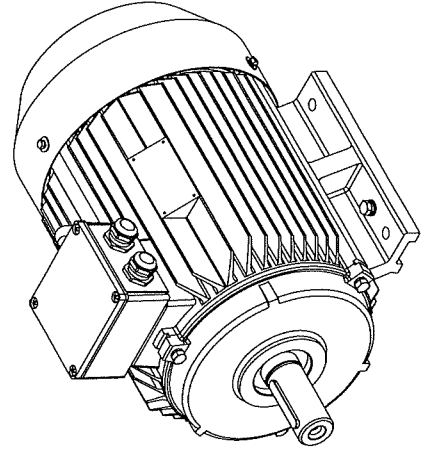
According to:
Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz



Remark:

1. Cable glands are 2xM25X1.5 and Blind cap is M20X1.5

2. Eye bolt is M10



HOYER PRECISION. EXCEPTIONAL.		Hoyer Motors Over Haldenvej 42, DK-8370 Hadsen Tel.: +45 86982111 Fax: +45 86982112 www.hoyermotors.com		REPAIR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION	01
DRAWN	NAME	SIGNATURE	DATE	TITLE: POS 8 GREY WATER DISCHARGE PUMP			
CHECKED	TH		2012/02/09	HMC2-132M-B3			
APPROVED				DWG NO. 70521			
AWG				MATERIAL: cast iron			
Q.A.				SCALE: 1:5			
				WEIGHT: Kg			
				SHEET 1 OF 1			

Data Sheet

HOYER

MOTORS

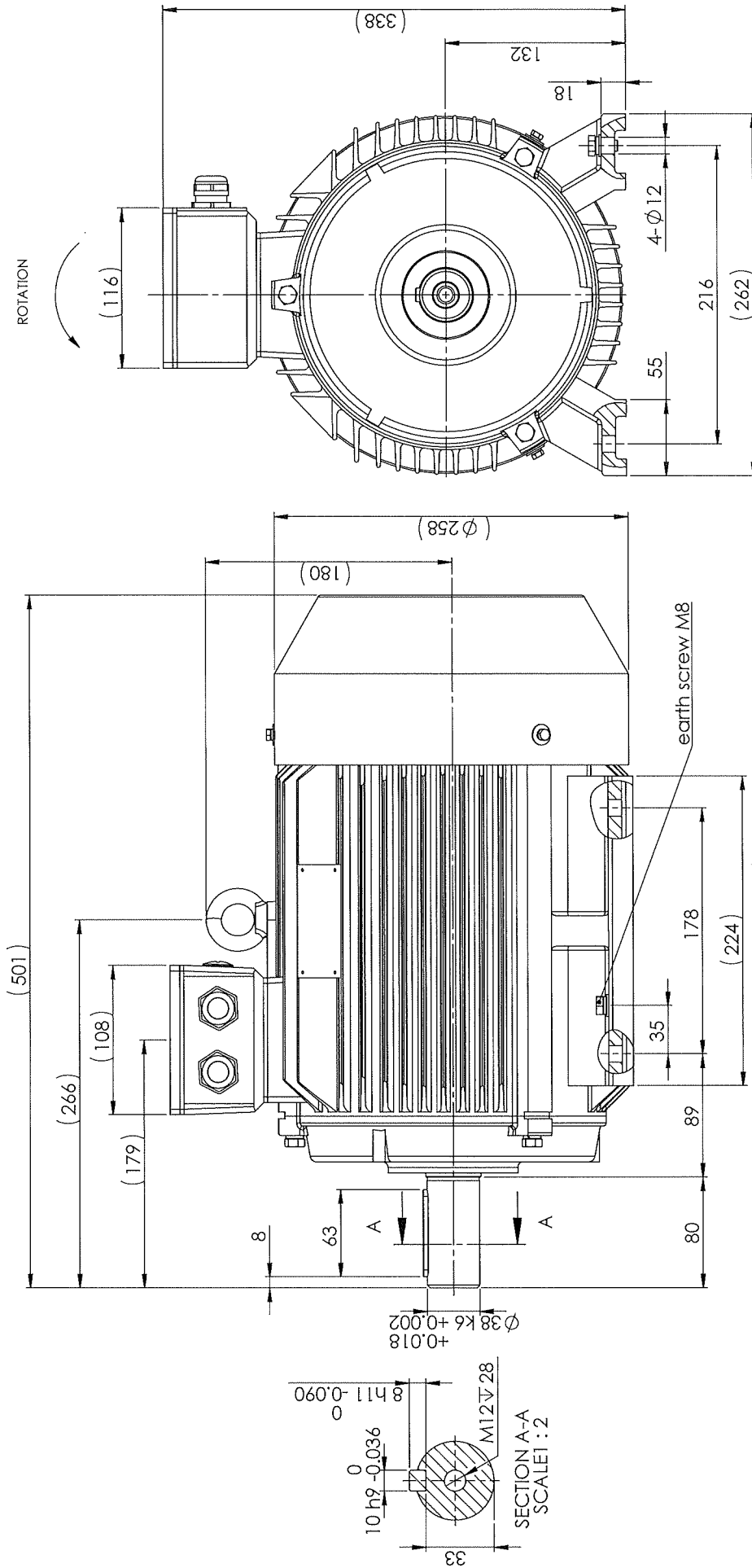
POS.9 SEWAGE DISCHARGE PUMP

Type		: HMC2 132M1-6
Mounting		: B3
Protection class		: IP55
Insulations class		: F
Temperature rise		: B
Standard		: IEC 60034-1
Test standard		: IEC 60034-2-1
Altitude above s.l. max.(m)	..	: 1000
Cooling method		: IC411/TEFC
Poles		: 6
Painting		: Munsell 2.5 G 8/4
DE. Bearing		: 6308-ZZ/C3
NDE. Bearing		: 6308-ZZ/C3
Bearing fixed in		: D-end
Weight (kg)		: 66.0

Rotor inertia (kgm ²)	...	0,044
Noise level (dB(A))	...	59
Vibration class	...	Grade-A
No load current (A)	...	5,35
Winding resistance (ohm)	...	3,62
Starting time (sec.)	...	0,05
Temperature rise winding	...	77
Temperature rise surface	...	84
Tbox position	...	Top
Cable entry main	...	2 x M25x1,5 Ø11-17mm
Cable entry accessory	...	1 x M20x1,5
Terminal board thread	...	6-M5
Thermal protection	...	
Space heater	...	
Starting method	...	DOL

Rated power (kW)	3,70
Frequency (Hz)	60
Voltage (V)	440
Connection	D
Full load current (A)	7,70
Speed (rpm)	1150
Power factor cos(phi)	0,78
Efficiency, 100/75/50 (%)	83,5 83,0 81,2
Ist/In	7,15
Full load torque (Nm)	31,0
Tst/Tn	2,90
Tmax /Tn	2,90
Duty	S1
Ambient temp. (°C)	45

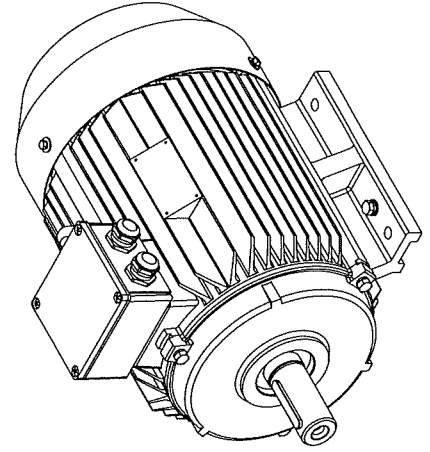
According to:
Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz



Remark:

1. Cable glands are 2xM25X1.5 and Blind cap is M20X1.5

2. Eye bolt is M10



HOYER		Hoyer Motors Over Hadsenvej 42, DK-8370 Hadsen Tel.: +45 86982111 Fax: +45 86981779 www.hoyermotors.com		DO NOT SCALE DRAWING	REVISION	01
DESIGN APPROVAL		BREAK AND BREAK SHARP EDGES		POS.9 SEWAGE DISCHARGE PUMP		
DRAWN	TH	SIGNATURE	DATE	TITLE		
CHECK			20/26/20	HMC2-132M-B3		
APP'D				DWG NO.		
WFG				70521		
G.A.				SCALE: 1:5		
MATERIAL: cast iron		WEIGHT: Kg		SHEET 1 OF 1		
				A3		

Data Sheet

Itemnumber....: TK61120117

POS.10 E/R SEWAGE DISCHARGE PUMP

Version

Type.....: HMC2 112M-6
Design.....: Induction motor
Standard series.....: IEC 60034-1
Phase / Voltage range.....: 3~ / Low

Electrical design

Efficiency.....: IE2
Pole.....: 6
Power at 50 Hz (kW).....: 7,50
Hz.....: 60 (50 Hz output)
Voltage.....: 440VD
Winding voltage.....: 400VD/690VY 50 Hz
Power output (kW).....: 7,50
Duty.....: S1
Insulation class.....: F
Temperature rise.....: B

Motor protection

Thermal protection main.....: None
Thermal protection second.....: None
Space heater.....: None
Temperature detector.....: No
SPM.....: No
IR wire.....: No
Tropical insulation.....: Yes

Explosion protection

According to.....: None

General

Starting method.....: DOL
Direction of rotation.....: CCW
Painting.....: Munsell 2.5 G 8/4
Nameplate.....: P2 50Hz Marine
Special packing.....: No
Special requirements.....: No

Mechanical design

Frame size.....: 112
Mounting.....: B3
Rain cap.....: No
Protection class.....: IP55
Cooling method.....: IC411/TEFC
External grounding.....: Yes
Drain hole.....: Yes
Frame material.....: Cast Iron
Material approval.....: None
Shaft.....: IEC standard
Key.....: Closed key
Balancing.....: Half key balancing
Vibration class.....: Grade-A
Weight (kg).....: 41,0

Environment condition

Ambient temp. min. (°C)...: -20
Ambient temp. max. (°C)...: 45
Altitude (mtr up to).....: 1000

Bearing

DE Bearing.....: 6306-ZZ/C3
NDE Bearing.....: 6306-ZZ/C3
Fixed bearing.....: D-end

Terminal box

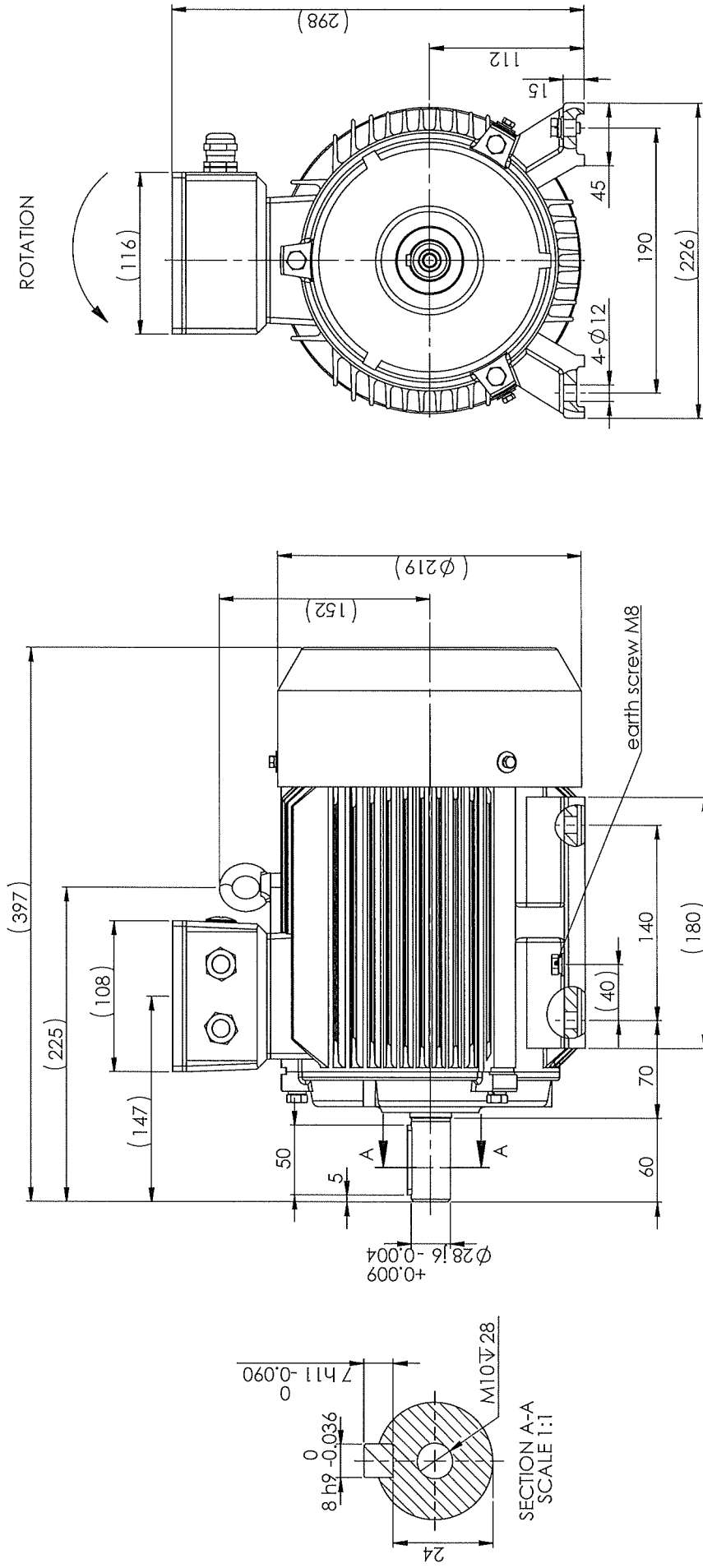
Tbox position.....: Top
Cable entry direction.....: Right (from DE)
Cable entry
Main.....: 2xM20x1,5
blindcaps
Accessory.....: 1 x M20x1,5
blindcaps
Terminal board thread.....: 6-M5

Test values

Rotor inertia (kgm²).....: 0,021
Noise level (dB(A)).....: 60
No load current (A).....: 3,20
Winding resist. (ohm).....: 1,98
Starting time (sec.).....: 0,05
Temp. rise winding (K).....: 66
Temp. rise surface (K).....: 74

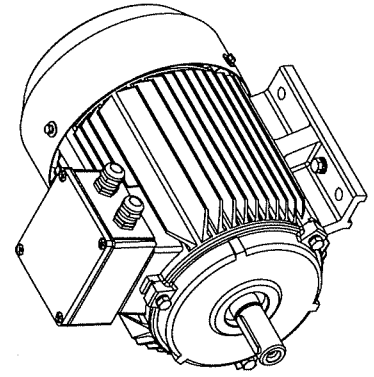
Rated power (kW): 2,20
Frequency (Hz): 60
Voltage (V): 440
Connection: Y
Full load current (A): 4,85
Speed (rpm): 1160
Power factor cos(phi): 0,74
Efficiency, 100/75/50 (%) : 80,5/ 79,8 /76,9
Ist/In: 5,25
Full load torque (Nm): 18,2
Tst/Tn: 2,60
Tmax/Tn: 2,65
Duty: S1
Ambient temp. (°C): 45

Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz



Remark:

1. Eye bolt is M8
2. Cable glands are 2XM20X1.5 and Blind cap is M20X1.5



HOYER		Hoyer Motors Over Hadslevvej 42, DK-8370 Halskø Tel.: +45 86982111 Fax: +45 86981779 www.hoyermotors.com		DO NOT SCALE DRAWING	REVISION	01
DRAWN: TH		DATE: 20/06/20		TITLE: POS.10 E/R SEWAGE DISCHARGE PUMP		
CHKD:		SIGNATURE:		HMC2-112M-B3		
APP'D:				DWG NO. 70524		
MFG:				MATERIAL: cast iron		
Q.A:				SCALE: 1:5		
				SHEET 1 OF 1		

HOYER

MOTORS

Data sheet

POS.11 HOT GLYCOL WATER CIRCULATING PUMP

Item no. / Configuration 02 / 2102160-18.5-020211-B022042-S-B00165
Description HMCX 160L-2 B5 18.5kW IE2 400/690@50

Ex db IIBT4

Date

25-11-2024

General data	
Standard series	IEC 60034
Type	HMCX 160L-2
Design	400 VD / 690 VY 50 Hz + 480 VD / 830 VY 60 Hz
Direction of rotation	CCW
Installation max. altitude (m)	1.000
Poles	2

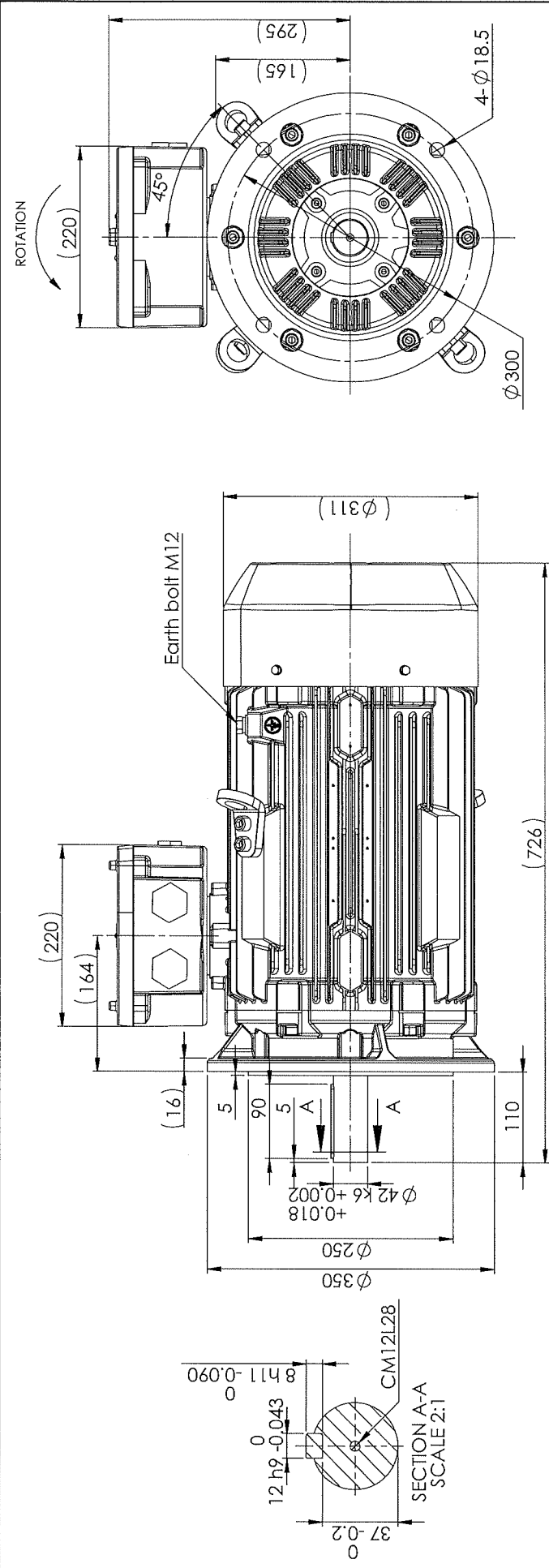
Motor protection	
Winding temp. detector	No
Temp. detector type	None
Bearing detector	No
Winding PTC main	PTC 3 x 145°C
Winding PTC spare	None
Space heater	230 V 40 W
Terminal board thread	M6
Electrical insulation	1300 V @ 1 µs (suitable for VFD at D-connection)
Tropical insulation	Yes
SPM	N/A
External grounding	
Drain hole	Yes
Rain cap	No

Classification	
Class society	
Shaft requirement acc. to	None
Ex. code	ATEX & IECEx: Ex II 2G Ex db IIB T4 Gb / Ex db IIB T4 Gb

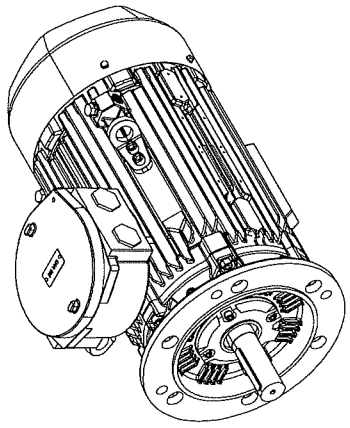
Mechanical data	
Frame / Shaft height	IEC 160
Type of construction	B5
Cooling method	IC411
Protection class	IP55
Temp. insulation class	F
Frame material	Reinforced cast iron
Keyway	Closed
Balancing	Half key
IEC vibration class	A
Weight (kg)	206
Bearing (DE)	6309 2Z / C3 / DGBB
Bearing (NDE)	6307 2Z / C3 / DGBB
Fixed bearing	DE
Moment of inertia (kg×m²)	0.06700
Painting (color)	Munsell 2.5G 8/4
Corrosion protection	C3L
Main terminal box pos.	DE Top
Main cable entries	Blind cap 2xM32x1.5
Main cable direction	Right from DE
Main cable material	Nickel Plated Brass
Aux. cable entries	Blind cap 2xM20x1.5
Aux. cable direction	NDE
Aux. cable material	Nickel Plated Brass

Winding operation variants					
Rated power (kW)	18.5				
Frequency (Hz)	60				
Connection	D/Y				
Voltage (V)	440 /				
Full load current (A)	28.4 /				
No load current (A)	7.62				
Speed (rpm)	3540				
Power factor 100/75/50 (%)	0.94 / 0.94 / 0.90				
Efficiency 100/75/50 (%)	91.0 / 91.0 / 90.7				
Efficiency class	IE2				
Full load torque (Nm)	49.90				
Tst/Tn	2.39				
Tmax/Tn	4.23				
Ist/In	8.90				
Temp. rise class	B				
Duty	S1				
Sound pressure level (dBA)	77				
Min. ambient temp. (°C)	-20°C				
Max. ambient temp. (°C)	45°C				

Data are not binding. Hoyer reserves the right to implement changes without notice. The extent of data are not equivalent to nameplate on the motors.



Fa=331N



END PLAYS 0.5mm

Hoyer		Hoyer Motors		DESIGN AND DRAWING		REVISION	
NAME	QUL	SIGNATURE	DATE	DO NOT SCALE DRAWING	REVISION	02	
DRAWN			20210830	POS.11 HOT GLYCOL WATER CIRCULATING PUMP			
CHECKED				TITLE:			
APPROVED				HMCX-160L-B5			
MFG				Ex db IIB/C T4 Gb			
Q.A				MATERIAL: cast iron			
				DWG NO. 97234			
				SCALE: 1:10			
				SHEET 1 OF 1			

HOYER

MOTORS

POS.12 COLD GLYCOL WATER CIRCULATING PUMP

Data sheet

Item no. / Configuration 02 / 2102112-4.00-020211-B022042-S-B00085
HMCX 112M-2 B5 4kW IE2 400/690@50 Ex db IIBT4

DescriptionDate2024/11/28

General data	
Standard series	IEC 60034
Type	HMCX 112M-2
Design	400 VD / 690 VY 50 Hz + 480 VD / 830 VY 60 Hz
Direction of rotation	CCW
Installation max. altitude (m)	1,000
Poles	2

Motor protection	
Winding temp. detector	No
Temp. detector type	None
Bearing detector	No
Winding PTC main	PTC 3 x 145°C
Winding PTC spare	None
Space heater	230 V 30 W
Terminal board thread	M5
Electrical insulation	1300 V @ 1 µs (suitable for VFD at D-connection)
Tropical insulation	Yes
SPM	N/A
External grounding	
Drain hole	Yes
Rain cap	No

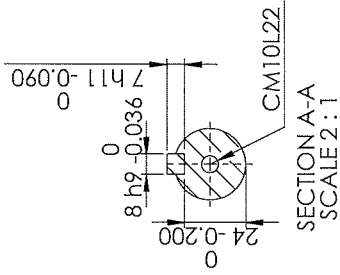
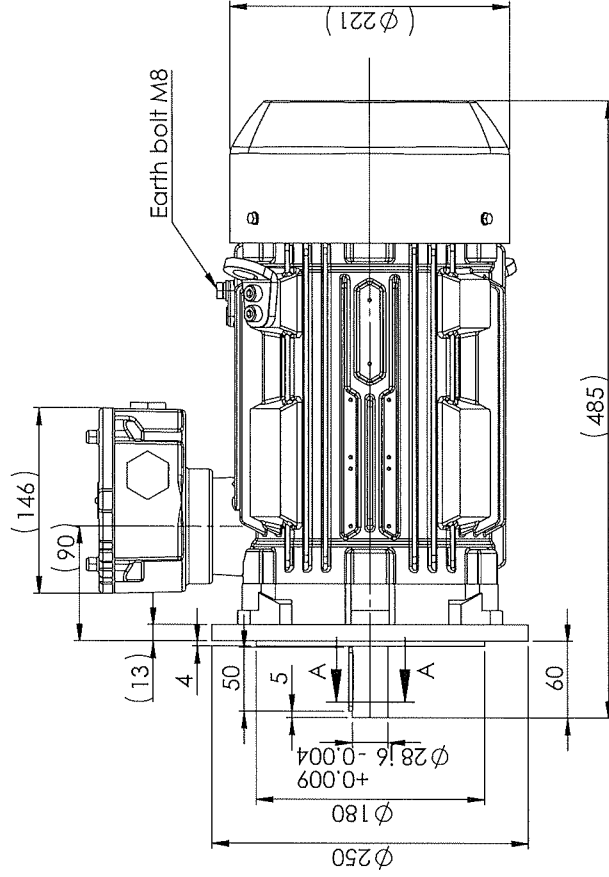
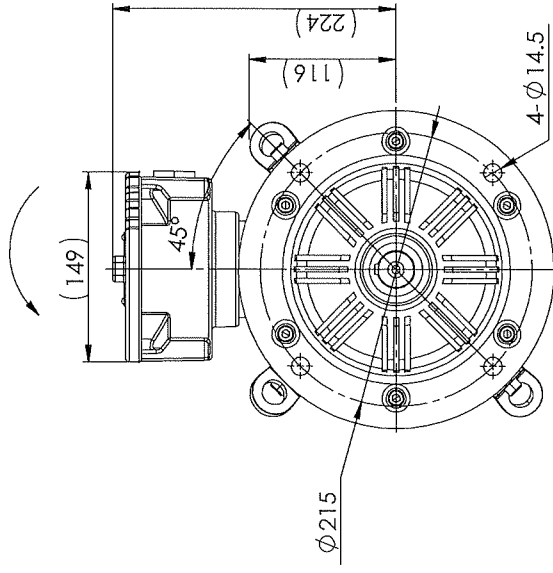
Classification	
Class society	
Shaft requirement acc. to	None
Ex. code	ATEX & IECEx: Ex II 2G Ex db IIB T4 Gb / Ex db IIB T4 Gb

Mechanical data	
Frame / Shaft height	IEC 112
Type of construction	B5
Cooling method	IC411
Protection class	IP55
Temp. insulation class	F
Frame material	Reinforced cast iron
Keyway	Closed
Balancing	Half key
IEC vibration class	A
Weight (kg)	70
Bearing (DE)	6206 2Z / C3 / DGBB
Bearing (NDE)	6206 2Z / C3 / DGBB
Fixed bearing	DE
Moment of inertia (kg×m²)	0.00760
Painting (color)	MUNSELL 2.5G 8/4
Corrosion protection	C3L
Main terminal box pos.	DE Top
Main cable entries	Blind cap 1xM25x1.5
Main cable direction	Right from DE
Main cable material	Nickel Plated Brass
Aux. cable entries	Blind cap 2xM20x1.5
Aux. cable direction	NDE
Aux. cable material	Nickel Plated Brass

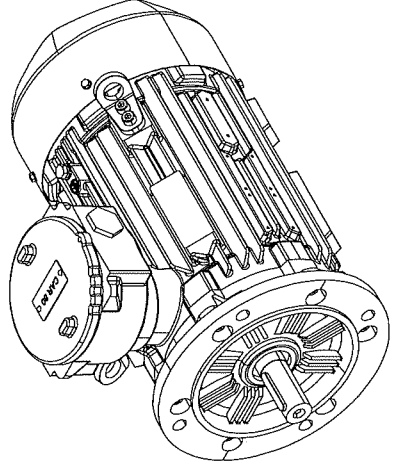
Winding operation variants					
Rated power (kW)	3.70				
Frequency (Hz)	60				
Connection	D/Y				
Voltage (V)	440 /				
Full load current (A)	6.10/				
No load current (A)	2.42				
Speed (rpm)	3505				
Power factor 100 (%)	0.91				
Efficiency 100/75/50 (%)	87.5 / 87.9 / 87.6				
Efficiency class	IE2				
Full load torque (Nm)	10.1				
Tst/Tn	2.44				
Tmax/Tn	4.00				
Ist/In	7.85				
Temp. rise class	B				
Duty	S1				
Sound pressure level (dBA)	79				
Min. ambient temp. (°C)	-20°C				
Max. ambient temp. (°C)	45°C				

Data are not binding. Hoyer reserves the right to implement changes without notice. The extent of data are not equivalent to nameplate on the motors.

ROTATION



F_{Pr}=108N



END PLAYS ≤ 0.5mm

HOYER RECEIVING APPROVATIONS				Hoyer Motors Over Hadsstenvej 42, DK-8370 Hadssten Tel: +45 86982111 Fax: +45 86981779 www.hoyer-motors.com				DEBUR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING REVISION 02	
POS.12 COLD GLYCOL WATER CIRCULATING PUMP											
NAME		SIGNATURE		DATE		TITLE:					
DRAWN	OUL			20210702		HMCX-112M-B5 Ex db IIB/C T4 Gb					
CHKD											
APPVD											
MFG											
G.A				MATERIAL:		97218		A3			
				cast iron							
						DWG NO.					
						SCALE:1:10		SHEET 1 OF 1			
						WEIGHT: kg					

Data Sheet

Itemnumber....: TK20901218

POS.13 NaOH TRANSFER PUMP

Version

Type.....: HMC2 90L-2
Design.....: Induction motor
Standard series.....: IEC 60034-1
Phase / Voltage range.....: 3~ / Low

Electrical design

Efficiency.....: IE2
Pole.....: 2
Power at 50 Hz (kW).....: 2,20
Hz.....: 60 (50 Hz output)
Voltage.....: 440VD
Winding voltage.....: 230VD/400VY 50 Hz
Power output (kW).....: 2,20
Duty.....: S1
Insulation class.....: F
Temperature rise.....: B

Motor protection

Thermal protection main.....: None
Thermal protection second.....: None
Space heater.....: None
Temperature detector.....: No
SPM.....: No
IR wire.....: No
Tropical insulation.....: Yes

Explosion protection

According to.....: None

General

Starting method.....: DOL
Direction of rotation.....: CCW
Painting.....: Munsell 2.5 G 8/4
Nameplate.....: P2 50Hz Marine
Special packing.....: No
Special requirements.....: No

Mechanical design

Frame size.....: 90
Mounting.....: B5
Rain cap.....: No
Protection class.....: IP55
Cooling method.....: IC411/TEFC
External grounding.....: Yes
Drain hole.....: Yes
Frame material.....: Cast Iron
Material approval.....: None
Shaft.....: IEC standard
Key.....: Closed key
Balancing.....: Half key balancing
Vibration class.....: Grade-A
Weight (kg).....: 33,0

Environment condition

Ambient temp. min. (°C)...: -20
Ambient temp. max. (°C)...: 45
Altitude (mtr up to).....: 1000

Bearing

DE Bearing.....: 6205-ZZ/C3
NDE Bearing.....: 6205-ZZ/C3
Fixed bearing.....: D-end

Terminal box

Tbox position.....: Top
Cable entry direction.....: Right (from DE)
Cable entry
Main.....: 1xM20x1,5
blindcaps
Accessory.....: 1xM20x1,5
blindcaps
Terminal board thread.....: 6-M4

Test values

Rotor inertia (kgm²).....: 0,0030
Noise level (dB(A)).....: 66
No load current (A).....: 1,94
Winding resist. (ohm).....: 2,15
Starting time (sec.).....: 0,05
Temp. rise winding (K).....: 54
Temp. rise surface (K).....: 39

Rated power (kW): 2,20
Frequency (Hz): 60
Voltage (V): 440
Connection: Y
Full load current (A): 4,00
Speed (rpm): 3440
Power factor cos(phi): 0,87
Efficiency, 100/75/50 (%) : 82,0/ 83,8 /83,6
Ist/In: 7,65
Full load torque (Nm): 5,80
Tst/Tn: 2,80
Tmax/Tn: 3,05
Duty: S1
Ambient temp. (°C): 45

Special features: Direction of rotation CCW - marked by red arrow IE2 efficiency @400V 50Hz

Pos.	Pump Name	Motor Qty	Motor Type	Spare Bearing										Spare Element Heater										
				6202-ZZ/C3	6203-ZZ/C3	6204-ZZ/C3	6205-ZZ/C3	6206-ZZ/C3	6306-ZZ/C3	6308-ZZ/C3	6309-ZZ/C3	6311/C3	6312-ZZ/C3	6313-ZZ/C3	6314/C3	6317/C3	6319/C3	6322/C3	220V/100W	220V/80W	220V/60W	220V/50W	220V/28W	220V/20W
01	M/E H.T.F.W. PUMP	2	HMC2 225M-4																					
02	M/E JACKET WATER PREHEATER PUMP	1	HMC2 132S1-2																					
03	L/T.C.W. BOOSTER PUMP	2	HMC2 225M-4																					
04	EMERGENCY FIRE PUMP	1	HMC2 250M-2																					
05	FRESH WATER PUMP	2	HMC2 132S2-2																					
06	HOT WATER CIRCULATING PUMP	2	HMC2 80 L-2			4																		
07	CHEMICAL CLEANING PUMP	1	HMC2 90L-2				2																	
08	GREY WATER DISCHARGE PUMP	1	HMC2 132M1-6																					
09	SEWAGE DISCHARGE PUMP	1	HMC2 132M1-6																					
10	E/R SEWAGE DISCHARGE PUMP	1	HMC2 112M2-6																					
13	NaOH TRANSFER PUMP	2	HMC2 90L-2				4																	
	Working	16		0	0	4	6	0	2	10	0	0	4	4	2	0	0	0	0	5	0	1	0	0
	Spare			0	0	1	2	0	1	3	0	0	1	1	1	0	0	0	0	1	0	1	0	0

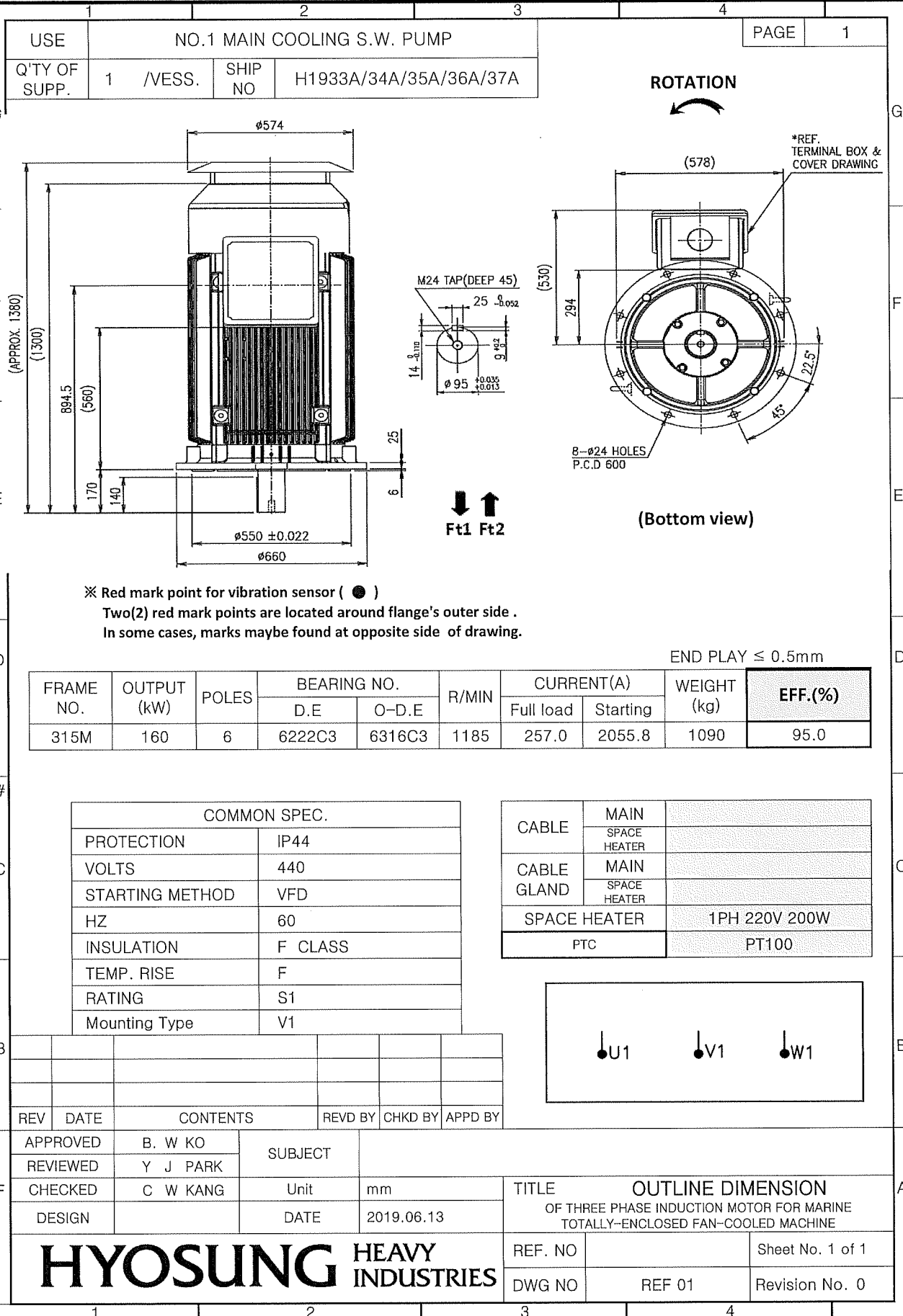
Terms of spare parts:
 -1 set of ball bearing for each 4 or less
 -1 set of heating element of space heater for each kind

			MESSRS: Hudong Zhonghua Shipbuilding(Group) Co., Ltd.		
			SHIP NO.: H1933A/34A/35A/36A/37A		
			SPECIFICATION OF THREE PHASE INDUCTION MOTOR		
			HYOSUNG HEAVY INDUSTRIES		
			APPROVED BY : BYUNG-WOOK, KO		
0	2024-11-22	FOR APPROVAL			
NO.	DATE	DESCRIPTION			
CHANGE OF ALTERATION					
NO.	DATE	REVD BY	CHKD BY	APPD BY	
REVISIONS					
		CLASS		LR	
		DATE		2024-11-22	

REV. HISTORY (Page 1 of 1)

[illegible]

INDEX													
SHIP NO.: H1933A/34A/35A/36A/37A				Rev.	0						Class:		LR
NO.	SERVICE NAME	Q'ty	Motor rating (kW)	RPM or Pole	Mounting Type (IEC)	Starting method	Efficiency (%)	Heater type	Winding Heating Capa. Or Space Heater Capa.			OUTLINE DWG NO.	
									Phase (φ)	Voltage (V)	Power (W)		
1	NO.1 MAIN COOLING S.W. PUMP	1	160	6	V1	VFD		EH	1	220		REF 01	PT100
2	NO.2&3 MAIN COOLING S.W. PUMP	2	160	6	V1	VFD		EH	1	220		REF 02	PT100
3	BALLAST WATER PUMP	2	145	4	V1	Y-D		EH	1	220		REF 03	
4	BILGE, FIRE & G/S PUMP	2	160	4	V1	Y-D		EH	1	220		REF 04	
5	L.T. COOLING F.W. PUMP	2	149	6	V1	Y-D		EH	1	220		REF 05	
6	M/E AIR COOLER L.T. COOLING F.W. PUMP	2	125	4	V1	Y-D		EH	1	220		REF 06	
7	MAIN L.O. PUMP	2	199	4	V1	Y-D		EH	1	220		REF 07	
8	C/H WATER SPRAY PUMP	1	249	4	V1	Y-D		EH	1	220		REF 08	
9	ICER COOLING SEA WATER PUMP	2	299	4	V1	Y-D		EH	1	220		REF 09	
INS.CLASS F / Temp. Rise F Paint color : Munsell No. 2.5 G 8/4 Terminal color : U(RED),V(WHITE),W(BLACK) Grease nipple : PIN type					B3: Horizontal B5: Horizontal Flange V1: Vertical Flange				1)The earth bolt shall be provided inside and outside of the terminal box 2)All motors are IE2				



USE	NO.2&3 MAIN COOLING S.W. PUMP			PAGE	2
Q'TY OF SUPP.	2 /VESS.	SHIP NO	H1933A/34A/35A/36A/37A		

(APPROX. 1380)
 894.5 (560)
 170
 140
 25
 6
 574
 550 ±0.022
 660

ROTATION

(578)
 (530)
 294
 22.5
 45
 8-24 HOLES P.C.D 600
 *REF. TERMINAL BOX & COVER DRAWING

↓ ↑

Ft1 Ft2

※ Red mark point for vibration sensor (●)
 Two(2) red mark points are located around flange's outer side .
 In some cases, marks maybe found at opposite side of drawing.

FRAME NO.	OUTPUT (kW)	POLES	BEARING NO.		R/MIN	CURRENT(A)		WEIGHT (kg)	EFF.(%)
			D.E	O-D.E		Full load	Starting		
315M	160	6	6222C3	6316C3	1185	257.0	2055.8	1090	95.0

END PLAY ≤ 0.5mm

COMMON SPEC.	
PROTECTION	IP44
VOLTS	440
STARTING METHOD	VFD
HZ	60
INSULATION	F CLASS
TEMP. RISE	F
RATING	S1
Mounting Type	V1

CABLE	MAIN	
	SPACE HEATER	
CABLE GLAND	MAIN	
	SPACE HEATER	
SPACE HEATER		1PH 220V 200W
PTC		PT100

REV	DATE	CONTENTS	REVD BY	CHKD BY	APPD BY

APPROVED	B. W KO	SUBJECT	TITLE
REVIEWED	Y J PARK		
CHECKED	C W KANG	Unit	mm
DESIGN		DATE	2019.06.13

HYOSUNG

HEAVY INDUSTRIES

REF. NO		Sheet No. 1 of 1
DWG NO	REF 02	Revision No. 0

USE	BALLAST WATER PUMP			PAGE	3
Q'TY OF SUPP.	2 /VESS.	SHIP NO	H1933A/34A/35A/36A/37A		

ROTATION

(Bottom view)

※ Red mark point for vibration sensor (●)

Two(2) red mark points are located around flange's outer side .

In some cases, marks maybe found at opposite side of drawing.

FRAME NO.	OUTPUT (kW)	POLES	BEARING NO.		R/MIN	CURRENT(A)		WEIGHT (kg)	EFF.(%)
			D.E	O-D.E		Full load	Starting		
280L	145	4	6319C3	6316C3	1780	235.1	1645.7	1090	95

END PLAY ≤ 0.5mm

COMMON SPEC.	
PROTECTION	IP44
VOLTS	440
STARTING METHOD	Y-D
HZ	60
INSULATION	F CLASS
TEMP. RISE	F
RATING	S1
Mounting Type	V1

CABLE	MAIN	
	SPACE HEATER	
CABLE GLAND	MAIN	
	SPACE HEATER	
SPACE HEATER		1PH 220V 160W

Y(START)		
U1	V1	W1
W2	U2	V2

Δ(RUN)		
U1	V1	W1
W2	U2	V2

REV	DATE	CONTENTS	REVD BY	CHKD BY	APPD BY

APPROVED	B. W KO	SUBJECT	TITLE
REVIEWED	Y J PARK		
CHECKED	C W KANG	Unit	mm
DESIGN		DATE	2019.06.13

HYOSUNG

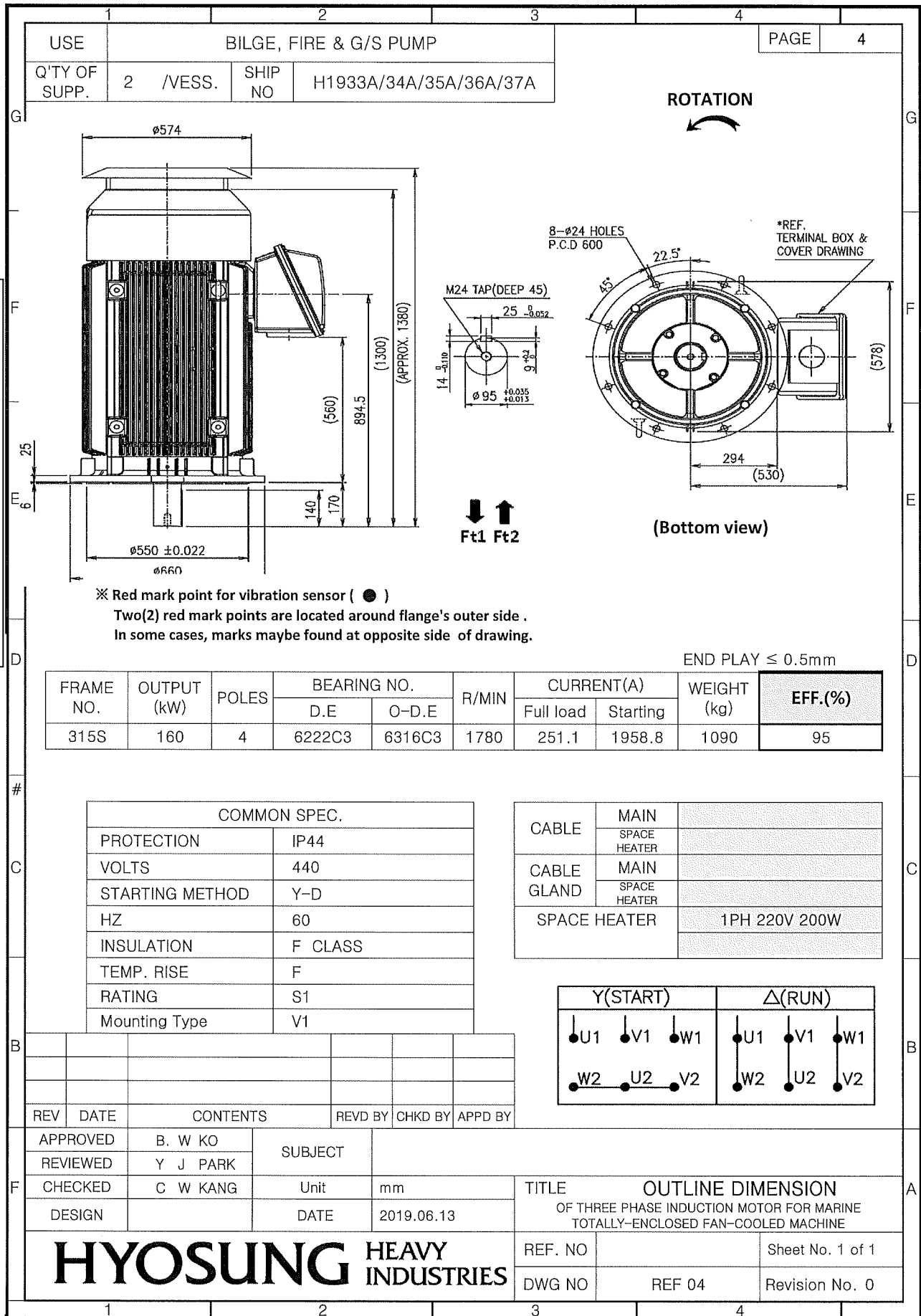
HEAVY INDUSTRIES

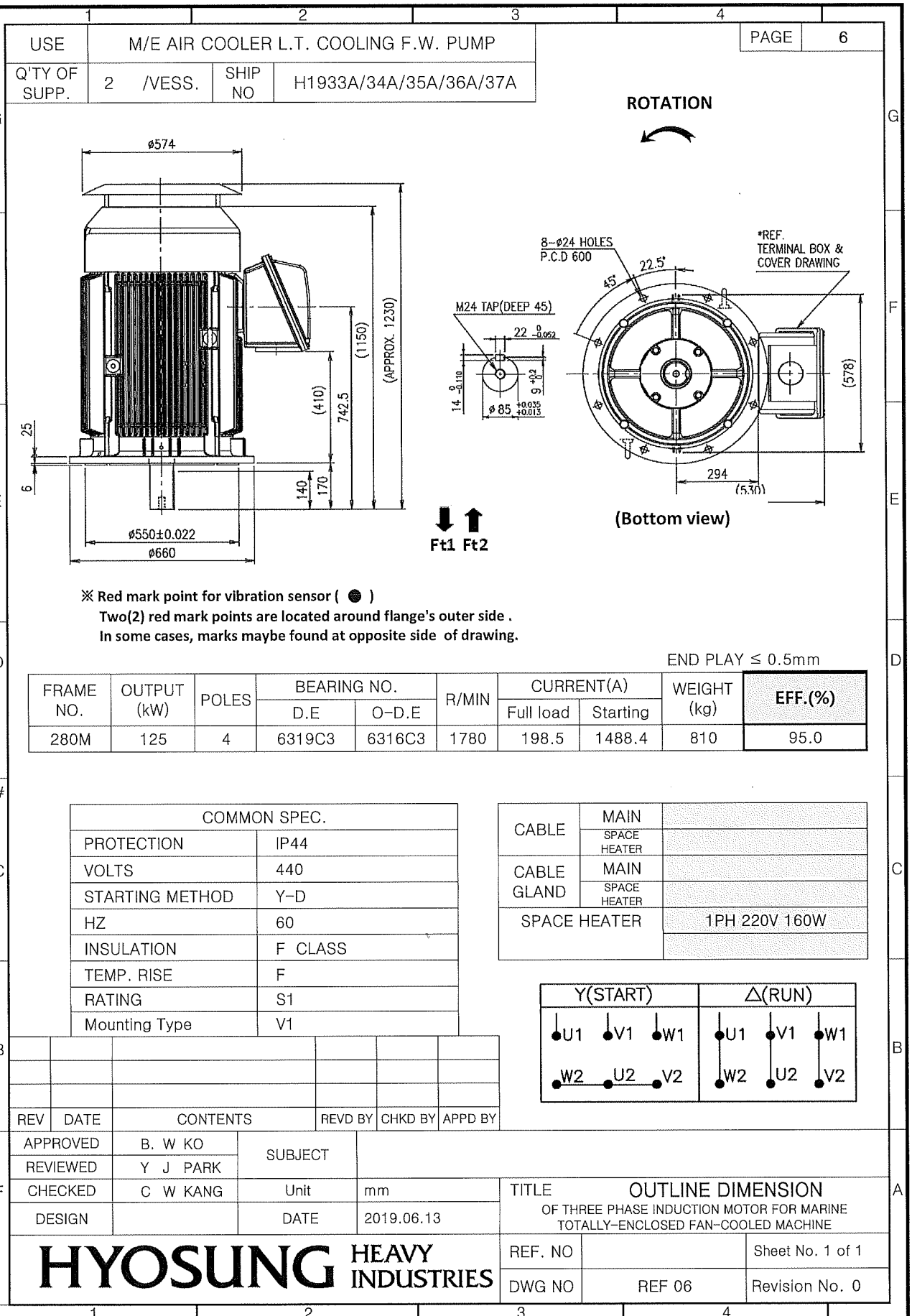
REF. NO

DWG NO

Sheet No. 1 of 1

Revision No. 0





USE	MAIN L.O. PUMP			PAGE	7
Q'TY OF SUPP.	2 /VESS.	SHIP NO	H1933A/34A/35A/36A/37A		

(APPROX. 1380)

894.5 (560)

170

140

25

6

Ø574

Ø550 ±0.022

Ø660

M24 TAP(DEEP 45)

14 25 9 9.5 0.035 0.013 0.052

*REF. TERMINAL BOX & COVER DRAWING

8-Ø24 HOLES P.C.D 600

22.5°

578

294

530

ROTATION

(Bottom view)

Ft1 Ft2

※ Red mark point for vibration sensor (●)
Two(2) red mark points are located around flange's outer side .
In some cases, marks maybe found at opposite side of drawing.

FRAME NO.	OUTPUT (kW)	POLES	BEARING NO.		R/MIN	CURRENT(A)		WEIGHT (kg)	EFF.(%)
			D.E	O-D.E		Full load	Starting		
315M	199	4	6222C3	6316C3	1780	310.6	2422.6	1090	95.0

END PLAY ≤ 0.5mm

COMMON SPEC.	
PROTECTION	IP44
VOLTS	440
STARTING METHOD	Y-D
HZ	60
INSULATION	F CLASS
TEMP. RISE	F
RATING	S1
Mounting Type	V1

CABLE	MAIN	
	SPACE HEATER	
CABLE GLAND	MAIN	
	SPACE HEATER	
SPACE HEATER		1PH 220V 200W

Y(START)	Δ(RUN)

REV	DATE	CONTENTS	REVD BY	CHKD BY	APPD BY

APPROVED	B. W KO	SUBJECT	Unit	mm	TITLE
REVIEWED	Y J PARK				
CHECKED	C W KANG	DATE	2019.06.13	OUTLINE DIMENSION OF THREE PHASE INDUCTION MOTOR FOR MARINE TOTALLY-ENCLOSED FAN-COOLED MACHINE	
DESIGN					

HYOSUNG

HEAVY INDUSTRIES

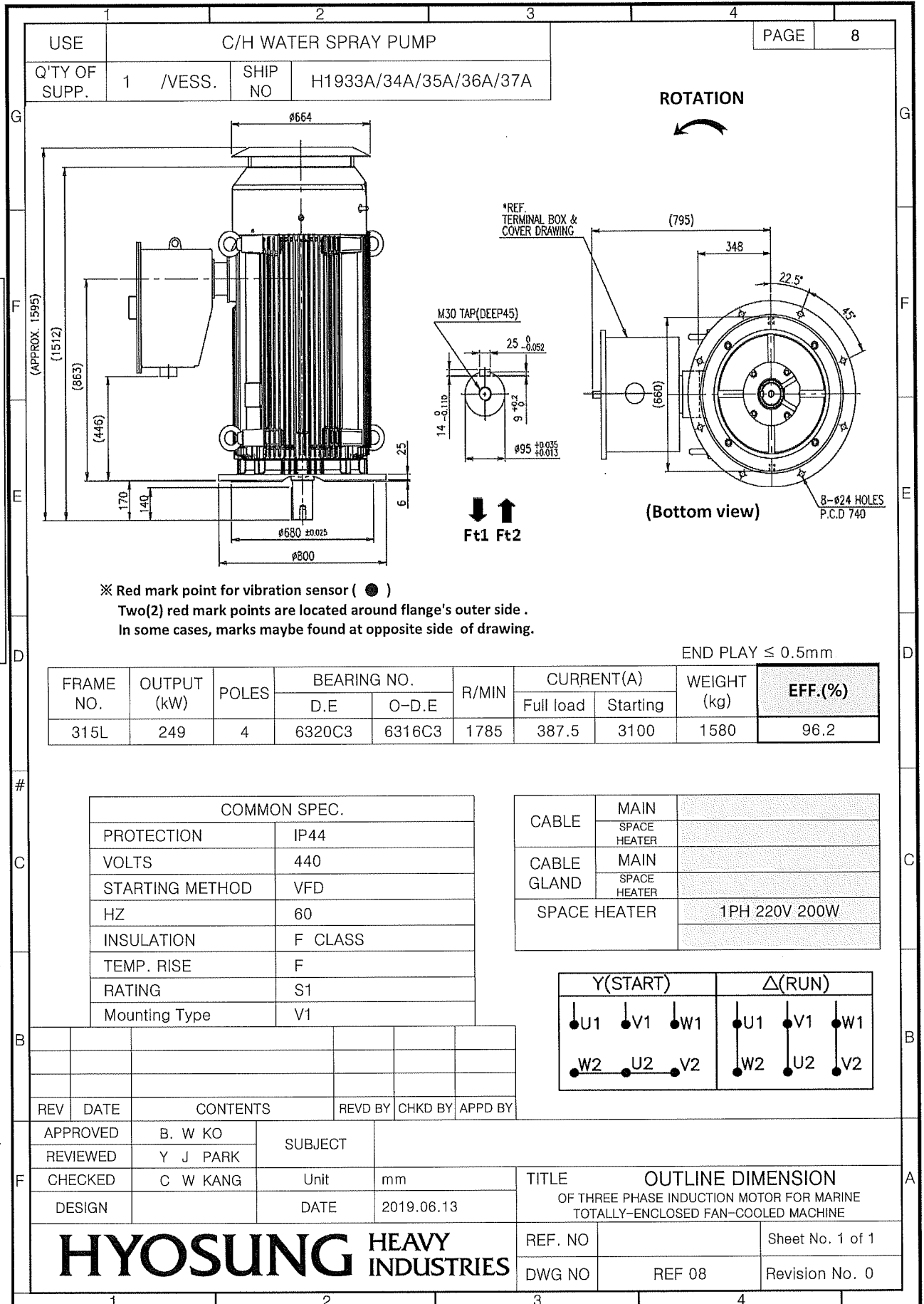
REF. NO

DWG NO

Sheet No. 1 of 1

Revision No. 0

The information written in this drawing is HYOSUNG's property, and may not be disclosed and/or reproduced without the prior official permission of HYOSUNG.



USE		ICER COOLING SEA WATER PUMP		PAGE 9	
Q'TY OF SUPP.	2 /VESS.	SHIP NO	H1933A/34A/35A/36A/37A		

ROTATION DIRECTION

Approx. (1325)

(659) (471.5)

22.5° 45°

8-ø22 HOLE

(Bottom view)

Ft1 Ft2

※ Red mark point for vibration sensor (●)
 Two(2) red mark points are located around flange's outer side .
 In some cases, marks maybe found at opposite side of drawing.

FRAME NO.	OUTPUT (kW)	POLES	BEARING NO.		R/MIN	CURRENT(A)		WEIGHT (kg)	EFF.(%)
			D.E	O-D.E		Full load	Starting		
355L	299	4	6322C3	6316C3	1785	475.9	3569.2	2450	

END PLAY ≤ 0.5mm

COMMON SPEC.	
PROTECTION	IP44
VOLTS	440
STARTING METHOD	Y-D
HZ	60
INSULATION	F CLASS
TEMP. RISE	F
RATING	S1
Mounting Type	V1

CABLE	MAIN	
	SPACE HEATER	
CABLE GLAND	MAIN	
	SPACE HEATER	
SPACE HEATER		1PH 220V 200W

Y(START)			Δ(RUN)		
U1	V1	W1	U1	V1	W1
W2	U2	V2	W2	U2	V2

REV	DATE	CONTENTS	REVD BY	CHKD BY	APPD BY

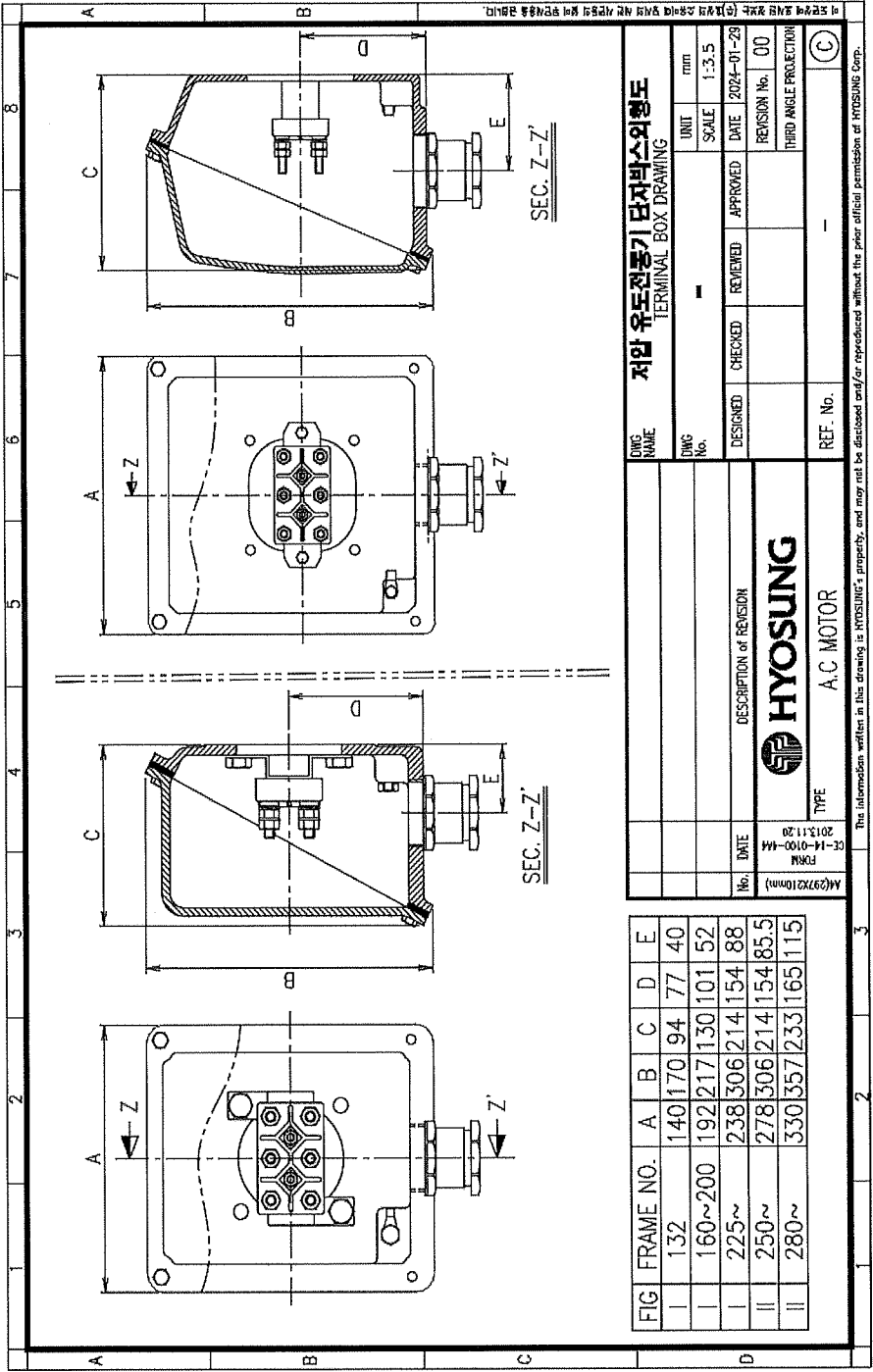
APPROVED	B. W KO	SUBJECT	Unit	mm	TITLE
REVIEWED	Y J PARK				
CHECKED	C W KANG	DATE	2019.06.13		
DESIGN					

HYOSUNG HEAVY INDUSTRIES		REF. NO		Sheet No. 1 of 1
		DWG NO	REF 09	Revision No. 0

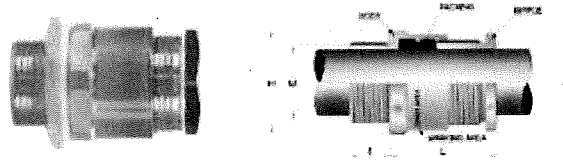
SPARE PARTS

SPARE PARTS LIST FOR				HULL NO.				USE				SETS PER VESSEL				CLASS			
LV MOTOR				HI933A/34A/35A/36A/37A				ENGINE ROOM PUMP				1				LR			
Name of Parts				Ball Bearing				Ball Bearing				Ball Bearing				Ball Bearing			
Size				6312C3				6312ZC3				6313C3				6316C3			
Outline				60x130x31				65x140x33				65x140x33				80x170x39			
Working				0				0				0				0			
Spare				0				0				0				0			
Motor Spec.				Service Name				Sets											
NO.				kW				Pole				M't							
1	160	6	V1	NO.1 MAIN COOLING S.W. PUMP				1											
2	160	6	V1	NO.2&3 MAIN COOLING S.W. PUMP				2											
3	145	4	V1	BALLAST WATER PUMP				2											
4	160	4	V1	BILGE, FIRE & G/S PUMP				2											
5	149	6	V1	L.T. COOLING F.W. PUMP				2											
6	125	4	V1	M/E AIR COOLER L.T. COOLING F.W. PUMP				2											
7	199	4	V1	MAIN L.O. PUMP				2											
8	249	4	V1	C/H WATER SPRAY PUMP				1											
9	299	4	V1	ICER COOLING SEA WATER PUMP				2											

Note: 25% of each type of bearing fitted



OSNU



I Application

: Non-Armoured Cable

- For use with non-armoured & braided armoured cable
- Weatherproof & Waterproof
- Single compression

Design Specification	BS-6171, IEC 62444, EN 60525
Ingress Protection (Metric)	IP 66 / IP 68 (depends on cable & PFA required)
(NPT)	NPT 1/2" to 4" (1" to 4")
	IP 66 (Supply with cable required)
	Cable price table: IP 66 (Supply with cable required)
Applicable Cable Specification	Non-armoured cable Braided armoured cable
Gland Material	Nylon (polyamide 66) (standard) Zinc (only for stainless steel)
Gasket (washer)	Nylon (standard), PTFE (Teflon) (optional)
Packing	Silicone
Accessories (optional)	Lock nut, Sealing washer, Earth tag, Shroud, Sealed washer

I Order Example

Part No.	Thread	Material	Accessories
OSNU 16	M16	Aluminum	Lock nut, Sealing washer, Earth tag, Shroud, Sealed washer

Part No. (Gland Size)	Entry Thread Size(M)			Thread Length(L) - Min			Cable Dia	Hexagon Dimensions	Weight (g)	Length (L)
	Metric	NPT	Option	M/PF	NPT	Option				
OSNU 16	M16	1/2"	M12/M20	15	15	15	3 ~ 7	H24 x P26	79/95	29
OSNU 16X	M16	1/2"	M20	15	15	15	5 ~ 10.3	H24 x P26	79/95	29
OSNU 20	M20	1/2"	3/4"	15	15	16	9 ~ 15.3	H27 x P29	95	31
OSNU 25	M25	3/4"	1"	15	16	18	13 ~ 20	H32 x P34	125	33
OSNU 32	M32	1"	1-1/4"	15	18	19	17 ~ 26.5	H39 x P41.5	195	36
OSNU 40	M40	1-1/4"	1-1/2"	15	19	21	22 ~ 32	H48 x P51	300	39
OSNU 50	M50	1-1/2"	2"	15	21	24	30 ~ 42	H58 x P62	425	41
OSNU 63	M63	2"		15	24		39 ~ 51	H73 x P78	680	44
OSNU 63X	M63		2-1/2"	15		27	46 ~ 55	H73 x P78	575	44
OSNU 75	M75	2-1/2"		20	27		54 ~ 61	H85 x P90	875	48
OSNU 75X	M75		3"	20		30	58 ~ 68	H94 x P99	860	49
OSNU 90	M90	3"	3-1/2"	25	30	32	66 ~ 78	H103 x P109	1690	53
OSNU 100	M100	3-1/2"	4"	25	32	32	77 ~ 89	H112 x P120	2250	59
OSNU 115	M115	4"		25	32		88 ~ 101	H125 x P135	2400	59

(L in mm)

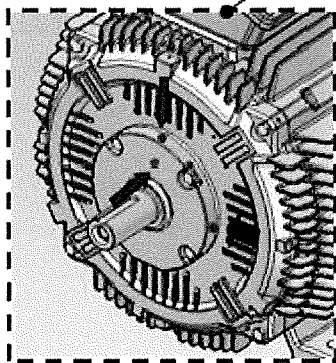
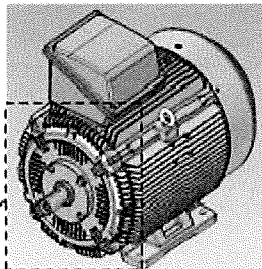
FRAME NO.	A	B
80M~132M	62	42
160M~315SM	93	63

Vibration Measurement Point

Foot Type Motor



Measurement Motor
Condition at No Load



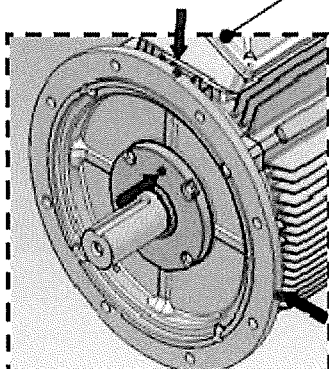
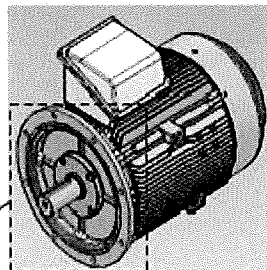
- Vertical
- Horizontal
- Axial

Flange Type Motor



Measurement Motor
Condition at No Load

On the driving motor
If you want to measure during motor driving,
Please measure at the square point.



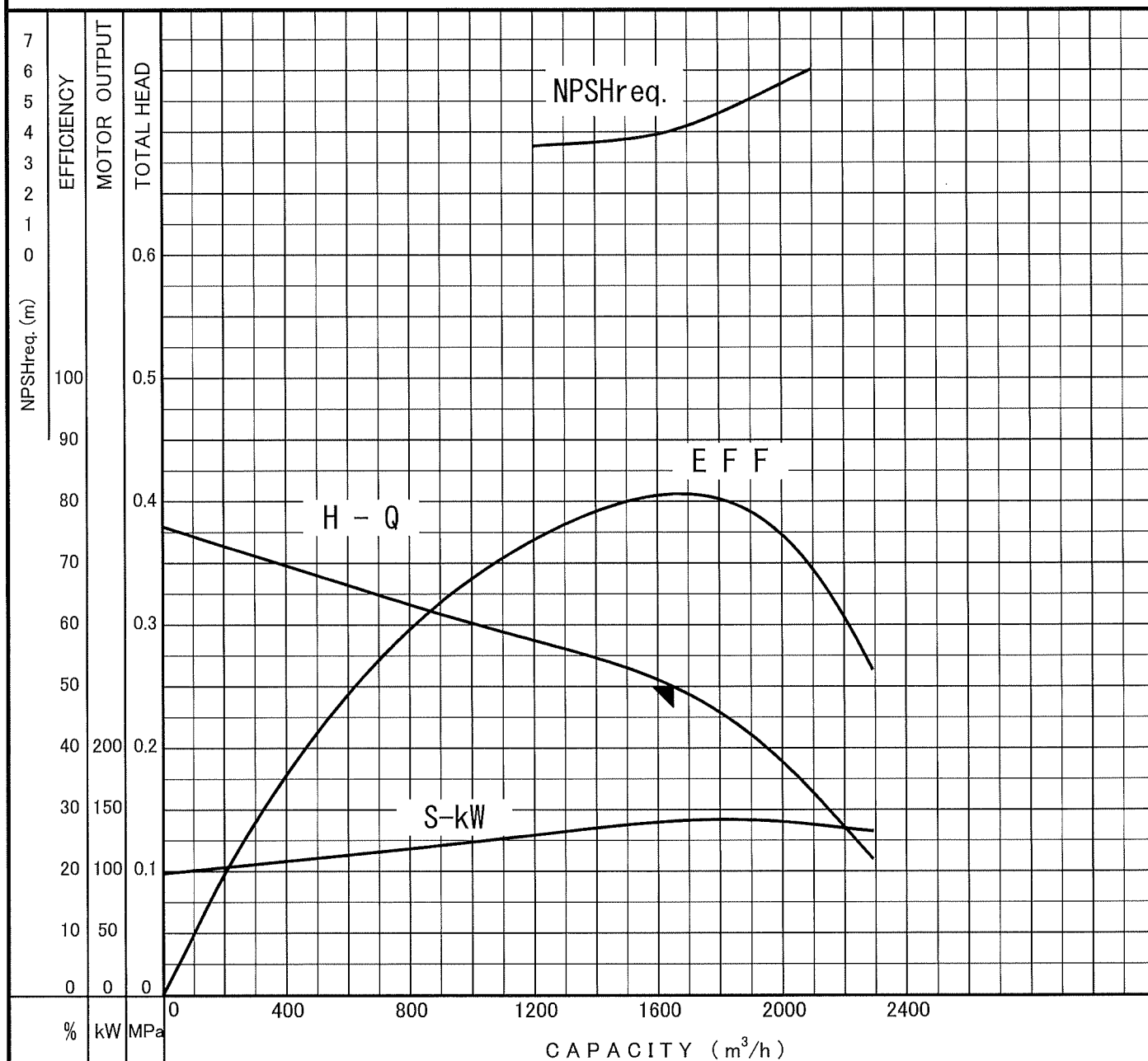
- Vertical
- Horizontal
- Axial



ESTIMATED PERFORMANCE CURVE

Page : 1,2
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	MAIN COOLING S.W.PUMP				
Pump Model	:	FEWV-450				
Bore	:	450×450				
Capacity	:	1650	m³/h	Total Press :	0.25	MPa
Output	:	160	kW	1150	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.6	MPa			
Remarks	:					

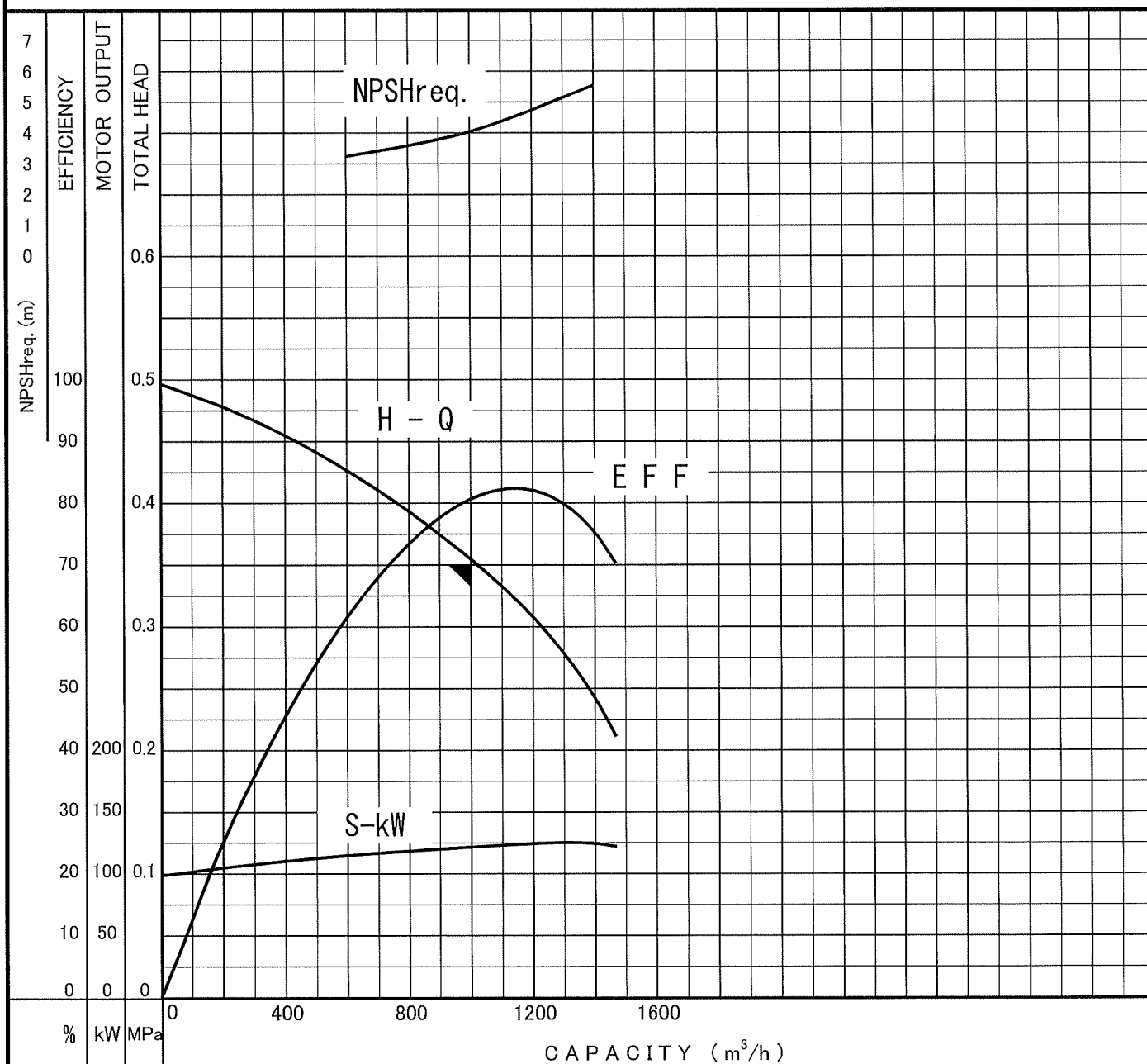




ESTIMATED PERFORMANCE CURVE

Page : 3
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	BALLAST WATER PUMP				
Pump Model	:	FEWV-350-3				
Bore	:	350×350				
Capacity	:	1000	m ³ /h	Total Press :	0.35	MPa
Output	:	145	kW	1750	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.8	MPa			
Remarks	:					

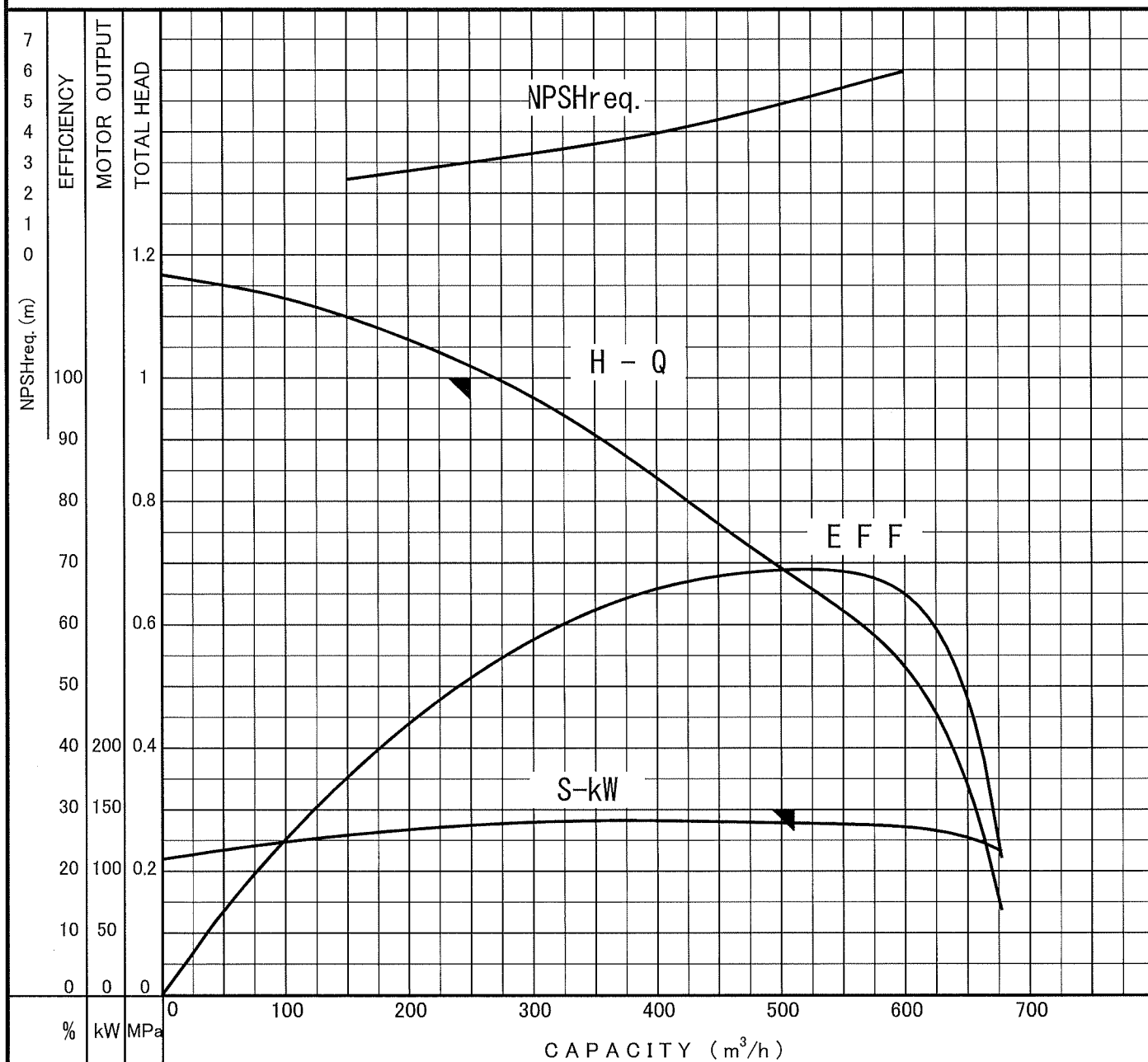




ESTIMATED PERFORMANCE CURVE

Page : 4
Date : 11 Nov. 2024

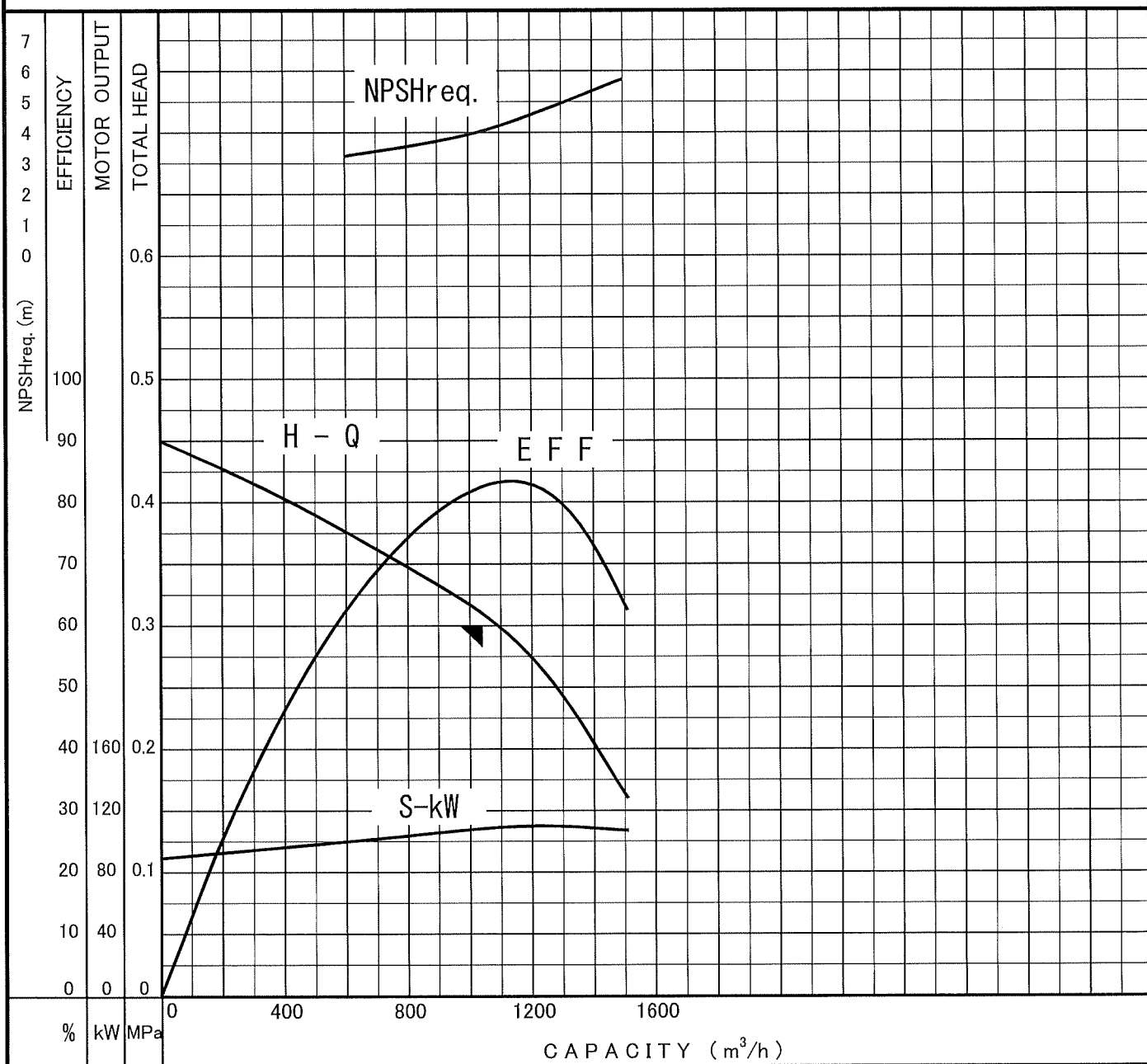
Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.					
Project	:	H1933A/1934A/1935A/1936A/1937A					
Service As	:	BILGE,FIRE & G/S PUMP					
Pump Model	:	FE2V-250-3					
Bore	:	250×250					
Capacity	:	250/510	m³/h	Total Press :	1.0/0.3	MPa	
Output	:	160	kW	1750	min-1		
		60	Hz	440	V		
Test Liquid	:	FRESH WATER					
H. Test Press.	:	1.8	MPa				
Remarks	:						





ESTIMATED PERFORMANCE CURVE

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	L.T.COOLING F.W.PUMP / M/E AIR COOLER L.T.COOLING F.W.PUMP				
Pump Model	:	FEWV-350				
Bore	:	350×350				
Capacity	:	1040	m³/h	Total Press :	0.3	MPa
Output	:	125	kW	1750	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.9	MPa			
Remarks	:					





ESTIMATED PERFORMANCE CURVE

Page :

7

Date :

7 Nov. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : M/E H.T.F.W.PUMP

Pump Model : FEV-200E1

Bore : 200×200

Capacity : 326 m³/h Total Press : 0.3 MPa

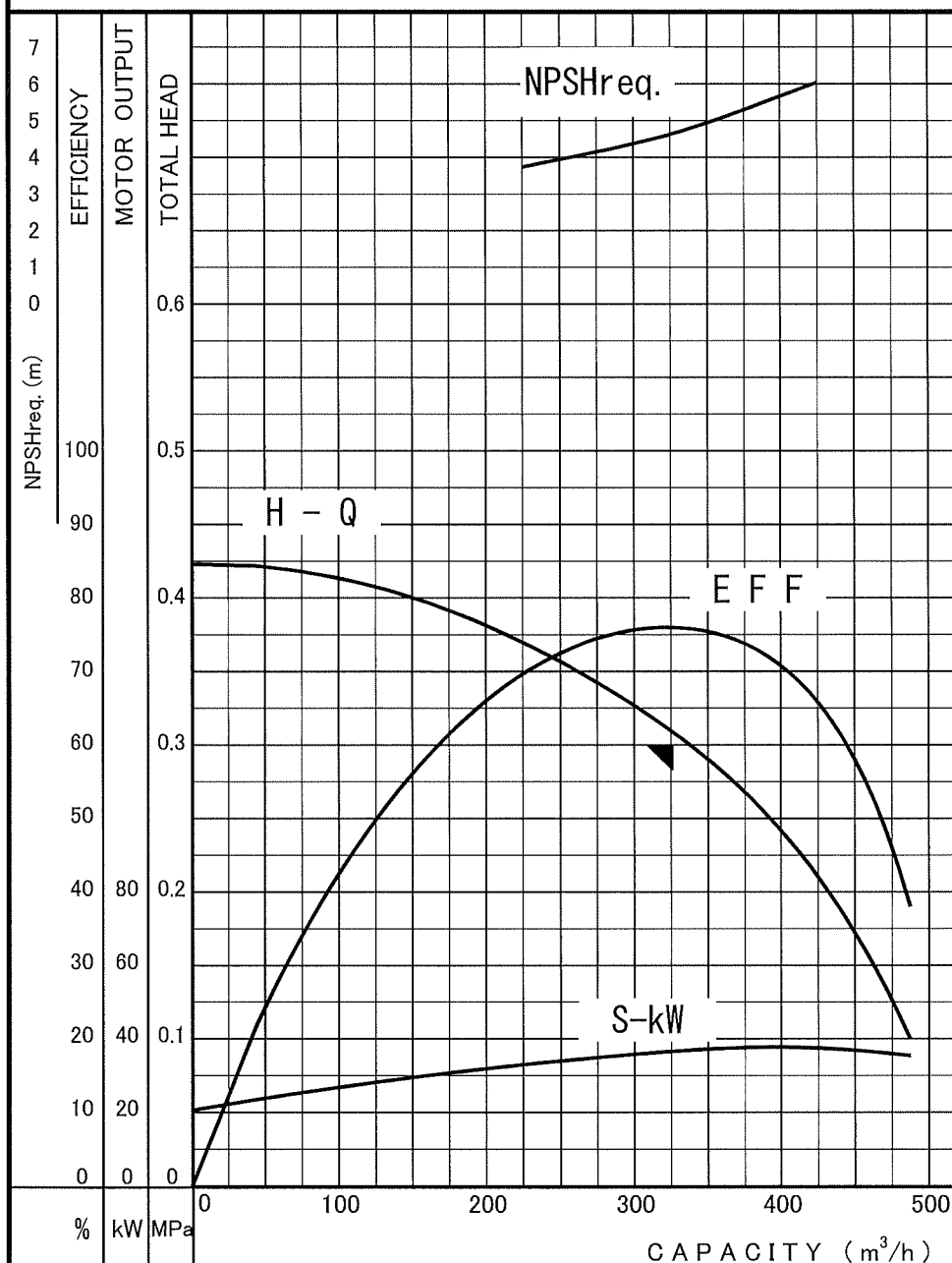
Output : 45 kW 1750 min-1

60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 0.8 MPa

Remarks :





ESTIMATED PERFORMANCE CURVE

Page : 8
Date : 7 Nov. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : M/E JACKET WATER PREHEATER PUMP

Pump Model : BHR-65

Bore : 65×65

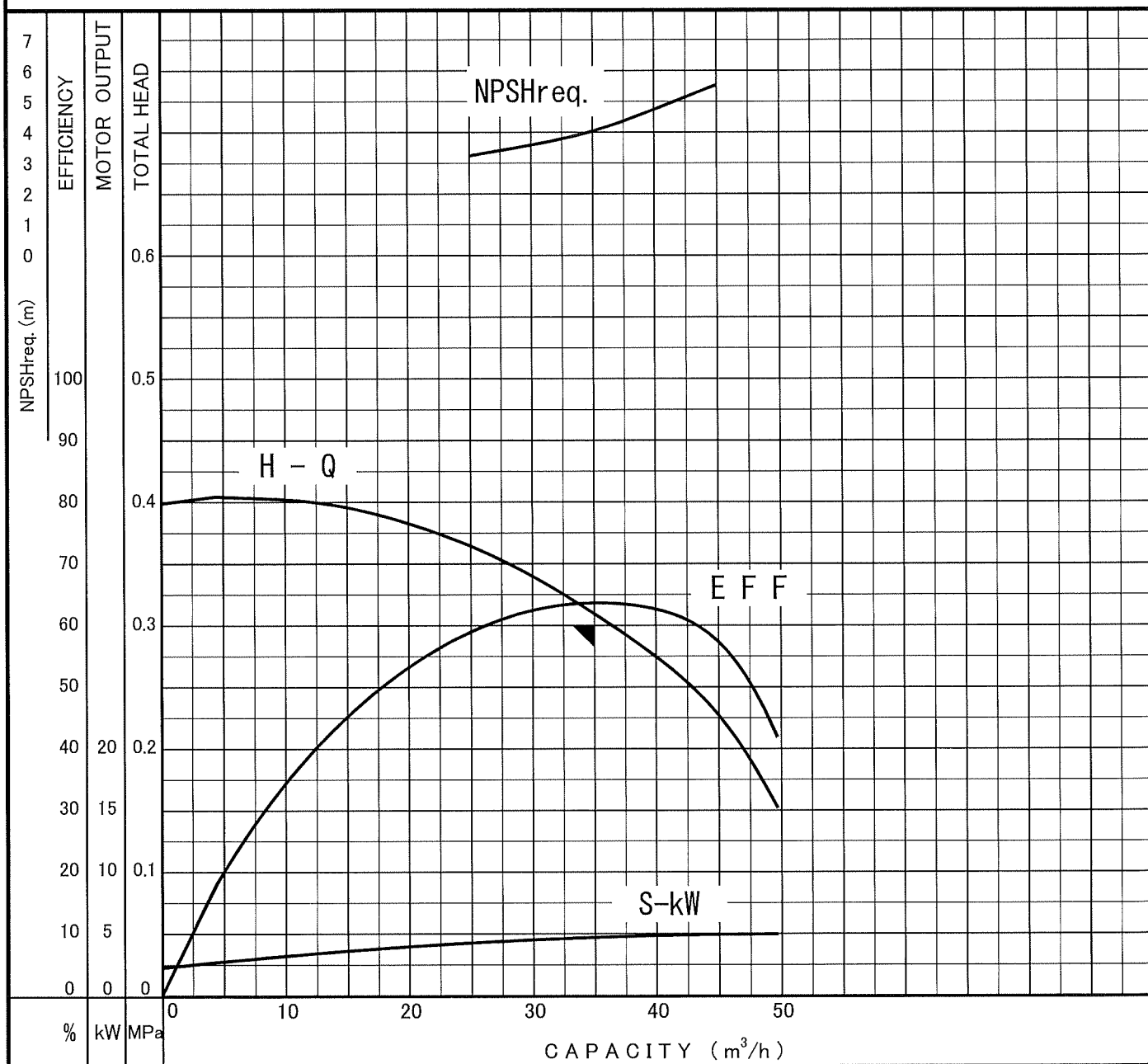
Capacity : 35 m³/h Total Press : 0.3 MPa

Output : 5.5 kW 3500 min⁻¹
60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 0.7 MPa

Remarks :





ESTIMATED PERFORMANCE CURVE

Page : 9

Date : 11 Dec. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : L/T.C.W.BOOSTER PUMP

Pump Model : FEV-200E1

Bore : 200×200

Capacity : 250 m³/h Total Press : 0.4 MPa

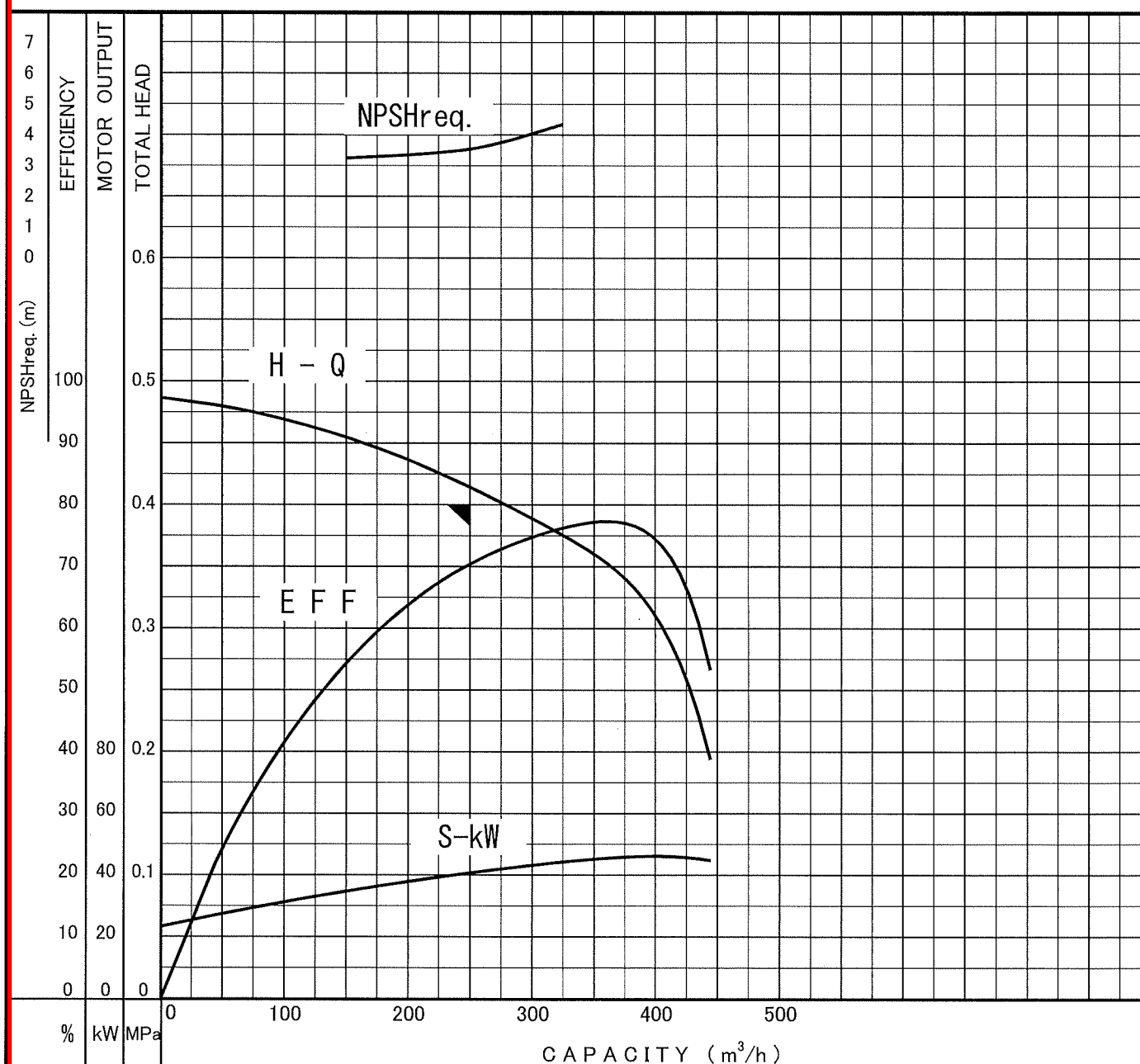
Output : 49 kW 1750 min-1

60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 0.9 MPa

Remarks :





ESTIMATED PERFORMANCE CURVE

Page : 10

Date : 11 Nov. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : EMERGENCY FIRE PUMP

Pump Model : FGV-100-2

Bore : 100×100

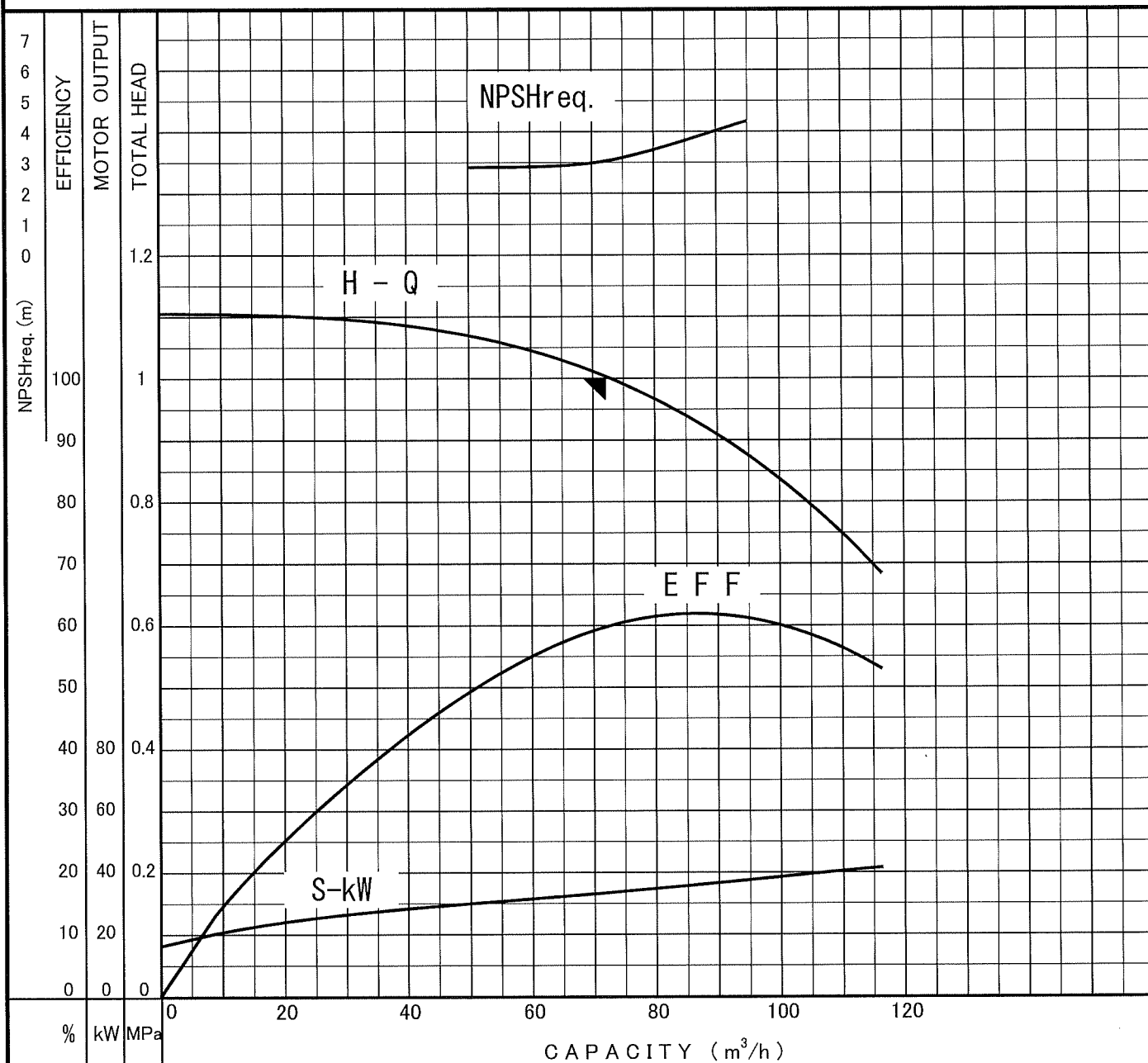
Capacity : 72 m³/h Total Press : 1.0 MPa

Output : 45 kW 3500 min⁻¹
60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 1.7 MPa

Remarks :



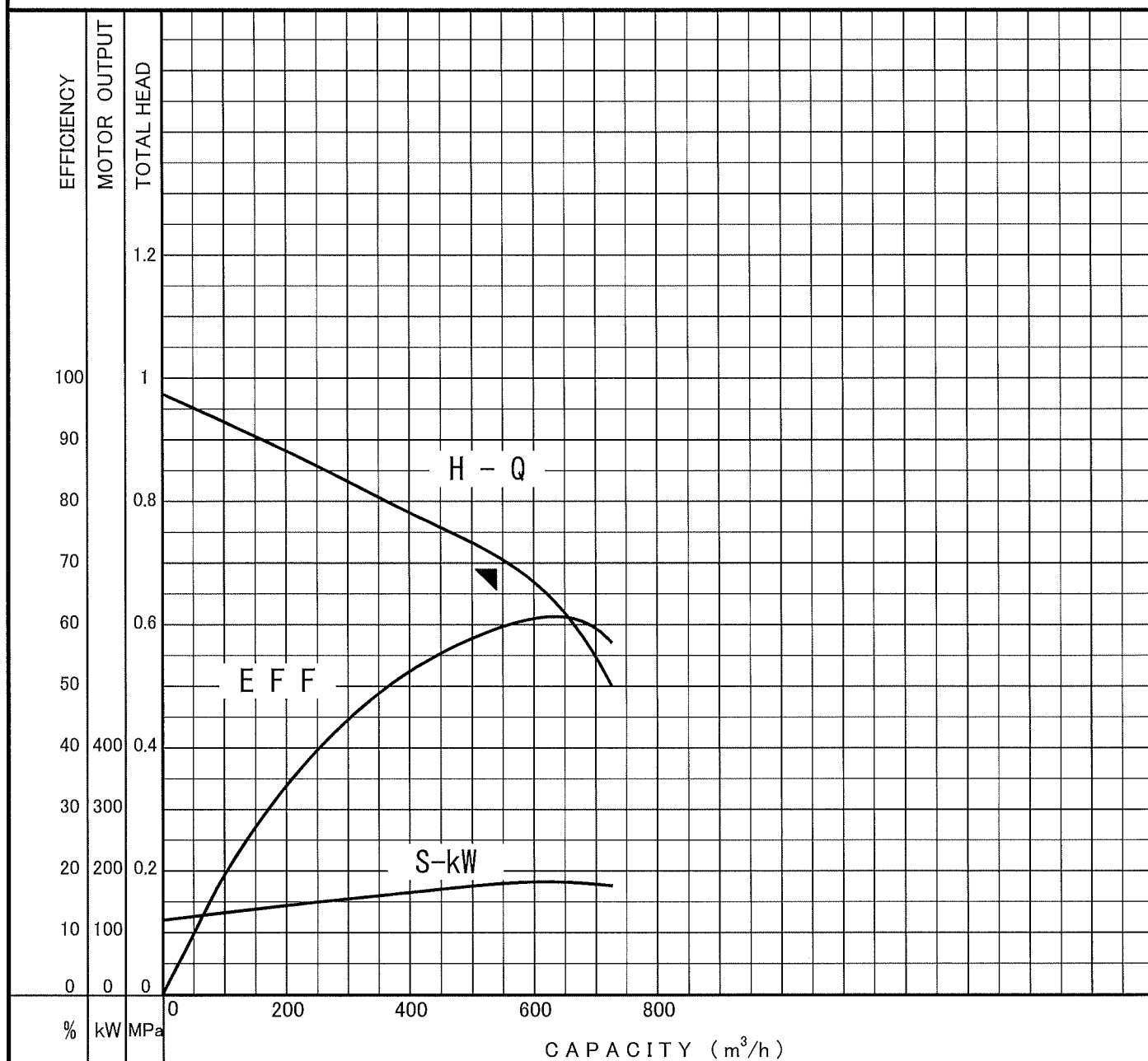


ESTIMATED PERFORMANCE CURVE

Page : 11

Date : 5 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	MAIN L.O.PUMP				
Pump Model	:	TOM-300-2				
Bore	:	DEL.300				
Capacity	:	540	m³/h	Total Press :	0.69	MPa
Output	:	199	kW	1750	min-1	
		60	Hz	440	V	
Test Liquid	:	L.O.				
H. Test Press.	:	1.5	MPa			
Remarks	:					

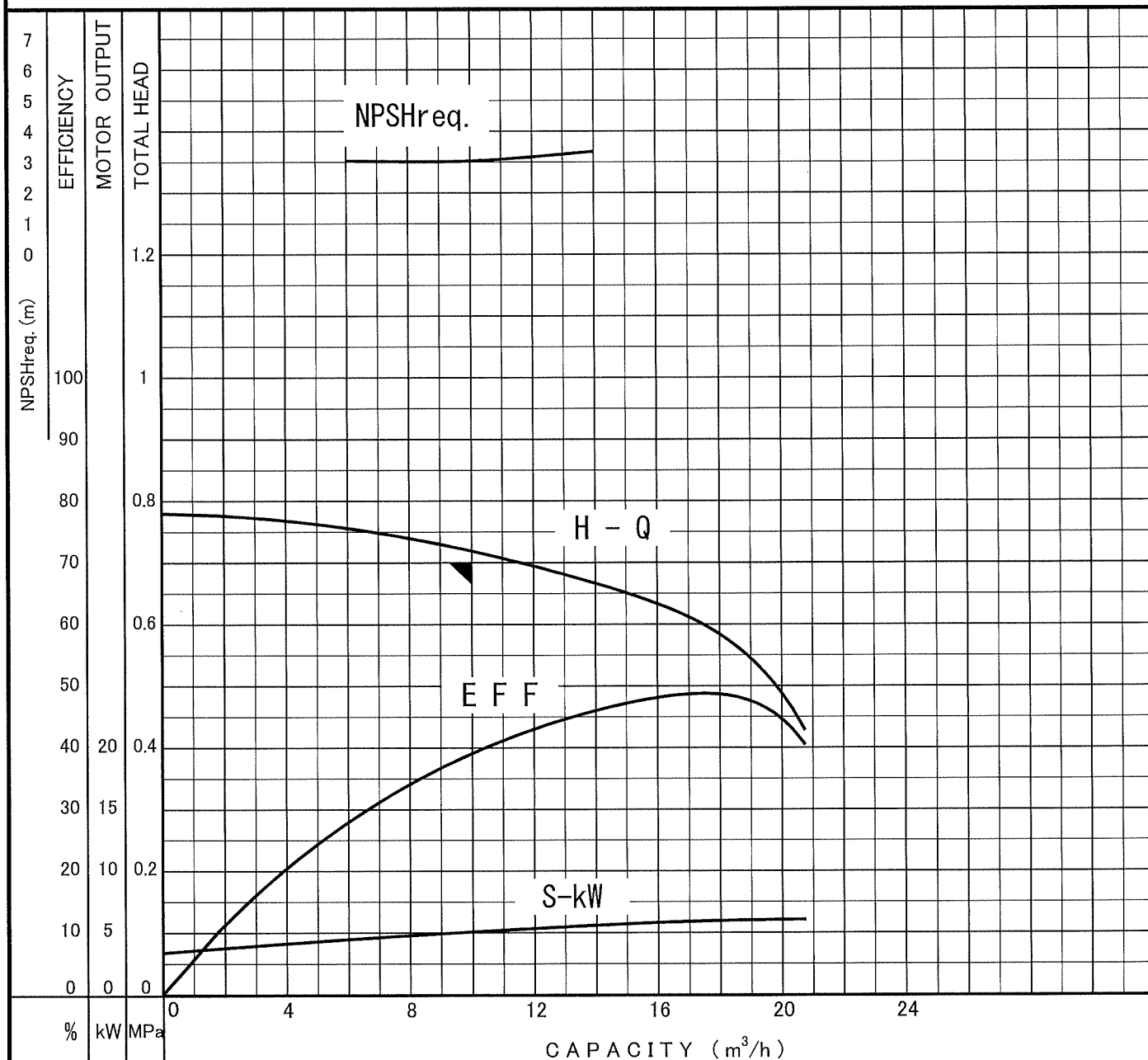




ESTIMATED PERFORMANCE CURVE

Page : 12
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.					
Project	:	H1933A/1934A/1935A/1936A/1937A					
Service As	:	FRESH WATER PUMP					
Pump Model	:	BHR-50-3					
Bore	:	50×50					
Capacity	:	10	m³/h	Total Press :	0.7	MPa	
Output	:	7.5	kW	3500	min-1		
		60	Hz	440	V		
Test Liquid	:	FRESH WATER					
H. Test Press.	:	1.2	MPa				
Remarks	:						

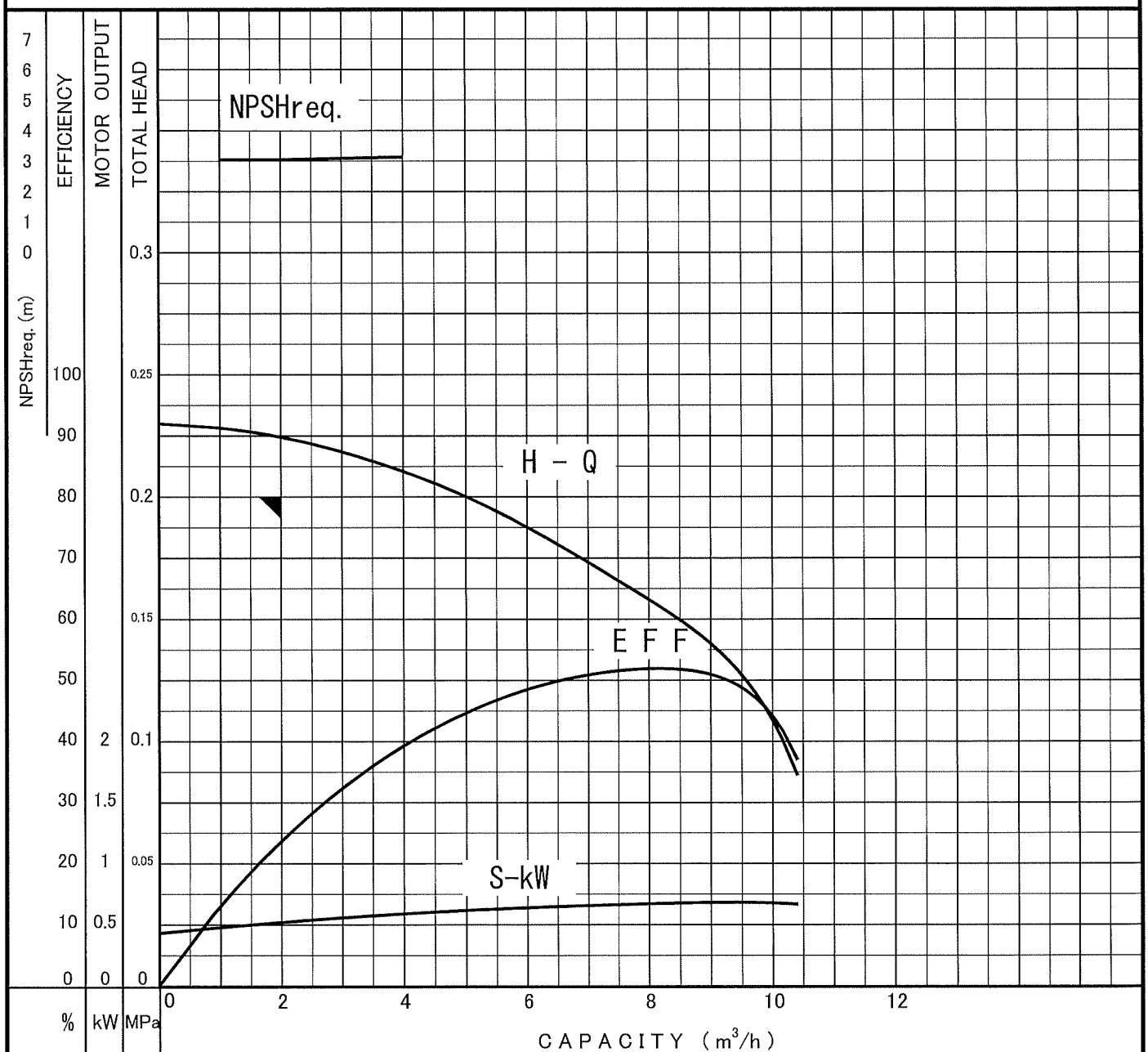




ESTIMATED PERFORMANCE CURVE

Page : 13
Date : 11 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	HOT WATER CIRCULATING PUMP				
Pump Model	:	BHR-32				
Bore	:	32×32				
Capacity	:	2	m³/h	Total Press :	0.2	MPa
Output	:	0.75	kW	3500	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	1.0	MPa			
Remarks	:					

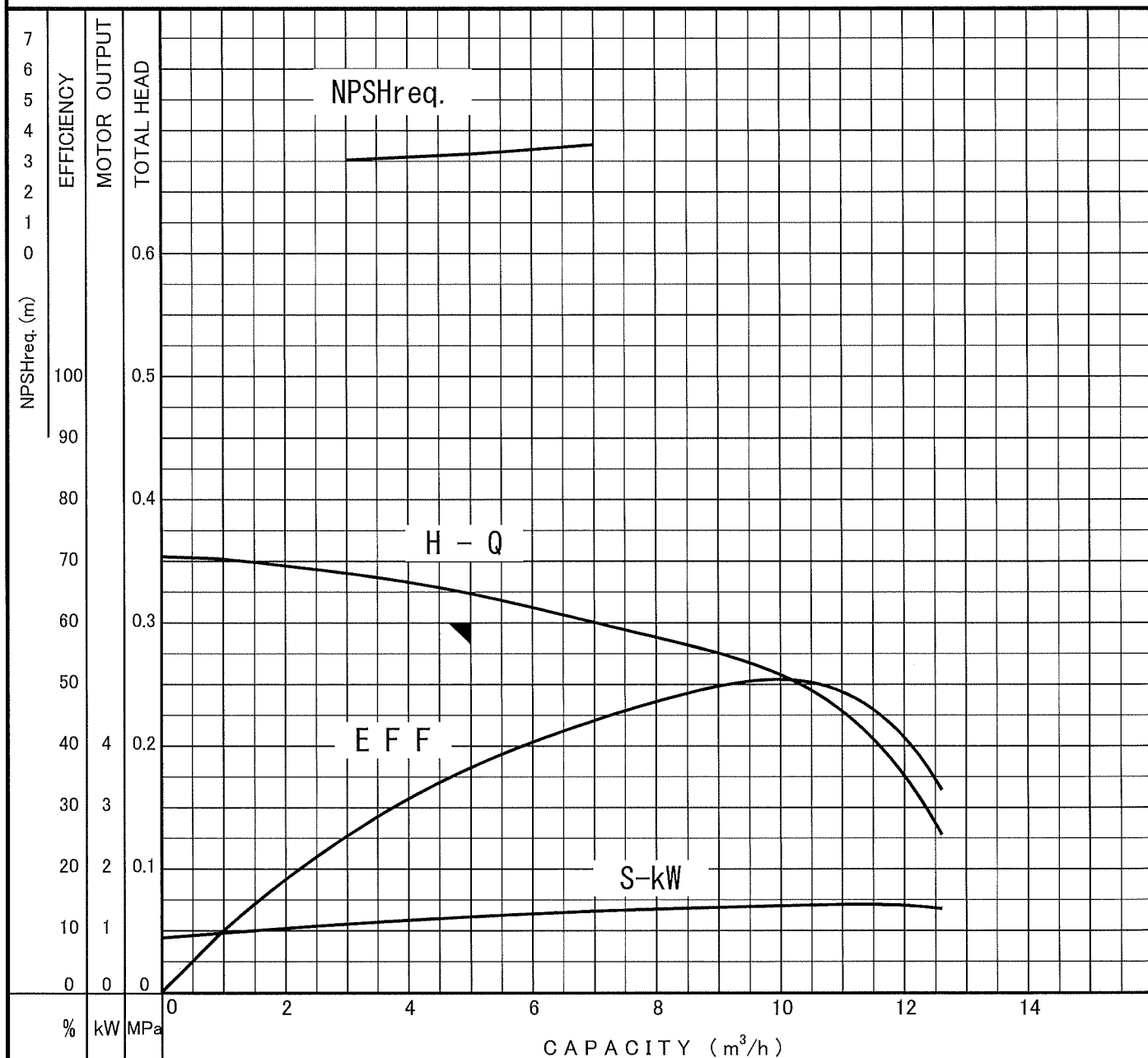




ESTIMATED PERFORMANCE CURVE

Page : 14
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	CHEMICAL CLEANING PUMP				
Pump Model	:	BHR-40				
Bore	:	40×40				
Capacity	:	5	m³/h	Total Press :	0.3	MPa
Output	:	2.2	kW	3500	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.6	MPa			
Remarks	:					

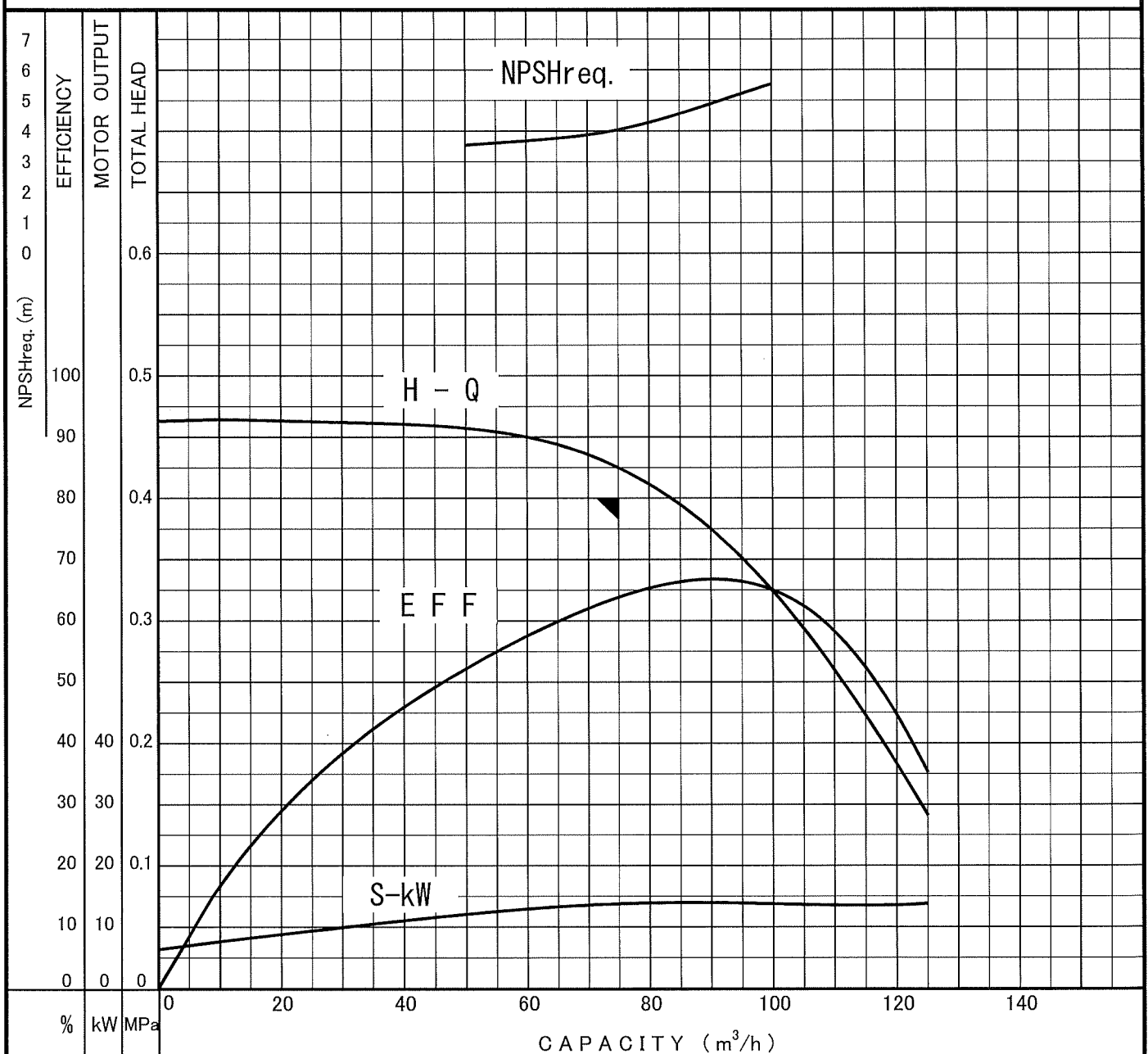




ESTIMATED PERFORMANCE CURVE

Page : 15
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	HOT GLYCOL WATER CIRCULATING PUMP				
Pump Model	:	BHR-100				
Bore	:	100×100				
Capacity	:	75	m³/h	Total Press :	0.4	MPa
Output	:	18.5	kW	3500	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.8	MPa			
Remarks	:					

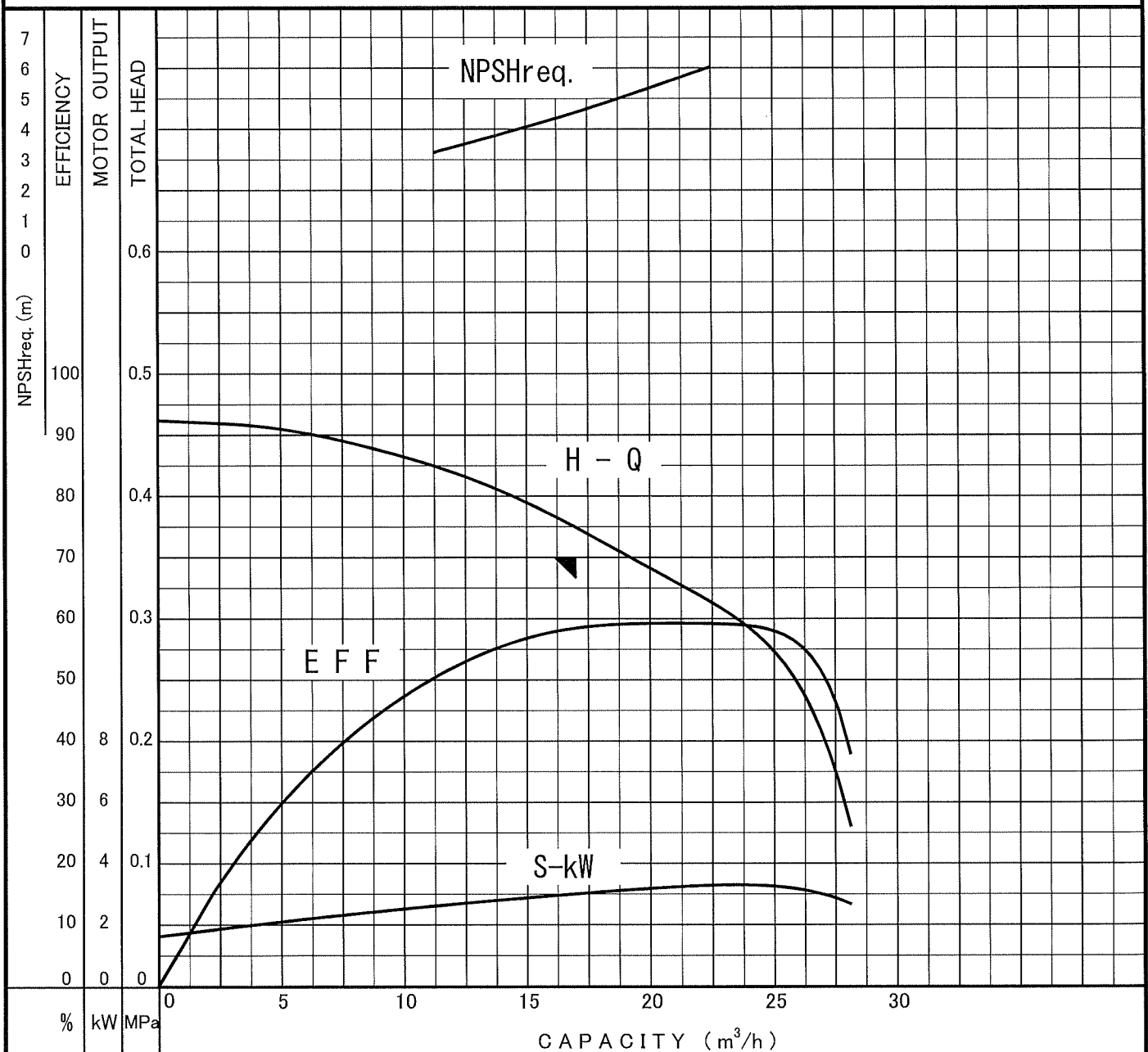




ESTIMATED PERFORMANCE CURVE

Page : 16
Date : 7 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	COLD GLYCOL WATER CIRCULATING PUMP				
Pump Model	:	BHR-50				
Bore	:	50×50				
Capacity	:	17	m³/h	Total Press :	0.35	MPa
Output	:	3.7	kW	3500	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.8	MPa			
Remarks	:					





ESTIMATED PERFORMANCE CURVE

Page : 17
Date : 7 Nov. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : C/H WATER SPRAY PUMP

Pump Model : FEWV-250-4D

Bore : 300×250

Capacity : 500 m³/h Total Press : 1.0 MPa

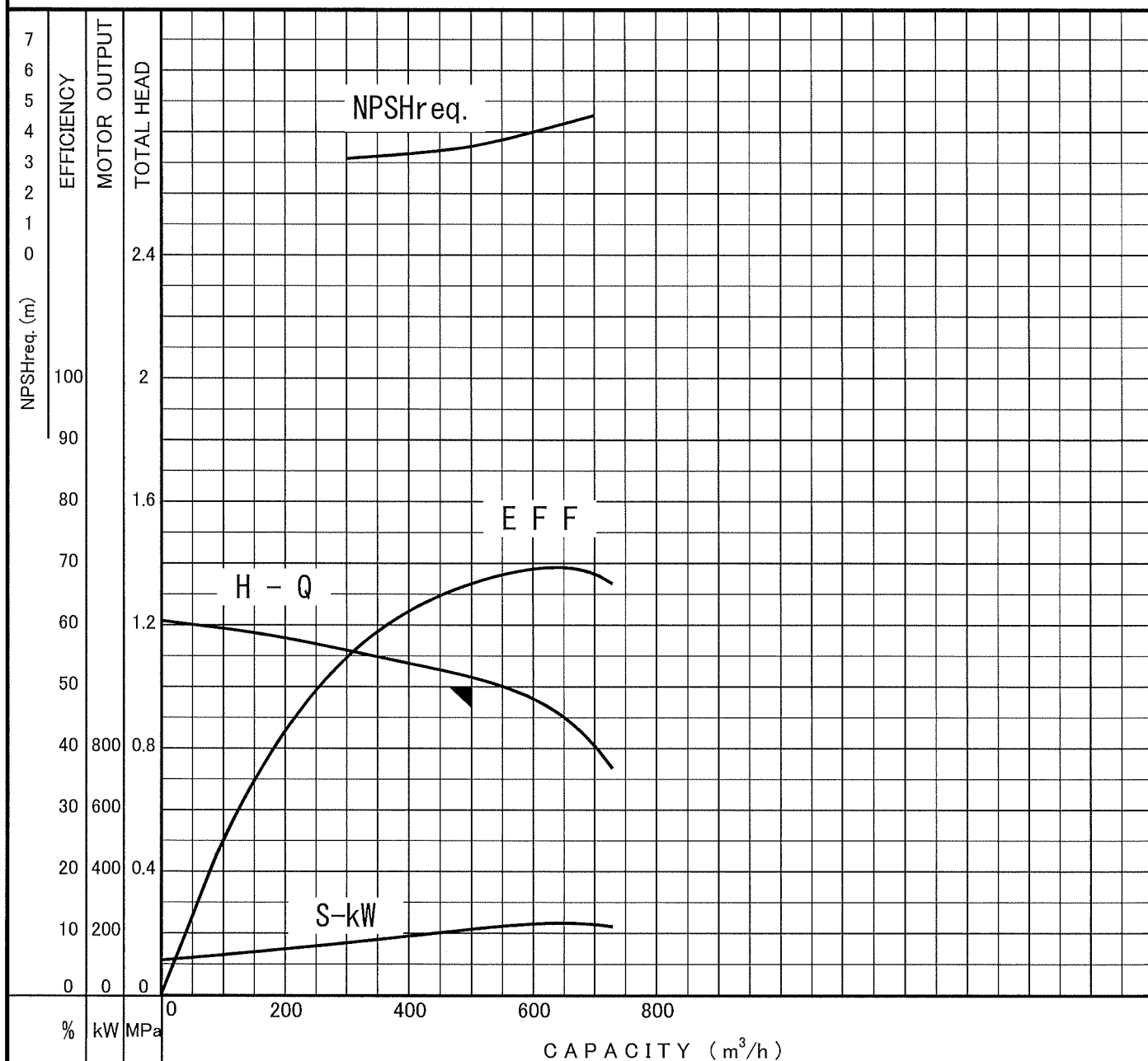
Output : 249 kW 1750 min-1

60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 1.9 MPa

Remarks :





ESTIMATED PERFORMANCE CURVE

Page : 18

Date : 7 Nov. 2024

Customer : Hudong Zhonghua Shipbuilding (Group) Co., Ltd.

Project : H1933A/1934A/1935A/1936A/1937A

Service As : ICER COOLING SEA WATER PUMP

Pump Model : FEWV-350-3

Bore : 350×350

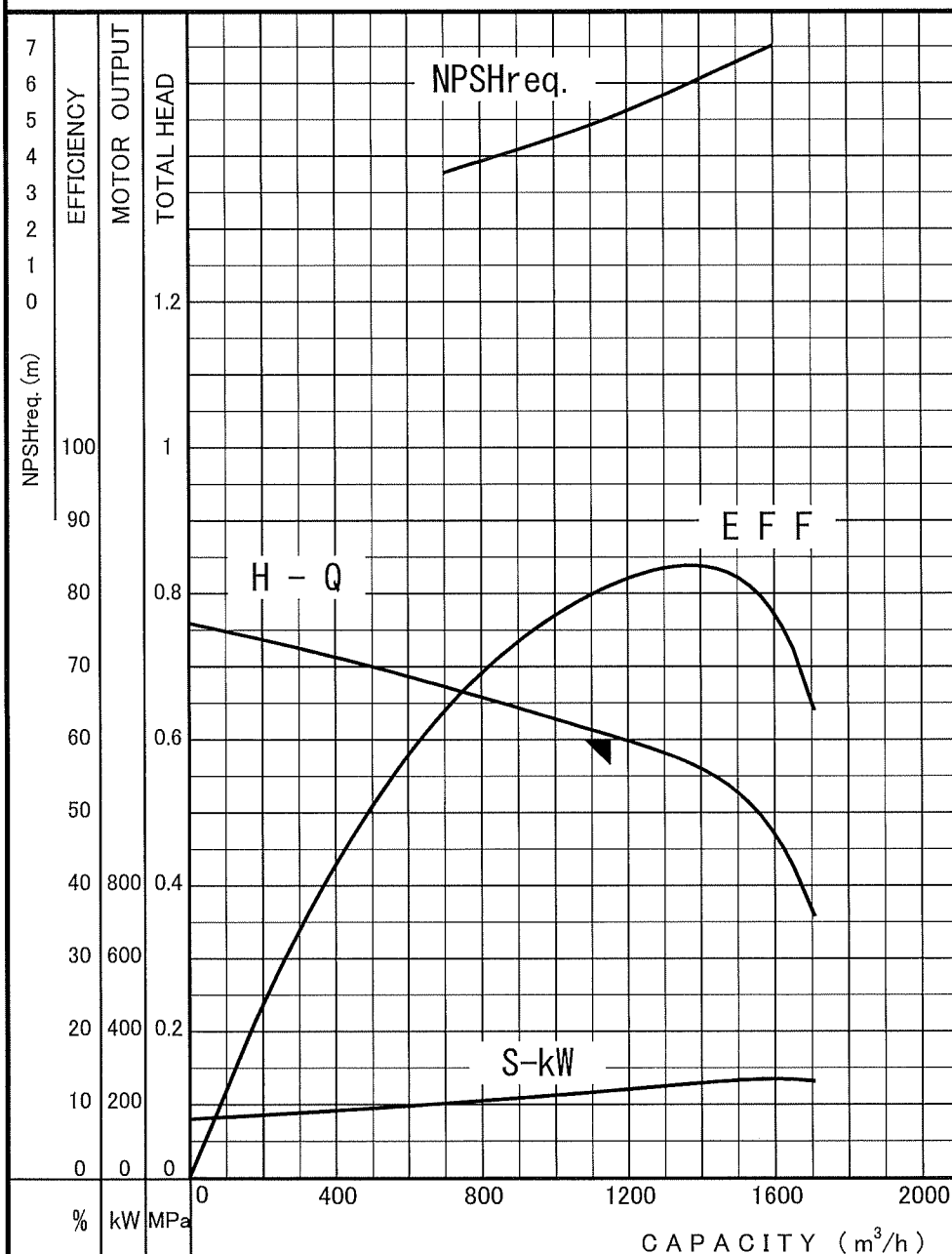
Capacity : 1150 m³/h Total Press : 0.6 MPa

Output : 299 kW 1750 min⁻¹
60 Hz 440 V

Test Liquid : FRESH WATER

H. Test Press. : 1.2 MPa

Remarks :





ESTIMATED PERFORMANCE CURVE

Page : 19
Date : 11 Nov. 2024

Customer	:	Hudong Zhonghua Shipbuilding (Group) Co., Ltd.				
Project	:	H1933A/1934A/1935A/1936A/1937A				
Service As	:	NaOH TRANSFER PUMP				
Pump Model	:	BHR-40				
Bore	:	40×40				
Capacity	:	1	m³/h	Total Press :	0.35	MPa
Output	:	2.2	kW	3500	min-1	
		60	Hz	440	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.6	MPa			
Remarks	:					

